Euler's Constant

Arithmetic, Monodromy, and Gamma-Related Structures

Complete integrated research collection

Compiled for KEIO

Eight manuscripts, with source provenance and original proof status

19 October 2026

Abstract

This single-source volume collects all six Euler/Gamma research manuscripts produced in the present conversation and the two newly supplied manuscripts. It preserves their mathematical statements, proofs, failed attempts, limitations, exact checks, appendices, and bibliographies. The separate 61-page Bessel–periods–tableaux–banana volume is excluded in full. The material covers asymptotic coefficient arithmetic, special-function inequalities, stochastic stopping boundaries, conservative matrix fields, moment determinants, differential and difference algebra, monodromy, primitive lattices, rational positivity certificates, logarithmic Meijer identities, complete-monotonicity degrees, and Dickman-related asymptotics. The editorial integration provides one pagination, one contents list, part-qualified cross-references, and a source-to-part concordance. It is a content-preserving compilation, not a new claim of arithmetic irrationality or transcendence. Source-level conjecture counts are not added together: several results overlap, some are restricted cases, and some are historical or classical consequences. No proof of the irrationality or transcendence of Euler's constant is claimed.

Edition. Original source versions dated 18–19 October 2026. Internal verification and scope notes are retained; external peer review and comprehensive priority certification are not asserted.

Editorial scope and source concordance

The phrase "this conversation" is interpreted as the entire conversation, not only the immediately preceding 29-page manuscript. The six generated Euler/Gamma manuscripts are distinct continuations, not interchangeable versions. In particular, *Arithmetic Transfers II* does not contain or replace *Arithmetic Transfers Around Euler's Constant*. Both are included, together with the scalar-convergence, monodromy, rational-certificate, and fractional-degree manuscripts.

The two attached TeX manuscripts are included in full as Parts IV and VII. The attached *Twelve further attempts* version is byte-identical to the earlier attachment used in the research; it is included once. The source PDFs total 236 pages. Re-typesetting and the shared front matter change the final page count without defining the scope of the contents.

The 61-page *Bessel Periods, Tableaux, and Banana* integrated volume is excluded as a whole. No chapter or appendix is extracted from that volume. The Bessel stopping results and inverse Bessel-polynomial recurrence already contained in the eight Euler/Gamma manuscripts remain included: they belong to the requested sources and are not chapters of the excluded banana volume.

Part	Source manuscript / topic	Source pages	This volume
I	Arithmetic transfers around Euler's constant	36	7–41
II	Arithmetic transfers II: units, parameter jets, and source assertions	35	42–75
III	Scalar convergence, moment problems, and differential algebra	37	76–110
IV	Twelve further irrationality attempts and their failure points	14	111–125
V	Monodromy, primitive lattices, and positive rational forms	30	126–153
VI	Rational certificates, ramification, and arithmetic obstructions	27	154–180
VII	A logarithmic Meijer identity and exact arithmetic structure	28	181–207
VIII	Fractional degrees and Dickman-related asymptotics	29	208–233

How to read the integration

Statements and proofs keep their source wording. References to "this paper", "this manuscript", or "earlier manuscripts" inside a part retain their source-local meaning. Internal labels and citations are qualified by part; for example, a reference prefixed VI belongs to Part VI. Bibliographies remain with their respective parts so that the original citation scope is preserved. Abstracts and preparation credits are retained; duplicate title pages and repeated local contents lists are replaced by this shared front matter. Reproducibility code is embedded directly in this one TeX file.

Some arguments recur in later parts in a stronger form or under different hypotheses. They are retained because dropping them would also drop their local proof context. They are not counted as independent new resolutions. The limits and corrections in later parts must be read together with the earlier statements; combining manuscripts does not upgrade a conditional, restricted, or unsuccessful argument into an unconditional arithmetic proof.

Version-sensitive correction note

Part VII's discussion of corrections to the supplied attempts was written against earlier formulations. The Part IV attachment included here already states the nonzero target-constant restriction, the change in a recurrence under a Gamma twist, and the qualification about modification of the final segment of a continuation path. Thus the historical correction discussion does

not assert that those particular qualifications are absent from the current Part IV version. Its original reasoning is retained for context.

The broader "discarded" verdict about arithmetic holonomy in Part IV is preserved as an assessment of that attempted route, not endorsed as a universal impossibility theorem. Proposition IV.4.2 has an identifiability hypothesis. The affine cancellation-locus analysis and the logical boundary explained in Section VI.4, together with Part VII's correction discussion, delimit its valid use. In particular, an obstruction to an unconditional cancellation formula does not by itself exclude a contradiction constructed under an algebraicity hypothesis.

Proof status and scope of this edition

This task is editorial integration. It does not add a new global theorem, rerun a comprehensive literature search, or certify every earlier claim anew. The proofs and source audits have the status stated in their own parts. In particular, source-level ledgers containing ten entries are not a claim of ten independent presently open major conjectures newly solved, either within that part or across this volume. The arithmetic nature of γ and of the Euler–Gompertz constant remains unresolved by these manuscripts.

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Part I
Arithmetic transfers around Euler's constant

Source: `KEIO_Euler_Arithmetic_Transfers_20261018.tex`.

Original PDF: 36 pages. Mathematical content and source-level qualifications are retained.

Abstract

We investigate transfers from the Euler–Gompertz and E-function setting to asymptotic arithmetic, orthogonal polynomials, special-function inequalities, and optimal stopping. For the coefficients d_k in the Brent–Glasser–Guttmann expansion, we obtain the exact reduced denominator $2^{6k-s_2(k)}$, where $s_2(k)$ is the binary digit sum, strengthening the recovered integrality result. We prove both all-order coefficient sign observations in that paper, using explicit analytic tail bounds and exact finite certificates. A radial-oscillator transfer also gives exact Gevrey order two and the Borel radius. We prove divergence of the factorially cleared Euler–Gompertz approximants discussed by Murabayashi–Yoshida and a uniform obstruction for all nonnegative integer-polynomial filters under that normalization. A polynomial–Gamma monotonicity question receives sharp answers with the coefficient assumptions made explicit. For rational dimension and reward, Bessel-bridge stopping boundaries are algebraic exactly on the lines $q - d \in 2\mathbb{Z}_{\geq 0}$; an exact parameter sensitivity recovers $\mathrm{Ein}(1) = \gamma + \delta/e$. Every finite dimension jet of the quadratic optimal value at dimension two is algebraically independent. Further results concern gamma-cancelling linear forms and the limitations of moment positivity and Gevrey zero removal. The manuscript distinguishes recovered results, extensions, and unproved targets. It does not prove irrationality or transcendence of γ and does not establish ten new resolutions of published open problems.

Keywords: Euler–Mascheroni constant; Euler–Gompertz constant; dyadic denominator; Kummer function; optimal stopping; positive polynomial; arithmetic Gevrey series.

I.1 Scope, source ledger, and the arithmetic question

We use

$$\gamma = \lim_{n\to\infty}(H_n - \log n), \qquad \delta = \int_0^\infty \frac{e^{-t}}{1+t}\,dt, \qquad \eta = \mathrm{Ein}(1) = \gamma + \delta/e.$$

The intended research target was ten genuinely new solutions to previously published open questions. That target is *not met by this manuscript*. The number of theorem statements below must not be mistaken for that target count. Parameter instances, a stronger form of an already recovered result, and a new problem formulated during the investigation are not independent published conjectures.

The published statements examined are listed below. "Proved here" refers to the argument supplied in this manuscript, not to completed external peer review or established historical priority. The older project records [I.19, I.20] already contain BGG integrality and its mod-32 refinement; both are identified as recovered, not first discoveries of this research round. An unnumbered numerical observation is kept separate from a numbered conjecture.

ID	Published target and location	Outcome in this manuscript
A1	BGG [I.1], Conjecture 10: $64^k d_k \in \mathbb{Z}$.	Recovered proof; strengthened to an exact reduced-denominator formula.
A2	BGG, Remark 11: $64^k d_k \equiv \binom{2k}{k} \pmod{32}$ for $k \geq 3$.	Recovered all-order proof; not a new discovery in this round.
A3	BGG, Remark 11: $d_k < 0$ for every $k \geq 3$.	Proved for all indices by an analytic tail bound and an exact finite certificate.

ID	Published target and location	Outcome in this manuscript
A4	BGG, Remark 8: $c_m < 0$ except $m = 0, 4$.	Proved for all indices by a second dominance argument and an exact finite certificate.
A5	Murabayashi–Yoshida [I.4], before Theorem 7.1: factorially cleared error tends to infinity.	Proved with explicit bounds. An unnumbered prediction, not Conjecture 7.5.
A6	Same paper, Conjecture 7.5: monotone fractional-part subsequences.	Not proved; the needed direction of monotonicity is not supplied by the monotone-subsequence theorem.
A7	Bulboacă–Zayed [I.5], Remark 3(2): polynomial-numerator shifts.	Fixed-polynomial answer and sharp nonnegative-coefficient answer; unrestricted degree-only version is false. One source question.
A8	Same remark: sharp constant-numerator shift.	Reproved; a prior public claim exists [I.18], so excluded from new resolutions.
A9	Lagarias [I.11], Conjecture 1.0.1: $\gamma \notin \mathbb{Q}$.	Not proved.
A10	Lagarias, Conjecture 1.0.2: γ is not a Kontsevich–Zagier period.	Not proved, nor is its transcendence consequence.
A11	Fischler–Rivoal [I.12], Conjecture 1: $\mathcal{E} \cap \mathcal{D} = \overline{\mathbb{Q}}$.	Not proved.
A12	Same paper, Conjecture 2: pure Gevrey zero removal.	Not proved.
A13	Same paper, Conjecture 3: mixed Gevrey zero removal.	Not proved; an explicit restriction is equivalent to the arithmetic target.
A14	Diamond–Ford [I.15]: lower edge of the generalized Euler spectrum.	Not proved; the stated equality is equivalent to RH.
A15	Ford–Luca–Moree [I.16], Conjecture 1: cyclotomic Euler–Kronecker limit points.	Not proved.
A16	Dixit–Murty [I.17], introduction: infinitely many negative cyclotomic Euler–Kronecker constants.	Not proved unconditionally here.

Four source-level targets receive proofs in this round: the two BGG sign observations (A3–A4), the unnumbered divergence observation (A5), and the polynomial shift question (A7), with its interpretations distinguished. This is neither ten new conjecture resolutions nor a claim of verified historical priority. The bounded literature search found no earlier proof of the two sign assertions; that search is not an exhaustive priority audit.

The polynomial question A7 lacks a coefficient-sign specification in its original remark. Adding nonnegativity silently would change the problem; the three interpretations are therefore treated separately. A5 is a modest analytic observation whose proof clarifies the denominator obstruction. Neither should be advertised as an arithmetic breakthrough for γ.

Two useful boundaries organize the investigation. A result about the *function* Γ is not automatically a result about the arithmetic of its logarithmic derivative at one. Likewise, an algebraic-independence theorem for finite E-function values does not automatically pass to the singular limits in which γ appears. The following sections make these distinctions mathematically explicit while retaining the transfers that do work.

I.2 Exact arithmetic of the BGG product coefficients

Let $M(a, b, x) = {}_1F_1(a; b; x)$ and let $U(a, b, x)$ be Tricomi's confluent hypergeometric function. Brent–Glasser–Guttmann [I.1] define Taylor coefficients a_n, b_n by

$$e^{z/(1-z)} = \sum_{n\geq 0} a_n z^n, \qquad e^{1/(1-z)} E_1\left(\frac{1}{1-z}\right) = \sum_{n\geq 0} b_n z^n.$$

For $n \geq 1$ their exact formulas are

$$a_n = e^{-1}M(n+1,2,1), \quad b_n = -\Gamma(n)U(n,0,1), \quad \rho_n = a_n b_n.$$

Their product has the asymptotic expansion

$$-2n^{3/2}\rho_n \sim \sum_{k\geq 0} d_k n^{-k}, \qquad d_0 = 1.$$

The integrality of $r_k = 64^k d_k$ is their Conjecture 10. Remark 11 records both a mod-32 pattern and the negative signs from $k = 3$ onward. We first recover the integrality mechanism and sharpen it to an exact denominator formula. The sign requires the independent analytic argument of Section I.3.

I.2.1 An exact differential identity and its normalization

For $x > 0$ and $n > 0$, set

$$u = e^{-x/2}xM(n+1,2,x), \qquad v = e^{-x/2}U(n,0,x),$$
$$H_n = \Gamma(n+1)uv, \qquad F(n,x) = 2\sqrt{n/x}\, H_n(x).$$

Both u, v satisfy $y'' = (n/x + 1/4)y$. At zero, $u \sim x$ and $v \to 1/\Gamma(n+1)$, so their constant Wronskian is $-1/\Gamma(n+1)$. The logarithmic term allowed in v' contributes $O(x\log x)$ to this limit and vanishes. Direct multiplication gives

$$(H_n')^2 - 2H_nH_n'' + 4\left(\frac{n}{x} + \frac{1}{4}\right)H_n^2 = 1.$$

After inserting the definition of F, this becomes

$$F^2 + \frac{1}{4n}\left[\left(x + \frac{3}{4x}\right)F^2 - FF' + x((F')^2 - 2FF'')\right] = 1. \tag{I.2.1}$$

At $x = 1$, $H_n(1) = -n\rho_n$ and $F(n,1) = -2n^{3/2}\rho_n$.

The heat-kernel representation proved in Section I.3 supplies $F(n,x) \sim \sum_{k\geq 0} d_k(x)n^{-k}$, $d_0(x) = 1$, locally uniformly for $x > 0$, together with any fixed number of x-derivatives. Indeed the Bessel expansion in the explicit kernel can be truncated uniformly on compact positive x-intervals and differentiated a fixed number of times; Watson's lemma then applies to the differentiated Laplace integrals. This justifies coefficient comparison in (I.2.1); no differentiation of a merely pointwise asymptotic assertion is being assumed. In particular $d_k(1) = d_k$. Define the formal series

$$R(t,x) = \sum_{k\geq 0} r_k(x)t^k, \qquad r_k(x) = 64^k d_k(x).$$

Substitution of $n^{-1} = 64t$ into (I.2.1) gives

$$R^2 = 1 + 32txRR_{xx} + 16tRR_x - 16txR_x^2 - (16x + 12x^{-1})tR^2. \tag{I.2.2}$$

Its constant term is $R(0,x) = 1$, and its coefficient of degree $k \geq 1$ determines $2r_k$ from preceding coefficients. Thus it has a unique formal solution over $\mathbb{Q}[x, x^{-1}]$.

I.2.2 Exact dyadic denominators and exclusion of integral dyadic roots

Write

$$R(x,t)=\sum_{n\ge 0} r_n(x)t^n, \qquad r_n(x)=64^n d_n(x), \qquad r_0(x)=1.$$

Throughout this subsection, a prime denotes differentiation with respect to x. We use the scaled identity (I.2.2) as an identity of formal power series; no convergence of the asymptotic expansion is required. For a nonnegative integer n, let $s_2(n)$ denote the sum of its binary digits. The notation v_2 means the additive valuation normalized by $v_2(2)=1$, also on $\overline{\mathbb{Q}}_2$.

Theorem I.2.1 (Exact dyadic valuation)**.** *For every $n\ge 0$, define*

$$A_n(x)=\frac{n!}{2^n}\,r_n(x).$$

Then

$$A_n(x)\in\mathbb{Z}[x,x^{-1}], \qquad A_n(x)\equiv x^{-n} \pmod{2\mathbb{Z}[x,x^{-1}]}. \tag{I.2.3}$$

More precisely,

$$P_n(x):=x^nA_n(x)\in\mathbb{Z}[x^2], \qquad P_n(x)\equiv 1 \pmod{2\mathbb{Z}[x^2]}. \tag{I.2.4}$$

In particular, for every $n\ge 1$,

$$v_2\big(r_n(1)\big)=s_2(n), \qquad \operatorname{den}\big(d_n(1)\big)=2^{6n-s_2(n)}, \tag{I.2.5}$$

where den *denotes the positive denominator in lowest terms.*

Proof. Put $V(x)=16x+12/x$. Extracting the coefficient of t^n from (I.2.2) gives, for $n\ge 1$,

$$\begin{aligned}2r_n=&\sum_{i+j=n-1}\left(32xr_ir_j''+16r_ir_j'-16xr_i'r_j'-Vr_ir_j\right)\\&-\sum_{i=1}^{n-1}r_ir_{n-i}.\end{aligned} \tag{I.2.6}$$

This triangular recurrence first shows that

$$r_n\in 2\mathbb{Z}[x,x^{-1}] \qquad (n\ge 1). \tag{I.2.7}$$

Indeed, all coefficients in the first sum of (I.2.6) are divisible by 4; the second sum has the same divisibility if the preceding positive-index r_i are even. Dividing by 2 completes the induction, including the base case with an empty second sum. The same recurrence shows inductively that the exponents occurring in r_n belong to

$$\{-n,-n+2,\dots,n-2,n\}.$$

After substituting $r_i=2^iA_i/i!$ and multiplying (I.2.6) by $n!/2^{n+1}$, we obtain

$$\begin{aligned}A_n=&\,n\sum_{i+j=n-1}\binom{n-1}{i}\left[8xA_iA_j''+4A_iA_j'-4xA_i'A_j'-\left(4x+\frac{3}{x}\right)A_iA_j\right]\\&-\frac{1}{2}\sum_{i=1}^{n-1}\binom{n}{i}A_iA_{n-i}.\end{aligned} \tag{I.2.8}$$

The last half-sum has integer coefficients whenever the preceding A_i do. To see this, pair its terms with indices i and $n-i$. Each pair contributes $\binom{n}{i}A_iA_{n-i}$. If $n=2m$, the remaining middle term has integer coefficient

$$\frac{1}{2}\binom{2m}{m}=\binom{2m-1}{m-1}.$$

Since $A_0=1$, recurrence (I.2.8) therefore proves $A_n\in\mathbb{Z}[x,x^{-1}]$ for every n.

It remains to determine the parity, and it is essential here to reduce the half-sum only after the preceding pairing. The first positive-index term is

$$A_1(x)=-4x-\frac{3}{x}\equiv x^{-1}\pmod 2.$$

Suppose now that $n\geq 2$ and that $A_i\equiv x^{-i}\pmod 2$ for $i<n$. All derivative terms and the $4x$ term in the first sum of (I.2.8) vanish modulo 2. Its remaining contribution is

$$-\frac{3n}{x}\sum_{i+j=n-1}\binom{n-1}{i}x^{-i-j}=-3n\,2^{n-1}x^{-n}\equiv 0\pmod 2.$$

In the paired half-sum, each product A_iA_{n-i} may be replaced modulo 2 by x^{-n}. The sum of the resulting integer coefficients is

$$\frac{1}{2}\sum_{i=1}^{n-1}\binom{n}{i}=\frac{2^n-2}{2}=2^{n-1}-1,$$

which is odd. Thus $A_n\equiv x^{-n}\pmod 2$, proving (I.2.3). The exponent range already established implies (I.2.4).

In particular $A_n(1)$ is an odd integer. Since $r_n(1)=2^nA_n(1)/n!$, Legendre's formula gives

$$v_2(r_n(1))=n-v_2(n!)=s_2(n).$$

Finally, (I.2.7) ensures that $r_n(1)$ is an integer, so $d_n(1)=r_n(1)/2^{6n}$ has no odd denominator factor. Cancelling its exact power of 2 proves (I.2.5). □

Corollary I.2.2 (Dyadic nonvanishing)**.** *Let $n\geq 1$ and let $0\neq\alpha\in\overline{\mathbb{Q}}_2$ satisfy $v_2(\alpha)\geq 0$. Then*

$$v_2(d_n(\alpha))=s_2(n)-6n-nv_2(\alpha). \tag{I.2.9}$$

Consequently d_n has no nonzero algebraic root that is integral at a place above 2. Equivalently, every algebraic root of d_n, under every embedding into $\overline{\mathbb{Q}}_2$, has strictly negative 2-adic valuation.

Proof. By (I.2.4), write $P_n=1+2Q_n$ with $Q_n\in\mathbb{Z}[x^2]$. Integrality of α gives $v_2(Q_n(\alpha))\geq 0$, and therefore $v_2(P_n(\alpha))=0$. The identity

$$d_n(\alpha)=\frac{2^{-5n}}{n!}\,\alpha^{-n}P_n(\alpha)$$

now yields (I.2.9). Its right-hand side is finite, proving the asserted nonvanishing and the statement about algebraic roots. □

The theorem establishes the exact denominator and a family of nonvanishing statements. It does not determine the real sign of $d_n(1)$; in particular, the negativity assertion for $n\geq 3$ is proved by the separate argument in Section I.3.

I.2.3 Recovered congruence and its Laurent refinement

The following all-order congruence is recovered from the earlier project record [I.20]. Its proof is included to make the present arithmetic account self-contained. Congruences are coefficientwise in $\mathbb{Z}[x, x^{-1}][[t]]$. Since positive-index $r_k(x)$ are even, write $R = 1 + 2A$.

Theorem I.2.3 (BGG congruence, with its two initial exceptions)**.** *For the normalized BGG sequence $r_k = 64^k d_k$, one has*

$$r_k \equiv \binom{2k}{k} \pmod{32} \qquad (k \geq 3).$$

More precisely,

$$R(t, 1) \equiv (1 - 4t)^{-1/2} + 16t + 16t^2 \pmod{32}.$$

Proof. Substitute $R = 1 + 2A$ into (I.2.2), subtract 1, and divide the resulting integer identity by 4. Reduction modulo 2 gives

$$A + A^2 \equiv t/x \pmod{2}.$$

The unique solution with zero constant coefficient is

$$A \equiv \sum_{j \geq 0} (t/x)^{2^j} \pmod{2}.$$

Indeed, the square shifts this sum by one index, and uniqueness follows coefficientwise from the equation. Differentiation with respect to x is legitimate for Laurent series and yields

$$A_x \equiv t/x^2 \pmod{2},$$

since the derivative of every summand with $j \geq 1$ vanishes in characteristic two.

Because $R_x = 2A_x$ and $R_{xx} = 2A_{xx}$, reduction of (I.2.2) modulo 64 now gives

$$\big(1 + (16x + 12x^{-1})t\big)R^2 \equiv 1 + 32tA_x \equiv 1 + 32t^2x^{-2} \pmod{64}.$$

Specializing to $x = 1$, and inverting the denominator of constant coefficient 1, we obtain

$$R(t, 1)^2 \equiv \frac{1 + 32t^2}{1 + 28t} \pmod{64}. \tag{I.2.10}$$

Set

$$C(t) = (1 - 4t)^{-1/2} = \sum_{k \geq 0} \binom{2k}{k} t^k, \qquad S(t) = C(t) + 16t + 16t^2.$$

For $k \geq 1$, the central binomial coefficient is even, so $C \equiv 1$ modulo 2. Consequently

$$S^2 \equiv \frac{1}{1 - 4t} + 32t + 32t^2 \pmod{64}.$$

Multiplication by $1 + 28t$ gives $1 + 32t^2$ modulo 64, since

$$\frac{1 + 28t}{1 - 4t} = 1 + \frac{32t}{1 - 4t} \equiv 1 + 32t \pmod{64}.$$

Together with (I.2.10), this proves $R(t, 1)^2 - S(t)^2 \in 64\mathbb{Z}[[t]]$. Both $R(t, 1)$ and $S(t)$ are congruent to 1 modulo 2. Therefore

$$U(t) := \frac{R(t, 1) + S(t)}{2} \in \mathbb{Z}[[t]], \qquad U(0) = 1.$$

It is an integral unit. Factoring the difference of squares and cancelling this unit gives

$$(R(t, 1) - S(t))U(t) \in 32\mathbb{Z}[[t]], \qquad R(t, 1) - S(t) \in 32\mathbb{Z}[[t]].$$

Extracting coefficients proves the theorem. □

The two corrections are necessary: $r_1 = -14 \equiv 2 + 16$ and $r_2 = 86 \equiv 6 + 16$ modulo 32. This theorem proves the all-order congruence observed numerically in BGG Remark 11, using the normalization and integrality proved above. The sign observation is established separately in Section I.3.

Proposition I.2.4 (Laurent refinement)**.** *Put $u = t/x$ and $h = x + x^{-1}$. Then*

$$R(t,x) \equiv (1-4u)^{-1/2} - 8th(1-4u)^{-3/2} + 16u^2(1-4u)^{-1/2} \pmod{32}.$$

Proof. Denote the displayed right-hand side by S_x. Its coefficients are integral Laurent polynomials: those of $(1-4u)^{-3/2}$ are $(2k+1)\binom{2k}{k}$. Squaring modulo 64 yields

$$S_x^2 \equiv \frac{1+32u^2}{1-4u} - \frac{16th}{(1-4u)^2} \pmod{64}.$$

Multiply by $1 + (16x + 12x^{-1})t = (1-4u) + 16th$. The cross terms cancel modulo 64, leaving $1 + 32u^2$. Thus $S_x^2 \equiv R^2 \pmod{64}$. Both series are congruent to 1 modulo 2, so $(S_x + R)/2$ is an integral unit, and the same factorization argument proves $S_x \equiv R \pmod{32}$. □

I.3 The full sign assertion for the Gompertz asymptotic coefficients

Brent, Glasser and Guttmann [I.1], Remark 11, reported $d_k < 0$ for $3 \le k \le 1000$ by computation. The following argument establishes the infinite assertion. It combines their special-function product with the heat kernel of a radial oscillator. The finite part uses exact integer arithmetic, with a self-contained verification program below.

Theorem I.3.1. *For the coefficients defined by*

$$-2n^{3/2}\rho_n \sim \sum_{k\ge 0} d_k n^{-k}, \qquad \rho_n = -e^{-1}\Gamma(n)M(n+1,2,1)U(n,0,1),$$

one has $d_k < 0$ for every integer $k \ge 3$.

I.3.1 A heat-kernel coefficient identity

Let

$$\mathscr{H} = -\frac{d^2}{dr^2} + \frac{r^2}{16} + \frac{3}{4r^2}$$

be the positive self-adjoint radial-oscillator operator on $(0,\infty)$ with its regular realization at zero. Its heat kernel is

$$K(s;r,z) = \frac{\sqrt{rz}}{4\sinh(s/2)} \exp\left(-\frac{r^2+z^2}{8}\coth(s/2)\right) I_1\left(\frac{rz}{4\sinh(s/2)}\right). \tag{I.3.1}$$

This is the Hermite-type Laguerre kernel with parameter $\alpha = 1$, after the changes of variables $x = r/2$, $y = z/2$, and $t = s/4$; see [I.2], Section 2.2 and equation (1). The factor $1/2$ contributed by dilation of the Lebesgue integration variable is included in (I.3.1).

To connect this kernel to the coefficients above, put $x = r^2/4$ and

$$u(x) = e^{-x/2}xM(n+1,2,x), \qquad v(x) = e^{-x/2}U(n,0,x), \qquad H_n(x) = \Gamma(n+1)u(x)v(x).$$

Both u, v solve $y'' = (n/x + 1/4)y$, and $W_x(u,v) = -1/\Gamma(n+1)$. The functions $u(r^2/4)/\sqrt{r}$ and $v(r^2/4)/\sqrt{r}$ are respectively the regular and decaying solutions of $(\mathscr{H} + n)\psi = 0$. Their r-Wronskian is $-1/(2\Gamma(n+1))$. The Green-function formula therefore gives the diagonal resolvent

$$G_n(r,r) := (\mathscr{H}+n)^{-1}(r,r) = \frac{2H_n(x)}{r} = \frac{H_n(x)}{\sqrt{x}} = \int_0^\infty e^{-ns}K(s;r,r)\,ds. \tag{I.3.2}$$

At $x = 1$, $H_n(1) = -n\rho_n$. Thus

$$2\sqrt{n}\, G_n(2,2) = -2n^{3/2}\rho_n. \tag{I.3.3}$$

For s near zero, set

$$S(s) = \frac{\sinh(s/2)}{s/2}, \qquad B_l(s) = S(s)^{l-1/2}\exp(-\tanh(s/4)),$$

where the square root is the analytic branch equal to one at zero. The standard large-positive-argument expansion of I_1 [I.3, 10.40.1] is

$$I_1(z) \sim \frac{e^z}{\sqrt{2\pi z}} \sum_{l\geq 0} b_l z^{-l}, \qquad b_l = \frac{(-1/2)_l (3/2)_l}{2^l l!}.$$

Here $b_0 = 1$ and $b_l < 0$ for every $l \geq 1$. Substitution in (I.3.1) at $r = 2$ gives

$$K(s;2,2) \sim \frac{1}{\sqrt{4\pi s}}\, S(s)^{-1/2} e^{-\tanh(s/4)} \sum_{l\geq 0} b_l \sinh(s/2)^l. \tag{I.3.4}$$

Indeed $\coth(s/2) - 1/\sinh(s/2) = \tanh(s/4)$. Only terms with $l \leq k$ affect the coefficient of s^k. Applying Watson's lemma to (I.3.2) and using (I.3.3) yields the exact coefficient formula

$$d_k = (1/2)_k \left([s^k]B_0(s) - \sum_{l=1}^{k} a_l [s^{k-l}] B_l(s) \right), \qquad a_l = -\frac{(-1/2)_l (3/2)_l}{4^l l!} > 0. \tag{I.3.5}$$

The use of the divergent Bessel expansion here is purely asymptotic: for each fixed k it is truncated after finitely many terms, with a remainder of the requisite order as $s \downarrow 0$. The heat-kernel tail away from zero is exponentially small in n, so it does not change any coefficient. Thus no convergence of the infinite formal sum is asserted or needed.

I.3.2 Uniform dominance of the negative term

Fix an integer $k \geq 144$ and take $r = 3/\sqrt{k} \leq 1/4$ as a Cauchy radius in the complex s-plane. For $|s| \leq r$,

$$|S(s) - 1| \leq \frac{r^2}{20}. \tag{I.3.6}$$

For example, the first nonconstant term of its absolutely convergent series is $r^2/24$, and for $r \leq 1$ the ratio of each subsequent term to its predecessor is at most $1/80$. In particular the branch defining $S^{-1/2}$ is analytic on the disk. Writing $u = r^2/20 \leq 1/2$ and using $-\frac{1}{2}\log(1-u) \leq u$ gives

$$|S(s)^{l-1/2}| \leq \exp((k+1)r^2/20) \qquad (0 \leq l \leq k).$$

Also $|\tanh(s/4)| \leq r/3$ on this disk. To see this explicitly, put $z = s/4$, so $|z| \leq 1/16$. Then

$$|\sinh z| \leq |z|\cosh|z|, \quad \cosh|z| \leq e^{|z|} \leq \frac{16}{15}, \quad |\cosh z| \geq 2 - \cosh|z| \geq \frac{14}{15},$$

whence $|\tanh z| \leq (8/7)|z| < r/3$. Consequently

$$|B_l(s)| \leq \exp\left(\frac{(k+1)r^2}{20} + \frac{r}{3}\right) \leq \exp\left(\frac{9}{20} + \frac{1}{320} + \frac{1}{12}\right) < 2.$$

The last strict inequality follows, for instance, because the exponent is less than $2/3$, while $e < 11/4$ implies $e^2 < 8$. Cauchy's estimate now gives

$$|[s^m]B_l(s)| \leq 2r^{-m} \qquad (m \geq 0,\ 0 \leq l \leq k). \tag{I.3.7}$$

The coefficients a_l satisfy $a_1 = 3/16$ and

$$\frac{a_{j+1}}{a_j} = \frac{(j-1/2)(j+3/2)}{4(j+1)} \geq \frac{j-1/2}{4} \qquad (j \geq 1).$$

Concavity of the logarithm gives the midpoint bound

$$\sum_{j=1}^{k-1} \log(j-1/2) \geq \int_0^{k-1} \log t \, dt = (k-1)\log(k-1) - (k-1).$$

Since the terms $\log(j-1/2)$ are increasing, the mean of a terminal segment is at least the mean of the whole segment. Hence, for $1 \leq l < k$,

$$\frac{a_l}{a_k} \leq \left(\frac{4e}{k-1}\right)^{k-l}. \tag{I.3.8}$$

Put

$$b = \frac{4e}{(k-1)r} = \frac{4e\sqrt{k}}{3(k-1)}.$$

The last expression decreases for $k > 1$; using $e < 11/4$ shows that

$$b < \frac{11\sqrt{144}}{3(144-1)} = \frac{4}{13}.$$

By (I.3.7) and (I.3.8), all the terms with $1 \leq l < k$, divided by a_k, have total absolute value at most

$$2\sum_{m=1}^{k-1} b^m < \frac{2b}{1-b} \leq \frac{8}{9}. \tag{I.3.9}$$

The term with $l = 0$ is bounded separately. Since $a_1 = 3/16$,

$$\begin{aligned}\frac{|[s^k]B_0(s)|}{a_k} &\leq \frac{32}{3r}b^{k-1} = \frac{32\sqrt{k}}{9}b^{k-1} \\ &< \frac{32\sqrt{k}}{9}\left(\frac{1}{3}\right)^{k-1} \leq \frac{128}{3}\,3^{-143} < \frac{1}{9}.\end{aligned}$$

The sequence $\sqrt{k}\,3^{-(k-1)}$ decreases for $k \geq 1$. Finally $B_k(0) = 1$, so the $l = k$ term in (I.3.5) is exactly $-a_k$. The remaining terms have combined absolute value strictly less than a_k by (I.3.9) and the last estimate. Therefore $d_k < 0$ for every $k \geq 144$.

I.3.3 Finite exact certificate

The remaining values $3 \leq k \leq 143$ can be verified using the scalar recurrence of [I.1], Lemma 14 and Corollary 15. The following program uses only integer arithmetic. Here $D_k = 64^k d_k$, and $T_{j,r} = 4^r(j+1/2)_r/r!$ is an integer. The four terms defining `beta` are exactly $64^{k-j}\alpha_{j,k}$ in that recurrence. The assertions verify divisibility at every step and all required strict signs. The first seven terms provide a normalization check.

```
K = 143
T = []
for j in range(K):
    row = [1]
    for r in range(1, K-j+3):
        v, rem = divmod(row[-1]*(4*j+4*r-2), r)
        assert rem == 0
        row.append(v)
    T.append(row)
```

```
p2 = [2**m for m in range(K+1)]
p3 = [3**m for m in range(K+1)]
D = [1]
for k in range(1, K+1):
    total = 0
    for j in range(k):
        m = k-j
        row = T[j]
        beta = ((-1+3*p2[m-1]-2*p3[m])*row[m-1]) << (4*m+2)
        beta += ((7-17*p2[m]+17*p3[m])*row[m]) << (4*m)
        beta += ((-13+38*p2[m]-33*p3[m])*row[m+1]) << (4*m-2)
        beta += (6*(1-4*p2[m]+3*p3[m])*row[m+2]) << (4*m-4)
        total += beta*D[j]
    val, rem = divmod(total, 8*k)
    assert rem == 0
    D.append(val)
assert D[:7] == [1, -14, 86, -3660, -1042202,
                 -247948260, -108448540420]
assert all(D[k] < 0 for k in range(3, K+1))
print("Exact finite certificate verified: 3 <= k <= 143")
```

This finite certificate, together with the analytic argument for all $k \geq 144$, proves Theorem I.3.1. The same scalar calculation was also run through $k = 1000$ as a consistency check; that larger range is not needed by the proof.

I.3.4 Uniform eventual sign and late-coefficient growth

The same argument also gives information not limited to evaluation at $x = 1$. Define $d_k(x)$ by

$$2\sqrt{n/x}\, H_n(x) \sim \sum_{k\geq 0} d_k(x) n^{-k} \qquad (x > 0).$$

The kernel derivation gives the analogue of (I.3.5) with

$$B_{l,x}(s) = S(s)^{l-1/2} e^{-x\tanh(s/4)}, \qquad a_l \text{ replaced by } a_l x^{-l}.$$

Corollary I.3.2. *For every $x > 0$ and every integer $k \geq 144\max(1, x^2)$, one has $d_k(x) < 0$. For each fixed $x > 0$, as $k \to \infty$,*

$$\begin{aligned} d_k(x) &= \frac{(-1/2)_k (1/2)_k (3/2)_k}{4^k k! x^k}\bigl(1 + O_x(k^{-1})\bigr) \\ &= -\frac{(k!)^2}{\pi^{3/2}(4x)^k k^{3/2}}\bigl(1 + O_x(k^{-1})\bigr). \end{aligned} \tag{I.3.10}$$

Proof. Keep $r = 3/\sqrt{k}$. Under the stated restriction, $xr/3 = x/\sqrt{k} \leq 1/12$, so the same bound $|B_{l,x}| < 2$ holds on the Cauchy circle. The geometric ratio is now

$$b_x = \frac{4ex\sqrt{k}}{3(k-1)} < \frac{11}{3}\frac{x}{\sqrt{k}}\frac{k}{k-1} \leq \frac{4}{13}.$$

The terms with $1 \leq l < k$ have total absolute value, relative to the dominant $-a_k x^{-k}$, less than $8/9$. The $l = 0$ term is bounded by

$$\frac{32x\sqrt{k}}{9}\, b_x^{k-1} \leq \frac{8k}{27}\, 3^{-(k-1)} < \frac{1}{9}.$$

This proves the sign assertion.

For fixed x, $b_x = O_x(k^{-1/2})$. Separate the $l = k-1$ term and use $[s]B_{k-1,x}(s) = -x/4$. Its relative absolute value is

$$\frac{x^2}{4}\frac{a_{k-1}}{a_k} = \frac{x^2 k}{(k-3/2)(k+1/2)} = O_x(k^{-1}).$$

The remaining terms with $1 \le l \le k-2$ have total relative absolute value at most $2b_x^2/(1-b_x) = O_x(k^{-1})$; the $l=0$ term is smaller still. This proves the first formula in (I.3.10). The second follows from the gamma-ratio form of Stirling's formula and

$$\Gamma(-1/2)\Gamma(1/2)\Gamma(3/2) = -\pi^{3/2}.$$

□

Corollary I.3.3. *For fixed $x > 0$, the formal series $\sum_{k\ge0} d_k(x)z^k$ has exact Gevrey order 2: there are constants $C, A > 0$ with $|d_k(x)| \le CA^k(k!)^2$, and no analogous bound with $(k!)^s$ for any $s < 2$. Its second-factorial Borel transform*

$$\mathcal{B}_x(z) = \sum_{k\ge0} \frac{d_k(x)}{(k!)^2} z^k$$

has radius of convergence exactly $4x$. The value $\mathcal{B}_x(4x)$ exists as an absolutely convergent series, but

$$\mathcal{B}_x'(z) \longrightarrow -\infty \quad \text{as } z \uparrow 4x.$$

In particular $z = 4x$ is a singular point of this transform.

Proof. All growth and radius assertions follow directly from (I.3.10). At $z = 4x$ the series terms are asymptotic to $-\pi^{-3/2}k^{-3/2}$ and hence are absolutely summable. The differentiated terms are asymptotic to $-(4x\pi^{3/2})^{-1}k^{-1/2}$. They are negative from some point onward, so monotone convergence of their absolute values gives the one-sided derivative limit. An analytic continuation across $4x$ would have a finite derivative there, which is impossible. □

I.4 The individual Kummer coefficients: resolving BGG Remark 8

Brent–Glasser–Guttmann (BGG) [I.1], Remark 8 of *A conjectured integer sequence arising from the exponential integral* (arXiv:1812.00316v4, p. 9), observed that the coefficients c_m in their Theorem 5 appear negative except for c_0 and c_4, and reported numerical verification through $m = 1000$. This section proves that sign statement. The coefficients here are exactly their c_m, not the Hadamard-product coefficients d_m.

Define the analytic function

$$\mu(z) = z^{-1} - (e^z - 1)^{-1} - \frac{1}{2}, \qquad A(z) = e^{\mu(z)} = \sum_{\ell\ge0} a_\ell z^\ell.$$

The apparent singularity at zero is removable. Both functions are analytic in $|z| < 2\pi$. Reindexing BGG Theorem 5, equation (12), gives the exact finite formula

$$(-1)^m c_m = \sum_{\ell=0}^{m} a_\ell T_{m,\ell}, \qquad T_{m,\ell} = \frac{(2\ell - m + 3/2)_{2m-2\ell}}{4^{m-\ell}(m-\ell)!}. \tag{I.4.1}$$

The subscript on the parenthesized expression denotes the rising factorial. In particular $a_0 = 1$.

Lemma I.4.1 (A uniform bound for the analytic multiplier)**.** *For every $\ell \ge 0$, $|a_\ell| < 2^{1-\ell}$.*

Proof. Using the Bernoulli expansion of μ and the usual alternating signs of the even Bernoulli numbers, on $|z| = 2$ one has

$$|\mu(z)| \le \sum_{j\ge1} \frac{|B_{2j}|2^{2j-1}}{(2j)!} = \frac{1}{2} - \frac{1}{2}\cot 1 < \frac{1}{2}.$$

The identity follows from the power series of $\cot(z/2)$, and the last inequality uses $0 < 1 < \pi/2$. Hence $|A(z)| < e^{1/2} < 2$ on this circle. Cauchy's coefficient estimate gives the result. □

Lemma I.4.2 (Uniform suppression away from the leading Bessel term)**.** *If $m \geq 10$ and $1 \leq \ell \leq m$, then*

$$\left|\frac{T_{m,\ell}}{T_{m,0}}\right| \leq \left(\frac{216}{m}\right)^{\ell}.$$

Proof. Cancellation of rising factorials, or the gamma-function expression for them, gives

$$\left|\frac{T_{m,\ell}}{T_{m,0}}\right| = \frac{4^{\ell}\, m!/(m-\ell)!}{|(3/2-m)_{2\ell}|}. \tag{I.4.2}$$

None of the denominator factors is zero. If $\ell \leq 3m/8$, all its 2ℓ factors are negative, and their absolute values are at least

$$m - 2\ell - \frac{1}{2} \geq \frac{m}{4} - \frac{1}{2} \geq \frac{m}{5}.$$

Since $m!/(m-\ell)! \leq m^{\ell}$, the ratio is at most $(100/m)^{\ell}$.

If $\ell > 3m/8$, the denominator consists of 2ℓ consecutive half-integers. Each positive absolute half-integer can occur at most twice, so

$$|(3/2-m)_{2\ell}| \geq \big((1/2)_{\ell}\big)^2.$$

Concavity of log and the midpoint rule on each unit interval imply

$$\log((1/2)_{\ell}) = \sum_{j=0}^{\ell-1} \log(j+1/2) \geq \int_0^{\ell} \log u\, du = \ell \log \ell - \ell.$$

Consequently the ratio in (I.4.2) is at most

$$\left(\frac{4me^2}{\ell^2}\right)^{\ell} < \left(\frac{256e^2}{9m}\right)^{\ell} < \left(\frac{216}{m}\right)^{\ell},$$

where $e < 11/4$ suffices for the final numerical bound. □

Theorem I.4.3 (All-order sign of the BGG coefficients c_m)**.** *One has*

$$c_0 = 1 > 0, \qquad c_4 = \frac{710401}{127401984} > 0, \qquad c_m < 0 \quad (m \geq 1,\ m \neq 4).$$

Proof. For $m \geq 325$, the preceding lemmas give

$$\left|\sum_{\ell=1}^{m} a_{\ell} \frac{T_{m,\ell}}{T_{m,0}}\right| < 2\sum_{\ell=1}^{\infty} \left(\frac{108}{m}\right)^{\ell} = \frac{216}{m-108} < 1.$$

Thus the sum in (I.4.1) has the same sign as $T_{m,0}$. Its numerator $(3/2-m)_{2m}$ has precisely $m-1$ negative factors for $m \geq 1$. Therefore $(-1)^m T_{m,0} < 0$, proving $c_m < 0$ for every $m \geq 325$.

The remaining finite range $1 \leq m \leq 324$ is checked in exact rational arithmetic by the certificate algorithm below. It gives only the exception $m = 4$; no floating-point comparisons are used. The six initial outputs are

$$1, -\frac{5}{48}, -\frac{479}{4608}, -\frac{15313}{3317760}, \frac{710401}{127401984}, -\frac{3532731539}{214035333120},$$

which agree with the published normalization. □

I.4.1 Reproducible finite certificate

The following Python program uses only the standard library. It implements the Bernoulli recurrence, the coefficient recurrence for e^{μ}, and the exact finite sum (I.4.1). To avoid repeatedly forming large rising factorials, it uses

$$\frac{T_{m+1,0}}{T_{m,0}} = \frac{4-(2m+1)^2}{16(m+1)}, \qquad \frac{T_{m,\ell+1}}{T_{m,\ell}} = \frac{16(m-\ell)}{(3-2m+4\ell)(5-2m+4\ell)}.$$

All divisors appearing here are nonzero. Execution verifies all required signs and produces a canonical text certificate with rows m, numerator, denominator. Its SHA-256 digest is

```
7be8eb1341cb2153157d2859c6719dc48
e1e6288a0daf5f76d7f62ac26dd36fc
```

where the digest is the concatenation of the two displayed lines.

```
from fractions import Fraction as Q
from math import factorial, comb
from hashlib import sha256

N = 324
B = [Q(1)]
for n in range(1, N + 2):
    B.append(-sum((comb(n+1,j)*B[j]
                   for j in range(n)), Q(0))/(n+1))
mu = [Q(0)] * (N+1)
for j in range(1, N+1, 2):
    mu[j] = -B[j+1] / factorial(j+1)
a = [Q(1)]
for n in range(1, N+1):
    a.append(sum((j*mu[j]*a[n-j]
                  for j in range(1,n+1,2)), Q(0))/n)
c = []
t0 = Q(1)
for m in range(N+1):
    if m:
        t0 *= Q(4-(2*m-1)**2, 16*m)
    term, value = t0, Q(0)
    for ell in range(m+1):
        value += a[ell] * term
        if ell < m:
            term *= Q(16*(m-ell),
                (3-2*m+4*ell)*(5-2*m+4*ell))
    c.append((-1)**m * value)
assert c[:6] == [Q(1), Q(-5,48), Q(-479,4608),
    Q(-15313,3317760), Q(710401,127401984),
    Q(-3532731539,214035333120)]
assert all(c[m] < 0 for m in range(1,N+1) if m != 4)
assert c[4] > 0
cert = '\n'.join(f'{m} {v.numerator} {v.denominator}'
                 for m,v in enumerate(c)) + '\n'
print(sha256(cert.encode()).hexdigest())
```

I.5 Euler–Gompertz approximants and a positive-filter obstruction

Put

$$\delta = \int_0^{\infty} \frac{e^{-t}}{1+t}\,dt.$$

In this section $p_m = P_m(1)$ and $q_m = Q_m(1)$ retain the normalization of Murabayashi and Yoshida [I.4]. To avoid confusion between an index and its even subsequence, write

$$\ell_j = q_{2j}, \qquad h_j = p_{2j} \qquad (j \ge 0).$$

The source's Conjecture 7.5 concerns certain infinite monotone subsequences of fractional-part expressions. We do not prove that conjecture. We do prove the separate, unnumbered numerical prediction stated immediately before its Theorem 7.1: the factorially cleared even linear forms diverge in absolute value. We then establish a uniform obstruction for a much larger class of positive polynomial filters.

I.5.1 Exact normalization of the remainder

Lemma I.5.1. *For every integer $j \geq 0$,*

$$R_j := \ell_j\delta - h_j = \int_0^\infty e^{-t}\frac{t^j}{(1+t)^{j+1}}\,dt > 0. \tag{I.5.1}$$

Moreover $j!\ell_j, j!h_j \in \mathbb{Z}$.

Proof. Eliminating the odd indices in the continuant recurrence, whose partial quotients at $x = 1$ are $c_{2j} = 1/j$ and $c_{2j+1} = 1$, gives

$$(j+1)u_{j+1} = (2j+2)u_j - ju_{j-1} \qquad (j \geq 1)$$

for both even subsequences, with

$$(\ell_0, \ell_1) = (1, 2), \qquad (h_0, h_1) = (0, 1).$$

Multiplication by $j!$ turns this into

$$U_{j+1} = (2j+2)U_j - j^2U_{j-1},$$

which proves the integrality assertion.

Denote the integral on the right of (I.5.1) temporarily by I_j. Its first two values are $I_0 = \delta$ and $I_1 = 2\delta - 1$. For $j \geq 1$ the expression $(j+1)I_{j+1} - (2j+2)I_j + jI_{j-1}$ equals

$$\int_0^\infty \frac{e^{-t}t^{j-1}}{(1+t)^{j+2}}(j - 2t - t^2)\,dt = \int_0^\infty \frac{d}{dt}\left(\frac{e^{-t}t^j}{(1+t)^{j+1}}\right)dt = 0.$$

The boundary terms vanish. Equality with $\ell_j\delta - h_j$ follows from the recurrence and initial values. Positivity follows from the integrand. □

Theorem I.5.2 (The published divergence observation)**.** *The even approximants in [I.4] satisfy*

$$|j!q_{2j}\delta - j!p_{2j}| \longrightarrow +\infty.$$

More quantitatively, for $j \geq 1$,

$$R_j \geq \frac{e^{-2\sqrt{j}-1}}{\sqrt{j}+2}, \tag{I.5.2}$$

and for $j \geq 4$,

$$R_j \leq \sqrt{j}\,e^{3-2\sqrt{j}}. \tag{I.5.3}$$

Consequently

$$\log(j!R_j) = j\log j - j - 2\sqrt{j} + O(\log j).$$

Proof. Restrict the integral in (I.5.1) to $[\sqrt{j}, \sqrt{j}+1]$. On this interval,

$$e^{-t} \geq e^{-\sqrt{j}-1}, \qquad \left(\frac{t}{1+t}\right)^j = e^{-j\log(1+1/t)} \geq e^{-\sqrt{j}}, \qquad \frac{1}{1+t} \geq \frac{1}{\sqrt{j}+2}.$$

This proves (I.5.2); multiplying by $j!$ already proves the claimed divergence.

For the upper bound, $\log(1+1/t) \geq 1/(t+1)$ for $t > 0$. Set $\varepsilon = j^{-1/2}$ and $a = 1-\varepsilon$. By the arithmetic–geometric mean inequality,

$$t + \frac{j}{t+1} = \varepsilon t + a(t+1) + \frac{j}{t+1} - a \geq \varepsilon t + 2\sqrt{aj} - a.$$

It follows that

$$R_j \leq e^{a-2\sqrt{aj}} \int_0^\infty \frac{e^{-\varepsilon t}}{1+t}\, dt \leq \sqrt{j}\, e^{a-2\sqrt{aj}} \leq \sqrt{j}\, e^{3-2\sqrt{j}}.$$

In the last step we used $a \leq 1$ and $\sqrt{aj} \geq \sqrt{j}-1$. The two estimates give $\log R_j = -2\sqrt{j} + O(\log j)$; Stirling's formula finishes the proof. □

This establishes a numerical prediction explicitly left unproved in the source. Its significance is the failure of the stated factorial normalization, not a resolution of the arithmetic nature of δ. It should not be reported as a proof of the source's numbered Conjecture 7.5.

I.5.2 All nonnegative integer-polynomial filters

The substitution $y = t/(1+t)$ transforms the remainder into a moment:

$$R_j = \int_0^1 y^j w(y)\, dy, \qquad w(y) = \frac{\exp(-y/(1-y))}{1-y} > 0. \tag{I.5.4}$$

A polynomial filter changes the moment index through a finite linear combination. The following theorem allows its degree and every coefficient to vary with the approximation index.

Theorem I.5.3 (Uniform positive-filter obstruction)**.** *Let $N \geq 0$ and let*

$$P(y) = \sum_{j=0}^{d} c_j y^j \in \mathbb{Z}[y] \setminus \{0\}, \qquad d \leq N,$$

be nonnegative on $[0,1]$. Define

$$A_N = N! \sum_{j=0}^{d} c_j \ell_j, \qquad B_N = N! \sum_{j=0}^{d} c_j h_j.$$

Then $A_N, B_N \in \mathbb{Z}$ and

$$A_N\delta - B_N = N! \int_0^1 P(y) w(y)\, dy \geq \frac{N!}{2e(2N+1)8^N}. \tag{I.5.5}$$

In particular, these positive integer linear forms tend to $+\infty$ uniformly over all the permitted polynomial filters as $N \to \infty$.

Proof. The integrality follows from Lemma I.5.1, and the integral identity follows from (I.5.4).

Write $a_d = c_d$ for the leading coefficient; then $|a_d| \geq 1$. Let $\mathcal{L}_d$ denote the Legendre polynomial normalized by $\mathcal{L}_d(1) = 1$. On $I = [0, 1/2]$, the shifted polynomial $\mathcal{L}_d(4y-1)$ satisfies

$$|\mathcal{L}_d(4y-1)| \leq 1, \qquad \int_I \mathcal{L}_d(4y-1)^2\, dy = \frac{1}{2(2d+1)}.$$

It is orthogonal on I to all polynomials of degree less than d. Since its leading coefficient is $2^d\binom{2d}{d}$, subtracting its appropriate multiple from P gives

$$\int_I P(y)\mathcal{L}_d(4y-1)\, dy = \frac{a_d}{2^{d+1}(2d+1)\binom{2d}{d}}.$$

Nonnegativity of P and the bound on $\mathcal{L}_d$ imply

$$\int_I P(y)\,dy \geq \left|\int_I P(y)\mathcal{L}_d(4y-1)\,dy\right| \geq \frac{1}{2^{d+1}(2d+1)\binom{2d}{d}} \geq \frac{1}{2(2d+1)8^d}.$$

The weight satisfies $w(y) \geq e^{-1}$ on I. The contribution from $[1/2, 1)$ is nonnegative, so

$$\int_0^1 P(y)w(y)\,dy \geq \frac{1}{2e(2d+1)8^d} \geq \frac{1}{2e(2N+1)8^N}.$$

Multiplication by $N!$ proves (I.5.5). Its lower bound tends to infinity by Stirling's formula. □

Corollary I.5.4 (Unbounded orders of finite differences)**.** *For all integers $n, m \geq 0$,*

$$\sum_{j=0}^{m}(-1)^j\binom{m}{j}R_{n+j} = \int_0^\infty e^{-t}\frac{t^n}{(1+t)^{n+m+1}}\,dt > 0.$$

For every path of pairs (n, m) with $n + m \to \infty$,

$$(n+m)!\sum_{j=0}^{m}(-1)^j\binom{m}{j}R_{n+j} \longrightarrow +\infty.$$

Proof. Apply Theorem I.5.3 to $P(y) = y^n(1-y)^m$, of degree $N = n+m$. Alternatively, integrating only over $t \in [1, 2]$ gives the direct lower bound

$$(n+m)!\int_0^\infty e^{-t}\frac{t^n}{(1+t)^{n+m+1}}\,dt \geq \frac{e^{-2}(n+m)!}{3^{n+m+1}} \longrightarrow +\infty.$$

□

Precise scope. Theorem I.5.3 concerns the specified factorial normalization. It does not exclude exceptional savings from common divisors of (A_N, B_N), smaller denominator-clearing factors, sign-changing polynomial filters, or other approximation families. Indeed, if G_N is a positive common divisor and $(A_N\delta - B_N)/G_N \to 0$, then (I.5.5) forces the necessary condition

$$\frac{G_N}{N!/[2e(2N+1)8^N]} \longrightarrow +\infty.$$

This last condition is necessary, not sufficient. The theorem is a newly formulated structural extension in this work, not an additional previously published conjecture counted as solved.

I.6 Polynomial–Gamma thresholds: a sharp answer with explicit scope

Bulboacă and Zayed [I.5], Remark 3(2), ask for shifts ensuring decrease of a polynomial divided by $\Gamma(s+\rho)\Gamma(s)$ on $[1, \infty)$. The remark does not specify signs of the polynomial coefficients. We separate a fixed positive polynomial, a uniform nonnegative-coefficient class, and an unrestricted uniform interpretation. These are three readings of one published question. The following results do not establish irrationality or transcendence of γ.

Write $\psi = \Gamma'/\Gamma$ and let $\chi : \mathbb{R} \to (0, \infty)$ be the inverse of ψ. This inverse exists because

$$\psi'(x) = \sum_{k=0}^{\infty}\frac{1}{(x+k)^2} > 0, \qquad \psi(0+) = -\infty, \qquad \psi(\infty) = \infty.$$

For a real polynomial P positive on $[1, \infty)$ and $\rho > -1$, put

$$Q_{P,\rho}(s) = \frac{P(s)}{\Gamma(s+\rho)\Gamma(s)}.$$

Theorem I.6.1 (Exact threshold for a fixed positive polynomial)**.** *The quantity*

$$B_P = \max_{s\geq 1}\left\{\chi\Big(\frac{P'(s)}{P(s)} - \psi(s)\Big) - s\right\}$$

is finite, the maximum is attained, and $B_P > -1$. The function $Q_{P,\rho}$ is strictly decreasing on $[1,\infty)$ if and only if $\rho \geq B_P$.

Proof. The expression being maximized is continuous. If $d = \deg P$, then

$$\frac{P'(s)}{P(s)} = \frac{d}{s} + O(s^{-2}), \qquad \psi(s) = \log s + O(s^{-1}).$$

Its argument inside χ tends to $-\infty$, so χ tends to zero. The expression therefore equals $-s + o(1)$ at infinity. A maximum exists, and its value is at least $\chi(P'(1)/P(1) + \gamma) - 1 > -1$.

The logarithmic derivative is

$$L_{P,\rho}(s) := \frac{Q'_{P,\rho}(s)}{Q_{P,\rho}(s)} = \frac{P'(s)}{P(s)} - \psi(s+\rho) - \psi(s).$$

By strict increase of ψ, $L_{P,\rho}(s) \leq 0$ for every $s \geq 1$ is equivalent to $\rho \geq B_P$. If $\rho < B_P$, the derivative is positive at some point, or by continuity immediately to the right of 1, and decrease is impossible.

Suppose $\rho \geq B_P$. The function $L_{P,\rho}$ is nonpositive and real analytic on $(1,\infty)$. Its behavior $L_{P,\rho}(s) = -2\log s + o(\log s)$ prevents it from vanishing identically. The analytic identity theorem implies that it cannot vanish on an interval. Its integral over every nondegenerate interval is strictly negative. Hence $Q_{P,\rho}$ is strictly decreasing, also at the critical shift. □

This is an attained variational characterization, rather than an elementary closed formula for an arbitrary polynomial. For nonnegative coefficients a uniform closed formula is available.

Theorem I.6.2 (Sharp shift uniform over nonnegative coefficients)**.** *Let*

$$\mathcal{P}_m = \left\{ s^m + \sum_{j=1}^{m-1} a_j s^j + 1 : a_j \geq 0 \right\}, \qquad c_m = \begin{cases} \frac{1}{2}, & m = 1, \\ m-1, & m \geq 2. \end{cases}$$

The smallest shift $\rho > -1$ for which $Q_{P,\rho}$ is strictly decreasing on $[1,\infty)$ for every $P \in \mathcal{P}_m$ is

$$R_m = \chi(\gamma + c_m) - 1.$$

Proof. For $m = 1$, $P'/P = 1/(s+1) \leq 1/2$, and the increasing function $\psi(s+\rho) + \psi(s)$ has left endpoint value $\psi(1+\rho) - \gamma$. This proves both sufficiency and necessity.

For $m \geq 2$, put $A_m(s) = (m-1)(s^m+1) - ms^{m-1}$. Then

$$A_m(1) = m - 2 \geq 0, \qquad A'_m(s) = m(m-1)s^{m-2}(s-1) \geq 0 \quad (s \geq 1).$$

Every internal monomial contributes

$$a_j\big((m-1)s^j - js^{j-1}\big) = a_j s^{j-1}\big((m-1)s - j\big) \geq 0.$$

Therefore $P'/P \leq m-1$. If $\rho \geq R_m$, then for $s > 1$,

$$\psi(s+\rho) + \psi(s) > \psi(1+\rho) - \gamma \geq m - 1 \geq \frac{P'(s)}{P(s)},$$

which proves strict decrease.

For $m = 2$, every $P \in \mathcal{P}_2$ satisfies $P'(1)/P(1) = 1$. Thus $\rho < R_2$ gives a positive derivative at 1. For $m \geq 3$, choose $P_A(s) = s^m + As^{m-1} + 1$. Then

$$\frac{P_A'(1)}{P_A(1)} = \frac{m + (m-1)A}{A+2} \longrightarrow m - 1.$$

If $\rho < R_m$, a sufficiently large finite A makes this ratio exceed $\psi(1+\rho) - \gamma$, again giving a positive derivative. This proves sharpness. □

Proposition I.6.3 (No uniform threshold without coefficient restrictions)**.** *For each $m \geq 2$ and every fixed $\rho > -1$, some monic degree-m real polynomial with constant coefficient 1, positive on $[1, \infty)$, makes $Q_{P,\rho}$ fail to decrease. Thus no finite threshold depending only on m exists for this unrestricted class.*

Proof. For $m = 2$, take $P_t(s) = (s-1)^2 + t^2 s$ with $t > 0$. At $s = 1 + t$,

$$P_t(1+t) = t^2(2+t), \quad P_t'(1+t) = t(2+t), \quad \frac{P_t'(1+t)}{P_t(1+t)} = \frac{1}{t}.$$

The quantity $\psi(1+t+\rho) + \psi(1+t)$ stays bounded as $t \downarrow 0$, so $Q'_{P_t,\rho}(1+t) > 0$ for small t. For $m \geq 3$, multiply P_t by $s^{m-2} + 1$. The product retains the required normalization and positivity, while its logarithmic derivative only increases. □

The proposition refutes the uniform unrestricted reading. It does not refute a reading in which the shift may depend on the coefficients; Theorem I.6.1 covers that case.

I.6.1 An interior transition for sparse numerators

Let $S_m = B_{P_m}$ for $P_m(s) = s^m + 1$, and define

$$r_m = \chi(\gamma + m/2) - 1.$$

The value r_m makes the logarithmic derivative vanish at $s = 1$. It is not always the optimal shift.

Theorem I.6.4 (The transition occurs at degree four)**.** *For integers $m = 1, 2, 3$, $S_m = r_m$. For every integer $m \geq 4$, $S_m > r_m$, and the maximum defining S_m is attained at an interior point $s > 1$.*

Proof. Put

$$J_m(s) = \frac{ms^{m-1}}{1+s^m}, \qquad G_{m,\rho}(s) = \psi(s+\rho) + \psi(s) - J_m(s).$$

For $m = 1, 2$, J_m is nonincreasing on $[1, \infty)$, hence $G_{m,\rho}$ strictly increases and its endpoint determines its sign. For $m = 3$,

$$J_3'(s) = \frac{3s(2-s^3)}{(1+s^3)^2} < \frac{1}{s} < \psi'(s), \qquad s \geq 1.$$

The first inequality follows from positivity of $f(s) = s^6 + 3s^5 + 2s^3 - 6s^2 + 1$: $f(1) = 1$ and $f'(s) \geq 15s > 0$ for $s \geq 1$. The second follows by comparing the trigamma series with its integral. Thus $G'_{3,\rho} > 0$ as well. This gives $S_m = r_m$ for these three degrees.

For $m \geq 4$ and $\rho = r_m$, one has $G_{m,r_m}(1) = 0$ and

$$G'_{m,r_m}(1) = \psi'(1 + r_m) + \zeta(2) - \frac{m(m-2)}{4}.$$

Because $\psi(4) = 11/6 - \gamma < \gamma + m/2$, $1 + r_m > 4$. Since ψ' is decreasing,

$$\psi'(1 + r_m) < \psi'(4) = \zeta(2) - \frac{49}{36}.$$

Also $\zeta(2) < 5/3$; for instance the first five terms plus the integral tail give $\zeta(2) < 5989/3600 < 5/3$. Hence

$$G'_{m,r_m}(1) < 2\zeta(2) - \frac{49}{36} - 2 < -\frac{1}{36}.$$

Thus $G_{m,r_m} < 0$ just to the right of 1, where the quotient increases. Consequently $S_m > r_m$. Since the endpoint value in the maximizing formula equals r_m, its attained maximum must be interior. □

Theorem I.6.5 (Euler's constant as a sharp-threshold scale)**.** *As $m \to \infty$ through integers,*

$$S_m = \frac{e^{m+\gamma-1}}{m}\left(1 + O\left(\frac{(\log m)^2}{m}\right)\right),$$

and

$$\frac{R_m}{S_m} = m\left(1 + O\left(\frac{(\log m)^2}{m}\right)\right).$$

In particular, $\lim_{m\to\infty} me^{-m}S_m = e^{\gamma-1}$.

Proof. For $m \geq 2$, J_m attains its maximum on $[1, \infty)$ at

$$s_m = (m-1)^{1/m}, \qquad M_m = J_m(s_m) = (m-1)^{1-1/m}.$$

The threshold formula and $\psi(s) \geq -\gamma$ give the two-sided bound

$$\chi(M_m - \psi(s_m)) - s_m \ \leq S_m \ \leq \chi(M_m + \gamma) - 1.$$

Elementary expansion gives

$$s_m = 1 + O\left(\frac{\log m}{m}\right), \qquad \psi(s_m) = -\gamma + O\left(\frac{\log m}{m}\right),$$

and

$$M_m = m - 1 - \log m + O\left(\frac{(\log m)^2}{m}\right).$$

The standard digamma expansion implies $\chi(y) = e^y + \frac{1}{2} + O(e^{-y})$ as $y \to \infty$. Applying it to both bounds yields

$$S_m = e^{M_m+\gamma}\left(1 + O\left(\frac{\log m}{m}\right)\right) = \frac{e^{m+\gamma-1}}{m}\left(1 + O\left(\frac{(\log m)^2}{m}\right)\right).$$

For $m \geq 2$, $R_m = \chi(m - 1 + \gamma) - 1 = e^{m+\gamma-1} - \frac{1}{2} + O(e^{-m})$, which proves the ratio formula. □

The last two theorems are extensions developed here, not additional independently posed open problems. They connect γ to a change from endpoint to interior optimization and to its exponential growth scale. Recovering e^γ as a limit gives no arithmetic classification of that number.

Prior-art qualification. The constant-numerator threshold $\chi(\gamma) - 1$ is immediate from the same endpoint argument. A public repository already records this result as APP-0009 [I.18], dated May 23, 2026, so it is not counted as a new solution here. That public claim is distinguished from peer-reviewed validation. No matching polynomial-family result was found in the searches made for this note; that limited search does not establish priority. The polynomial analysis above should be counted at most once against the published question.

I.7 Arithmetic of Bessel-bridge stopping boundaries

The analytic stopping problem in this section is already solved: for reflected Brownian bridges see Ekström–Wanntorp [I.7], and for arbitrary positive-dimensional Bessel bridges see Hobson–Liu [I.8]. We prove an arithmetic classification of their boundary constants. No explicit published arithmetic conjecture was identified, so this classification is not counted as the resolution of a previously stated open problem. Its novelty relative to the full literature remains unverified.

I.7.1 A Kummer arithmetic lemma

For positive rational numbers a, b, put

$$F_{a,b}(z) = M(a, b, z) = \sum_{n=0}^{\infty} \frac{(a)_n}{(b)_n} \frac{z^n}{n!}.$$

Here $(a)_0 = 1$ and $(a)_n = a(a+1)\cdots(a+n-1)$.

Lemma I.7.1 (Strictly increasing elasticity)**.** *For $a, b > 0$, the function $zF'_{a,b}(z)/F_{a,b}(z)$ increases strictly from zero to infinity on $(0, \infty)$. Consequently, for every $c > 0$,*

$$zF'_{a,b}(z) = cF_{a,b}(z) \tag{I.7.1}$$

has exactly one positive solution.

Proof. Write $A_n = (a)_n/((b)_n n!) > 0$ and define

$$\mathbb{P}_z(N = n) = \frac{A_n z^n}{F_{a,b}(z)}.$$

Termwise differentiation of the entire series yields

$$\frac{zF'_{a,b}(z)}{F_{a,b}(z)} = \mathbb{E}_z N, \qquad \frac{\mathrm{d}}{\mathrm{d}z}\mathbb{E}_z N = \frac{\mathrm{Var}_z(N)}{z} > 0.$$

The variance is positive since both $n = 0$ and $n = 1$ have positive probability. The limit at zero is zero. For each fixed integer K, the polynomial $\sum_{n \le K} A_n z^n$ has smaller degree than $A_{K+1} z^{K+1}$, so $\mathbb{P}_z(N \le K) \to 0$ as $z \to \infty$. Thus $\liminf_{z\to\infty} \mathbb{E}_z N \ge K + 1$ for every K. □

Lemma I.7.2 (Arithmetic growth of the coefficients)**.** *If a, b are positive rational numbers, then $F_{a,b}$ and $F'_{a,b}$ are E-functions. They form a system with no nonzero finite singularity:*

$$\frac{\mathrm{d}}{\mathrm{d}z}\begin{pmatrix} F_{a,b} \\ F'_{a,b} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ a/z & 1 - b/z \end{pmatrix} \begin{pmatrix} F_{a,b} \\ F'_{a,b} \end{pmatrix}. \tag{I.7.2}$$

Proof. Write $a = r/s$, $b = u/v$ in lowest positive terms. The factorial-normalized coefficients are

$$C_n = \frac{(a)_n}{(b)_n} = \left(\frac{v}{s}\right)^n \frac{\prod_{j=0}^{n-1}(r + sj)}{\prod_{j=0}^{n-1}(u + vj)}, \qquad |C_n| \le \max(1, a/b)^n.$$

We verify the denominator condition. Fix an integer $L \geq u + v$. For a prime $p \nmid sv$, each progression has precisely one root modulo p^k. The counts of roots in n consecutive indices differ by at most one. Since denominator factors for $n \leq N$ are at most LN, the negative p-adic valuation of any C_n, $n \leq N$, is at most $\lfloor \log_p(LN) \rfloor$. All these denominator prime powers therefore divide $\operatorname{lcm}(1, \ldots, LN)$. For each of the finitely many primes dividing sv, progression valuations and the prefactor contribute at most $O_{a,b}(N)$: a progression with step prime to p contributes at most $\sum_{k \geq 1}(N/p^k + \mathbf{1}_{p^k \leq L'N}) = O_{a,b}(N)$ for a fixed L', whereas a progression whose step is divisible by p has initial term prime to p and contributes zero. The elementary Chebyshev estimate $\log \operatorname{lcm}(1, \ldots, m) = O(m)$ now gives a common denominator $\exp(O_{a,b}(N))$. The derivative has coefficients C_{n+1} and obeys the same estimates. Finally, direct substitution of the series gives $zF'' + (b - z)F' - aF = 0$, hence (I.7.2). □

Lemma I.7.3 (The only first-order cases)**.** *For $a, b > 0$, the functions $F_{a,b}$ and $F'_{a,b}$ are linearly dependent over $\mathbb{C}(z)$ if and only if $a - b \in \mathbb{Z}_{\geq 0}$. In that case, with $m = a - b$,*

$$F_{a,b}(z) = e^z P_m(z), \qquad P_m(z) = \sum_{j=0}^{m} \frac{m!}{(m-j)!\,(b)_j\,j!} z^j. \tag{I.7.3}$$

Proof. Suppose $R = F'_{a,b}/F_{a,b}$ is rational. Because $F_{a,b}$ is entire and $F_{a,b}(0) = 1$, the finite poles of R are simple, with positive integer residues equal to zero multiplicities. If P is the polynomial formed from these finitely many zeros, then $R - P'/P$ is a polynomial. The Riccati equation

$$z(R' + R^2) + (b - z)R - a = 0$$

excludes any positive-degree polynomial part, by comparing degrees at infinity. Its constant part c satisfies $c^2 - c = 0$. Hence $F_{a,b} = Ce^{cz}P$ with $c = 0$ or 1. If $c = 0$, leading coefficients in Kummer's equation give $\deg P + a = 0$, impossible. If $c = 1$, the polynomial satisfies

$$zP'' + (b + z)P' + (b - a)P = 0,$$

so $\deg P = a - b \in \mathbb{Z}_{\geq 0}$. Conversely, the polynomial in (I.7.3) satisfies this last equation, and $e^z P_m(z)$ is the unique analytic solution of Kummer's equation with value 1 at zero. This proves the converse and the formula. □

Theorem I.7.4 (Arithmetic classification of Kummer elasticity roots)**.** *Let a, b be positive rational numbers and let c be a positive algebraic number. The unique positive solution z_c of (I.7.1) is algebraic if $a - b \in \mathbb{Z}_{\geq 0}$, and is transcendental otherwise.*

Proof. In the first case, (I.7.3) reduces the boundary equation to the nonzero polynomial equation

$$z(P_m + P'_m) - cP_m = 0.$$

In the second case, Lemmas I.7.2 and I.7.3 allow the use of Beukers's Corollary 1.4 [I.6]: linearly independent E-functions in a rational first-order system take linearly independent values over $\overline{\mathbb{Q}}$ at every nonzero algebraic ordinary point of the system. If z_c were algebraic, the relation $z_c F'_{a,b}(z_c) - cF_{a,b}(z_c) = 0$ would contradict this corollary. □

I.7.2 Application to Bessel bridges

Let $d > 0$. Define the squared Bessel bridge Y pinned at zero at time 1 through

$$\mathrm{d}Y_s = \left(d - \frac{2Y_s}{1-s}\right)\mathrm{d}s + 2\sqrt{Y_s}\,\mathrm{d}W_s, \qquad R_s = \sqrt{Y_s}.$$

This formulation also covers $0 < d < 1$, where a naive radial SDE at zero is unsuitable. For $q > 0$, write

$$V_{d,q}(t,r) = \sup_{t \le \tau \le 1} \mathbb{E}_{t,r}[R_\tau^q], \qquad F(z) = M(q/2, d/2, z).$$

Let $z_{d,q}$ be the unique positive root of

$$2z_{d,q}F'(z_{d,q}) = qF(z_{d,q}), \qquad B_{d,q} = \sqrt{2z_{d,q}}. \tag{I.7.4}$$

The analytic solution of Hobson–Liu [I.8, Theorem 3.1] is

$$V_{d,q}(t,r) = \begin{cases} (1-t)^{q/2} B_{d,q}^q \dfrac{F(r^2/[2(1-t)])}{F(z_{d,q})}, & r < B_{d,q}\sqrt{1-t}, \\ r^q, & r \ge B_{d,q}\sqrt{1-t}. \end{cases} \tag{I.7.5}$$

Their parameters are $\alpha = d$, $n = q$, and their squared boundary constant is $Z_{\alpha,n} = B_{d,q}^2 = 2z_{d,q}$. Indeed their series $\psi(y)$ has coefficient ratio $(q+2k)/[2(k+1)(d+2k)]$, so $\psi(y) = M(q/2, d/2, y/2)$. This checks the factor of two in (I.7.4).

Theorem I.7.5 (Boundary dichotomy for rational dimension and reward)**.** *For $d, q \in \mathbb{Q}_{>0}$,*

$$B_{d,q} \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad q - d \in 2\mathbb{Z}_{\ge 0}.$$

In every other case $B_{d,q}$ is transcendental.

Proof. Apply Theorem I.7.4 with $a = q/2$, $b = d/2$, $c = q/2$. Algebraicity of $B_{d,q}$ is equivalent to algebraicity of $z_{d,q}$. □

Proposition I.7.6 (A strict boundary estimate and verification)**.** *For all real $d, q > 0$,*

$$B_{d,q}^2 > \frac{q+d-2}{2}.$$

For integer $d \ge 1$, formula (I.7.5) can be verified directly by the following argument.

Proof. Put $h(y) = F(y^2/2)$ and $\lambda(y) = yh'(y)/h(y)$. Lemma I.7.1 gives $\lambda'(y) > 0$ for $y > 0$. The radial differential equation is

$$h'' + \left(\frac{d-1}{y} - y\right) h' - qh = 0.$$

At $B = B_{d,q}$, where $\lambda(B) = q$, it follows that

$$0 < \lambda'(B) = \frac{q}{B}(2B^2 + 2 - d - q).$$

More precisely, for the probability law in Lemma I.7.1 at $z = B^2/2$, $\lambda'(B) = 4\,\mathrm{Var}(N)/B$, and hence

$$B^2 = \frac{q+d-2}{2} + \frac{2}{q}\,\mathrm{Var}(N).$$

This proves the strict estimate and identifies its exact positive gap. In particular, it strengthens the nonstrict bound in [I.8, Proposition 3.1], which is an already-proved proposition rather than an open conjecture.

To verify the value formula, set $v(y) = B^q h(y)/h(B)$ for $y < B$ and $v(y) = y^q$ for $y \ge B$. Because $\lambda(y) < q$ for $0 < y < B$, the ratio $h(y)/y^q$ decreases there. Thus $v(y) \ge y^q$. The

candidate $(1-t)^{q/2}v(r/\sqrt{1-t})$ satisfies the continuation PDE, and the boundary equation gives continuous and smooth fit. In the stopping region its generator is

$$qr^{q-2}\left(\frac{q+d-2}{2}-\frac{r^2}{1-t}\right)\le 0.$$

Applying Itô's formula to the squared radius avoids an issue at $r=0$, since the continuation function is analytic in $Y=r^2$. Smooth fit removes the boundary local-time term. The candidate along the bridge is therefore a nonnegative local supermartingale, bounding all expected stopping rewards from above. Up to its first boundary hit it is a local martingale. For integer dimension, the supremum of the bridge radius has all positive moments, and the candidate is bounded by a constant plus R_s^q; localization and dominated convergence give equality at the first boundary hit (or at time 1). This proves optimality. The result for arbitrary positive dimension is supplied by the squared-Bessel verification in [I.8, Theorem 3.1]. □

Corollary I.7.7 (Arithmetic of the value in the algebraic-boundary cases)**.** *Suppose $d,q\in\mathbb{Q}_{>0}$ and $q=d+2m$, $m\in\mathbb{Z}_{\ge 0}$. Then $V_{d,q}(0,0)$ is transcendental. More generally, if t,r are algebraic, $0\le t<1$, $r\ge 0$, and $r<B_{d,q}\sqrt{1-t}$, then $V_{d,q}(t,r)$ is transcendental. On and outside the stopping boundary its value is algebraic.*

Proof. Let $z=z_{d,q}$ and $w=r^2/[2(1-t)]$. Formula (I.7.3) rewrites the continuation value as

$$(1-t)^{q/2}\frac{B_{d,q}^q P_m(w)}{P_m(z)}e^{w-z}.$$

The prefactor is nonzero algebraic; all coefficients of P_m are positive. The exponent is nonzero algebraic because $0\le w<z$. Lindemann's theorem proves transcendence. Outside the continuation region the value is the algebraic number r^q. □

Corollary I.7.8 (Reflected bridges and excursions)**.** *For reflected one-dimensional Brownian bridges, $B_{1,q}$ is algebraic exactly when q is a positive odd integer. In particular,*

$$B_{1,1}=1,\qquad B_{1,3}=3^{1/4},\qquad B_{1,5}^6+5B_{1,5}^4-15B_{1,5}^2-15=0.$$

The corresponding first two initial values are

$$V_{1,1}(0,0)=e^{-1/2},\qquad V_{1,3}(0,0)=\frac{3^{3/4}}{1+\sqrt{3}}e^{-\sqrt{3}/2}.$$

A Brownian excursion is a three-dimensional Bessel bridge. Its linear reward boundary $B_{3,1}$ is transcendental. More generally, for rational $q>0$, its boundary is algebraic exactly for $q\in\{3,5,7,\dots\}$. At $q=3$,

$$B_{3,3}=\sqrt{3},\qquad V_{3,3}(0,0)=3^{3/2}e^{-3/2}.$$

Scope of the transfer. These results transfer an established E-function theorem to a family of free-boundary constants. They do not imply irrationality or transcendence of Euler's constant. The individual examples are instances of one classification theorem, not separate resolutions of published conjectures.

I.7.3 Euler–Gompertz constants as a dimension sensitivity

The previous arithmetic classification does not use γ directly. There is, however, an exact sensitivity identity involving the classical combination $\eta = \mathrm{Ein}(1) = \gamma + \delta/e$. Here $\delta = \int_0^\infty e^{-u}\log(1+u)\,\mathrm{d}u$ and

$$\mathrm{Ein}(z) = \int_0^z \frac{1-e^{-u}}{u}\,\mathrm{d}u.$$

Let $J(d,q) = V_{d,q}(0,0)$, with the dimension d treated as a positive real parameter in the squared-Bessel model.

Theorem I.7.9 (Arithmetic independence of value and dimension sensitivity)**.** *The function J is real analytic for $d, q > 0$. At $(d,q) = (2,2)$,*

$$\begin{aligned} J(2,2) &= \frac{2}{e}, \\ \left.\frac{\partial J}{\partial d}\right|_{(2,2)} &= \frac{\mathrm{Ein}(1)}{e} = \frac{\gamma}{e} + \frac{\delta}{e^2}, \\ \left.\frac{\partial J}{\partial q}\right|_{(2,2)} &= \frac{\log 2 - \mathrm{Ein}(1)}{e}. \end{aligned}$$

Moreover, $J(2,2)$ and $\partial J/\partial d|_{(2,2)}$ are algebraically independent over $\overline{\mathbb{Q}}$.

Proof. The Kummer function is analytic in its positive parameters. Strict elasticity and the implicit function theorem show that $z = z_{d,q}$ is locally real analytic, hence so is

$$J(d,q) = \frac{(2z)^{q/2}}{M(q/2, d/2, z)}.$$

At $d = q = 2$, one has $M(1,1,z) = e^z$ and $z = 1$. For $H_n = \sum_{k=1}^n 1/k$ and $H_0 = 0$, termwise parameter differentiation gives

$$\partial_a M(a,1,z)|_{a=1} = \sum_{n=0}^{\infty} H_n \frac{z^n}{n!} = e^z\,\mathrm{Ein}(z).$$

The last identity follows by writing $H_n = \int_0^1 (1-t^n)/(1-t)\,\mathrm{d}t$ and summing the exponential series. Differentiating $M(a,a,z) = e^z$ along the diagonal gives

$$\partial_b M(1,b,z)|_{b=1} = -e^z\,\mathrm{Ein}(z).$$

Now

$$\log J(d,q) = \frac{q}{2}\log(2z) - \log M(q/2, d/2, z).$$

Its coefficient of every derivative of z vanishes, since smooth fit is precisely $M_z/M = q/(2z)$. At $(2,2)$ the remaining terms yield

$$\partial_d \log J = \frac{1}{2}\,\mathrm{Ein}(1), \qquad \partial_q \log J = \frac{1}{2}(\log 2 - \mathrm{Ein}(1)),$$

which prove the identities.

Use the established algebraic independence of e and $\eta = \mathrm{Ein}(1)$; see [I.9, Section 5(ii)], where the independent pair $e^{-1}, E(-1)$ equals $e^{-1}, -\eta$ in the notation used there. Set $U = J(2,2)$ and $D = \partial_d J(2,2)$. The displayed identities give

$$U = 2/e, \quad D = \eta/e, \qquad e = 2/U, \quad \eta = 2D/U.$$

Thus $\overline{\mathbb{Q}}(U,D) = \overline{\mathbb{Q}}(e,\eta)$, of transcendence degree two. □

The independence theorem concerns an optimal value together with its response to a change in the Bessel dimension. It leaves the separate arithmetic nature of γ and δ unresolved. In particular, no transcendence assertion about the reward-exponent derivative $(\log 2 - \eta)/e$ is made here.

I.7.4 Algebraic independence of every finite dimension jet

Define $\mathcal{J}(d) = J(d, 2)$ and, for integers $n \geq 1$, put

$$H_n(z) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} z^k}{k^n k!}, \qquad H_0(z) = 1 - e^{-z}.$$

Thus $H_1 = \mathrm{Ein}$ and $H_n' = H_{n-1}/z$ for $n \geq 1$, with the quotient interpreted analytically at zero.

Theorem I.7.10 (All finite dimension jets are algebraically independent)**.** *For every integer $N \geq 0$, the $N + 1$ real numbers*

$$\mathcal{J}(2), \mathcal{J}'(2), \ldots, \mathcal{J}^{(N)}(2)$$

are algebraically independent over $\overline{\mathbb{Q}}$. More precisely,

$$\overline{\mathbb{Q}}(\mathcal{J}(2), \ldots, \mathcal{J}^{(N)}(2)) = \overline{\mathbb{Q}}(e, H_1(1), \ldots, H_N(1)). \tag{I.7.6}$$

Proof. Let $\varepsilon = (d - 2)/2$. An exact parameter expansion is

$$M(1, 1 + \varepsilon, z) = e^z S(\varepsilon, z), \qquad S(\varepsilon, z) = 1 + \sum_{n \geq 1} (-1)^n H_n(z) \varepsilon^n. \tag{I.7.7}$$

To verify it first take $\varepsilon > 0$ and use

$$e^{-z} M(1, 1 + \varepsilon, z) = \varepsilon \int_0^1 e^{-zt} t^{\varepsilon - 1} \, \mathrm{d}t = 1 + \varepsilon \sum_{k \geq 1} \frac{(-z)^k}{k!(k + \varepsilon)}.$$

Expanding $\varepsilon/(k + \varepsilon)$ proves (I.7.7); analytic continuation supplies a neighborhood of $\varepsilon = 0$.

Write $z(\varepsilon) = z_{2+2\varepsilon,2}$, so $z(0) = 1$ and

$$\mathcal{J}(2 + 2\varepsilon) = \frac{2 z(\varepsilon) e^{-z(\varepsilon)}}{S(\varepsilon, z(\varepsilon))}.$$

The stationary boundary equation is

$$\frac{1}{z(\varepsilon)} - 1 - \frac{S_z(\varepsilon, z(\varepsilon))}{S(\varepsilon, z(\varepsilon))} = 0. \tag{I.7.8}$$

Its derivative with respect to z at $(\varepsilon, z) = (0, 1)$ is -1, so it determines a unique convergent local power series for $z(\varepsilon)$.

Put $\eta_n = H_n(1)$ and $K_n = \overline{\mathbb{Q}}(e, \eta_1, \ldots, \eta_n)$, with $K_0 = \overline{\mathbb{Q}}(e)$. Every positive-order z-derivative of H_n at $z = 1$ belongs to K_{n-1}: differentiate $H_n' = H_{n-1}/z$ repeatedly, using that all derivatives of H_0 at 1 belong to K_0. In particular, induction on the coefficient of ε^n in (I.7.8) shows that

$$[\varepsilon^n] z(\varepsilon) \in K_{n-1} \qquad (n \geq 1). \tag{I.7.9}$$

Indeed the coefficient involving the new root coefficient is -1 times that coefficient, whereas every other term involves derivatives of H_k with $k \leq n$, at least one derivative when $k = n$, and lower root coefficients.

Next write

$$L(\varepsilon) = \log \frac{\mathcal{J}(2 + 2\varepsilon)}{2/e} = \log z(\varepsilon) - z(\varepsilon) + 1 - \log S(\varepsilon, z(\varepsilon)).$$

The function $\log z - z + 1$ has zero first derivative at $z = 1$. For each $n \geq 1$, the coefficient of ε^n in L is therefore

$$[\varepsilon^n] L(\varepsilon) = (-1)^{n+1} \eta_n + R_n, \qquad R_n \in K_{n-1}. \tag{I.7.10}$$

For completeness, the unique term containing $H_n(1)$ comes from the linear term $-S+1$ in $-\log S$; nonlinear terms use only H_k with $k<n$. Substituting $z(\varepsilon)$ adds only the lower-field terms specified in (I.7.9). The zero derivative of $\log z - z + 1$ prevents a new unrestricted root coefficient from entering. Exponentiating (I.7.10) gives

$$[\varepsilon^n]\mathcal{J}(2+2\varepsilon) = \frac{2}{e}((-1)^{n+1}\eta_n + \widetilde{R}_n), \qquad \widetilde{R}_n \in K_{n-1}.$$

On the other hand this coefficient is $2^n\mathcal{J}^{(n)}(2)/n!$. Starting with $\mathcal{J}(2) = 2/e$, the displayed triangular relations recursively recover every η_n from the dimension jet and vice versa. This proves (I.7.6).

Finally, the established algebraic independence of $e, H_1(1), \dots, H_N(1)$ follows from [I.10, Theorem 2]; its constants are $1-e^{-1}$ and $n!H_n(1)$. It is also the fixed-$\alpha=0$, $z=1$ case of the integral family discussed in [I.9, Section 7], since

$$H_n(1) = \frac{1}{n!}\int_0^1 (-\log t)^n e^{-t}\,\mathrm{d}t.$$

The common field in (I.7.6) consequently has transcendence degree $N+1$. □

This is one theorem about an infinite family of stochastic-optimization sensitivities. Taking ten or more derivatives does not turn it into ten independently posed historical questions. The first nontrivial member is the exact Euler–Gompertz sensitivity proved above.

Corollary I.7.11 (No arithmetic algebraic differential equation)**.** *The function $\mathcal{J}(d) = V_{d,2}(0,0)$ is differentially transcendental over $\overline{\mathbb{Q}}(d)$: for no $N \geq 0$ and nonzero polynomial $P \in \overline{\mathbb{Q}}(d)[X_0, \dots, X_N]$ does*

$$P(d, \mathcal{J}(d), \mathcal{J}'(d), \dots, \mathcal{J}^{(N)}(d)) = 0$$

hold on an interval.

Proof. Clear the rational-function denominators. Divide the resulting polynomial by the highest common power of $d-2$ dividing all its coefficient polynomials. The relation still holds and its specialization at $d=2$ is a nonzero polynomial relation over $\overline{\mathbb{Q}}$ among a finite dimension jet, contradicting Theorem I.7.10. Real analyticity extends any interval relation throughout the connected domain $d>0$ before this evaluation. □

The coefficient field in this corollary is expressly $\overline{\mathbb{Q}}(d)$. No claim over $\mathbb{C}(d)$ is inferred from this specialization argument.

I.8 Prime valuations, cyclotomic units, and rank certificates

This section derives linear-independence certificates for generalized Euler–Briggs constants from their character decomposition and Baker's theorem. The inputs are classical. The results are presented as proved consequences, without a claim of priority or a claim to have resolved a published open conjecture. In particular, the argument does not determine the arithmetic nature of Euler's constant itself.

I.8.1 The arithmetic decomposition

Fix a finite set Ω of rational primes and write

$$M_\Omega = \prod_{p\in\Omega} p, \qquad d_\Omega = \prod_{p\in\Omega}\left(1-\frac{1}{p}\right), \qquad C_\Omega = \sum_{p\in\Omega}\frac{\log p}{p-1}.$$

Empty products and sums have their usual values 1 and 0. For $q > 1$, $(q, M_\Omega) = 1$, and $1 \le a \le q$ with $(a, q) = 1$, define

$$G_\Omega(a,q) = \lim_{x\to\infty} \left(\sum_{\substack{n\le x \\ n\equiv a \pmod q \\ (n,M_\Omega)=1}} \frac{1}{n} - \frac{d_\Omega}{q}\log x \right), \qquad T_\Omega(a,q) = \frac{q}{d_\Omega} G_\Omega(a,q). \tag{I.8.1}$$

The limiting constant exists: the indicator of the summation set is periodic, has mean d_Ω/q, and has bounded discrepancy from this mean. Partial summation then gives convergence with error $O_{q,M_\Omega}(x^{-1})$.

Lemma I.8.1 (Character decomposition)**.** *With the preceding notation,*

$$T_\Omega(a,q) = \gamma + C_\Omega + \sum_{p\mid q} \frac{\log p}{p-1} + U_\Omega(a,q), \tag{I.8.2}$$

where

$$U_\Omega(a,q) = \frac{q}{d_\Omega \varphi(q)} \sum_{\substack{\chi \bmod q \\ \chi\ne\chi_0}} \overline{\chi}(a) L(1,\chi) \prod_{p\in\Omega} \left(1 - \frac{\chi(p)}{p}\right). \tag{I.8.3}$$

In particular, $U_\Omega(a,q)$ is an algebraic linear combination of nonprincipal Dirichlet L-values at 1.

Proof. This is the normalization of formula (2) in [I.13]. For completeness, character orthogonality gives, for $\Re s > 1$,

$$\sum_{\substack{n\ge 1 \\ n\equiv a \pmod q \\ (n,M_\Omega)=1}} n^{-s} = \frac{1}{\varphi(q)} \sum_{\chi \bmod q} \overline{\chi}(a) L(s,\chi) \prod_{p\in\Omega} (1 - \chi(p)p^{-s}).$$

The principal-character contribution equals

$$\frac{\zeta(s)}{\varphi(q)} \prod_{p\mid q} (1-p^{-s}) \prod_{p\in\Omega} (1-p^{-s}).$$

Its residue at $s = 1$ is d_Ω/q, and its constant term is

$$\frac{d_\Omega}{q} \left(\gamma + C_\Omega + \sum_{p\mid q} \frac{\log p}{p-1} \right).$$

The other character contributions are holomorphic at 1 and give the terms in (I.8.3) after normalization. The bounded discrepancy noted above identifies the constant term of the Dirichlet series with the limit in (I.8.1), for example by partial summation. This proves the formula. □

Lemma I.8.2 (Prime logarithms modulo unit logarithms)**.** *Let S be a finite set of distinct rational primes, and let $u_1, \dots, u_m$ be algebraic units with chosen complex logarithms. If the coefficients b_p, c_j, c_0 are algebraic and at least one b_p is nonzero, then*

$$\sum_{p\in S} b_p \log p + \sum_{j=1}^{m} c_j \log u_j + c_0\pi \tag{I.8.4}$$

is transcendental. Consequently, if this expression vanishes, every b_p vanishes.

Proof. Use $\log(-1) = i\pi$ to incorporate $c_0\pi$ into the unit-logarithm sum. Choose a basis $\ell_1, \dots, \ell_r$ over $\mathbb{Q}$ for the span of these finitely many unit logarithms, choosing the basis from the logarithms themselves. We first show that

$$\{\log p : p \in S\} \cup \{\ell_1, \dots, \ell_r\}$$

is linearly independent over $\mathbb{Q}$. Clear denominators in a proposed rational relation and exponentiate. The result has the form

$$\prod_{p \in S} p^{m_p}\, v = 1,$$

where $m_p \in \mathbb{Z}$ and v is an algebraic unit. Place all these numbers in a number field K. At a prime ideal $\mathfrak{p}$ of K above a fixed $p \in S$, valuation of the displayed identity gives $m_p v_{\mathfrak{p}}(p) = 0$. The other rational primes and every algebraic unit have zero valuation there. Hence every $m_p = 0$, and the remaining relation vanishes by the choice of the ℓ_j.

Baker's theorem lifts the resulting rational linear independence of logarithms of algebraic numbers to linear independence over $\overline{\mathbb{Q}}$ [I.14]. Thus (I.8.4) is nonzero whenever some $b_p \neq 0$. The zero-or-transcendental part of the same theorem then makes it transcendental. One may equivalently use the formulation of Baker's theorem recorded in [I.13, Theorem 14]. □

Lemma I.8.3 (The unit contribution of nonprincipal characters)**.** *For every nonprincipal Dirichlet character χ, the number $L(1, \chi)$ belongs to the $\overline{\mathbb{Q}}$-linear span of π and logarithms of algebraic units.*

Proof. First suppose that χ is primitive of conductor f. For odd χ, the finite formula for $L(1, \chi)$ expresses it as an algebraic multiple of π. For even χ, the finite Fourier formula expresses it as an algebraic multiple of

$$\sum_{a \in (\mathbb{Z}/f\mathbb{Z})^*} \overline{\chi}(a) \log|1 - \zeta_f^a|, \qquad \zeta_f = e^{2\pi i/f}.$$

If f has at least two distinct prime factors, then $\Phi_f(1) = 1$, so the algebraic integers $1 - \zeta_f^a$ are units. Their absolute values are also algebraic units: the square of such an absolute value is the product of a unit with its complex conjugate, and both the positive square root and its inverse are algebraic integers.

If f is a prime power, use

$$\log|1 - \zeta_f^a| = \log|1 - \zeta_f| + \log\left|\frac{1 - \zeta_f^a}{1 - \zeta_f}\right|.$$

The quotients are cyclotomic units. Since χ is nonprincipal, $\sum_a \overline{\chi}(a) = 0$, and the common nonunit logarithm disappears from the character sum. This proves the primitive case.

Finally, if χ modulo q is induced by a primitive character χ^* of conductor f, then χ^* is nonprincipal and

$$L(1, \chi) = L(1, \chi^*) \prod_{\substack{p \mid q \\ p \nmid f}} \left(1 - \frac{\chi^*(p)}{p}\right).$$

The finite factor is algebraic, which proves the general case. These are the same standard cyclotomic-unit inputs used in [I.13, Section 3]. □

I.8.2 Coprime moduli and transcendental differences

Theorem I.8.4 (A coprime-modulus certificate)**.** *Let $q_1, \dots, q_n > 1$ be pairwise coprime, with $(q_i, M_\Omega) = 1$, and let a_i be any reduced residue modulo q_i. Put $T_i = T_\Omega(a_i, q_i)$. If $c_1, \dots, c_n \in \overline{\mathbb{Q}}$ are not all zero and $\sum_i c_i = 0$, then*

$$\sum_{i=1}^{n} c_i T_i$$

is transcendental.

Proof. By Lemma I.8.1,

$$T_i = \gamma + C_\Omega + \sum_{p \mid q_i} \frac{\log p}{p-1} + U_\Omega(a_i, q_i).$$

The condition $\sum_i c_i = 0$ cancels $\gamma + C_\Omega$. Lemma I.8.3 puts all remaining U-terms into a finite algebraic linear combination of unit logarithms and π. Because the moduli are pairwise coprime, a prime $p \mid q_i$ occurs in no other modulus. Its coefficient is therefore $c_i/(p-1)$. Some c_i is nonzero, and that $q_i > 1$ has a prime divisor. Lemma I.8.2 proves the assertion. □

Corollary I.8.5 (Dimension and the relation space)**.** *Under the hypotheses of Theorem I.8.4,*

$$\dim_{\overline{\mathbb{Q}}} \operatorname{span}_{\overline{\mathbb{Q}}} \{G_\Omega(a_i, q_i) : 1 \le i \le n\} \ge n - 1. \tag{I.8.5}$$

The space of algebraic linear relations among $T_1, \dots, T_n$ has dimension at most one. Moreover,

$$1,\ T_2 - T_1, \dots, T_n - T_1$$

are linearly independent over $\overline{\mathbb{Q}}$. An infinite family of pairwise coprime admissible moduli consequently gives an infinite-dimensional span over $\overline{\mathbb{Q}}$.

Proof. Let

$$R = \left\{ c \in \overline{\mathbb{Q}}^{\,n} : \sum_i c_i T_i = 0 \right\}.$$

The map $R \longrightarrow \overline{\mathbb{Q}}$ given by $c \longmapsto \sum_i c_i$ is injective by Theorem I.8.4. Hence $\dim R \le 1$, and rank–nullity proves the bound for the T_i. Multiplication by the nonzero rational factors d_Ω/q_i gives (I.8.5).

For the stronger affine statement, suppose $b + \sum_{i=2}^{n} b_i(T_i - T_1) = 0$ with algebraic coefficients. If any $b_i \neq 0$, the sum of differences is transcendental by Theorem I.8.4, whereas the equation makes it the algebraic number $-b$. Thus every $b_i = 0$ and then $b = 0$. Applying the rank bound to arbitrarily long finite subfamilies proves the infinite-dimensional assertion. □

For a concrete normalization check, when Ω is empty one has

$$2G_\varnothing(1,2) = \gamma + \log 2, \qquad 3G_\varnothing(1,3) = \gamma + \frac{1}{2}\log 3 + \frac{\pi}{2\sqrt{3}}.$$

Their difference is therefore

$$3G_\varnothing(1,3) - 2G_\varnothing(1,2) = \frac{1}{2}\log 3 - \log 2 + \frac{\pi}{2\sqrt{3}},$$

a transcendental number by the theorem. This is an illustration of the certificate, not a new claim about a previously unresolved named constant.

I.8.3 A prime-divisibility incidence matrix

Theorem I.8.6 (Incidence rank without coprimality)**.** *Let $q_1, \dots, q_n > 1$ satisfy $(q_i, M_\Omega) = 1$, without assuming pairwise coprimality. Choose reduced residues a_i and put $T_i = T_\Omega(a_i, q_i)$. Let S be the set of primes dividing $q_1 \cdots q_n$, and define the matrix*

$$A_{pi} = \frac{\mathbf{1}_{p|q_i}}{p-1} \quad (p \in S,\ 1 \le i \le n), \qquad H = \left\{ c \in \overline{\mathbb{Q}}^{\,n} : \sum_i c_i = 0 \right\}.$$

For every $c \in H$ with $Ac \neq 0$, the number $\sum_i c_i T_i$ is transcendental. Consequently,

$$\dim_{\overline{\mathbb{Q}}} \left\{ \sum_i c_i T_i : c \in H \right\} \ge \operatorname{rank}_{\overline{\mathbb{Q}}}(A|_H). \tag{I.8.6}$$

Thus a finite prime-divisibility rank calculation supplies an arithmetic linear-independence certificate.

Proof. For $c \in H$, the decomposition (I.8.2) cancels $\gamma + C_\Omega$ and leaves the coefficient vector Ac for the rational-prime logarithms. All other terms are unit logarithms and π by Lemma I.8.3. If $Ac \neq 0$, Lemma I.8.2 makes the resulting number transcendental.

Write $r = \operatorname{rank}(A|_H)$ and choose $c^{(1)}, \dots, c^{(r)} \in H$ such that their images under A form a basis of $A(H)$. Any nonzero algebraic linear combination of these vectors has nonzero image under A, and its corresponding combination of the T_i is therefore nonzero. Their r values are linearly independent, proving (I.8.6). □

The restriction to coefficients cancelling γ is substantive. The method identifies prime-logarithm components by valuations, but it supplies no corresponding independence assertion for a linear form retaining γ. For this reason, even the infinite-dimensional family conclusion does not establish the transcendence of a specified individual constant.

I.8.4 Position relative to the external questions

The character decomposition above is established prior work [I.13]; the proofs combine it with Baker's theorem [I.14] and elementary valuations. In particular, [I.13, Theorems 8–9] already studies algebraic-multiple classes and infinite-dimensional spaces of these constants. No assertion that the present rank formulations were previously open is made here.

The analytic questions in the accompanying ledger remain separate. Diamond and Ford identify the spectral equality $\inf_\Omega \gamma(\Omega) = e^{-\gamma}$ with the Riemann hypothesis, and already disprove monotonicity along initial prime sets [I.15]. The cyclotomic Euler–Kronecker limit-point conjecture of Ford, Luca, and Moree [I.16], and the unconditional infinitude of negative cyclotomic Euler–Kronecker constants discussed by Dixit and Murty [I.17], are not resolved by any result in this section.

I.9 Arithmetic Gevrey transfer: an exact obstruction

Put

$$\mathrm{Ein}(z) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} z^n}{n\, n!}, \qquad \eta = \mathrm{Ein}(1) = \gamma + \frac{\delta}{e}, \qquad \delta = \int_0^\infty \frac{e^{-t}}{1+t}\, dt.$$

The algebraic independence of e and η is established in Rivoal's 2012 paper [I.9, Section 5]; it is not a result first proved here. We use $\overline{\mathbb{Q}}$ for the algebraic numbers.

The relevant open statements are recorded in the source ledger; we use the mixed-function framework of [I.12]. The coefficient-comparison argument below specializes the mechanism of

that paper's Proposition 2. It is included as an exact obstruction, not as an independently new resolution.

I.9.1 Definitions needed for the obstruction

An E-function has an expansion $\sum_{n\ge 0} a_n z^n/n!$ with algebraic coefficients satisfying the standard exponential conjugate and common- denominator bounds, and satisfies a linear differential equation over $\overline{\mathbb{Q}}(z)$. An $\mathfrak{E}_1$-function is a formal series $\sum_{n\ge 0} n! a_n w^n$ whose associated $\sum a_n w^n$ is a G-function. A mixed function is a formal sum

$$A(z) + \sum_{n\ge 1} b_n z^{-n},$$

where A is an E-function and the negative-power part is an $\mathfrak{E}_1$-function in $1/z$. In particular, all b_n are algebraic. Equality of mixed functions means coefficientwise equality; multiplication by $z-1$ is consequently unambiguous.

Define

$$\widehat{D}(z) = \sum_{n\ge 0} (-1)^n n! z^{-n-1}, \qquad W_r(z) = e^z\big(\mathrm{Ein}(z) - r\big) - \widehat{D}(z) \quad (r \in \overline{\mathbb{Q}}).$$

The series $\widehat{D}$ is of the required arithmetic Gevrey type. Indeed, its ordinary Borel transform is $1/(1+t)$; alternatively the coefficient sequence after dividing the coefficient of z^{-m} by $m!$ is $(-1)^{m-1}/m$, a G-function sequence. The positive part of W_r is an E-function because this class is closed under products and algebraic linear combinations.

Proposition I.9.1 (Arithmetic nondivisibility)**.** *For every algebraic r, there is no mixed function V satisfying*

$$W_r(z) = (z-1)V(z).$$

Proof. Suppose $V = A(z) + \sum_{n\ge 1} b_n z^{-n}$ is mixed. Comparing the nonnegative powers of z gives

$$e^z\big(\mathrm{Ein}(z) - r\big) = (z-1)A(z) + b_1.$$

Since A is entire, evaluation at $z=1$ implies $e(\eta - r) = b_1 \in \overline{\mathbb{Q}}$. This contradicts the algebraic independence of e and η: the polynomial $X(Y-r) - b_1$ is nonzero in $\overline{\mathbb{Q}}[X,Y]$. □

Proposition I.9.2 (An exact threshold for a restricted zero-removal rule)**.** *The following statements are equivalent:*

1. *γ is transcendental.*
2. *For every $r \in \overline{\mathbb{Q}}$, the implication*

$$(W_r)_0(1) = 0 \quad \Longrightarrow \quad W_r = (z-1)V \text{ for some mixed } V$$

holds.

Replacing $\overline{\mathbb{Q}}$ by $\mathbb{Q}$ in statement (2) makes it equivalent to the irrationality of γ.

Proof. On the positive real axis the Borel–Laplace sum is

$$D_0(z) = \int_0^\infty \frac{e^{-zt}}{1+t}\,dt = e^z E_1(z), \qquad E_1(z) = \int_z^\infty \frac{e^{-u}}{u}\,du.$$

The classical identity $\mathrm{Ein}(z) = \gamma + \log z + E_1(z)$, valid for $z > 0$, yields

$$(W_r)_0(z) = e^z(\gamma + \log z - r), \qquad (W_r)_0(1) = e(\gamma - r).$$

If γ is transcendental, the antecedent in (2) is false for every algebraic r, so the implication is true. Conversely, if γ is algebraic, choose $r = \gamma$. Its antecedent then holds, but its conclusion contradicts the preceding proposition. The rational version is identical. □

This establishes an exact limitation of the proposed analytic-to-arithmetic transfer. A removable analytic zero does not automatically preserve arithmetic coefficient conditions. For this explicit family, establishing that preservation is equivalent to establishing the desired arithmetic nature of γ. This is an equivalence and obstruction theorem, not a proof of the conjecture. It must not be counted as a solved previously open problem.

I.9.2 A field-theoretic constraint on coefficient searches

Let K denote the algebraic closure of $\mathbb{Q}(e)$ inside $\mathbb{C}$. The known algebraic independence of e, η implies $\eta \notin K$.

Proposition I.9.3 (At most one exceptional coefficient)**.** *There is at most one $a \in K$ for which $\gamma + a\delta \in K$. Moreover $a = 1/e$ is not exceptional. If an exceptional a_0 exists, then $\delta \notin K$ and, for every $a \in K \setminus \{a_0\}$, $\gamma + a\delta$ is transcendental over K.*

Proof. If $a \neq b$ and both combinations belong to K, subtraction gives $\delta \in K$, and then $\gamma \in K$. Consequently $\eta = \gamma + \delta/e \in K$, a contradiction. The identity $\gamma + \delta/e = \eta \notin K$ excludes $1/e$. If an exceptional a_0 exists and δ were in K, the same contradiction would follow. For $a \neq a_0$, the first assertion excludes membership in K; since K is algebraically closed, this is equivalent to transcendence over K. □

This elementary consequence considerably narrows the possible exceptional coefficient set, but does not identify it. In particular, the method does not rule out $a_0 = 0$, which is exactly the obstruction to deciding whether γ is algebraic over $\mathbb{Q}(e)$. No novelty or open-problem resolution is asserted for this field-theoretic deduction.

I.10 What moment positivity can and cannot decide

The Gumbel distribution provides a direct probability model for γ. The following exact translation calculation explains a limitation of an arithmetic strategy that fixes the higher cumulants and tests only moment positivity.

Let T have density e^{-t} on $(0, \infty)$ and put $X = -\log T$. For real $t < 1$,

$$\mathbb{E}e^{tX} = \Gamma(1-t), \qquad \log\Gamma(1-t) = \gamma t + \sum_{k\geq 2} \frac{\zeta(k)}{k} t^k \quad (|t| < 1).$$

Thus its cumulants are $\kappa_1 = \gamma$ and $\kappa_k = (k-1)!\zeta(k)$ for $k \geq 2$. For an arbitrary real number g define $X_g = X + g - \gamma$ and $\mu_j(g) = \mathbb{E}X_g^j$. Its first cumulant is g, while all higher cumulants are unchanged.

Theorem I.10.1 (Translation invariance of the leading Hankel determinants)**.** *For $N \geq 0$, put $M_N(g) = (\mu_{i+j}(g))_{0\leq i,j\leq N}$. Every $M_N(g)$ is positive definite, for every $g \in \mathbb{R}$. Moreover,*

$$\det M_N(g) = \det M_N(\gamma) > 0.$$

Consequently the moment-positivity conditions with these fixed higher cumulants cannot distinguish a rational, algebraic irrational, or transcendental value of the first cumulant.

Proof. Set $h = g - \gamma$ and let $T_N(h)$ be the lower triangular matrix whose (i, j) entry is $\binom{i}{j}h^{i-j}$ for $j \leq i$, and zero otherwise. The binomial theorem gives

$$(1, X_g, \ldots, X_g^N)^{\mathsf{T}} = T_N(h)(1, X, \ldots, X^N)^{\mathsf{T}}.$$

Taking Gram matrices yields $M_N(g) = T_N(h)M_N(\gamma)T_N(h)^{\mathsf{T}}$. All diagonal entries of $T_N(h)$ equal one, so its determinant is one, which proves determinant invariance. For a nonzero real polynomial p of degree at most N, the continuous everywhere-positive density of X_g implies $\mathbb{E}p(X_g)^2 > 0$. This proves positive definiteness. The construction works for every real g, proving the final assertion within the stated class of tests. □

For example, the first two nontrivial determinants are

$$\det M_1 = \zeta(2), \qquad \det M_2 = 2\zeta(2)^3 + 6\zeta(2)\zeta(4) - 4\zeta(3)^2 > 0.$$

The statement is not that every minor is invariant: the one-entry minor $\mu_2(g) = g^2 + \zeta(2)$ already varies with g. Nor does it exclude arithmetic information from integral representations, denominator control, or additional identities tying the mean to the higher cumulants. It excludes only an inference from positivity alone after free translation of the first cumulant.

I.11 What remains to reach the arithmetic target

The strongest transfers in this manuscript use structures that survive exactly: a Wronskian identity survives as an integral Laurent recurrence; a radial heat kernel exposes the dominant sign of asymptotic coefficients; and a stationary optimization formula transports algebraic independence of E-values to parameter derivatives. Each transfer includes the relevant normalization and its proof.

None produces a nonzero integer linear form in $1, \gamma$ tending to zero. The positive-filter theorem identifies a quantitative denominator obstruction for one broad approximation family. The mixed Gevrey theorem identifies a restricted zero-removal statement that is already equivalent to the arithmetic target. The Gumbel calculation shows why moment positivity alone cannot supply the missing arithmetic input.

For future work, three concrete mathematical gaps remain useful targets:

1. Find and prove denominator cancellation beyond factorial clearing for sign-changing or multivariate Euler–Gompertz filters, while retaining a strictly nonzero, sufficiently small remainder.
2. Determine whether the exact BGG coefficient recurrences admit additional congruences or root-location theorems that constrain rational approximants, rather than only the asymptotic series coefficients.
3. Extend the stochastic parameter-jet correspondence to other E-function families with established functional independence, and separately determine whether any singular parameter limit preserves enough arithmetic information to reach γ itself.

These are research directions, not claims that the remaining conjectures have been settled. The present document is a research draft with complete arguments for its stated results and reproducible finite certificates; it has not received external peer review. No journal-level or prize-level assessment follows from the number of its theorems.

I.11.1 Verification and reproducibility

All source targets were checked against the primary text, with the coefficient conventions retained. The finite sign certificates use exact integer or rational arithmetic and are embedded in this document, so the single accompanying `.tex` file is self-contained. For the product coefficients, the scalar certificate was independently compared with the nonlinear Laurent recurrence through degree 143. Both methods gave identical integers. Symbolic recurrence proofs, rather than finite tests, establish the denominator and congruence assertions at every degree.

The mathematical arguments were separately checked in parallel research passes. Such checks are useful error detection, not external validation. The source ledger deliberately separates numbered conjectures, unnumbered observations, recovered project results, and new extensions. The requested ten new solutions of existing open questions have not been established.

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Part II
Arithmetic transfers II: units, parameter jets, and source assertions

Source: `KEIO_Euler_Arithmetic_Frontiers_20261018.tex`.

Original PDF: 35 pages. Mathematical content and source-level qualifications are retained.

Abstract

We classify which positive rational-parameter Kummer values at positive algebraic arguments are units in the ring of E-values, using the recent phase morphism of Fischler–Rivoal. At argument one the only cases are the exponential-polynomial family and $M(2,3,1) = 2$. The same morphism gives a denominator obstruction for $\mathrm{Ein}(1) = \gamma + \delta/e$. Independent transfers give algebraically independent dimension derivatives of a Bessel stopping boundary and differential transcendence over $\mathbb{C}(d)$. We also supply proofs or counterexamples for ten distinct assertions left unproved in precisely identified source versions: a Whittaker real-zero branch conjecture, Vargas's exterior-zero conjecture, Görgens' local-minimum conjecture and two stated endpoint expectations, a sharp digamma exponent question, two Gurland-ratio questions, Bolbachan's digamma-limit conjecture, and an odd-totient sign question. The last limit conjecture fails exponentially, with growth rate $1/\log 2$. Further results sharpen periodic-Dirichlet descent bounds. The source count includes minor questions and preprints; it is not a claim of ten major or historically first breakthroughs. Current priority is not certified, especially where a revised journal version could not be inspected. Neither irrationality nor transcendence of γ is proved.

Keywords: Euler–Mascheroni constant; Euler–Gompertz constant; E-function; Kummer function; optimal stopping; differential transcendence; digamma inequality; exponential-integral sections; periodic Dirichlet series.

II.1 What is proved, and what the count means

Write

$$\gamma = \lim_{n\to\infty}\left(\sum_{k=1}^{n}\frac{1}{k} - \log n\right), \qquad \delta = \int_0^\infty \frac{e^{-t}}{1+t}\,dt, \qquad \eta = \mathrm{Ein}(1) = \gamma + \delta/e.$$

This second research stage starts from the arithmetic of these constants and transfers to special-function differential equations, stochastic stopping, complex zeros, convexity, and periodic Dirichlet series. The strongest arithmetic statements concern the *ring* containing η and the finite jets of optimized values. They do not separate the arithmetic of γ from that of δ.

The following ledger contains ten distinct mathematical targets stated as conjectures, open questions, or unproved expectations in the cited source versions. For each target the manuscript gives a full proof or counterexample. Two equivalent convergence formulations in one source are counted once. Parameter cases and endpoints of one conjecture count once; corollaries count only when separately posed in a source. No result from the preceding research stage is reused to increase this count.

ID	Exact source target	Answer and proof location
Q1	Boussaada–Mazanti–Niculescu [II.4, p. 6, after Fig. 1]: maximal domain and endpoint limits of a real Whittaker zero branch.	Proved by a Sturm identity; a sharper Bessel-scaled endpoint is obtained. Theorem II.4.1.
Q2	Vargas [II.5, Conjecture 5.1]: all finite limits of normalized Taylor-section zeros lie on the proposed curves.	False for the stated signed/complex-integrand class. Any prescribed point outside the unit disk can occur. Theorem II.5.1.
Q3	Görgens [II.6, Section 4]: the observed local minimum occurs exactly at $\alpha = 1/2$.	Proved by a two-sided expansion with a positive quadratic coefficient. Theorem II.6.1.

ID	Exact source target	Answer and proof location
Q4	Görgens [II.6, Section 4]: the optimal stopping time tends to 1 almost surely as $\alpha \downarrow 0$.	Proved under the natural common-Brownian coupling. Theorem II.6.3. This is an elementary stated expectation.
Q5	Görgens [II.6, Introduction]: the value tends to zero as $\alpha \to \infty$.	Proved, with the stronger rate $V(\alpha) \sim \sqrt{\log \alpha/\alpha}$ and second-order squared asymptotics. Theorems II.6.3 and II.6.5. One expectation, not two targets.
Q6	Qi–Lim–Nantomah [II.7, Open Problem 3]: the largest exponent range for a reciprocal digamma inequality.	Exactly $[-1, B)$, where $B \approx 5.97283686$ is characterized by a proved unique tangency point. Theorem II.9.1.
Q7	Wiśniewska [II.9, Open Problems 1 and 2]: global convergence and equality of the Gurland-ratio series.	Both equivalent assertions hold for all positive arguments, with a geometric remainder bound. Section II.11; counted once.
Q8	Wiśniewska [II.9, Open Problem 3]: location of the mean-value parameter and its distance from the midpoint.	Sharp bound $t > g + \frac{2}{3}(s - g)$, plus a sharper ratio-dependent bound and exact scaling limits. Theorem II.11.2.
Q9	Bolbachan [II.10, Conjecture 1.4]: a Bernoulli–Stirling limit represents the digamma function.	False as printed: the approximants diverge, with exponential rate $1/\log 2$. Theorem II.12.1.
Q10	Bharadwaj [II.11, Section 8, pp. 24–25]: existence of an odd N with a signed totient sum equal to zero.	Explicit nontrivial answer $N = 105$, also the least odd $N > 1$. Section II.14.

The distinction between a proof and a priority claim. The ledger is a count of resolved *source assertions*, not a certification that ten problems were still globally open immediately before this work. The proofs were checked internally against the stated hypotheses; they have not undergone external peer review or formal proof-assistant verification. Searches and source comparisons did not locate earlier resolutions of the listed exact assertions, but cannot establish historical priority. In particular, the revised journal text of Bharadwaj's paper was not accessible; our question match and numerical comparison concern its 2022 arXiv version. Bolbachan's source is a 2011 preprint prepared for a high-school mathematical conference. Görgens' two endpoint expectations and the totient example are modest results. Thus this document does not certify the stronger request for ten new major breakthroughs, or an Annals/Fields-level achievement.

Additional results excluded from that count. The complete positive Kummer unit classification (Theorem II.2.1) is a family-wide verification of a larger units conjecture, not a solution of the full conjecture. The rational-denominator obstruction (Theorem II.3.1), boundary-jet algebraic independence, and differential-transcendence statements are additional transfers. The adaptive descent bound improves an explicitly suggested direction rather than closing the full Erdős conjecture. The higher-polygamma range in Section II.10 has an earlier public claim and is excluded. The value-jet proposition is repeated only to make the later proof self-contained.

How the fields connect. The phase morphism converts a special value into a finite exponential polynomial, so algebraic factorization detects forbidden reciprocals. Kummer differential equations also encode stopping boundaries; parameter expansion then exposes a tower of arithmetically independent constants. Sturm identities control real zeros, while reversing Taylor sections reveals complex zeros missed by a proposed limiting curve. Positivity and curvature settle the sharp inequality questions. In the periodic Dirichlet setting, mean zero cancels γ and leaves exact prime-logarithm obstructions. Each transfer has a different conclusion; none is an implicit proof of the original irrationality problem.

II.2 Units throughout a positive Kummer family

Let $\mathcal{E}$ be the ring of values of Siegel E-functions at algebraic arguments. Fischler and Rivoal, *The ring of values of E-functions is not a field*, arXiv:2610.16603v1 (12 October 2026; manuscript dated 14 October 2026), Conjecture 1.2, propose

$$\mathcal{E}^\times = \{\alpha e^\beta : \alpha, \beta \in \overline{\mathbb{Q}},\ \alpha \neq 0\}.$$

Their Theorem 1.1 supplies nonunits $M(a,b,1)$ when $0 < a < 1$ and $0 < b - a < 1$ are rational. Here

$$M(a,b,z) = {}_1F_1(a;b;z) = \sum_{j\geq 0} \frac{(a)_j}{(b)_j j!} z^j.$$

We classify units in this family for *every* positive rational a, b and every positive algebraic argument. This is a restricted-family result concerning Conjecture 1.2, not a proof of that conjecture. Rivoal–Roques, *E-functions of order 2 and units of E-values* (2015), Theorem 5, does not already give this exclusion: it requires the function and the witness for its inverse value to be components of one joint two-dimensional differential system. Here an inverse is excluded even if its possible E-function witness has arbitrarily large order.

Theorem II.2.1 (Positive Kummer unit classification)**.** *Let $a, b \in \mathbb{Q}_{>0}$ and $x \in \overline{\mathbb{Q}} \cap \mathbb{R}_{>0}$. For $n \geq 1$ put*

$$A_{n,b}(z) = \frac{1}{(n-1)!} \sum_{j=0}^{n-1} \frac{(b-n)_j (1-n)_j}{j!} z^{n-1-j}.$$

Then $M(a,b,x)$ is a unit of $\mathcal{E}$ if and only if at least one of the following conditions holds:

1. *$a - b \in \mathbb{Z}_{\geq 0}$;*
2. *$a = n \in \mathbb{Z}_{>0}$ and $A_{n,b}(x) = 0$.*

In the first case the value is a nonzero algebraic multiple of e^x. In the second case it is a positive algebraic number. In particular,

$$M(a,b,1) \in \mathcal{E}^\times \quad \Longleftrightarrow \quad a - b \in \mathbb{Z}_{\geq 0} \quad \text{or} \quad (a,b) = (2,3).$$

The exceptional value is $M(2,3,1) = 2$.

Proof. We use the injective phase morphism of Fischler–Rivoal, Proposition 3.1. It sends an E-value $f(1)$ to the exponential polynomial

$$\Phi_\theta(f(1))(X) = \sum_\rho f_\rho(1) e^{\rho X},$$

where $f(z) = \sum_\rho f_\rho(z) e^{\rho z}$ is its canonical large-sector, Gevrey-order-one decomposition, in a direction just above θ. It fixes algebraic constants and sends e^α to $e^{\alpha X}$. A unit has only one nonzero phase, by their Corollary 3.3 (or by the units of the ring of exponential polynomials). We apply it to the E-function $z \mapsto M(a,b,xz)$.

The standard Kummer connection formula (DLMF 13.2.41), on the upper side of the positive real axis, is

$$M(a,b,z) = \frac{\Gamma(b) e^{i\pi a}}{\Gamma(b-a)} U(a,b,z) + \frac{\Gamma(b) e^{-i\pi(b-a)}}{\Gamma(a)} e^z U(b-a, b, e^{-i\pi} z). \tag{II.2.1}$$

Both summands are the canonical terms with respective phases 0 and 1: this follows from the Gevrey asymptotic expansion $U(c,b,z) \sim z^{-c} \sum_{j\geq 0} (-1)^j (c)_j (1+c-b)_j / (j! z^j)$ in the usual sectors, together with uniqueness of the large-sector decomposition. For clarity, a sector bisected by a

sufficiently small positive angle has opening greater than π and lies in both relevant U-asymptotic sectors; hence the choice matches Φ_0. After replacing z by xz, write

$$\Phi_0(M(a,b,x))(X) = C_0 + C_1 e^{xX}.$$

Its constant coefficient is

$$C_0 = \frac{\Gamma(b)e^{i\pi a}}{\Gamma(b-a)} U(a,b,x). \tag{II.2.2}$$

The integral (DLMF 13.4.4)

$$U(a,b,x) = \frac{1}{\Gamma(a)} \int_0^\infty e^{-xt} t^{a-1} (1+t)^{b-a-1}\, dt$$

is strictly positive for the parameters in the theorem.

First suppose $a-b = m \in \mathbb{Z}_{\geq 0}$. Kummer's transformation gives $M(a,b,x) = e^x M(-m,b,-x)$, an algebraic multiple of e^x. It is nonzero because the defining series for $M(a,b,x)$ is strictly positive. Such values and their reciprocals belong to $\mathcal{E}$.

Suppose henceforth that $b-a \notin \mathbb{Z}_{\leq 0}$, so that $C_0 \neq 0$. If a is not an integer, C_0 has nonzero imaginary part, since $\Gamma(b)/\Gamma(b-a)$ is real and nonzero and $\sin(\pi a) \neq 0$. But $M(a,b,x)$ is real and $M(a,b,x) = C_0 + C_1 e^x$. Thus $C_1 \neq 0$. The two nonzero phases exclude a unit.

It remains to consider $a = n \in \mathbb{Z}_{>0}$. The phase-1 formal series terminates because $(1-n)_j = 0$ for $j \geq n$; its sum is therefore exact. Equivalently, substituting the terminating U-expansion into (II.2.1) gives

$$C_1 = \Gamma(b) x^{1-b} A_{n,b}(x). \tag{II.2.3}$$

Consequently $A_{n,b}(x) \neq 0$ implies two nonzero phases and excludes a unit.

To treat $A_{n,b}(x) = 0$ without making an unsupported arithmetic claim about U-values, use the contiguous identity (DLMF 13.3.1)

$$nM(n+1,b,z) = (b-n)M(n-1,b,z) + (2n-b+z)M(n,b,z).$$

It gives polynomials $B_{n,b} \in \mathbb{Q}[z]$ such that

$$M(n,b,z) = A_{n,b}(z)M(1,b,z) + B_{n,b}(z). \tag{II.2.4}$$

Here the initial data are $A_{0,b} = 0$, $A_{1,b} = 1$, $B_{0,b} = 1$, $B_{1,b} = 0$, and both sequences satisfy

$$nY_{n+1,b} = (b-n)Y_{n-1,b} + (2n-b+z)Y_{n,b}.$$

The displayed explicit formula for $A_{n,b}$ satisfies this recurrence; alternatively it follows by comparing phase-1 coefficients with $M(1,b,z)$, whose phase-1 coefficient is $\Gamma(b)z^{1-b}$. If $A_{n,b}(x) = 0$, equation (II.2.4) shows that the value is $B_{n,b}(x) \in \overline{\mathbb{Q}}$. Positivity again makes it nonzero. This proves the general classification.

For $x = 1$, the polynomial

$$P_n(b) = (n-1)!A_{n,b}(1) = \sum_{j=0}^{n-1} (-1)^j \binom{n-1}{j} (b-n)_j$$

has integer coefficients and leading coefficient $(-1)^{n-1}$. For $n = 1$ it is the constant 1. For $n \geq 2$ its rational roots are integers. Thus a zero with rational $b > 0$ requires $b = m \in \mathbb{Z}_{>0}$. The case $m \leq n$ is already covered by the first alternative, so assume $q = m - n \geq 1$, and put $k = n-1$. If T_q has the gamma density $e^{-t}t^{q-1}/\Gamma(q)$ on $(0,\infty)$, then

$$P_n(m) = \mathbb{E}(1-T_q)^k.$$

For even k this is strictly positive. For odd k, the function $t \mapsto (1-t)^k$ is strictly decreasing. Coupling $T_q = T_1 + T_{q-1}$ with independent positive summands proves, for $q > 1$,

$$\mathbb{E}(1-T_q)^k < \mathbb{E}(1-T_1)^k = \sum_{j=0}^{k} (-1)^j \binom{k}{j} j! = (-1)^k\, !k,$$

where $!k$ is the number of derangements of k elements. For odd $k \geq 3$, $!k > 0$; the same formula handles $q = 1$. For $k = 1$ the expectation is $1 - q$ and vanishes exactly when $q = 1$. Hence the only further zero is $n = 2, m = 3$. Finally $M(2, b, z) = (z + 2 - b)M(1, b, z) + b - 1$, giving $M(2, 3, 1) = 2$. □

Remark II.2.2 (A genuine exceptional curve at variable argument). For every rational $b > 2$ the algebraic point $x = b - 2$ satisfies

$$M(2, b, b-2) = b - 1.$$

Thus a blanket statement that non-polynomial Kummer values are nonunits would be false. The second alternative in Theorem II.2.1 is essential.

II.3 A denominator obstruction involving Euler's constant

Write

$$\mathrm{Ein}(z) = \sum_{n \geq 1} \frac{(-1)^{n-1} z^n}{n\, n!}, \qquad \eta = \mathrm{Ein}(1) = \gamma + \frac{\delta}{e}, \qquad \delta = \int_0^\infty \frac{e^{-t}}{1+t}\, dt.$$

Thus $\eta \in \mathcal{E}$ and $\gamma, \delta > 0$. The identity on the positive real axis

$$\mathrm{Ein}(z) = \gamma + \log z + E_1(z), \qquad E_1(z) = \int_z^\infty \frac{e^{-t}}{t}\, dt$$

and the Gevrey expansion $e^z E_1(z) \sim \sum_{j \geq 0} (-1)^j j! z^{-j-1}$ show that its canonical phases are 0 and -1. In particular, Fischler–Rivoal's morphism gives

$$\Phi_0(\eta)(X) = \gamma + \delta e^{-X}. \tag{II.3.1}$$

The logarithm in the phase-0 term vanishes when its coefficient function is evaluated at $z = 1$. Both coefficients in (II.3.1) are nonzero, so η is not a unit of $\mathcal{E}$. More generally:

Theorem II.3.1 (Rational denominators in Ein(1)). *Let $P, Q \in \overline{\mathbb{Q}}[Y]$ be relatively prime, $P \neq 0$, $Q \neq 0$, and assume $Q(\eta) \neq 0$. If*

$$\frac{P(\eta)}{Q(\eta)} \in \mathcal{E},$$

then either Q is constant or $Q(Y) = c(Y - \gamma)^m$ for some $c \neq 0$ and $m \geq 1$. In particular,

$$\frac{1}{\gamma + \delta/e} \notin \mathcal{E}.$$

More generally, $1/(\eta + \alpha) \notin \mathcal{E}$ for every nonnegative real algebraic α. If γ is transcendental, the exact intersection is

$$\overline{\mathbb{Q}}(\eta) \cap \mathcal{E} = \overline{\mathbb{Q}}[\eta].$$

The final assertion is explicitly conditional; no irrationality or transcendence claim for γ is being made.

Proof. Let $\xi = P(\eta)/Q(\eta) \in \mathcal{E}$. Apply the ring morphism to $Q(\eta)\xi = P(\eta)$ and use (II.3.1). As an identity of meromorphic functions,

$$\Phi_0(\xi)(X) = \frac{P(\gamma + \delta e^{-X})}{Q(\gamma + \delta e^{-X})}.$$

We use the elementary fact that a rational function of $T = e^{-X}$ which is a finite exponential polynomial in X must be a Laurent polynomial in T. Indeed the rational expression is $2\pi i$-periodic. If its exponential-polynomial expression is $\sum_\rho c_\rho e^{\rho X}$ with distinct phases and nonzero coefficients, periodicity and linear independence of exponentials give $e^{2\pi i\rho} = 1$ for every ρ. Hence $\rho \in \mathbb{Z}$, which proves the assertion.

Consequently $P(\gamma + \delta T)/Q(\gamma + \delta T)$ has no finite pole except possibly at $T = 0$. The numerator and denominator remain relatively prime under the invertible affine substitution $Y = \gamma + \delta T$ because $\delta \neq 0$. Therefore every root of Q is γ. This gives the stated shape of Q. For $Q(Y) = Y + \alpha$, its root $-\alpha$ differs from the positive number γ, proving the explicit exclusions. If γ is transcendental, a nonconstant polynomial over $\overline{\mathbb{Q}}$ cannot have it as a root; hence Q must be constant. The reverse inclusion in the intersection statement follows from $\eta \in \mathcal{E}$ and the ring property. □

Exact scope and provenance. The Kummer classification is a new extension derived here from the published phase morphism and classical Kummer identities. The Ein result is a new application of the same morphism. Neither is the full units conjecture, nor Fischler–Rivoal's $\mathcal{E} \cap \mathcal{D} = \overline{\mathbb{Q}}$ conjecture, nor a resolution of the irrationality of γ or δ. Multiple parameter choices or denominators are not counted as independent published open questions.

Primary references. S. Fischler and T. Rivoal, *The ring of values of E-functions is not a field*, https://arxiv.org/abs/2610.16603, Proposition 3.1, Corollary 3.3, Conjecture 1.2 and Theorem 1.1. The phase convention is also displayed explicitly in their Section 4.3 (printed page 12). Compare T. Rivoal and J. Roques, *E-functions of order 2 and units of E-values*, https://rivoal.perso.math.cnrs.fr/articles/unitse.pdf, Theorem 5 (printed page 3). The relevant special-function formulas are https://dlmf.nist.gov/13.2.E39, https://dlmf.nist.gov/13.2.E41, https://dlmf.nist.gov/13.3.E1, https://dlmf.nist.gov/13.4.E4, and https://dlmf.nist.gov/13.7.E3.

II.4 The real-zero branch conjecture for Whittaker functions

Boussaada, Mazanti and Niculescu, *Some remarks on the location of non-asymptotic zeros of Whittaker and Kummer hypergeometric functions*, Bulletin des Sciences Mathématiques **174** (2022), 103093, arXiv:2106.03378, conjecture immediately after Figure 1 that the real-zero branch of $M_{k,l}$ passing through $(k, z) = (l + 3/2, 2l + 1)$ has maximal parameter interval $(l + 1/2, \infty)$, and tends to infinity and zero at its respective endpoints. The following proof establishes this statement for every $l > -1/2$, and gives a sharper right-endpoint limit.

Theorem II.4.1. *Fix $b > 0$. For every $\lambda > 0$, the function $y_\lambda(x) = {}_1F_1(-\lambda; b; x)$ has a first positive zero $\rho_b(\lambda)$. This zero is simple, depends real analytically on λ, and satisfies*

$$\rho_b'(\lambda) = -\frac{\displaystyle\int_0^{\rho_b(\lambda)} x^{b-1}e^{-x}y_\lambda(x)^2\,dx}{\rho_b(\lambda)^b e^{-\rho_b(\lambda)} y_\lambda'(\rho_b(\lambda))^2} < 0.$$

Moreover,

$$\lim_{\lambda\downarrow 0} \rho_b(\lambda) = \infty, \qquad \lim_{\lambda\to\infty} \lambda\rho_b(\lambda) = \frac{j_{b-1,1}^2}{4},$$

where $j_{\nu,1}$ is the first positive zero of J_ν. Consequently, for $l > -1/2$, the branch in the cited conjecture is

$$z_l(k) = \rho_{2l+1}(k - l - 1/2), \qquad k > l + 1/2.$$

Proof. Set $p(x) = x^b e^{-x}$ and $w(x) = x^{b-1}e^{-x}$. Kummer's equation is

$$(py'_\lambda)' + \lambda w y_\lambda = 0, \qquad y_\lambda(0) = 1,$$

and integration from zero gives

$$p(x)y'_\lambda(x) = -\lambda \int_0^x w(t)y_\lambda(t)\,dt.$$

If y_λ had no positive zero, it would remain positive. Its integral on any fixed initial interval would then be a positive constant, and for all sufficiently large x the last identity would imply $y'_\lambda(x) \le -Ce^x x^{-b}$ with $C > 0$. Integration would force y_λ to become negative, a contradiction. At the first zero the same identity gives $y'_\lambda < 0$, so that zero is simple.

The Kummer series is jointly analytic in λ and x for fixed $b > 0$. The implicit function theorem, together with positivity on each closed subinterval before the first zero, shows that the first-zero function is locally real analytic. These local branches agree and cover $(0, \infty)$.

Write $v = \partial_\lambda y_\lambda$. Differentiating the equation and forming the Lagrange identity gives

$$[p(y_\lambda v' - v y'_\lambda)]' = -w y_\lambda^2.$$

The boundary term at zero vanishes because $b > 0$ and the functions and their derivatives are bounded there. At $\rho = \rho_b(\lambda)$ the identity therefore reads $p(\rho)v(\rho)y'_\lambda(\rho) = \int_0^\rho w y_\lambda^2$. Implicit differentiation of $y_\lambda(\rho) = 0$ proves the displayed strictly negative derivative formula.

As $\lambda \downarrow 0$, $y_\lambda \to 1$ uniformly on every compact interval, proving the first limit. For the second limit, the defining series gives, uniformly on compact subsets of the complex t-plane,

$$y_\lambda(t/\lambda) \longrightarrow {}_0F_1(;b;-t) = \Gamma(b)t^{(1-b)/2}J_{b-1}(2\sqrt{t}).$$

For completeness, when $\lambda \ge 1$, $|(-\lambda)_m|/\lambda^m \le m!$, so the summands are dominated on $|t| \le T$ by $T^m/(b)_m$, a summable sequence. The limiting function is positive before $j_{b-1,1}^2/4$ and changes sign at that simple zero. Uniform convergence therefore locates the first zero of the approximants between any two sufficiently close points on its opposite sides. This proves the asserted scaled limit.

Finally, $M_{k,l}(x) = e^{-x/2}x^{l+1/2}{}_1F_1(l + 1/2 - k; 2l + 1; x)$ for $x > 0$. At $\lambda = 1$ the Kummer function is $1 - x/b$, so the branch passes through the specified point. For $\lambda \le 0$ all its series coefficients are nonnegative and its constant term is one; there are no positive zeros. Since the branch diverges as $\lambda \downarrow 0$, its maximal interval as a finite real zero branch is exactly the one stated. □

Status and scope. This is one conjecture, with two endpoint assertions, and should be counted as one problem. The proof uses classical Sturm–Liouville identities. Recent related spectral results include R. Vanlaere, arXiv:2511.20025v4 (2026), Theorem 1.6, for zeros in the first Kummer parameter when $b \ge 1$. They do not state the branch conjecture above. No claim of priority beyond the checked sources is implicit here.

II.5 An exterior-zero counterexample to a conjecture on exponential integrals

Vargas, *Limit curves for zeros of sections of exponential integrals*, Constructive Approximation **40** (2014), 219–239, arXiv:1302.3695v4, Conjecture 5.1, asserts that every finite limit point of the

zeros of $s_n[F](nz)$ lies on $D_{a,b} \cup D_{\text{imag}}$. Here $F(z) = \int_{-a}^{b} \phi(t) e^{zt}\, dt$ belongs to the class defined in Section 2 of that paper, and $s_n[F]$ denotes its Taylor polynomial of degree n. Signed and complex-valued integrands are expressly allowed. For $a = b = 1$ the proposed set is contained in $\{|z| \le 1\}$.

Theorem II.5.1. *For every complex c with $|c| > 1$, let*

$$F_c(z) = \int_{-1}^{1} (1 + ct) e^{zt}\, dt.$$

There are zeros z_n of $s_n[F_c](nz)$, indexed by all sufficiently large even integers, such that $z_n \to -c$. Thus any prescribed point strictly outside the unit disk can occur as a zero limit. In particular, $c = 3/2$ is a real signed-integrand counterexample to Vargas's Conjecture 5.1.

Proof. The integrand is integrable and analytic, with nonzero endpoint values $1 - c$ and $1 + c$. It satisfies the source's hypotheses with $a = b = 1$, $\mu = \nu = 0$, $f_1(u) = 1 - c + cu$ and $f_2(u) = 1 + c - cu$. The extra noncancellation condition (5) of that paper is also satisfied: its expression equals 2 for even indices and $2c$ for odd indices.

Its moments are

$$m_k = \int_{-1}^{1} (1 + ct) t^k\, dt = \begin{cases} 2/(k+1), & k \text{ even}, \\ 2c/(k+2), & k \text{ odd}. \end{cases}$$

For even n, reverse the normalized Taylor polynomial:

$$Q_n(z) := \frac{n!}{m_n (nz)^n} s_n[F_c](nz) = \sum_{j=0}^{n} \frac{m_{n-j}}{m_n} \frac{n!}{(n-j)! n^j} z^{-j}.$$

For fixed j, the coefficient tends to 1 if j is even, and to c if j is odd. Also,

$$0 < \frac{n!}{(n-j)! n^j} \le 1, \qquad \left| \frac{m_{n-j}}{m_n} \right| \le \max(1, |c|)(j+1),$$

where the latter inequality follows from $(n+1)/(n-j+1) \le j+1$ for $0 \le j \le n$. Dominated convergence of the geometric majorant shows that, locally uniformly on $|z| > 1$,

$$Q_n(z) \longrightarrow \sum_{h \ge 0} z^{-2h} + c \sum_{h \ge 0} z^{-2h-1} = \frac{1 + c/z}{1 - z^{-2}}.$$

The last function has a simple zero at $z = -c$, lying strictly outside the unit disk. Choose a small disk about $-c$ contained in $|z| > 1$. Rouché's theorem gives exactly one zero of Q_n in that disk for all sufficiently large even n, and these zeros converge to $-c$. The normalizing factor is nonzero there, so they are zeros of $s_n[F_c](nz)$ as well. This contradicts the stated conjecture. □

Scope. This disproves the conjecture for the signed-integrand class actually specified in its source. It does not settle the source's separate Conjecture 5.2 on positive angular densities. Nor is it an assertion about nonnegative integrands. The construction produces a tunable isolated exterior zero, a phenomenon not detected by the limiting normalized zero-counting measure.

II.6 A conjecture about the scaling of a Brownian bridge

This section addresses an independently posed conjecture. In [II.6, Section 4], Görgens explicitly conjectures that the optimal expected reward of an α-Brownian bridge has a local minimum at $\alpha = 1/2$, and reports that he could not prove it. The same paper states two further limiting

expectations, in its introduction and Section 4. We prove the local-minimum assertion and both limits below. The latter are elementary accompanying results; they should not be presented as two major conjectures. A search locating no later proof is not a proof of priority, and the newness of these arguments remains to be checked by specialists.

Let W be a standard Brownian motion and, for $\alpha > 0$, let

$$dX_t^{(\alpha)} = dW_t - \frac{\alpha X_t^{(\alpha)}}{1-t}\,dt, \qquad X_0^{(\alpha)} = X_1^{(\alpha)} = 0.$$

Set

$$V(\alpha) = \sup_{0 \le \tau \le 1} \mathbb{E} X_\tau^{(\alpha)}.$$

Görgens' established solution has an optimal threshold $B(\alpha)\sqrt{1-t}$. Write $\varepsilon = 2\alpha - 1$, and normalize the positive increasing continuation solution by $h_\varepsilon(0) = 1$. Then

$$h_\varepsilon''(y) - \varepsilon y h_\varepsilon'(y) - h_\varepsilon(y) = 0, \qquad B h_\varepsilon'(B) = h_\varepsilon(B), \qquad V(\alpha) = \frac{B}{h_\varepsilon(B)}. \tag{II.6.1}$$

At $\varepsilon = 0$ these are $h_0(y) = e^y$, $B = 1$, and $V(1/2) = e^{-1}$.

Theorem II.6.1 (The local-minimum conjecture)**.** *For real $u \to 0$ from either side,*

$$B(\tfrac{1}{2} + u) = 1 - \frac{u}{2} + \frac{5u^2}{8} + O(|u|^3), \tag{II.6.2}$$

$$V(\tfrac{1}{2} + u) = \frac{1}{e} + \frac{u^2}{12e} + O(|u|^3). \tag{II.6.3}$$

In particular $\alpha = 1/2$ is a strict local minimum of V.

Proof. The initial slope obtained from the explicit Kummer formula in [II.6, Theorem 1] is

$$h_\varepsilon'(0) = \begin{cases} \sqrt{2\varepsilon}\,\dfrac{\Gamma(x + 1/2)}{\Gamma(x)}, & \varepsilon > 0, \quad x = (2\varepsilon)^{-1}, \\[2mm] \sqrt{-2\varepsilon}\,\dfrac{\Gamma(x+1)}{\Gamma(x+1/2)}, & \varepsilon < 0, \quad x = -(2\varepsilon)^{-1}. \end{cases} \tag{II.6.4}$$

Stirling's formula with remainder gives, as $x \to +\infty$,

$$\log \frac{\Gamma(x+1/2)}{\Gamma(x)} = \frac{1}{2}\log x - \frac{1}{8x} + O(x^{-3}),$$

and hence

$$\frac{\Gamma(x+1/2)}{\Gamma(x)\sqrt{x}} = 1 - \frac{1}{8x} + \frac{1}{128x^2} + O(x^{-3}).$$

Substitution in the first case of (II.6.4), and inversion of the same expansion in the second case using $\Gamma(x+1) = x\Gamma(x)$, yield the identical two-sided expansion

$$h_\varepsilon'(0) = 1 - \frac{\varepsilon}{4} + \frac{\varepsilon^2}{32} + O(|\varepsilon|^3). \tag{II.6.5}$$

The uniform remainder here is essential: the continuation solution is selected by a condition at $y = -\infty$, so merely substituting a formal power series in its differential equation would not suffice.

On any fixed compact interval containing $[0, 1]$, standard continuous dependence for the first-order system associated with (II.6.1) shows that h_ε and its first derivative converge uniformly to e^y. Thus $r_\varepsilon = h_\varepsilon'/h_\varepsilon$ is well defined there for small $|\varepsilon|$ and satisfies

$$r_\varepsilon' + r_\varepsilon^2 - \varepsilon y r_\varepsilon - 1 = 0.$$

Let

$$p_1(y) = \frac{y}{2} - \frac{1}{4}, \qquad p_2(y) = \frac{y^2}{8} - \frac{y}{8} + \frac{1}{32}.$$

They obey

$$p_1' + 2p_1 = y, \qquad p_2' + 2p_2 = yp_1 - p_1^2,$$

with $p_1(0) = -1/4$ and $p_2(0) = 1/32$. Subtracting the Riccati equations for r_ε and $1 + \varepsilon p_1 + \varepsilon^2 p_2$, and applying the integral Gronwall inequality, proves uniformly on that compact interval that

$$r_\varepsilon(y) = 1 + \varepsilon p_1(y) + \varepsilon^2 p_2(y) + O(|\varepsilon|^3). \tag{II.6.6}$$

Indeed the approximate solution has residual $O(|\varepsilon|^3)$, its initial discrepancy has that size by (II.6.5), and all coefficients remain bounded.

At $(\varepsilon, y) = (0, 1)$ the function $yr_\varepsilon(y) - 1$ has y-derivative 1. There is consequently a root near 1 for small $|\varepsilon|$, and its y-derivative remains bounded away from zero. This is the unique positive stopping threshold in Görgens' solution. Substitution of $B = 1 + b_1\varepsilon + b_2\varepsilon^2 + O(|\varepsilon|^3)$ in $Br_\varepsilon(B) = 1$ gives $b_1 = -1/4$ and $b_2 = 5/32$. More formally, substitution of this quadratic polynomial leaves residual $O(|\varepsilon|^3)$, and the mean-value theorem gives the asserted error for the actual root.

Since $h_\varepsilon(0) = 1$,

$$\log V(\alpha) = \log B - \int_0^B r_\varepsilon(y)\, dy.$$

The necessary integrals are

$$\int_0^1 p_1(y)\, dy = 0, \qquad p_1(1) = \frac{1}{4}, \qquad \int_0^1 p_2(y)\, dy = \frac{1}{96}.$$

With $B - 1 = -\varepsilon/4 + O(\varepsilon^2)$ this yields

$$\begin{aligned} \log V(\alpha) &= -1 - \frac{1}{2}(B-1)^2 - \varepsilon p_1(1)(B-1) - \varepsilon^2 \int_0^1 p_2(y)\, dy + O(|\varepsilon|^3) \\ &= -1 + \frac{\varepsilon^2}{48} + O(|\varepsilon|^3). \end{aligned}$$

Exponentiating and setting $\varepsilon = 2u$ proves both expansions. The positive quadratic term gives strict local minimality. □

Remark II.6.2. Differentiating Stirling's expansion with its differentiated remainder also gives the ordinary derivatives $V'(1/2) = 0$ and $V''(1/2) = 1/(6e)$. The strict-minimum conclusion above requires only the explicitly proved two-sided value expansion, so it does not depend on this additional regularity assertion.

Theorem II.6.3 (Two stated limiting expectations)**.** *Under the coupling by a common Brownian motion, the optimal stopping times in [II.6, equation (10)] satisfy*

$$\tau^*(\alpha) \longrightarrow 1 \quad \text{almost surely as } \alpha \downarrow 0.$$

Furthermore

$$\sup_{0 \le t \le 1} |X_t^{(\alpha)}| \longrightarrow 0 \quad \text{almost surely and in every } L^p, \quad 0 < p < \infty, \qquad \alpha \to \infty.$$

In particular $V(\alpha) \to 0$ *as* $\alpha \to \infty$.

Proof. Integration by parts gives a common pathwise version, for $t < 1$,

$$X_t^{(\alpha)} = W_t - \alpha(1-t)^\alpha \int_0^t \frac{W_s}{(1-s)^{\alpha+1}}\, ds. \tag{II.6.7}$$

The positive kernel in the integral has total mass $1-(1-t)^\alpha \le 1$. Therefore, writing $M = \sup_{0\le s\le 1}|W_s|$, we have simultaneously for every $\alpha > 0$ and $t \in [0,1]$,

$$|X_t^{(\alpha)}| \le 2M. \tag{II.6.8}$$

The known limit $B(\alpha) \to \infty$ from [II.6, Theorem 2(i)] also follows directly from the ODE. As $\alpha \downarrow 0$, its parameter ε tends to -1 and its initial slope in (II.6.4) tends to $\sqrt{\pi/2}$. Thus the normalized solutions converge on compact y-intervals to a solution h of $h'' + yh' - h = 0$, with $h(0) = 1$. The residual $Q = h - yh'$ satisfies $Q' = -yQ$ and $Q(0) = 1$, hence $Q(y) = e^{-y^2/2} > 0$. Uniform compact convergence therefore precludes a bounded subsequence of roots $B(\alpha)$ of $h_\varepsilon - yh'_\varepsilon = 0$, proving that limit. At any threshold hit strictly before 1, continuity gives $B(\alpha)\sqrt{1-\tau^*(\alpha)} = X^{(\alpha)}_{\tau^*(\alpha)}$. Consequently, also when $\tau^*(\alpha) = 1$,

$$0 \le 1 - \tau^*(\alpha) \le \frac{4M^2}{B(\alpha)^2},$$

which proves the first assertion and, additionally, convergence in every L^p.

For the second assertion fix $\delta \in (0,1)$ and set

$$\omega_W(\delta) = \sup\{|W_t - W_s| : s,t \in [0,1],\ |t-s| \le \delta\}.$$

Rewrite (II.6.7) as

$$X_t^{(\alpha)} = (1-t)^\alpha W_t + \alpha(1-t)^\alpha \int_0^t \frac{W_t - W_s}{(1-s)^{\alpha+1}}\, ds.$$

The part of the integral with $s \ge (t-\delta)^+$ is bounded in absolute value by $\omega_W(\delta)$. For $t \ge \delta$, the kernel mass of the other part is at most

$$\left(\frac{1-t}{1-t+\delta}\right)^\alpha \le (1+\delta)^{-\alpha}.$$

The first term is at most $\omega_W(\delta)$ if $t < \delta$, and at most $(1-\delta)^\alpha M$ otherwise. Thus

$$\sup_t |X_t^{(\alpha)}| \le 2\omega_W(\delta) + M(1-\delta)^\alpha + 2M(1+\delta)^{-\alpha}.$$

First let $\alpha \to \infty$ and then $\delta \downarrow 0$. Uniform continuity of every Brownian path gives the almost-sure limit. The domination (II.6.8) and the finiteness of every moment of M give all the L^p limits. Finally, $0 \le V(\alpha) \le \mathbb{E}\sup_t |X_t^{(\alpha)}|$. □

Remark II.6.4 (Scope and arithmetic connection). These results concern a drift parameter α, not the dimension parameter of the Bessel bridge considered below. The common mechanism is parameter sensitivity of a special-function free boundary. The gamma function enters (II.6.4), and the critical value and curvature are e^{-1} and $(6e)^{-1}$. None of this proves irrationality of the Euler–Mascheroni constant. The local-minimum result resolves the precise assertion made in the cited preprint; it neither establishes a global minimum nor locates the other numerically observed critical point.

II.6.1 A sharp form of the large-drift limit

The limiting expectation in Theorem II.6.3 admits the following quantitative strengthening. It is part of the same endpoint analysis, not an additional historical question.

Theorem II.6.5 (Large-drift asymptotics)**.** *Put $\varepsilon = 2\alpha - 1$. As $\alpha \to \infty$,*

$$\varepsilon B(\alpha)^2 = 2\log\varepsilon - \log\log\varepsilon - \log(4\pi) + o(1), \tag{II.6.9}$$

$$\varepsilon V(\alpha)^2 = 2\log\varepsilon - \log\log\varepsilon - \log(4\pi) - 2 + o(1). \tag{II.6.10}$$

In particular $B(\alpha) \sim V(\alpha) \sim \sqrt{\log\alpha/\alpha}$.

Proof. For $\varepsilon > 0$ put $a = 1/\varepsilon$ and define

$$I_a(u) = \int_0^\infty s^{a-1}\exp(us - s^2/2)\,ds, \qquad F_a(u) = I_a'(u) = \int_0^\infty s^a \exp(us - s^2/2)\,ds.$$

Integration by parts in s shows $I_a'' - uI_a' - aI_a = 0$. Hence $h_\varepsilon(y) = I_a(\sqrt{\varepsilon}y)/I_a(0)$ solves (II.6.1). Its initial slope is $\sqrt{2\varepsilon}\,\Gamma((a+1)/2)/\Gamma(a/2)$, which agrees with (II.6.4); uniqueness for the ordinary initial-value problem identifies it with the normalized continuation solution. With $u = \sqrt{\varepsilon}B(\alpha)$ its threshold equation and value are therefore

$$uF_a(u) = I_a(u), \qquad V(\alpha) = B(\alpha)\frac{I_a(0)}{I_a(u)}. \tag{II.6.11}$$

Also

$$I_a(0) = 2^{a/2-1}\Gamma(a/2) \sim \frac{1}{a} \qquad (a \downarrow 0).$$

For u in any fixed bounded positive interval, $F_a(u)$ is bounded uniformly for $0 < a \le 1$, since $s^a \le 1 + s$. On the other hand $I_a(u) \ge I_a(0) \to \infty$. The root in (II.6.11) must consequently satisfy $u \to \infty$ as $a \downarrow 0$.

We require two estimates, uniform for $0 < a \le 1$ as $u \to \infty$:

$$F_a(u) = \sqrt{2\pi}\,u^a e^{u^2/2}(1 + o(1)), \qquad \int_0^u F_a(v)\,dv = \frac{F_a(u)}{u}(1 + o(1)). \tag{II.6.12}$$

For the first, completion of the square gives

$$\frac{F_a(u)}{u^a e^{u^2/2}} = \int_{-u}^\infty (1 + t/u)^a e^{-t^2/2}\,dt.$$

On bounded t-intervals the factor converges to 1 uniformly for $a \in (0, 1]$. For $t \ge 0$ it is bounded by $1 + t$ when $u \ge 1$; for $-u < t < 0$ it is bounded by 1. Gaussian domination gives the first estimate uniformly.

For completeness, the endpoint integral used in the second estimate has uniform asymptotics

$$\int_A^u v^a e^{v^2/2}\,dv \sim u^{a-1}e^{u^2/2} \qquad (0 < a \le 1).$$

Indeed integration by parts writes this integral as

$$u^{a-1}e^{u^2/2} - A^{a-1}e^{A^2/2} + (1 - a)\int_A^u v^{a-2}e^{v^2/2}\,dv.$$

For $A > 1$ the final integral is between zero and A^{-2} times the original integral. After division by the upper-end term, letting $u \to \infty$ and then $A \to \infty$ squeezes the ratio to 1, uniformly in a. The first estimate of (II.6.12), applied above A, transfers this conclusion to F_a. The integral below A is uniformly bounded and negligible. This proves both estimates.

Let $J_a(u) = \int_0^u F_a(v)\,dv$, so $I_a(u) = I_a(0) + J_a(u)$. At the threshold,

$$\frac{J_a(u)}{uF_a(u)} = \frac{1+o(1)}{u^2}, \qquad uF_a(u) = I_a(0)(1+o(1)) \sim \frac{1}{a}.$$

Writing $L = \log(1/a) = \log \varepsilon$ and taking logarithms in the first asymptotic estimate for F_a now gives

$$\frac{u^2}{2} + (a+1)\log u + \frac{1}{2}\log(2\pi) = L + o(1). \tag{II.6.13}$$

Since $u \to \infty$, this first implies $u^2 \le 2L + O(1)$; substitution back implies $u^2 = 2L + O(\log L)$. Consequently $\log u^2 = \log(2L) + o(1)$ and $a \log u^2 = o(1)$. Equation (II.6.13) therefore yields

$$u^2 = 2L - \log L - \log(4\pi) + o(1),$$

which is (II.6.9).

Finally, (II.6.11) gives the exact relation

$$\frac{V(\alpha)}{B(\alpha)} = 1 - \frac{J_a(u)}{uF_a(u)} = 1 - \frac{1+o(1)}{u^2}.$$

It follows that $\varepsilon V(\alpha)^2 = u^2 - 2 + o(1)$, proving (II.6.10) and the stated equivalences. □

II.7 Positive-order jets of the Bessel stopping boundary

We give all definitions and inputs so that this section can be read independently. In a Bessel bridge of dimension $d > 0$, the squared radius solves

$$dY_t = \left(d - \frac{2Y_t}{1-t}\right)dt + 2\sqrt{Y_t}\,dW_t, \qquad Y_0 = 0.$$

For the reward Y_τ, the solution of [II.12, Theorem 3.1] has a radius threshold $B(d,2)\sqrt{1-t}$ and initial value $\mathcal{J}(d)$, given by

$$F_d(z) = M(1, d/2, z), \qquad z_d F_d'(z_d) = F_d(z_d), \qquad B(d,2) = \sqrt{2z_d}, \qquad \mathcal{J}(d) = \frac{2z_d}{F_d(z_d)}. \tag{II.7.1}$$

Here $M(a,b,z) = \sum_{k\ge 0}(a)_k z^k/((b)_k k!)$. The root $z_d > 0$ is unique and depends real analytically on $d > 0$. To see this directly, write $F_d(z) = \sum_{k\ge 0} c_k z^k$ with $c_k > 0$ and give the integer k probability $c_k z^k / F_d(z)$. Then zF_d'/F_d is its mean and its derivative is its strictly positive variance divided by z. The mean increases from zero to infinity: for any fixed m, the mass on $\{0,\dots,m\}$ tends to zero as $z \to \infty$, by comparison with a higher coefficient. The analytic implicit function theorem gives the last assertion.

Define the entire functions

$$H_0(z) = 1 - e^{-z}, \qquad H_n(z) = \sum_{k\ge 1} \frac{(-1)^{k+1} z^k}{k^n k!} \quad (n \ge 1), \qquad \eta_n = H_n(1).$$

They satisfy $H_n' = H_{n-1}/z$ for $n \ge 1$, with the quotient understood by continuity at zero. We use the established arithmetic input that, for every $N \ge 1$, the numbers

$$e, H_1(1), \dots, H_N(1) \quad \text{are algebraically independent over } \overline{\mathbb{Q}}. \tag{II.7.2}$$

One explicit source is [II.14, Theorem 2]; its constants are $1 - e^{-1}$ and $n!H_n(1)$. The same finite-family statement is the $\alpha = 0$, $z = 1$ case of the integral family discussed in [II.13, Section 7], using $H_n(1) = \frac{1}{n!}\int_0^1 (-\log t)^n e^{-t}\,dt$.

Put $\varepsilon = (d-2)/2$, $z(\varepsilon) = z_{2+2\varepsilon} = B(2+2\varepsilon, 2)^2/2$, and $K_n = \overline{\mathbb{Q}}(e, \eta_1, \ldots, \eta_n)$, with $K_0 = \overline{\mathbb{Q}}(e)$. The parameter identity

$$M(1, 1+\varepsilon, z) = e^z S(\varepsilon, z), \qquad S(\varepsilon, z) = 1 + \sum_{n\geq 1} (-1)^n H_n(z)\varepsilon^n \tag{II.7.3}$$

holds analytically near $\varepsilon = 0$. Indeed for $\varepsilon > 0$, Euler's integral gives

$$e^{-z} M(1, 1+\varepsilon, z) = \varepsilon \int_0^1 e^{-zt} t^{\varepsilon-1}\, dt = 1 + \varepsilon \sum_{k\geq 1} \frac{(-z)^k}{k!(k+\varepsilon)}.$$

Expansion in ε proves the identity, and analytic continuation supplies a neighborhood of zero.

The next theorem is an additional transfer result, not an answer to a separately identified historical open question.

Theorem II.7.1 (Boundary-jet independence)**.** *For every integer $N \geq 1$,*

$$\overline{\mathbb{Q}}(\partial_d B(2,2), \ldots, \partial_d^N B(2,2)) = \overline{\mathbb{Q}}(e, H_1(1), \ldots, H_{N-1}(1)).$$

In particular the N positive-order dimension derivatives on the left are algebraically independent over $\overline{\mathbb{Q}}$. The zero-order value $B(2,2) = \sqrt{2}$ is algebraic and is deliberately excluded.

Proof. Write $z(\varepsilon) = 1 + \sum_{n\geq 1} z_n \varepsilon^n$. As before,

$$S(\varepsilon, z) = 1 + \sum_{n\geq 1} (-1)^n H_n(z)\varepsilon^n, \qquad \frac{1}{z} - 1 - \frac{S_z}{S} = 0.$$

The first coefficient gives $z_1 = H_0(1) = 1 - e^{-1}$. We show, for $n \geq 2$, the sharpened triangular formula

$$z_n = (-1)^{n+1} e^{-1} \eta_{n-1} + R_n, \qquad R_n \in K_{n-2}. \tag{II.7.4}$$

In the coefficient of ε^n in S_z/S at $z = 1$, the only terms containing η_{n-1} are the numerator term $(-1)^n H_n'(1) = (-1)^n \eta_{n-1}$ and the product of $-H_1'(1) = -H_0(1)$ with the term $(-1)^n \eta_{n-1}$ in $[\varepsilon^{n-1}]S^{-1}$. Their sum is

$$(-1)^n (1 - H_0(1))\eta_{n-1} = (-1)^n e^{-1} \eta_{n-1}.$$

Every other term lies in K_{n-2}, since positive-order z-differentiation lowers the index of the H constants. Substitution of $z = z(\varepsilon)$ does not change that leading coefficient: terms resulting from the substitution involve either lower coefficients $z_j \in K_{j-1}$ or a further positive derivative of some H_k, and hence lie in K_{n-2}. The coefficient z_n appears in $1/z - 1$ with multiplier -1. This proves (II.7.4) inductively.

Now $e = (1 - z_1)^{-1}$, and (II.7.4) recursively recovers $\eta_1, \ldots, \eta_{N-1}$ from $z_1, \ldots, z_N$ and vice versa. The conversion $B = \sqrt{2z}$ is triangular with nonzero algebraic diagonal coefficient, since $z(0) = 1$. The change of variables $d = 2 + 2\varepsilon$ only adds nonzero rational factors. This proves the field equality. The known algebraic independence (II.7.2) gives the last assertion (for $N = 1$, only the transcendence of e is needed). □

Proposition II.7.2 (Value jets, restated from the preceding research stage)**.** *For every $N \geq 0$,*

$$\overline{\mathbb{Q}}(\mathcal{J}(2), \mathcal{J}'(2), \ldots, \mathcal{J}^{(N)}(2)) = \overline{\mathbb{Q}}(e, H_1(1), \ldots, H_N(1)).$$

Thus the $N+1$ displayed values of $\mathcal{J}$ and its derivatives are algebraically independent. This proposition repeats a prior-stage result to make the differential-transcendence argument self-contained; it is not counted as a new result of this stage.

Proof. Equations (II.7.1) and (II.7.3) give $z(0) = 1$, $\mathcal{J}(2) = 2/e$, and

$$\mathcal{J}(2 + 2\varepsilon) = \frac{2z(\varepsilon)e^{-z(\varepsilon)}}{S(\varepsilon, z(\varepsilon))}.$$

Every positive-order derivative of H_n at 1 belongs to K_{n-1}, by repeated differentiation of $H_n' = H_{n-1}/z$. The root-coefficient argument above gives $z_n \in K_{n-1}$. Now set

$$L(\varepsilon) = \log \frac{\mathcal{J}(2 + 2\varepsilon)}{2/e} = \log z(\varepsilon) - z(\varepsilon) + 1 - \log S(\varepsilon, z(\varepsilon)).$$

For each $n \geq 1$, its coefficient has the triangular form

$$[\varepsilon^n]L(\varepsilon) = (-1)^{n+1}\eta_n + R_n, \qquad R_n \in K_{n-1}.$$

Indeed the only occurrence of $H_n(1)$ is the linear term $-(-1)^n H_n(1)\varepsilon^n$ in $-\log S$. All its nonlinear terms use lower H indices. Substitution of $z(\varepsilon)$ uses positive derivatives of the H functions and lower root coefficients, which belong to the indicated lower field. The derivative of $\log z - z + 1$ at $z = 1$ is zero, so the newest root coefficient does not appear in that term. Exponentiation consequently gives

$$\frac{2^n \mathcal{J}^{(n)}(2)}{n!} = \frac{2}{e}((-1)^{n+1}\eta_n + \widetilde{R}_n), \qquad \widetilde{R}_n \in K_{n-1}.$$

Starting from $e = 2/\mathcal{J}(2)$, these relations recover the constants η_n successively from the derivatives and vice versa. This proves the field identity, and (II.7.2) proves independence. □

Remark II.7.3 (The first Euler-constant sensitivity). The first value derivative is

$$\mathcal{J}'(2) = \frac{H_1(1)}{e} = \frac{\gamma}{e} + \frac{\delta}{e^2}, \qquad \delta = \int_0^\infty \frac{e^{-t}}{1+t}\, dt,$$

where $H_1(1) = \gamma + \delta/e$ is the classical Euler–Gompertz identity. Thus these jets do encode an exact combination involving Euler's constant. The independence theorem does not separate the arithmetic of γ from that of δ. The first boundary derivative is $\partial_d B(2, 2) = (1 - e^{-1})/(2\sqrt{2})$; Euler's constant enters the next derivative through $z_2 = e^{-1}(1 - e^{-1} - H_1(1))$.

II.8 The coefficient field in differential transcendence

The earlier conclusion over $\overline{\mathbb{Q}}(d)$ in fact implies the usual conclusion over $\mathbb{C}(d)$. This is a correction and strengthening of its formulation, using the standard transitivity of differential algebraicity; it is not a separate arithmetic discovery. Here is a self-contained version of the needed fact.

Lemma II.8.1 (Elimination of finitely many constant coefficients)**.** *An analytic germ which is differentially algebraic over $\mathbb{C}(t)$ is differentially algebraic over $\mathbb{Q}(t)$.*

Proof. An algebraic differential equation uses only finitely many complex constants, say $c_1, \ldots, c_m$. Put $K = \mathbb{Q}(c_1, \ldots, c_m)(t)$, a differential field with the usual derivative and with constant c_j. Choose a nonzero differential polynomial relation for the germ f having minimal order r, and then minimal degree in its highest derivative. Its separant, the partial derivative with respect to the indeterminate representing $f^{(r)}$, does not vanish after substitution: otherwise it would give a nonzero relation of lower order or of smaller degree at the same order, in characteristic zero. Differentiating the chosen relation therefore expresses $f^{(r+1)}$ rationally in $K(f, f', \ldots, f^{(r)})$. This field is stable under differentiation and contains every derivative of f.

As an ordinary field it has finite transcendence degree over $\mathbb{Q}(t)$, at most $m+r+1$. Sufficiently many derivatives of f must therefore be algebraically dependent over $\mathbb{Q}(t)$. That is an algebraic differential equation with rational-function coefficients over $\mathbb{Q}$, as asserted. □

Corollary II.8.2 (Differential transcendence over complex rational functions)**.** *Both* $d \mapsto V_{d,2}(0,0)$ *and* $d \mapsto B(d,2)$ *are differentially transcendental over* $\mathbb{C}(d)$*.*

Proof. Proposition II.7.2 and specialization at $d = 2$ exclude an algebraic differential equation over $\overline{\mathbb{Q}}(d)$ for the value function. In detail, clear rational-function denominators in an alleged relation and divide its polynomial coefficients by their largest common power of $d - 2$. Their specialization at 2 is then a nonzero polynomial relation over $\overline{\mathbb{Q}}$ among a finite jet, a contradiction. For the boundary, Theorem II.7.1 gives the same exclusion for $d \mapsto \partial_d B(d,2)$: its complete finite jets at 2 consist of the algebraically independent positive-order derivatives of B. If B were differentially algebraic, so would be its derivative; this follows also from the derivative-stable finite-transcendence- degree field constructed in Lemma II.8.1. Finally that lemma excludes equations over $\mathbb{C}(d)$ for both functions. Analytic continuation on $d > 0$ permits any asserted interval relation to be evaluated near $d = 2$. □

II.9 A sharp exponent range for a reciprocal digamma inequality

Qi, Lim, and Nantomah [II.7, Open Problem 3] ask for the largest range of real β such that

$$F_\beta(x) = x^\beta h(x) - P(x) > 0 \quad (x > 0), \qquad h(x) = \psi(x) + x\psi'(x), \quad P(x) = \psi(x)\psi(1/x).$$

The following theorem gives the range and specifies its non-elementary endpoint by one equation having exactly one solution. Its proof does not rely on numerical optimization or interval subdivision.

Theorem II.9.1 (Exact range for the digamma problem)**.** *Let* r *be the unique positive zero of* ψ*, and, for* $1/r < x < 1$*, put*

$$L(x) = \log \frac{\psi(x)\psi(1/x)}{\psi(x) + x\psi'(x)}, \qquad G(x) = x \log x\, L'(x) - L(x).$$

There is exactly one $\xi \in (1/r, 1)$ *with* $G(\xi) = 0$*. Define*

$$B = \frac{L(\xi)}{\log \xi}.$$

Then

$$\boxed{F_\beta(x) > 0 \text{ for every } x > 0 \quad \Longleftrightarrow \quad -1 \le \beta < B.}$$

Moreover $B > 0$*, and*

$$(x\partial_x)^2 L(x) < -\frac{22}{81} \qquad (1/r < x < 1).$$

Numerically, for orientation only, $\xi \approx 0.76859976$ *and* $B \approx 5.97283686$*.*

Proof. We give the elementary estimates used below explicitly. The standard series and integral representations imply

$$\psi(x) = -\gamma - \frac{1}{x} + \sum_{k=1}^{\infty} \frac{x}{k(k+x)}, \tag{II.9.1}$$

$$h(x) = -\gamma + \sum_{k=1}^{\infty} \frac{x(2k+x)}{k(k+x)^2}, \tag{II.9.2}$$

$$h'(x) = 2\sum_{k=1}^{\infty} \frac{k}{(x+k)^3} = \int_0^\infty e^{-xt} \left(\frac{t}{2\sinh(t/2)} \right)^2 dt > 0, \tag{II.9.3}$$

$$h''(x) = -6\sum_{k=1}^{\infty} \frac{k}{(x+k)^4} < 0. \tag{II.9.4}$$

All differentiations are locally uniform on $x > 0$. We use the elementary rational bounds

$$0 < \gamma < \frac{29}{50}, \qquad \log 2 < \frac{7}{10}, \qquad \frac{8}{5} < \zeta(2) < \frac{5}{3}, \qquad \zeta(3) > \frac{6}{5}.$$

For completeness, here are finite certificates for these constants. Integral comparison gives

$$\zeta(2) > \sum_{k=1}^{3} \frac{1}{k^2} + \frac{1}{4} = \frac{29}{18} > \frac{8}{5}, \quad \zeta(2) < \sum_{k=1}^{5} \frac{1}{k^2} + \frac{1}{5} = \frac{5989}{3600} < \frac{5}{3},$$

and

$$\zeta(3) > \sum_{k=1}^{6} \frac{1}{k^3} + \frac{1}{98} = \frac{1411783}{1176000} > \frac{6}{5}.$$

The positive series $\log[(1+z)/(1-z)] = 2\sum_{j\geq 0} z^{2j+1}/(2j+1)$ gives $\log 2 < 2/3 + 1/36 = 25/36 < 7/10$ by taking $z = 1/3$ and bounding the denominators of all terms after the first by 3. The midpoint integral comparison for $1/t$ yields $\gamma < H_4 - \log(9/2)$. The same logarithm series, now at $z = 7/11$, gives

$$\log(9/2) > 2\sum_{j=0}^{5} \frac{(7/11)^{2j+1}}{2j+1} = \frac{451}{300} + \frac{213407380807}{2824585539048900} > \frac{451}{300}.$$

Since $H_4 = 25/12$, this proves $\gamma < 29/50$. Finally $\gamma \geq 1 - \log 2 > 0$. No decimal approximation is needed. Since $\psi(1) = -\gamma < 0$ and $\psi(3/2) = 2 - \gamma - 2\log 2 > 0$, we have $1 < r < 3/2$. The first three terms of (II.9.2) give

$$h(1/3) > \frac{7}{16} + \frac{13}{98} + \frac{19}{300} - \frac{29}{50} = \frac{629}{11760} > 0.$$

Also

$$h(1/2) = -\gamma - 2\log 2 + \tfrac{3}{2}\zeta(2) > \frac{21}{50} > \gamma^2, \qquad \frac{h(1)}{2} = \frac{\zeta(2) - \gamma}{2} > \frac{51}{100} > \gamma^2. \tag{$*$}$$

We shall need the slightly more precise, still rational, estimate

$$h(2/3) > \frac{7}{10}. \tag{$**$}$$

Indeed, for $x = 2/3$ the summand $f(t) = x(2t+x)/[t(t+x)^2] = 1/t - t/(t+x)^2$ is decreasing on $t > 0$. Thus its tail starting at $k = 7$ exceeds

$$\int_7^\infty f(t)\,dt = \log(1 + x/7) + \frac{x}{7+x} > \frac{2x}{7+x} = \frac{4}{23}.$$

The sum of its first six terms is $1523252659/1370784800$, whence

$$h(2/3) > \frac{889267245}{1261122016} = \frac{7}{10} + \frac{32409169}{6305610080} > \frac{7}{10}.$$

We next record two curvature estimates. For $z < r$ define

$$u(z) = \frac{z\psi'(z)}{-\psi(z)}, \qquad R(z) = \frac{-z\psi''(z)}{\psi'(z)}, \qquad K_\psi(z) = (z\partial_z)^2 \log[-\psi(z)].$$

The polygamma series give $R(z) < 2$, because

$$\frac{R(z)}{2} = \frac{\sum_{k\geq 0} z(z+k)^{-3}}{\sum_{k\geq 0}(z+k)^{-2}} < 1.$$

Direct differentiation therefore yields

$$K_\psi(z) = u(z)(R(z) - 1) - u(z)^2 < u(z) - u(z)^2. \tag{†}$$

If $1/r < z < r$, then $z > 2/3$, so $h(z) > 0$ and $u(z) > 1$. Consequently $K_\psi(z) < 0$ throughout that interval. In particular, $t \mapsto \log P(e^t)$ is even and strictly concave for $|t| < \log r$. It attains its maximum at zero. Outside $(1/r, r)$ the product P is nonpositive, so in fact

$$P(x) \le \gamma^2 \quad \text{for every } x > 0. \tag{‡}$$

When $1 \le z < r < 3/2$, strict convexity in the trapezoidal comparison for $\sum_{k\ge0}(z+k)^{-2}$ gives

$$\psi'(z) > \frac{1}{z} + \frac{1}{2z^2}.$$

Using $-\psi(z) \le \gamma < 3/5$, we obtain $u(z) > 20/9$. It follows from (†) that

$$K_\psi(z) < -\frac{220}{81} \qquad (1 \le z < r). \tag{††}$$

Let $2/3 \le x \le 1$ and set $v = xh'(x)/h(x)$. The function $K(t) = [t/(2\sinh(t/2))]^2$ is decreasing on $(0, \infty)$. After $u = xt$ in (II.9.3),

$$xh'(x) = \int_0^\infty e^{-u} K(u/x)\, du$$

is increasing in x. Therefore

$$xh'(x) \le h'(1) = 2\zeta(2) - 2\zeta(3) < \frac{14}{15}, \qquad 0 < v < \frac{4}{3},$$

where (∗∗) was used. The series for h' and h'' also give

$$\frac{-xh''(x)}{h'(x)} < \frac{3x}{x+1} \le \frac{3}{2}.$$

Hence

$$K_h(x) := (x\partial_x)^2 \log h(x) = v\left(1 + \frac{xh''(x)}{h'(x)}\right) - v^2 > -\frac{v}{2} - v^2 > -\frac{22}{9}. \tag{⋆}$$

For $1/r < x < 1$, both arguments in the product defining P are below r, and $1/x \ge 1$. Combining the curvature bounds gives

$$(x\partial_x)^2 L(x) = K_\psi(x) + K_\psi(1/x) - K_h(x) < 0 - \frac{220}{81} + \frac{22}{9} = -\frac{22}{81}. \tag{⋆⋆}$$

We now prove the admissible lower endpoint directly. If $0 < x < 1/3$, put $w = -\psi(x) > 0$. From (II.9.1), $w < 1/x + \gamma$, while $\psi'(x) > 1/x^2$. Thus

$$\frac{\psi'(x)}{w} > \frac{1}{x(1+\gamma x)} > \frac{1}{x} - \gamma.$$

Since $\psi(1/x) > \psi(3) = 3/2 - \gamma > \gamma$, it follows that

$$F_{-1}(x) = w\left(\frac{\psi'(x)}{w} + \psi(1/x) - \frac{1}{x}\right) > 0.$$

For $1/3 \le x \le 1/2$, one has $h(x) > 0$ and $P(x) < 0$. For $1/2 \le x \le 1$, (∗) and (‡) give $h(x)/x \ge h(1/2) > \gamma^2 \ge P(x)$. For $1 \le x \le r < 2$, they give $h(x)/x > h(1)/2 > \gamma^2 \ge P(x)$. Finally, for $x \ge r$, we have $h(x) > 0$ and $P(x) \le 0$. Consequently $F_{-1}(x) > 0$ for every $x > 0$.

Let $a \in (0, 1/3)$ be the unique zero of h. If $\beta \geq -1$, then $F_\beta \geq F_{-1} > 0$ on $(0, a)$, because $x < 1$ and $h(x) < 0$; the same comparison holds on $[1, \infty)$, where $h > 0$. On $[a, 1/r]$, the two terms have opposite signs, and $F_\beta > 0$. The only remaining interval is $(1/r, 1)$, where

$$F_\beta(x) > 0 \quad \Longleftrightarrow \quad \beta < Q(x) := \frac{L(x)}{\log x}. \tag{♣}$$

On this interval $h > 7/10$ and $P \leq \gamma^2$, so $L < 0$ and $Q > 0$. Moreover $Q(x) \to +\infty$ at both endpoints: at $1/r$ the numerator P has a simple zero, while at 1 the finite limit $L(1) = \log[\gamma^2/(\zeta(2) - \gamma)]$ is strictly negative. By (⋆⋆),

$$G'(x) = \frac{\log x}{x}(x\partial_x)^2 L(x) > 0.$$

The simple zero of P gives $G(x) \to -\infty$ as $x \downarrow 1/r$; as $x \uparrow 1$, $G(x) \to -L(1) > 0$. Thus G has precisely one zero ξ, and

$$Q'(x) = \frac{G(x)}{x(\log x)^2}$$

shows that Q decreases up to ξ and increases thereafter. Its minimum is $B = Q(\xi) > 0$. Relation (♣) yields exactly $-1 \leq \beta < B$, including the strict exclusion of B by equality at ξ. Finally, if $\beta < -1$, then as $x \downarrow 0$,

$$h(x) = -\gamma + O(x), \qquad P(x) = \frac{\log x}{x} + O(|\log x| + 1),$$

so $F_\beta(x) \sim -\gamma x^\beta < 0$. This proves necessity of the lower endpoint and completes the proof. □

Interpretation and provenance. The endpoint is a unique tangency slope for the strictly concave function $t \mapsto L(e^t)$; the quantitative curvature bound prevents additional contacts. This is an analytic optimization result, not a claim about the irrationality or transcendence of γ. A targeted literature search on 18 October 2026 found the original problem and the 2026 alternative proofs of its $\beta = 0$ case [II.8], but no previous solution of this exponent range. This is not an exhaustive priority claim.

II.10 The higher derivative exponent range: an independent proof

The following is independently proved here, but a public repository [II.18] already claims the corresponding classification outside $n = 2$. Its priority is unresolved, and it is excluded from any strict count of newly resolved open questions. All orders below are parts of the single Open Problem 4 in [II.7].

Theorem II.10.1 (Odd orders and all even orders at least four)**.** *For an integer $n \geq 1$, let*

$$C_n(x) = \psi^{(n)}(x) + x\psi^{(n+1)}(x), \qquad P_n(x) = \psi^{(n)}(x)\psi^{(n)}(1/x).$$

If n is odd, $x^\beta C_n(x) - P_n(x) < 0$ for every real β and every $x > 0$. If $n \geq 4$ is even, this inequality holds for every $x > 0$ exactly when $n \leq \beta \leq n + 1$. For those admissible exponents,

$$0 < \frac{x^\beta C_n(x)}{P_n(x)} < \frac{n^2 + 2n - 1}{n!} \leq \frac{23}{24} \qquad (x > 0).$$

Proof. Write $Z_s(x) = \sum_{k\geq 0}(x+k)^{-s}$ and $A_n(x) = (n+1)xZ_{n+2}(x) - Z_{n+1}(x)$. Then

$$C_n = (-1)^n n! A_n, \qquad P_n = (n!)^2 Z_{n+1}(x) Z_{n+1}(1/x).$$

Integration by parts in the Hurwitz-zeta integral gives

$$A_n(x) = \frac{1}{n!}\int_0^\infty e^{-xt}\frac{t^n}{1-e^{-t}}\left(n - \frac{t}{e^t-1}\right)dt > 0.$$

The boundary terms vanish even when $n = 1$. Thus the odd case is immediate. For $s > 1$, integral comparison gives

$$\frac{x^{1-s}}{s-1} < Z_s(x) < x^{-s} + \frac{x^{1-s}}{s-1}, \qquad Z_s(x) > x^{-s}.$$

Consequently

$$A_n(x) < (n+1)x^{-n-1} + \frac{n-1}{n}x^{-n}, \quad P_n(x) > \frac{(n!)^2}{nx}, \quad P_n(x) > \frac{(n!)^2 x}{n}.$$

For even $n \geq 4$ these imply, respectively,

$$\frac{x^n C_n(x)}{P_n(x)} < \frac{n(n+1)+(n-1)x}{n!} \leq \frac{n^2+2n-1}{n!} \quad (0 < x \leq 1),$$

and

$$\frac{x^{n+1} C_n(x)}{P_n(x)} < \frac{n(n+1)/x+(n-1)}{n!} \leq \frac{n^2+2n-1}{n!} \quad (x \geq 1).$$

The last coefficient equals $23/24$ at $n = 4$ and decreases thereafter. The two displayed estimates prove sufficiency for every $\beta \in [n, n+1]$. For necessity, the standard integral comparison asymptotics yield

$$\frac{x^\beta C_n(x)}{P_n(x)} \sim \frac{n}{(n-1)!}x^{\beta-n} \quad (x \downarrow 0), \qquad \frac{x^\beta C_n(x)}{P_n(x)} \sim \frac{n-1}{n!}x^{\beta-n-1} \quad (x \to \infty).$$

The first diverges for $\beta < n$, and the second for $\beta > n+1$. □

II.11 Gurland's ratio: global convergence and a sharp location theorem

Wiśniewska [II.9, Section 2, Open problems 1–3] asks about global convergence of a Hurwitz-zeta expansion and the position of its mean-value parameter. Problems 1 and 2 are equivalent through the exact finite expansion and are treated as one mathematical question here. The location question has the following quantitative answer.

Put $s = (x+y)/2$, $d = |x-y|/2$, and $g = \sqrt{xy} = \sqrt{s^2-d^2}$. Throughout this section $x, y > 0$. Define

$$L = \log\frac{\Gamma(1+x)\Gamma(1+y)}{\Gamma(1+s)^2}, \qquad S_m = \sum_{k=1}^{m-1}\frac{d^{2k}}{k}\zeta(2k, 1+s).$$

Theorem II.11.1 (Unconditional convergence of the published expansion)**.** *For every $x, y > 0$,*

$$L = \sum_{k=1}^{\infty}\frac{d^{2k}}{k}\zeta(2k, 1+s).$$

Writing $q = d/(1+s) < 1$, the remainder satisfies, for $m \geq 2$,

$$0 \leq L - S_m \leq \frac{d^{2m}\zeta(2m, 1+s)}{m(1-q^2)} \leq \frac{q^{2m}}{m(1-q^2)}\left(1 + \frac{1+s}{2m-1}\right).$$

In particular it tends to zero, without the extra sufficient condition imposed in [II.9, Corollary 1].

Proof. The Weierstrass product, with the linear terms cancelling, gives

$$L = \sum_{j=1}^{\infty} -\log\left(1 - \frac{d^2}{(j+s)^2}\right).$$

Since $d < s < 1+s$, every summand can be expanded into its positive logarithmic series. Tonelli's theorem interchanges the two sums. For $0 \le u \le q^2$, $\sum_{k=m}^{\infty} u^k/k \le u^m/[m(1-q^2)]$. Summing this estimate proves the first bound. The integral test gives

$$\zeta(2m, 1+s) \le (1+s)^{-2m} + \frac{(1+s)^{1-2m}}{2m-1},$$

which proves the second. The diagonal case $d = 0$ is immediate. □

For $d > 0$ let t be the unique number satisfying $L = d^2\psi'(1+t)$. Uniqueness follows because ψ' is strictly decreasing on $(0, \infty)$. For $0 < r < 1$ set

$$h(r) = (1+r)\log(1+r) + (1-r)\log(1-r), \qquad H(s,d) = \frac{sr^2}{h(r)}, \quad r = d/s.$$

Theorem II.11.2 (Sharp uniform location and distance from the midpoint)**.** *For $x \neq y$,*

$$\frac{2s+g}{3} < H(s,d) < t < s.$$

Thus the parameter always lies strictly in the upper third of (g, s). In particular its distance from the midpoint is positive and obeys

$$\frac{s-g}{6} < H(s,d) - \frac{s+g}{2} < t - \frac{s+g}{2} < \frac{s-g}{2}.$$

The best possible constant c in the uniform inequality $t > g + c(s-g)$ is $c = 2/3$. More precisely, for each fixed $r \in (0,1)$, with $d = rs$,

$$\lim_{s\to\infty} \frac{t}{s} = \frac{r^2}{h(r)}, \qquad \lim_{s\downarrow 0} \frac{t}{s} = 1.$$

Consequently both the shape-dependent lower bound $H(s,d)/s$ and the upper bound 1 are best possible when d/s is fixed.

Proof. Let U have probability density $1 - |u|$ on $[-1, 1]$, and put $X = s + dU$. Twice integrating the second derivative of $\log\Gamma$ gives the exact identity

$$\frac{L}{d^2} = \mathbb{E}\, f(X), \qquad f(v) = \psi'(1+v).$$

Since f is strictly convex and decreasing, $\mathbb{E}f(X) > f(s)$ and hence $t < s$.

We next prove that $F(w) = f(1/w)$ is strictly concave for $w > 0$. The derivative test reduces this assertion to

$$2\zeta(3, 1+v) - 3v\zeta(4, 1+v) > 0 \quad (v > 0).$$

For completeness, write $a(u) = 1/(e^u - 1)$ and use the Laplace integral for Hurwitz zeta. Integration by parts, whose boundary terms vanish, yields

$$2\zeta(3, 1+v) - 3v\zeta(4, 1+v) = \frac{1}{2}\int_0^{\infty} e^{-vu}u^2[-a(u) - ua'(u)]\, du > 0.$$

Indeed $u/(e^u-1)$ is strictly decreasing on $(0,\infty)$: its derivative has numerator $e^u(1-u)-1<0$. Equivalently, $f''(v)/f'(v)-(-2/v)=2/v-3\zeta(4,1+v)/\zeta(3,1+v)>0$, which gives $F''<0$ by the chain rule. Strict Jensen inequality now gives

$$\mathbb{E}f(X)=\mathbb{E}F(1/X)<F(\mathbb{E}(1/X))=f\big((\mathbb{E}(1/X))^{-1}\big).$$

Since f decreases, $t>(\mathbb{E}(1/X))^{-1}$. Direct integration of the triangular density shows that this last reciprocal is exactly $H(s,d)$.

To prove the remaining explicit bound, put $q=\sqrt{1-r^2}$ and $A(r)=3r^2/(2+q)-h(r)$. Direct differentiation gives

$$A''(r)=\frac{(1-q)^2(6+5q-2q^2)}{q^3(q+2)^3}>0,\qquad A(0)=A'(0)=0.$$

Thus $h(r)<3r^2/(2+q)$, or $H(s,d)>(2s+g)/3$. Subtracting the midpoint proves the asserted distance bounds.

For the limiting assertions use $\psi'(1+v)=v^{-1}+O(v^{-2})$ as $v\to\infty$. Because $r<1$ is fixed, this is uniform over $v=s(1+ru)$, $|u|\le 1$. Therefore

$$\mathbb{E}f(X)=\frac{h(r)}{sr^2}+O(s^{-2}).$$

The proved inequalities bound t/s away from zero, so the same asymptotic for $f(t)$ proves the first limit. For $s\downarrow 0$, Taylor expansion around zero, using $\mathbb{E}X=s$ and $\mathbb{E}(X-s)^2=d^2/6$, gives

$$t=s-\frac{\zeta(4)}{4\zeta(3)}d^2+O(s^3),$$

which proves the second limit. Finally, $r^2/h(r)=1-r^2/6+O(r^4)$ and $\sqrt{1-r^2}=1-r^2/2+O(r^4)$. First taking $s\to\infty$ and then $r\downarrow 0$ makes $(t-g)/(s-g)$ tend to $2/3$. No larger uniform constant is possible. □

Interpretation and scope. The convergence result answers both of the first two source questions; it is one closure, not two independent conjectures. The location theorem answers the third by a sharp universal fraction and sharper bounds depending on the ratio of the arguments. These are special-function results, not an irrationality theorem for γ. Their connection with γ is through logarithmic derivatives of the gamma function; the constant term cancels in the symmetric ratio.

II.12 A counterexample to Bolbachan's digamma limit conjecture

Bolbachan's preprint [II.10, Conjecture 1.4] proposes the following limit for every $u>0$:

$$\psi(u)=\log u+\lim_{m\to\infty}L_m(u),\qquad L_m(u)=\frac{1}{m!}\sum_{k=1}^{m}(-1)^k\binom{m}{k}\frac{A_{k,m+1}}{k!}\int_0^\infty x^{k-1}e^{-x}\log(1+x/u)\,dx, \tag{II.12.1}$$

where

$$A_{k,n}=\sum_{t=2}^{n}\left\{{n\atop t}\right\}\sum_{w=1}^{t-1}(-k)^{t-w}\sum_{j=1}^{w}(-1)^jB_j\left[{w\atop j}\right].$$

Here the Stirling numbers of the first kind in square brackets are unsigned. We first use the convention $B_1=-1/2$.

Theorem II.12.1. *For every fixed real $u > 0$, the sequence in (II.12.1) satisfies*

$$L_m(u) \longrightarrow +\infty, \qquad \lim_{m\to\infty} L_m(u)^{1/m} = \frac{1}{\log 2}.$$

Consequently Conjecture 1.4 is false as printed. If instead Bernoulli numbers with $B_1 = +1/2$ are intended, then $L_m(u) \to -\infty$ and $|L_m(u)|^{1/m} \to 1/\log 2$; the limit assertion still fails.

II.12.1 Exact reduction to a coefficient integral

Put

$$c_w = \sum_{j=1}^{w} (-1)^j B_j \begin{bmatrix} w \\ j \end{bmatrix}, \qquad C(y) = \sum_{w\geq 1} c_w \frac{y^w}{w!}.$$

The Stirling exponential generating function and the Bernoulli generating function give, as convergent identities for $|y| < 1$,

$$1 + C(y) = \sum_{j\geq 0} \frac{B_j \log(1-y)^j}{j!} = \frac{-\log(1-y)}{y}.$$

In particular

$$c_w = \frac{w!}{w+1}, \qquad C(y) = -\frac{\log(1-y)}{y} - 1. \tag{II.12.2}$$

The apparent singularity at $y = 0$ is removable.

The expression $A_{k,m+1}/k$ extends polynomially to $k = 0$ and has degree at most $m-1$. Define

$$T_m(z) = \frac{1}{m!} \sum_{k=0}^{m} (-1)^k \binom{m}{k} \left.\frac{A_{k,m+1}}{k}\right|_{\text{polynomial extension}} z^k.$$

The mth finite difference of a polynomial of degree at most $m-1$ vanishes, so

$$T_m(1) = 0. \tag{II.12.3}$$

Writing $Y(v) = e^v - 1$, set

$$J_m(v, z) = \int_0^{Y(v)} C(w) \left(1 - z e^{w - Y(v)}\right)^m dw. \tag{II.12.4}$$

Then the following exact identity holds:

$$T_m(z) = -(m+1)[v^{m+1}] J_m(v, z). \tag{II.12.5}$$

For verification, let $H_t(k) = \sum_{w=1}^{t-1} (-k)^{t-w} c_w$, with $H_0 = H_1 = 0$. The recurrence $H_{t+1} = -kH_t - kc_t$ gives

$$\sum_{t\geq 0} H_t(k) \frac{y^t}{t!} = -k e^{-ky} \int_0^y e^{kw} C(w)\, dw.$$

Composition with $y = e^v - 1$ performs the Stirling transform, hence

$$\frac{A_{k,m+1}}{k} = -(m+1)![v^{m+1}] \int_0^{Y(v)} e^{k(w - Y(v))} C(w)\, dw.$$

Summing the binomial expansion proves (II.12.5), including $k = 0$ by polynomial extension.

The elementary identity

$$\frac{1}{k!} \int_0^\infty x^{k-1} e^{-x} \log(1 + x/u)\, dx = \frac{1}{k} \int_0^\infty \frac{e^{-ut}}{t} \left(1 - (1+t)^{-k}\right) dt$$

follows from the Laplace integral for the logarithm and Tonelli's theorem. Therefore, putting $z(t) = (1+t)^{-1}$, the original finite sum becomes

$$L_m(u) = -\int_0^\infty \frac{e^{-ut}}{t} \left(T_m(z(t)) - T_m(1)\right) dt. \tag{II.12.6}$$

Keeping the subtraction at $z = 1$ is essential for uniform estimates near $t = 0$.

II.12.2 A disk estimate with rational constants

Let $R = 7/10$ and $0 \le s \le 1$. We claim

$$|1 - \exp\{-s(e^v - 1)\}| \le \frac{318799}{320000} < 1 \qquad (|v| \le R). \tag{II.12.7}$$

Write $\mathcal{B}_n(x) = \sum_{j=0}^{n} \left\{ {n \atop j} \right\} x^j$ for the Touchard polynomials and $b_n = \mathcal{B}_n(1)$ for the Bell numbers. The exponential generating function is

$$e^{-s(e^v-1)} = \sum_{n \ge 0} \mathcal{B}_n(-s) \frac{v^n}{n!}.$$

For $1 \le n \le 5$, the Bernstein coefficients of $\mathcal{B}_n(-s)$ on $[0,1]$, in increasing order, are

n	Bernstein coefficients	M_n
1	$0, -1$	1
2	$0, -1/2, 0$	$1/2$
3	$0, -1/3, 1/3, 1$	1
4	$0, -1/4, 2/3, 5/4, 1$	$5/4$
5	$0, -1/5, 11/10, 7/5, 1/5, -2$	2

and their absolute values are bounded by the last column. For $n \ge 6$, $|\mathcal{B}_n(-s)| \le b_n$. Consequently the left side of (II.12.7) is at most

$$e^{e^R - 1} - 1 + \sum_{n=1}^{5} (M_n - b_n) \frac{R^n}{n!} < \frac{14}{5} - 1 + \sum_{n=1}^{5} (M_n - b_n) \frac{(7/10)^n}{n!} = \frac{318799}{320000}.$$

The elementary bounds $e^{7/10} < 101/50$ and $e^{51/50} < 14/5$ justify $e^{e^R - 1} < 14/5$; they can also be checked directly from the exponential series with its geometric remainder bound.

For $z \in [0,1]$, convexity of the closed unit disk now gives

$$|1 - z e^{-sY(v)}| \le 1 \qquad (|v| \le R,\ 0 \le s \le 1). \tag{II.12.8}$$

Indeed $1 - ze^{-sY} = (1 - z) + z(1 - e^{-sY})$.

II.12.3 The branch cut gives a positive dominant contribution

Put $\rho = \log 2 < R$. Use the principal branch of $\log(1 - w)$ in (II.12.2). Formula (II.12.4), with the straight path $w = \tau Y(v)$, defines an analytic function on

$$\{v : |v| < R\} \setminus [\rho, R).$$

There are no other cuts in this disk: since $|\operatorname{Im} v| < \pi$, $Y(v)$ is real and at least 1 exactly when v is real and at least ρ. For real $\rho < v < R$, its upper minus lower jump is

$$J_m(v + i0, z) - J_m(v - i0, z) = 2\pi i \int_1^{e^v - 1} \frac{(1 - z e^{w - (e^v - 1)})^m}{w}\, dw. \tag{II.12.9}$$

The positive sign follows because $\log(1 - w)$ has upper boundary value $\log(w - 1) - i\pi$ for $w > 1$.

There exists a finite constant M, independent of m and z, such that on the circle $|v| = R$, with either boundary value at its positive real point,

$$|J_m(v, z) - J_m(v, 1)| \le M m (1 - z). \tag{II.12.10}$$

To see this, factor the difference of the two mth powers in (II.12.4), use (II.12.8), and bound $|e^{w-Y(v)}|$ by $e^{|Y(v)|} \le e^{e^R-1}$. The remaining integral $\int_0^1 |Y(v)C(\tau Y(v))|\,d\tau$ is uniformly bounded: C is regular at zero and has only an integrable logarithmic singularity at 1. The same assertion holds on any smaller circle $|v| = r < \rho$.

Deform the coefficient contour in (II.12.5) to the circle of radius R and the two sides of the cut. Small indentations at ρ make no contribution because the logarithmic singularity is integrable. Equations (II.12.6)–(II.12.10) give

$$L_m(u) = (m+1)\int_\rho^R \frac{dv}{v^{m+2}} \int_1^{e^v-1} \frac{dw}{w} \int_0^\infty \frac{e^{-ut}}{t} \times \left[\left(1 - \frac{e^{w-(e^v-1)}}{1+t}\right)^m - \left(1 - e^{w-(e^v-1)}\right)^m\right] dt + O_u(m^2 R^{-m}). \tag{II.12.11}$$

All three integrations in the displayed main term have nonnegative integrands. The error follows from (II.12.10) and $\int_0^\infty e^{-ut}(1-z(t))\,dt/t = \int_0^\infty e^{-ut}/(1+t)\,dt < \infty$. These estimates also justify all contour–integral interchanges.

Choose $\varepsilon > 0$ with $\rho + 2\varepsilon < R$, put $h = e^\varepsilon - 1$, and restrict the main integral to

$$\rho + \varepsilon \le v \le \rho + 2\varepsilon, \qquad 1 \le w \le 1 + h, \qquad T \le t \le T + 1.$$

This rectangle lies inside its integration domain. On it, set

$$a = \frac{T}{T+1}, \qquad b = 1 - \exp(2 - 2e^{2\varepsilon}).$$

The bracket in (II.12.11) is at least $a^m - b^m$. Choose T sufficiently large that $a > b$. It follows that for all sufficiently large m,

$$L_m(u) \ge c_{u,\varepsilon,T}(m+1)\left(\frac{a}{\rho + 2\varepsilon}\right)^m - C_u m^2 R^{-m}, \qquad c_{u,\varepsilon,T} > 0. \tag{II.12.12}$$

By taking ε small and T large we can make $a/(\rho + 2\varepsilon)$ arbitrarily close to $1/\rho$ from below, and in particular strictly greater than $1/R$. Thus the positive term in (II.12.12) dominates the error, proving divergence and

$$\liminf_{m\to\infty} L_m(u)^{1/m} \ge \frac{1}{\rho}.$$

On the other hand, Cauchy's estimate on any circle $|v| = r < \rho$, together with the smaller-circle version of (II.12.10), gives $|L_m(u)| \le C_{u,r} m(m+1) r^{-m-1}$. Letting $r \uparrow \rho$ proves the matching upper bound.

Finally, replacing $B_1 = -1/2$ by $B_1 = +1/2$ changes c_w by $-(w-1)!$, hence changes $C(w)$ to

$$C_+(w) = \left(1 - \frac{1}{w}\right)\log(1-w) - 1.$$

Its upper minus lower jump for $w > 1$ is $-2\pi i(1 - 1/w)$. The same proof applies with a negative cut term and the positive weight $1 - 1/w$ in place of $1/w$. In the lower bound one may restrict w to $[1 + h/2, 1 + h]$, where this weight is bounded below by a positive constant. This proves the alternative-convention assertion and completes the theorem.

II.13 The fractional-part and gcd obstructions that remain

Murabayashi–Yoshida's Conjecture 7.5 asks for an infinite nondecreasing subsequence of $\delta\{q_{2k}\} - \{p_{2k}\}$ and an infinite nonincreasing subsequence of $\delta\{q_{2k-1}\} - \{p_{2k-1}\}$. Their Theorem 7.4 shows that either would imply irrationality of δ. We do not prove this conjecture. A general monotone-subsequence theorem cannot choose the required direction, and fixed-order Laguerre asymptotics do not control fractional parts.

Proposition II.13.1 (Finite-chain obstruction for rational constants)**.** *Let $\alpha = a/b > 0$ be rational in lowest terms, with a, b positive integers. If $r_n = \alpha q_n - p_n$ strictly decreases inside $(0, 1/b)$, every nondecreasing subsequence of $s_n = \alpha\{q_n\} - \{p_n\}$ has at most $a+b$ terms. If instead r_n strictly increases inside $(-1/b, 0)$, the same bound holds for nonincreasing subsequences of s_n.*

Proof. The number $h_n = bs_n - br_n = b\lfloor p_n \rfloor - a\lfloor q_n \rfloor$ is integral. In the first case it lies in $\{-b, \dots, a-1\}$ and strictly increases along the specified subsequence. In the second it lies in $\{-b+1, \dots, a\}$ and strictly decreases. Both sets have $a+b$ elements. □

For the integer Laguerre approximants $Q_n = n!q_{2n}$ and $P_n = n!p_{2n}$, we have

$$U_{n+1} = 2(n+1)U_n - n^2 U_{n-1}, \qquad (Q_0, Q_1) = (1, 2), \quad (P_0, P_1) = (0, 1).$$

Write $g_n = \gcd(Q_n, P_n)$ and $\Lambda_n = \operatorname{lcm}(1, \dots, n)$.

Proposition II.13.2 (Exact congruence and reset identities)**.** *For nonnegative integers n, r and any positive integer M,*

$$n \equiv r \pmod{M} \quad \Longrightarrow \quad Q_n \equiv Q_r \pmod{M}.$$

For $n \geq M$ one has $P_n \equiv P_M Q_{n \bmod M} \pmod{M}$. Moreover, for $n \geq 1$,

$$\gcd(Q_n, n) = 1, \qquad \gcd(Q_n, \Lambda_n^2) \mid g_n \mid ((n-1)!)^2.$$

Proof. Writing $(n)_{\underline{j}} = n(n-1)\cdots(n-j+1)$, the explicit Laguerre formula gives $Q_n = \sum_{j \geq 0} (n)_{\underline{j}}^2 / j!$, with all $j > n$ terms zero. For $n \equiv r \pmod{M}$, each difference of summands equals

$$((n)_{\underline{j}} - (r)_{\underline{j}}) \left(\binom{n}{j} + \binom{r}{j} \right)$$

and is divisible by M. Taking $r = 0, M = n$ gives $Q_n \equiv 1 \pmod{n}$. Modulo M, the recurrence at index M gives $P_{M+1} \equiv 2P_M$; uniqueness for the forward recurrence and periodicity of its coefficients give $P_{M+r} \equiv P_M Q_r \pmod{M}$ for $r \geq 0$.

For a prime power p^a, put $j = p^{\lceil a/2 \rceil}$. If $j \leq n$, then $j^2 \equiv 0 \pmod{p^a}$ and $Q_j \equiv 1 \pmod{p}$ is invertible modulo p^a. At indices $j, j+1$, the vectors (P_j, P_{j+1}) and (Q_j, Q_{j+1}) are proportional modulo p^a; the recurrence preserves this proportionality at every later index. Hence $p^{\min(a, v_p(Q_n))} \mid g_n$ whenever $j \leq n$. Taking $a = 2\lfloor \log_p n \rfloor$ for each prime $p \leq n$ proves the lower divisibility bound. Finally, the recurrence gives $Q_n P_{n-1} - P_n Q_{n-1} = -((n-1)!)^2$, yielding the upper bound. □

The gap between these bounds is substantial. They neither establish sufficient gcd cancellation to obtain an irrationality proof nor bound all cancellation strongly enough to exclude it. In particular, the factorial normalization in the positive-polynomial obstruction must not be replaced without proof by a reduced common denominator.

II.14 Sharper descent for periodic Dirichlet series

The argument below strengthens the explicit large-prime bound in Theorem 2 of the 2022 version of Bharadwaj's paper [II.11]. Section 8 of that version expressly suggests improving this bound. Our comparison is with that identified version; priority relative to subsequent work is not asserted. The full Erdős nonvanishing conjecture remains unresolved here.

For a rational-valued periodic function h of period K and mean zero, write

$$L(1, h) = \sum_{n \geq 1} \frac{h(n)}{n}, \qquad \mathcal{L}_K = \operatorname{span}_{\mathbb{Q}}\{\log r : r \text{ prime}, \ r \mid K\}.$$

Following [II.11], say that h has Property U if $L(1,h) \notin \mathcal{L}_K$. An Erdős function of period K has values ± 1 outside $K\mathbb{Z}$, is zero on $K\mathbb{Z}$, and has mean zero. Property U implies the nonvanishing sought in the conjecture, but is a stronger condition.

II.14.1 The conductor criterion and its integral form

All moduli in the descent argument are odd and squarefree. If $m \mid t$, let P_m^t denote averaging over the fibers of

$$(\mathbb{Z}/t\mathbb{Z})^* \longrightarrow (\mathbb{Z}/m\mathbb{Z})^*.$$

Thus P_m^t has operator norm at most one for the supremum norm. For r coprime to m, let $S_r v(x) = v(rx)$ on units modulo m. For each prime $r \mid m$, choose a generator τ_r of its prime-coordinate group in $(\mathbb{Z}/m\mathbb{Z})^* \cong \prod_{r\mid m}(\mathbb{Z}/r\mathbb{Z})^*$, and let it act by translation. Put

$$\Delta_m = \prod_{r\mid m}(I - \tau_r).$$

In the character basis, Δ_m is zero precisely on the characters imprimitive modulo m. Indeed, a character is primitive for a squarefree modulus exactly when every prime-coordinate character is nonprincipal, and a generator detects each such nonprincipal character.

For $d \mid K$, define the function h_d on units modulo K/d by $h_d(x) = h(dx)$. We use the following established conductor criterion from the proof of Theorem 4 in [II.11]:

$$L(1,h) \in \mathcal{L}_K \quad\Longleftrightarrow\quad \Delta_{K/D} \sum_{d\mid D} \frac{1}{d} \prod_{r\mid D/d} \left(I - \frac{S_r}{r}\right) P_{K/D}^{K/d} h_d = 0 \quad \text{for every proper } D \mid K. \tag{II.14.1}$$

Here and below products indexed by r are over primes. To explain the input's scope, the character expansion of $L(s,h) = \sum_{d\mid K} d^{-s} L(s, h_d)$ groups its nonprincipal terms by primitive conductor. The averaging operators select characters whose conductors divide K/D, and the displayed Euler factors remove the additional prime factors of K/d. The nonprincipal conductor contributions vanish exactly under the conditions on the right of (II.14.1); their separation is the cited theorem. The remaining principal contribution is a rational combination of prime logarithms, since the mean of h is zero. This is the Property U formulation of that criterion; no prime-logarithm vanishing equations are imposed in it. More formally, set $u_r(n) = 1 - r\mathbf{1}_{r\mid n}$ for each $r \mid K$. This is a mean-zero K-periodic function with $L(1,u_r) = \log r$, as follows by taking the limit at $s = 1$ in $(1 - r^{1-s})\zeta(s)$. Every $(u_r)_d$ is constant on the relevant unit group, so all its nonprincipal conductor expressions vanish. If $L(1,h) = \sum_{r\mid K} c_r \log r$, applying the cited vanishing criterion to $h - \sum_r c_r u_r$ proves necessity in (II.14.1). For sufficiency, grouping by primitive conductor leaves only the principal contribution, which belongs to $\mathcal{L}_K$.

For an integer-valued f of period N, define, for each proper $D \mid N$ and $m = N/D$,

$$G_D(f) = D\varphi(D)\Delta_m \sum_{d\mid D} \frac{1}{d} \prod_{r\mid D/d} \left(I - \frac{S_r}{r}\right) P_m^{N/d} f_d. \tag{II.14.2}$$

Lemma II.14.1 (Integrality and the exact product bound)**.** *If N is odd and squarefree and $\|f\|_\infty \le B$, then $G_D(f)$ is integer-valued and*

$$\|G_D(f)\|_\infty \le B\, 2^{\omega(N/D)} \varphi(D) \prod_{r\mid D}(r+2). \tag{II.14.3}$$

If r_0 is the least prime divisor of $N > 1$, the maximum of the factor multiplying B over proper divisors D is

$$C(N) = 2\varphi(N/r_0) \prod_{r\mid N/r_0} (r+2) = 2 \prod_{r\mid N/r_0} (r-1)(r+2). \tag{II.14.4}$$

Proof. The denominator introduced by $P_m^{N/d}$ divides $\varphi(D/d)$. The factor d^{-1} and the Euler operators have combined denominator dividing D, since D is squarefree. Thus multiplication by $D\varphi(D)$ makes each term integral. Translation differences preserve integrality.

The norm of Δ_m is at most $2^{\omega(m)}$, and $\|I - S_r/r\| \le 1 + 1/r$. Consequently,

$$\begin{aligned}\|G_D(f)\|_\infty &\le BD\varphi(D)2^{\omega(m)} \sum_{d|D} \frac{1}{d} \prod_{r|D/d} \left(1 + \frac{1}{r}\right) \\ &= B\varphi(D)2^{\omega(m)} \prod_{r|D} (r+2).\end{aligned}$$

The equality is the squarefree factorization $D \sum_{d|D} d^{-1} \prod_{r|D/d}(1 + 1/r) = \prod_{r|D}(r + 2)$. In the factor on the right of (II.14.3), a prime $r \mid D$ contributes $(r - 1)(r + 2)$, whereas a prime $r \mid N/D$ contributes 2. The ratio $(r - 1)(r + 2)/2$ exceeds one and increases with $r \ge 3$. Since D must be proper, the maximum is obtained by omitting just the least prime r_0 from D. This proves (II.14.4). □

Theorem II.14.2 (Adaptive and uniform large-prime descent)**.** *Let $N > 1$ be odd and squarefree. Suppose f is an integer-valued mean-zero function of period N with Property U. Define*

$$C_f = \min_{\substack{D|N,\ D<N \\ G_D(f)\neq 0}} \|G_D(f)\|_\infty.$$

The minimum is over a nonempty finite set. If $p \nmid N$ is prime and $p > 1 + C_f$, then every integer-valued mean-zero function g of period pN satisfying $g(pn) = f(n)$ has Property U. In particular, if $\|f\|_\infty \le B$, the uniform condition

$$p > 1 + BC(N) \tag{II.14.5}$$

suffices. No size bound on g is required.

Proof. The nonemptiness follows from (II.14.1). Suppose, towards a contradiction, that g fails Property U. Fix a proper $D \mid N$, put $m = N/D$, and define a function on the units modulo pm by

$$h_D = D\varphi(D) \sum_{d|D} \frac{1}{d} \prod_{r|D/d} \left(I - \frac{S_r}{r}\right) P_{pm}^{pN/d} g_d.$$

The same denominator argument as in Lemma II.14.1 shows that h_D is integer-valued; the averaging fibers have size $\varphi(D/d)$, independent of p. Apply the conductor criterion to g with divisor D. It gives

$$\Delta_m (I - \tau_p) h_D = 0.$$

Hence $\Delta_m h_D$ is invariant under a generator of the p-coordinate group and is constant along that coordinate. It follows that

$$F_D = \Delta_m P_m^{pm} h_D$$

is integer-valued: its average is simply the value at any chosen p-coordinate.

Next apply the conductor criterion to g with divisor pD. Split its divisor sum into $d \mid D$ and pd with $d \mid D$. In the second part, $g_{pd} = f_d$. In the first part, the additional Euler factor is $I - S_p/p$. Averaging is transitive and commutes with the translation operators. After multiplication by $pD\varphi(D)$, the resulting identity is therefore

$$(pI - S_p)F_D + G_D(f) = 0. \tag{II.14.6}$$

If $G_D(f) \neq 0$, then $F_D \neq 0$, so its integer values give $\|F_D\|_\infty \ge 1$. Since S_p permutes the coordinates, the reverse triangle inequality gives

$$\|G_D(f)\|_\infty = \|(pI - S_p)F_D\|_\infty \ge (p - 1)\|F_D\|_\infty \ge p - 1.$$

Choose D attaining C_f. This contradicts $p > 1 + C_f$. Finally, Lemma II.14.1 gives $C_f \le BC(N)$, proving the uniform statement. □

Corollary II.14.3 (Congruence and sign refinement)**.** *In the setting of Theorem II.14.2, suppose additionally that $g(n) \equiv a \pmod M$ whenever $pN \nmid n$, where $M \ge 1$ is an integer. Then $p > 1 + C_f/M$ suffices. In particular, for Erdős functions f and g, the sufficient uniform condition is*

$$p > 1 + \frac{C(N)}{2} = 1 + \prod_{r \mid N/r_0} (r-1)(r+2). \tag{II.14.7}$$

Proof. Every g_d occurring in h_D takes values congruent to a modulo M, because $d \mid D < N$. On expanding the averaging and Euler operators in the definition of h_D, their coefficients are integers after multiplication by $D\varphi(D)$, as established above. Their row sums are independent of the unit-group coordinate: a translation has row sum one, and a fiber sum has its fixed fiber size as row sum. Therefore h_D is constant modulo M. Since $m = N/D > 1$, the operator Δ_m contains a translation difference, so $\Delta_m h_D$ takes values in $M\mathbb{Z}$. Its constancy along the p-coordinate implies that F_D also takes values in $M\mathbb{Z}$. Thus, if $G_D(f) \ne 0$, equation (II.14.6) gives $\|G_D(f)\|_\infty \ge M(p-1)$. Taking a divisor attaining C_f proves the first assertion. For sign functions, use $M = 2$, $a = 1$, and $B = 1$. □

Corollary II.14.4 (A primitive-component shortcut)**.** *Under the hypotheses of Theorem II.14.2, suppose that $\Delta_N f_1 \ne 0$. Then $p > 1 + B2^{\omega(N)}$ is sufficient.*

Proof. Take $D = 1$ in (II.14.2). Then $G_1(f) = \Delta_N f_1 \ne 0$ and its supremum norm is at most $B2^{\omega(N)}$. □

For orientation, the sign-function thresholds in (II.14.7) are compared below with $2^{\omega(N)} \prod_{r \mid N} (r^2 + 2r + 2)$, the threshold in Theorem 2 of the cited 2022 version. These are sufficient bounds, not claimed optimal bounds.

N	sign-function condition	2022 condition
3	$p > 2$	$p > 34$
15	$p > 29$	$p > 2516$
105	$p > 1513$	$p > 327080$
195	$p > 5041$	$p > 991304$
1155	$p > 196561$	$p > 94853200$

II.14.2 Prime marginals and the precise remaining obstruction

Lemma II.14.5 (Prime marginal obstruction)**.** *Let K be squarefree and let h be a rational-valued mean-zero function of period K. For $r \mid K$, define*

$$S_r(h) = \sum_{1 \le a \le K,\ r \mid a} h(a).$$

In the decomposition of $L(1,h)$ into rational-prime logarithms and an algebraic combination of unit logarithms and π, the coefficient of $\log r$ is

$$-\frac{rS_r(h)}{K(r-1)}.$$

If any $S_r(h) \ne 0$, then $L(1,h)$ is transcendental and nonzero.

Proof. Put $M_d = \sum_{(a,K/d)=1} h(da)$ for $d \mid K$. The principal-character contributions to the constant term at 1 give

$$\frac{1}{K}\sum_{d\mid K} M_d\left(\gamma - \log d + \sum_{r\mid K/d}\frac{\log r}{r-1}\right).$$

The coefficient of γ vanishes because $\sum_d M_d = 0$. Squarefreeness gives $S_r(h) = \sum_{r\mid d} M_d$, and therefore the coefficient of $\log r$ is

$$\frac{1}{K}\left(\frac{\sum_{r\nmid d} M_d}{r-1} - \sum_{r\mid d} M_d\right) = -\frac{rS_r(h)}{K(r-1)}.$$

The other character terms are algebraic combinations of logarithms of algebraic units and π. Prime valuations and Baker's theorem separate the nonzero rational-prime component from these terms, and give the stated transcendence conclusion. □

For example, if every Erdős function of period N has Property U, the uniform bound proves nonvanishing for every Erdős function g of period pN. Indeed, put $f(n) = g(pn)$. If f has nonzero mean, the p-marginal in Lemma II.14.5 already excludes $L(1, g) = 0$. If its mean is zero, apply Theorem II.14.2. The known $N \equiv 3 \pmod 4$ argument of Murty and Saradha supplies the required Property U through a nonzero π component; see the proof of Corollary 1 in [II.11]. Thus this specialization yields periods $pN \equiv 1 \pmod 4$ when also $p \equiv 3 \pmod 4$.

II.14.3 An exact answer to a stated existence question

Section 8 of the 2022 version of [II.11] asks whether there exists an odd natural number N and signs $a_d \in \{\pm 1\}$ for $d \mid N$, $d > 1$, such that

$$a_1 = 0, \qquad \sum_{d\mid N} a_d\varphi(d) = 0. \tag{II.14.8}$$

For $N = 1$ the displayed condition is vacuous; we answer its nontrivial interpretation $N > 1$. The following supplies an explicit example and the least nontrivial odd period. The later journal version was identified but its full text was unavailable for comparison, so current priority is not asserted.

Proposition II.14.6 (An odd example and its minimality)**.** *The least odd integer $N > 1$ satisfying* (II.14.8) *is* 105*.*

Proof. For $N = 105$, take $a_5 = a_{105} = 1$ and all other $a_d = -1$ for $d > 1$. Then

$$\begin{aligned} &-\varphi(3) + \varphi(5) - \varphi(7) - \varphi(15) - \varphi(21) - \varphi(35) + \varphi(105) \\ &\quad = -2 + 4 - 6 - 8 - 12 - 24 + 48 = 0. \end{aligned}$$

If an odd integer $N > 1$ has at most two distinct prime factors, then

$$\frac{\varphi(N)}{N} \geq \left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right) = \frac{8}{15} > \frac{1}{2}.$$

The identity $\sum_{d\mid N}\varphi(d) = N$ shows that the $d = N$ term has absolute value greater than the sum of the absolute values of all other terms with $d > 1$: $\varphi(N) > N - 1 - \varphi(N)$. Cancellation is impossible. Every odd integer below $105 = 3 \cdot 5 \cdot 7$ has at most two distinct prime factors. This proves minimality. □

This example also shows why Property U cannot simply be replaced by nonvanishing in the descent proof. At period 105, define a function constant on gcd classes by the following table:

$d=(n,105)$	1	3	5	7	15	21	35	105
$f(n)$	1	−1	−1	−1	−1	1	−1	0

It has mean zero, since $48-24-12-8-6+4-2=0$. Its Dirichlet series factors as $L(s,f)=F(s)\zeta(s)$, with

$$F(s)=1-2\,3^{-s}-2\,5^{-s}-2\,7^{-s}+2\,15^{-s}+4\,21^{-s}+2\,35^{-s}-3\,105^{-s}.$$

This follows by checking the eight gcd classes, or by Möbius inversion on the divisors of 105. Direct evaluation gives $F(1)=0$ and

$$L(1,f)=F'(1)=\frac{13}{35}\log 3+\frac{5}{21}\log 5+\frac{1}{15}\log 7>0. \tag{II.14.9}$$

Thus it fails Property U despite having a positive transcendental L-value. This explicit example identifies a genuine limitation of iterating Property U descent.

Proposition II.14.7 (Elementary exclusion of a double zero)**.** *Let $N>1$ be odd and not a square. Every Erdős function of period N that is constant on gcd classes has nonzero $L(1,f)$. Equivalently, its Dirichlet-polynomial factor $F(s)$ has a simple zero at $s=1$. Nonvanishing here has an elementary proof that does not use Baker's theorem.*

Proof. Write $f(n)=a_t$ when $t=N/(n,N)$, so $a_1=0$ and $a_t\in\{\pm1\}$ for $t>1$. Partitioning by gcd gives

$$L(s,f)=F(s)\zeta(s),\qquad F(s)=\sum_{t\mid N}a_t(N/t)^{-s}\prod_{r\mid t}(1-r^{-s}).$$

The mean-zero condition gives $F(1)=N^{-1}\sum_t a_t\varphi(t)=0$. Differentiating and using this identity yields

$$F'(1)=\frac{1}{N}\sum_{r\mid N}B_r\log r,\qquad B_r=\sum_{t\mid N}a_t\varphi(t)\left(v_r(t)+\frac{\mathbf{1}_{r\mid t}}{r-1}\right). \tag{II.14.10}$$

Each B_r is an integer. Choose a prime r whose exponent $v=v_r(N)$ is odd. Terms with $r\nmid t$ make no contribution. For $t=r^je$, with $j\ge1$ and $(e,r)=1$, the unsigned weight is

$$\varphi(r^je)\left(j+\frac{1}{r-1}\right)=r^{j-1}((r-1)j+1)\varphi(e).$$

It is even when $e>1$, and odd when $e=1$. Hence $B_r\equiv v\equiv1\pmod 2$, independently of the signs. Thus at least one coefficient in (II.14.10) is nonzero. The logarithms of distinct rational primes are linearly independent over $\mathbb{Q}$: exponentiating an integer relation would contradict unique prime factorization. Therefore $F'(1)\neq0$, and $L(1,f)=F'(1)\neq0$. □

The source also asks for an approach to excluding a double zero of this Dirichlet-polynomial factor without Baker's theorem. The last proposition supplies such an approach for every odd nonsquare period, including every odd squarefree period; odd square periods are outside its conclusion. We distinguish the exact existence answer from this partial response to a broader method request, and from the quantitative descent improvement. Neither the full Erdős conjecture nor current novelty is claimed. Nor does the argument prove the irrationality or transcendence of γ: throughout, the mean-zero condition has cancelled its coefficient before the arithmetic separation is applied.

II.15 Remaining arithmetic barriers and excluded routes

The following central targets remain unproved by this manuscript: $\gamma \notin \mathbb{Q}$; transcendence of γ; irrationality of δ; the full description of the units of $\mathcal{E}$; the conjectured intersection of E-values with arithmetic Gevrey values; and the full Erdős nonvanishing conjecture. A finite family of E-values can be algebraically independent even though its singular limiting constants cannot currently be separated arithmetically. The condition on the intersection $\overline{\mathbb{Q}}(\eta)\cap\mathcal{E}$ in Theorem II.3.1 remains explicitly conditional on the transcendence of γ.

Murabayashi–Yoshida's Conjecture 7.5 [II.15] was examined again. It prescribes particular directions for monotone subsequences of fractional-part combinations arising from their even and odd Euler–Gompertz approximants. The general monotone-subsequence theorem does not supply those prescribed directions. Each proposed direction would already have strong arithmetic consequences for δ. Likewise, divergence after factorial denominator clearing does not rule out substantial cancellation by common divisors or by sign-changing filters. No such cancellation theorem was established.

Several seemingly promising historical questions were excluded after checking later sources. The incomplete-gamma question in a 2016 paper is answered by the same authors' 2017 sharp Gautschi result [II.16]. A rational gamma-bound parameter problem from 2017 has a 2020 solution [II.17]. Higher-order polygamma claims in a public 2026 archive [II.18] prevent assigning unqualified novelty to our independent proof. None of these results is silently counted as a new solution.

The manuscript therefore separates three issues: validity of each displayed argument, exact agreement with the source statement, and historical novelty or mathematical importance. The first two are addressed explicitly. The third requires further specialist comparison and review. In particular, resolving ten source assertions of mixed difficulty does not establish the ten major new breakthroughs requested as the original research goal.

II.A Exact checks used in the proofs

All principal theorems are analytic or algebraic proofs. Numerical values of tangency points are illustrations, not proof inputs. The following short Python 3 certificate verifies the rational arithmetic in the contour bound, the sharp-exponent proof, and the totient example. It uses only the standard library.

```
from fractions import Fraction as F
from math import comb, factorial

# Touchard polynomials and Bernstein coefficients, degrees 1,...,5.
S = [[1]]
for n in range(1, 6):
    row = [0]*(n+1)
    for j in range(1, n+1):
        row[j] = S[n-1][j-1]
        if j < n:
            row[j] += j*S[n-1][j]
    S.append(row)
maxima, bells = [], []
for n in range(1, 6):
    coeff = [(-1)**j*S[n][j] for j in range(n+1)]
    bern = [sum(F(comb(k,j),comb(n,j))*coeff[j]
                for j in range(k+1)) for k in range(n+1)]
    maxima.append(max(map(abs, bern)))
    bells.append(sum(S[n]))
assert maxima == [F(1), F(1,2), F(1), F(5,4), F(2)]
R = F(7,10)
bound = F(14,5)-1 + sum((a-b)*R**n/factorial(n)
           for n,(a,b) in enumerate(zip(maxima,bells),1))
assert bound == F(318799,320000) < 1
```

```
# Rigorous exponential-series upper bound.
def exp_upper(x, N):
    partial = sum(x**k/factorial(k) for k in range(N+1))
    tail = x**(N+1)/factorial(N+1)/(1-x/F(N+2))
    return partial + tail
assert exp_upper(F(7,10),20) < F(101,50)
assert exp_upper(F(51,50),20) < F(14,5)

x = F(2,3)
lower = sum(x*(2*k+x)/(k*(k+x)**2) for k in range(1,7))
lower += F(4,23)-F(29,50)
assert lower == F(889267245,1261122016) > F(7,10)
assert -2+4-6-8-12-24+48 == 0
print("All exact rational checks passed.")
```

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Part III
Scalar convergence, moment problems, and differential algebra

Source: `Keio_Euler_Constant_Structures_20261018.tex`.

Original PDF: 37 pages. Mathematical content and source-level qualifications are retained.

Abstract

We distinguish two asymptotic assertions that are combined in a recent conjecture on conservative matrix fields for Euler's constant. For every rank-three trajectory $(-A_1, -A_2; \beta, \beta, \beta + d)$ with $A_i \geq 0$ and $\beta, d \geq 1$ integral, we prove that almost every fixed real seed has dual projective limit $(1, -1, 0)^T$, contradicting the proposed full-state isolation. Nevertheless, for every nonzero fixed complex seed, the prescribed scalar ratio converges to γ and its denominator is eventually nonzero. The proof combines exact contiguity determinants, factorial separation, and concentration of positive finite sums.

We also prove the all-order coefficient pairing explicitly asked for by Xu and You, and evaluate the bordered factorial Hankel determinants conjectured by Ferguson; these two proofs are applications of classical Stieltjes and Laguerre theory. Further results determine exact differential and difference orders, a solvable differential Galois group, denominator obstructions, and rational intersections in the ring of E-function values. An application of a theorem of Bostan–Di Vizio–Raschel establishes differential transcendence in the parameter of fixed-cutoff incomplete gamma functions. We separate source-level resolutions, restricted-family results, and classical consequences. No irrationality or transcendence of Euler's constant or the Euler–Gompertz constant is established.

Status. Research manuscript prepared with AI assistance. Proofs received internal independent checks; external peer review and a comprehensive priority search have not been completed.

III.1 Results, source questions, and scope

Euler's constant and the Euler–Gompertz constant are

$$\gamma = \lim_{n\to\infty} (H_n - \log n), \qquad \delta = \int_0^\infty \frac{e^{-t}}{1+t}\, dt, \qquad H_n = \sum_{j=1}^{n} \frac{1}{j}.$$

Their approximation theory touches several distinct structures: polynomial recurrences, differential equations, positive moments, and rings of special values. The distinction between these structures matters. A function can have a completely factored differential operator and still have maximal minimal differential order. A dual vector can converge in a direction annihilated by both extraction forms while their ratio has a meaningful limit. Positive approximation can coexist with insufficient denominator control for an irrationality argument.

III.1.1 The principal recurrence result

Kalisch et al. [III.1, Definitions 20–21, Conjecture 1] prove full-state isolation and scalar convergence for a rank-three family with lower-parameter increments $(1, 1, 1)$. They conjecture both statements for all ranks and nonpositive upper increments, with positive lower increments whose minimum is repeated. Our main result treats every unequal rank-three lower tuple with a repeated minimum, up to permutation. For

$$t = (-A_1, -A_2; \beta, \beta, \beta + d), \quad A_1, A_2 \in \mathbb{Z}_{\geq 0}, \quad \beta, d \in \mathbb{Z}_{\geq 1},$$

let Φ_n be the source transition from the origin to nt, and put $p = (1, 1, 0)^T$, $q = (1, 1, 1)^T$. We prove the following two conclusions.

1. For almost every fixed real v, the projective limit of $\Phi_n^{-T} v$ is $u = (1, -1, 0)^T$, which is not the canonical Meijer state. This refutes the full-state part of Conjecture 1 (Theorem III.2.1).
2. For every fixed $v \in \mathbb{C}^3 \setminus \{0\}$, $q^T \Phi_n^{-T} v \neq 0$ eventually and

$$\frac{p^T \Phi_n^{-T} v}{q^T \Phi_n^{-T} v} \longrightarrow \gamma.$$

Thus the scalar conclusion survives on the whole stated family (Theorem III.3.1).

The mechanism is $p^T u = q^T u = 0$. Exact factorial estimates first separate the dominant dual direction. The ratio is then recovered from the subdominant coordinates using a positive weight distribution concentrated at an index of order $\sqrt{n}$. This gives a proof in the relevant quotient without relying on the false full-state extension.

III.1.2 What is resolved in the cited sources

The following accounting uses a source assertion as the unit. Parameter instances, corollaries, and algorithmic consequences are not counted as separate open problems.

Source assertion	Conclusion in this manuscript	Scope
Kalisch et al., Conjecture 1 [III.1]	Full-state extension refuted; scalar convergence proved on all unequal rank-three repeated-minimum trajectories.	One conjecture separated into a false assertion and a true restricted-family assertion.
Xu–You, Open problem after Theorem 1 [III.3]	$a_{2j+1} = -a_{2j}$ for every $j \geq 1$, with positive rational even coefficients and a determinant algorithm.	Exact answer by classical Stieltjes theory; first priority not asserted.
Ferguson, final example of Section 2 [III.4]	$P_N/Q_N = H_N$ and positive product formulas for both determinants.	Exact answer by classical Laguerre theory; first priority not asserted.
Kalisch et al., Section 9.2, item 1 [III.1]	Explicit $\gamma + \log z$ identity and a shifted-parameter extension.	The identity component only; the unshifted identity also follows directly from their equation and evaluation at one.
Rivoal, Problem 6 [III.16]; Fischler–Rivoal, Conjecture 1.2 [III.15]	Polynomial field intersections and exclusion of nonconstant polynomial units for specified exponential-integral values and combinations.	Restricted subfields; the full unit-group conjecture is not proved.

There are consequently two explicit source questions answered affirmatively and one general conjecture refuted as stated. The remaining entries are partial results or consequences, not additional full conjecture resolutions. The first three conclusions do not certify that the corresponding identities have no equivalent earlier proofs. In particular, the positive digamma continued fraction predates the Xu–You question.

III.1.3 Structural results and their logical limits

Sections III.4–III.5 connect Mellin–Barnes residues with gamma and logistic random variables, and determine a Picard–Vessiot group $\mathbb{G}_a^p \rtimes (\mathbb{G}_m \times \mathbb{G}_a)$ of dimension $p+2$. The minimal differential order of the selected Meijer solution is $p+1$. Section III.6 proves that Rivoal's fixed-parameter third-order recurrence is irreducible exactly when its parameter is nonzero. Its integral solution lattice vanishes; cumulative denominators of nonzero rational solutions grow exponentially along a subsequence, and the original CMF vector normalization has a factorial denominator barrier.

Sections III.8–III.9 classify a missing zeroth moment and show that logarithmic Gompertz moments do not satisfy polynomial-coefficient linear recurrences. A stronger functional differential-transcendence result follows from the established parabolic difference-equation theorem of Bostan–Di Vizio–Raschel. Section III.10 uses the phase morphism of Fischler–Rivoal to prove complete rational-intersection statements for the stated families.

These conclusions address different mathematical categories. In particular, functional differential transcendence is not numerical transcendence; a common denominator of a vector is not the height of a reduced scalar ratio; and minimal order of one recurrence does not exclude a different lower-order construction. These distinctions are used in the proofs and in the unresolved-target ledger at the end.

Conventions. $(x)_m = x(x+1)\cdots(x+m-1)$, with $(x)_0 = 1$; $H_n^{(r)} = \sum_{j=1}^{n} j^{-r}$, with $H_0^{(r)} = 0$; $\theta = z\,d/dz$. The symbols $\Gamma(s,a)$ and $\gamma(s,a)$ denote the upper and lower incomplete gamma functions; their two-argument notation distinguishes the latter from Euler's constant. Differential order is over $\mathbb{C}(z)$ unless specified; recurrence order is over $\mathbb{C}(n)$.

III.2 A counterexample to full-state isolation in Conjecture 1

Scope. This section concerns Conjecture 1 of Kalisch et al., arXiv:2610.13919v1 [III.1], specifically its assertion that Theorem 3 (full-state isolation) extends to every trajectory in $\mathcal{T}_r$. It does not refute the separate scalar-ratio conclusion of their Theorem 1. The distinction is essential: the two linear forms used to extract γ both annihilate the limiting direction exhibited below.

Write $\theta = z\,d/dz$. In rank three at the source point, the differential operator is

$$L_0 = \theta^3 + z(\theta+1)^2 = \theta^2(\theta+z).$$

For a germ f at $z=1$, put

$$J(f) = (f(1), \theta f(1), \theta^2 f(1)).$$

All matrix conventions below use row states, exactly as the source paper: $J(f_n) = J(f_0)\Phi_n$. Equivalently, the matrix Φ_n can be defined intrinsically by (III.2.1) on the three-dimensional solution space of L_0; the polynomial P_n is displayed below. This characterizes the transition without needing a separate list of elementary CMF matrices.

Theorem III.2.1 (Failure of full-state isolation on unequal rank-three trajectories)**.** *Let $A_1, A_2 \geq 0$ and $\beta, d \geq 1$ be integers, and consider the rank-three CMF trajectory*

$$t = (-A_1, -A_2; \beta, \beta, \beta + d).$$

Let $\Phi_n = M_{0,t}(0)\cdots M_{0,t}(n-1)$ be its transition matrix. Every matrix in this product is defined and invertible. For Lebesgue-almost every fixed $v \in \mathbb{R}^3$ there are nonzero real scalars $c_n(v)$ such that

$$c_n(v)\Phi_n^{-T} v \longrightarrow u := (1,-1,0)^T.$$

In particular, the conclusion of Theorem 3 in arXiv:2610.13919v1 cannot hold for all generic seeds on this trajectory. Consequently Conjecture 1, which explicitly includes that conclusion, is false as stated. The smallest example is $t = (0,0;1,1,2)$.

Proof. Set $K = \beta + d$ and

$$P_n(X) = (X+1)_{A_1 n}(X+1)_{A_2 n}(-X)^2_{\beta n}(-X)_{Kn},$$

where $(x)_m$ is the rising factorial. Upper-parameter decrements are implemented by $\theta - a_i + 1$, and lower-parameter increments by $-\theta + b_j$. All these scalar operators are polynomials in θ and commute. It follows from the contiguous relations, without any asymptotic argument, that

$$J(P_n(\theta)f) = J(f)\Phi_n \qquad (L_0 f = 0). \tag{III.2.1}$$

Indeed each contiguous matrix is obtained by reducing the operators $\theta^j D$ modulo the old monic differential equation, where D is the corresponding first-order contiguous operator. Thus the identity holds for every solution germ, not only the canonical Meijer G branch.

We first record the exact determinant. At arbitrary parameters, the rank-three scalar operator has polynomial in θ

$$\prod_{j=1}^{3}(\theta - b_j) + z\prod_{i=1}^{2}(\theta - a_i + 1).$$

At $z = 1$, an upper decrement has determinant $\prod_j(b_j - a_i + 1)$; a lower increment has determinant $\prod_i(b_j - a_i + 1)$. Along a path consisting first of upper decrements and then lower increments all these factors are positive integers. Multiplying them gives

$$\det \Phi_n = \prod_{i=1}^{2}\bigl(((A_i+\beta)n)!\bigr)^2((A_i+\beta+d)n)!. \tag{III.2.2}$$

Conservation gives the same transition for any admissible path. In particular, there is no singular-step issue on the stated trajectory.

Use the principal logarithm near $z = 1$, and set $H_0 = H_0^{(2)} = 0$. Three solutions of $L_0 f = 0$ are

$$\begin{aligned}
f_E(z) &= e\,e^{-z},\\
f_B(z) &= \sum_{k\ge 0}\frac{(-1)^k z^k}{k!}(\log z - H_k),\\
f_A(z) &= \sum_{k\ge 0}\frac{(-1)^k z^k}{k!}\left(\frac{(\log z)^2}{2} - H_k\log z + \frac{H_k^2 + H_k^{(2)}}{2}\right).
\end{aligned}$$

Indeed $(\theta+z)f_E = 0$, $(\theta+z)f_B = 1$, and $(\theta+z)f_A = \log z$. These identities follow either by termwise calculation, or by multiplying the integrals of e^z/z and $e^z\log z/z$ by e^{-z}. If $b = f_B(1)$ and $a = f_A(1)$, their initial jet matrix is

$$C_0 = \begin{pmatrix} 1 & -1 & 0\\ b & 1-b & -1\\ a & -a & 1\end{pmatrix}, \qquad \det C_0 = 1.$$

Let E_n, B_n, A_n be the corresponding rows after applying $P_n(\theta)$, and let U_n be the matrix with these three rows. Then

$$U_n = C_0\Phi_n, \qquad \det U_n = \det \Phi_n, \qquad \Phi_n^{-T} = C_0^T U_n^{-T}.$$

Put $s = A_1 + A_2 + 2\beta$. We establish bounds, with constants depending only on the fixed trajectory,

$$\|E_n\| \le C^n n^h (n!)^s, \qquad \|B_n\| + \|A_n\| \le C^n n^h (n!)^{s+d} \quad (n \ge 1). \tag{III.2.3}$$

The same letter $C > 1$ and integer h may be enlarged below.

For the elementary row there is an exact series

$$P_n(\theta)e^{-z} = z^{Kn}\frac{((K+A_1)n)!((K+A_2)n)!}{(dn)!^2}\sum_{j\ge 0}\frac{(-z)^j}{j!}\frac{((K+A_1)n+1)_j((K+A_2)n+1)_j}{(dn+1)_j^2}.$$

The quotient of rising factorials is at most R^j, where $R = (K+A_1)(K+A_2)/d^2$. Applying θ^m for $m = 0, 1, 2$ inserts $(Kn+j)^m$ at $z = 1$. The resulting absolute sum is $O(n^2)$ times the prefactor. The elementary multinomial bounds

$$(qn)! \ge (n!)^q, \qquad (qn)! \le q^{qn}(n!)^q \quad (q \in \mathbb{N})$$

then give the first assertion in (III.2.3).

Here are details for the other two rows. Put $Q_{m,n}(X) = X^m P_n(X)$. Termwise application of the finite-order differential operator is valid on a neighborhood of $z = 1$. The entries of B_n and A_n, for $m = 0, 1, 2$, are respectively

$$\sum_{k\ge 0}\frac{(-1)^k}{k!}(Q'_{m,n}(k) - H_k Q_{m,n}(k)),$$

$$\sum_{k\ge 0}\frac{(-1)^k}{k!}\left(\frac{Q''_{m,n}(k)}{2} - H_k Q'_{m,n}(k) + \frac{H_k^2 + H_k^{(2)}}{2}Q_{m,n}(k)\right).$$

For $0 \le k < \beta n$, P_n has a zero of multiplicity at least three, so every displayed summand vanishes. For $\beta n \le k < Kn$ the zero is simple. Writing $k = \beta n + j$ with $0 \le j < dn$, one obtains

$$\frac{|P'_n(k)|}{k!} = \frac{((\beta+A_1)n+j)!((\beta+A_2)n+j)!}{j!^2}(dn-1-j)!. \tag{III.2.4}$$

Also $|P''_n(k)| \le C\log(n+1)|P'_n(k)|$: remove the simple root and use the logarithmic derivative of the remaining linear factors. Their roots are integers, with multiplicity at most four, and their nonzero distances from k give harmonic sums of length $O(n)$. The factors X^m contribute only powers of n and logarithms.

To bound (III.2.4), put $m_i = (\beta + A_i)n$. Since $j < dn$,

$$(m_i + j)! \le m_i!\,((K+A_i)n)^j, \qquad (dn-1-j)! \le (dn)!.$$

The extra factor $(Rn^2)^j/(j!)^2$ is at most $\exp(2\sqrt{R}\,n)$, by the exponential series. Thus the sum of these dn terms is bounded by a polynomial in n times $C^n(n!)^{s+d}$.

For $k = Kn + j$, all roots of P_n are a nonzero integer distance from k. Logarithmic differentiation gives

$$|P'_n(k)| \le C\log(n+j+2)|P_n(k)|, \qquad |P''_n(k)| \le C\log^2(n+j+2)|P_n(k)|.$$

Moreover

$$\frac{|P_n(Kn+j)|}{(Kn+j)!} = \frac{((K+A_1)n)!((K+A_2)n)!}{(dn)!^2}\frac{1}{j!}\frac{((K+A_1)n+1)_j((K+A_2)n+1)_j}{(dn+1)_j^2}.$$

The tail is therefore at most the elementary prefactor times a polynomial in n and $\log(n+2)$, since $\sum R^j(j+1)^h/j!$ is finite for fixed h, R. This proves (III.2.3).

Let $w_{1,n}, w_{2,n}, w_{3,n}$ denote the rows of the cofactor matrix of U_n. Thus $w_{1,n} = B_n \times A_n$, and the other two rows are cross products involving E_n. The determinant identity and Cauchy–Schwarz give

$$\|w_{1,n}\| \ge \frac{\det U_n}{\|E_n\|}.$$

On the other hand,

$$\|w_{2,n}\| + \|w_{3,n}\| \le \|E_n\|(\|A_n\| + \|B_n\|).$$

By (III.2.2) and $(qn)! \ge (n!)^q$, $\det U_n \ge (n!)^{3s+2d}$. Combining the last inequalities yields

$$\frac{\|w_{2,n}\| + \|w_{3,n}\|}{\|w_{1,n}\|} \le \varepsilon_n := \frac{C^n n^h}{(n!)^d}. \tag{III.2.5}$$

In particular $\sum_n \sqrt{\varepsilon_n} < \infty$.

For completeness, the following elementary measure argument turns this operator estimate into a statement for fixed seeds. In any fixed bounded ball in $\mathbb{R}^3$, the set

$$\{v : |w_{1,n} \cdot v| < \sqrt{\varepsilon_n}\, \|w_{1,n}\|\}$$

is a slab of volume at most a constant times $\sqrt{\varepsilon_n}$. The first Borel–Cantelli lemma, followed by a countable union over balls, shows that almost every v belongs to only finitely many such slabs. For each such v,

$$\left|\frac{w_{i,n} \cdot v}{w_{1,n} \cdot v}\right| \le \|v\|\sqrt{\varepsilon_n} \longrightarrow 0 \qquad (i = 2, 3).$$

Since U_n^{-T} is its cofactor matrix divided by $\det U_n$, it follows that

$$\frac{\det U_n}{w_{1,n} \cdot v} \Phi_n^{-T} v = C_0^T \begin{pmatrix} 1 \\ (w_{2,n} \cdot v)/(w_{1,n} \cdot v) \\ (w_{3,n} \cdot v)/(w_{1,n} \cdot v) \end{pmatrix} \longrightarrow C_0^T e_1 = u.$$

The prefactor is nonzero for all sufficiently large n and may be chosen arbitrarily nonzero for the finitely many earlier indices.

Finally, the source's Definition 12 excludes a proper linear subspace of seeds when it requires a nonzero dominant-solution component. That subspace has Lebesgue measure zero. Hence the preceding almost-everywhere conclusion includes generic seeds in that definition. The canonical state $S_3(0)$ satisfies

$$q^T S_3(0) = 1, \qquad q = (1, 1, 1)^T,$$

by the source's Proposition 1, whereas $q^T u = 0$. These two nonzero vectors are not proportional. No scalar normalization of a sequence whose projective limit is u can converge to $S_3(0)$. This contradicts the full-state assertion on the stated trajectories. □

Remark III.2.2 (Why a convergent scalar ratio does not rescue full-state isolation). The two extraction vectors in rank three are $p = (1, 1, 0)^T$ and $q = (1, 1, 1)^T$. Both satisfy $p^T u = q^T u = 0$. Thus the preceding theorem does not determine $(p^T \Phi_n^{-T} v)/(q^T \Phi_n^{-T} v)$: it is a ratio of two subdominant components after their common dominant component cancels. Numerical convergence of this ratio to γ is fully compatible with failure of the asserted vector limit. Scalar convergence on these trajectories requires an argument in the quotient by the u-direction or an equivalent analysis of subdominant components; it cannot be justified by extending Theorem 3 unchanged. The next section supplies such a proof using positive finite sums.

III.3 Scalar convergence on unequal rank-three trajectories

Theorem III.3.1 (Scalar convergence survives on the same trajectories)**.** *Under the hypotheses of Theorem III.2.1, let $p = (1, 1, 0)^T$ and $q = (1, 1, 1)^T$. For every fixed nonzero $v \in \mathbb{C}^3$, the denominator below is nonzero for all sufficiently large n, and*

$$\lim_{n \to \infty} \frac{p^T \Phi_n^{-T} v}{q^T \Phi_n^{-T} v} = \gamma.$$

Thus the scalar conclusion of the source's Conjecture 1 is true on these rank-three trajectories, although its full-state conclusion is false.

Proof. Retain the basis, matrices, and notation from the preceding proof. Set $m_i = (\beta + A_i)n$, and, for $0 \le j < dn$, define

$$k_j = \beta n + j, \qquad W_{n,j} = \frac{(m_1 + j)!(m_2 + j)!(dn - 1 - j)!}{(j!)^2} > 0, \qquad Z_n = \sum_{j=0}^{dn-1} W_{n,j}.$$

Let $b(k) = (1, k, k^2)$ and $b'(k) = (0, 1, 2k)$. The simple-root calculation in (III.2.4), including its sign, gives

$$B_n = -\sum_{j=0}^{dn-1} W_{n,j} b(k_j) + T_{B,n}, \tag{III.3.1}$$

$$A_n = -\sum_{j=0}^{dn-1} W_{n,j}\{R_{n,j} b(k_j) + b'(k_j)\} + T_{A,n}, \tag{III.3.2}$$

where

$$R_{n,j} = H_{m_1+j} + H_{m_2+j} - 2H_j - H_{dn-1-j}.$$

To verify the latter formula, the logarithmic derivative of $P_n(X)/(X - k_j)$ at k_j gives $P_n''(k_j)/(2P_n'(k_j))$; subtracting H_{k_j} cancels the intermediate harmonic numbers. The derivative of the factor X^m supplies $b'(k_j)$. The remainders are precisely the terms $k \ge Kn$ in the preceding proof and satisfy

$$\|T_{B,n}\| + \|T_{A,n}\| \le C^n n^h (n!)^s.$$

Since $Z_n \ge W_{n,0} = m_1! m_2! (dn - 1)! \ge (n!)^{s+d}/(dn)$, these remainders are negligible relative to Z_n even after multiplication by any fixed power of n.

We need two elementary concentration facts for the probability weights $W_{n,j}/Z_n$. Put

$$c = \sqrt{\frac{(\beta + A_1)(\beta + A_2)}{d}} > 0.$$

Then

$$\frac{j}{\sqrt{n}} \longrightarrow c \quad \text{in probability}, \qquad \frac{1}{Z_n}\sum_j W_{n,j} |R_{n,j} + \gamma| \longrightarrow 0. \tag{III.3.3}$$

Here are direct estimates proving both assertions. The exact successive ratio is

$$\frac{W_{n,j+1}}{W_{n,j}} = \frac{(m_1 + j + 1)(m_2 + j + 1)}{(j + 1)^2 (dn - 1 - j)}.$$

For fixed $0 < \epsilon < c$, this ratio is uniformly greater than $1 + \delta$ below $(c - \epsilon/2)\sqrt{n}$, for a suitable $\delta > 0$ and all sufficiently large n. Hence the total weight below $(c - \epsilon)\sqrt{n}$, divided by Z_n, is $O(\sqrt{n}\, e^{-c_\epsilon \sqrt{n}})$. Choose $\eta > 0$ so small that

$$\frac{(\beta + A_1 + \eta)(\beta + A_2 + \eta)}{d - \eta} < (c + \epsilon/2)^2.$$

After an arbitrarily small enlargement of the numerator to absorb the $1/n$ terms, the same ratio is uniformly less than a constant $q < 1$ between $(c + \epsilon/2)\sqrt{n}$ and ηn. Thus the total weight from $(c + \epsilon)\sqrt{n}$ to ηn is $O(e^{-c_\epsilon \sqrt{n}}) Z_n$. For $j \ge \eta n$, use

$$\frac{W_{n,j}}{W_{n,0}} = \frac{(m_1 + 1)_j (m_2 + 1)_j}{(j!)^2} \frac{(dn - 1 - j)!}{(dn - 1)!} \le \frac{e^{Cn}}{j!}.$$

Indeed each rising-factorial quotient is at most $(Cn)^j/j! \le e^{Cn}$, and $(dn-1)!/(dn-1-j)! \ge j!$. The sum of this last range is at most $dn\, e^{Cn}/\lfloor \eta n \rfloor!$ times $W_{n,0}$, which is negligible. This proves concentration, with tails decreasing faster than any power of n.

Uniformly for $|j/\sqrt{n} - c| \le \epsilon$, the elementary asymptotic $H_N = \log N + \gamma + o(1)$ gives

$$R_{n,j} + \gamma = \log \frac{c^2}{(j/\sqrt{n})^2} + o(1).$$

Throughout the full range $0 \le j < dn$ one has $|R_{n,j}| \le C \log(n+2)$. The tail estimates, followed by $\epsilon \downarrow 0$, prove the second assertion in (III.3.3).

Next write $E_n = E_{0,n}(1, e_{1,n}, e_{2,n})$. The exact series for the elementary solution in the previous proof, together with dominated convergence, implies

$$E_{0,n} \sim e^{1-R} \frac{((K+A_1)n)!((K+A_2)n)!}{(dn)!^2}, \qquad R = \frac{(K+A_1)(K+A_2)}{d^2},$$
$$\frac{e_{1,n}}{n} \longrightarrow K, \qquad \frac{e_{2,n}}{n^2} \longrightarrow K^2.$$

For the moment estimates, domination follows from $\sum_{j \ge 0} R^j (j+1)^2/j! < \infty$; the leading exponential is nonzero. In particular $E_{0,n} \neq 0$ eventually.

For the fixed seed $v = (v_0, v_1, v_2)^T \neq 0$, set

$$h_n(k) = \frac{(E_n \times b(k)) \cdot v}{E_{0,n}} = v_0(e_{1,n}k^2 - e_{2,n}k) + v_1(e_{2,n} - k^2) + v_2(k - e_{1,n}).$$

Let $m = 3$ if $v_0 \neq 0$, $m = 2$ if $v_0 = 0, v_1 \neq 0$, and $m = 1$ otherwise. Uniformly on every fixed bounded range of $j/\sqrt{n}$,

$$\frac{h_n(k_j)}{n^m} \longrightarrow h_*(v) \neq 0,$$

where respectively

$$h_*(v) = v_0 K\beta(\beta - K), \qquad h_*(v) = v_1(K^2 - \beta^2), \qquad h_*(v) = v_2(\beta - K).$$

These constants do not vanish because $K > \beta > 0$. On the entire summation range, $|h_n(k_j)| \le C_v n^m$ and $|h_n'(k_j)| \le C_v n^{m-1}$. Consequently (III.3.3) yields

$$\frac{\sum_j W_{n,j} h_n(k_j)}{Z_n n^m} \longrightarrow h_*(v),$$
$$\frac{\sum_j W_{n,j}\{R_{n,j} h_n(k_j) + h_n'(k_j)\}}{Z_n n^m} \longrightarrow -\gamma h_*(v).$$

The remainders in (III.3.2) do not affect these limits: $\|E_n\|/|E_{0,n}| = O(n^2)$ and their factorial saving exceeds every polynomial factor.

Finally $p^T C_0^T = (0,1,0)$ and $q^T C_0^T = (0,0,1)$, so Cramer's rule gives

$$\frac{p^T \Phi_n^{-T} v}{q^T \Phi_n^{-T} v} = \frac{(A_n \times E_n) \cdot v}{(E_n \times B_n) \cdot v} = -\frac{(E_n \times A_n) \cdot v}{(E_n \times B_n) \cdot v} \longrightarrow \gamma.$$

The denominator is eventually nonzero by $h_*(v) \neq 0$ and the first weighted limit. This completes the proof. □

III.4 An exact variable-argument identity and a probability interpretation

Kalisch et al. [III.1, Proposition 1] evaluate two differential forms in a Meijer G-function at $z = 1$. Their Section 9.2 asks for an extension involving $\gamma + \log z$. The identity component has the exact extension below. It does not require a limiting argument in a matrix recurrence, and does not by itself prove the dynamical assertions or the identification with Rivoal's recurrences also discussed in that section.

Throughout, $\Omega = \mathbb{C} \setminus (-\infty, 0]$, Log is the principal logarithm, and $\theta = z\,d/dz$. For an integer $p \geq 1$ and a real number $a > -1$, define

$$\Phi_{p,a}(s) = \Gamma(a - s)\big(\Gamma(-s)\Gamma(1 + s)\big)^p, \tag{III.4.1}$$

$$G_{p,a}(z) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Phi_{p,a}(s) z^s \, ds, \qquad -1 < c < \min(0, a). \tag{III.4.2}$$

Here $z^s = \exp(s \operatorname{Log} z)$. This is $G^{p+1,p}_{p,p+1}(z \mid 0, \ldots, 0; 0, \ldots, 0, a)$ in the Mellin–Barnes convention of [III.5, §16.17]. In particular $G_{p,0}$ is the G_p of [III.1]. The value of the integral is independent of the allowed choice of c.

Put $\alpha = a + 1 > 0$, and define polynomials $P_{p,a,j}(x)$ by

$$\frac{\Gamma(\alpha - t)}{\Gamma(\alpha)} \left(\frac{\pi t}{\sin \pi t} \right)^p e^{xt} = \sum_{j \geq 0} P_{p,a,j}(x) t^j \quad (|t| < \min(1, \alpha)). \tag{III.4.3}$$

Theorem III.4.1 (Exact two-parameter identity)**.** *For $p \geq 1$, $a > -1$, $0 \leq k \leq p - 1$ and $z \in \Omega$,*

$$\Big[z(1 + \theta)^k + (-1)^p (\theta - a)\theta^k \Big] G_{p,a}(z) = \Gamma(a + 1) P_{p,a,p-k-1}(\operatorname{Log} z). \tag{III.4.4}$$

In particular, for $p \geq 2$,

$$\Big[z(1 + \theta)^{p-2} + (-1)^p (\theta - a)\theta^{p-2} \Big] G_{p,a}(z) = \Gamma(a + 1)\big(\operatorname{Log} z - \psi(a + 1)\big), \tag{III.4.5}$$

$$\Big[z(1 + \theta)^{p-1} + (-1)^p (\theta - a)\theta^{p-1} \Big] G_{p,a}(z) = \Gamma(a + 1). \tag{III.4.6}$$

Consequently the quotient of the two left sides is exactly $\operatorname{Log} z - \psi(a+1)$*, with a nonzero constant denominator. At $a = 0$ this quotient is $\gamma + \operatorname{Log} z$.*

Remark III.4.2 (A direct consequence of the existing equation)*.* For $a = 0$, the two displayed identities can alternatively be read off from the Meijer differential equation and the values at $z = 1$ already proved in [III.1]. Indeed, $\theta z = z(\theta + 1)$ gives

$$A_{p,k} = \theta^k\big(z + (-1)^p \theta\big), \qquad \theta^p\big(z + (-1)^p \theta\big) G_p = 0.$$

Thus $A_{p,p-1} G_p$ is constant and equals 1 at $z = 1$, whereas $\theta(A_{p,p-2} G_p) = 1$ and its value at $z = 1$ is γ. This proves the original variable-argument extension as a direct consequence of their existing formulas. The contour argument below also provides the shifted parameter a and every polynomial in (III.4.3).

Proof. Euler's reflection and recurrence formulas give

$$\Phi_{p,a}(s - 1) = (-1)^p (a - s) \Phi_{p,a}(s). \tag{III.4.7}$$

On any fixed vertical strip, away from its finitely many real poles, Stirling's estimate and the reflection formula imply

$$|\Phi_{p,a}(\sigma + iT)| = O\left((1 + |T|)^C e^{-(p+1/2)\pi|T|} \right).$$

Thus the integral in (III.4.2), its derivatives of every fixed order, and all contour shifts used below converge absolutely and locally uniformly on Ω. In particular, applying a polynomial in θ amounts to multiplication by that polynomial in s. Horizontal sides of the rectangles with real parts $c-1, c$ tend to zero: the additional factor z^s has modulus $|z|^\sigma e^{-T\arg z}$ and $|\arg z| < \pi$.

Let $Q(s) = (s+1)^k$ and $A(s) = Q(s)\Phi_{p,a}(s)z^s$. By (III.4.7), the left side of (III.4.4), divided by z, is

$$\frac{1}{2\pi i}\int_{\Re s=c}(A(s)-A(s-1))\,ds = \frac{1}{2\pi i}\left(\int_{\Re s=c} - \int_{\Re s=c-1}\right)A(s)\,ds.$$

Both vertical lines are directed upward. The difference is the positively oriented boundary integral of the intervening strip. The only pole in that strip is $s=-1$: the poles of $\Gamma(a-s)$ lie to the right of c, and the other integer poles are outside it. Therefore the expression is $\operatorname{Res}_{s=-1} A(s)$.

Writing $s = -1+t$, the reflection formula gives the exact local expression

$$A(-1+t) = z^{-1}t^{k-p}\Gamma(a+1-t)\left(\frac{\pi t}{\sin \pi t}\right)^p e^{t\operatorname{Log} z}.$$

Taking the coefficient of t^{-1} proves (III.4.4). The coefficients of t^0 and t^1 in (III.4.3) are 1 and $x-\psi(a+1)$. Finally $\Gamma(a+1)>0$ and $\psi(1) = -\gamma$, proving the last assertions. □

Remark III.4.3 (Parameters and source scope). For a nonnegative integer $a=m$, formula (III.4.5), divided by $m!$, gives $\gamma + \operatorname{Log} z - H_m$. For $a=-1/2$, division by $\sqrt{\pi}$ gives $\gamma + \operatorname{Log} z + 2\log 2$; this is $\gamma + \log(4z)$ on the positive axis. These special values of ψ are classical. The theorem supplies their exact occurrence in this family of differential forms. For fixed rational a, z the operator coefficients are rational, but no assertion about rationality of the resulting constant or convergence of a dual matrix system follows merely from this observation.

Proposition III.4.4 (Probability and Bell-polynomial interpretation)**.** *Let T have the gamma distribution of shape $a+1$ and rate 1, and let $L_1, \dots, L_p$ be independent standard logistic variables, also independent of T. Set $X_{p,a} = -\log T + L_1 + \cdots + L_p$. Then*

$$P_{p,a,j}(x) = \frac{1}{j!}\mathbb{E}[(x+X_{p,a})^j]. \tag{III.4.8}$$

The cumulants of $X_{p,a}$ are

$$\kappa_1 = -\psi(a+1), \qquad \kappa_m = (-1)^m\psi^{(m-1)}(a+1) + 2p\,\mathbf{1}_{2|m}(m-1)!\zeta(m) \quad (m\ge 2).$$

Equivalently, normalize the complete exponential Bell polynomials by

$$\exp\left(\sum_{m\ge1} u_m t^m/m!\right) = \sum_{j\ge0} B_j(u_1,\dots,u_j)t^j/j!.$$

Then

$$P_{p,a,j}(x) = \frac{B_j(x+\kappa_1, \kappa_2, \dots, \kappa_j)}{j!}.$$

For $a=0$ and $w=\gamma+x$, the first four polynomials are

$$1, \qquad w, \qquad \frac{w^2+(2p+1)\zeta(2)}{2}, \qquad \frac{w^3+3(2p+1)\zeta(2)w+2\zeta(3)}{6}.$$

The formula of index j occurs in Theorem III.4.1 whenever $j \le p-1$.

Proof. The gamma density gives $\mathbb{E}[e^{-t\log T}] = \Gamma(a+1-t)/\Gamma(a+1)$ for $t < a+1$. A standard logistic variable has density $e^{-v}/(1+e^{-v})^2$; substituting $u=e^{-v}$ and using Euler's beta integral yields

$$\mathbb{E}[e^{tL_i}] = \int_0^\infty \frac{u^{-t}}{(1+u)^2}\,du = \Gamma(1-t)\Gamma(1+t) = \frac{\pi t}{\sin \pi t} \qquad (|t|<1).$$

Independence identifies (III.4.3) as the moment generating function of $x + X_{p,a}$, proving (III.4.8). All moments exist because that generating function is finite in a neighborhood of zero. Its logarithm has the expansion

$$xt + \log\Gamma(a+1-t) - \log\Gamma(a+1) + p\sum_{\ell\geq 1}\frac{\zeta(2\ell)}{\ell}t^{2\ell}.$$

The cumulant and Bell formulas follow by Taylor expansion. At $a = 0$ use $\psi^{(m-1)}(1) = (-1)^m(m-1)!\zeta(m)$ for $m \geq 2$. □

Proposition III.4.5 (Classical incomplete-gamma endpoint)**.** *For $a > -1$ and $z \in \Omega$,*

$$G_{1,a}(z) = \int_0^\infty \frac{e^{-u}u^a}{u+z}\,du = \Gamma(a+1)z^a e^z\Gamma(-a,z).$$

In particular $G_{1,0}(z) = e^z E_1(z)$, where $E_1(z) = \Gamma(0,z)$.

Proof. First take $z > 0$. Mellin inversion for $(1+x)^{-1}$, followed by absolute integration on $\Re s = c$, gives the first equality; the inner integral is $\Gamma(a-s)$ and $c < a$. Alternatively its Mellin transform uniquely identifies the same absolutely convergent Barnes integral. To evaluate the real integral, write $(u+z)^{-1} = \int_0^\infty e^{-v(u+z)}\,dv$ and use Tonelli's theorem. The result is

$$\Gamma(a+1)\int_0^\infty e^{-zv}(1+v)^{-a-1}\,dv = \Gamma(a+1)z^a e^z\Gamma(-a,z),$$

by $w = z(1+v)$. All three expressions are holomorphic on Ω, so analytic continuation proves the stated domain. This is the classical incomplete-gamma integral representation; see [III.5, §8.6]. □

An exact finite telescoping remainder

The following certificate may be useful for checking computations. It is an identity of analytic expressions, not a rational approximation theorem: the residues themselves can contain γ and zeta values. For $a = 0$ put

$$F_{p,N}(z) = -\sum_{n=0}^{N}\operatorname{Res}_{s=n}\bigl(\Phi_{p,0}(s)z^s\bigr), \qquad A_{p,k} = z(1+\theta)^k + (-1)^p\theta^{k+1}.$$

For every $N \geq 0$, $0 \leq k \leq p-1$ and $z \in \Omega$,

$$P_{p,0,p-k-1}(\operatorname{Log} z) - A_{p,k}F_{p,N}(z) = z\operatorname{Res}_{s=N}\bigl((s+1)^k\Phi_{p,0}(s)z^s\bigr). \tag{III.4.9}$$

Indeed, applying $A_{p,k}/z$ turns each summand into the difference of the residues of $A(s)$ and $A(s-1)$ from the preceding proof. The finite sum telescopes to the residues at -1 and N, with exactly the signs in (III.4.9).

More explicitly, define $H_N^{(m)} = \sum_{\nu=1}^N \nu^{-m}$, $H_N = H_N^{(1)}$ (empty sums are zero), and $w_N = \operatorname{Log} z + \gamma - H_N$. The right side of (III.4.9) is exactly

$$\frac{(-1)^{(N+1)(p+1)}z^{N+1}}{N!}[t^p](N+1+t)^k\exp\left(w_N t + \sum_{m=2}^{p}\frac{(1+2p\mathbf{1}_{2|m})\zeta(m) + (-1)^m H_N^{(m)}}{m}t^m\right). \tag{III.4.10}$$

To verify this, use

$$\Gamma(-N-t) = \frac{(-1)^{N+1}\Gamma(1-t)}{N!\,t\prod_{\nu=1}^N(1+t/\nu)}$$

and expand the logarithm at $t = 0$; powers higher than p cannot affect $[t^p]$. For fixed p, k, z this gives the rigorous bound

$$P_{p,0,p-k-1}(\operatorname{Log} z) - A_{p,k}F_{p,N}(z) = O_{p,k,z}\left(\frac{|z|^{N+1}(N+1)^k(1+\log(N+1))^p}{N!}\right).$$

Here $H_N = O(\log(N+1))$, while $H_N^{(m)} \le \zeta(m)$ for $m \ge 2$; the coefficient in (III.4.10) is a fixed finite polynomial in these quantities. Thus no unproved asymptotic summation or interchange of infinite residue sums is needed for this remainder bound.

III.5 Exact differential order and the full solvable Galois group

The identities in the preceding section expose a particularly simple differential module behind the Meijer functions. This section concerns differentiation in z, not the difference equations in the trajectory index. Reducibility in one of these settings does not imply reducibility in the other.

Put $\theta = z\,d/dz$, $\epsilon = (-1)^p$, and retain the function $G_p = G_{p,0}$ defined by the Mellin–Barnes integral above. The standard Meijer equation factors in the Ore ring $\mathbb{C}(z)[\theta]$:

$$\mathcal{L}_p = z(1+\theta)^p + (-1)^p\theta^{p+1} = \theta^p(z + (-1)^p\theta). \tag{III.5.1}$$

The identity uses $\theta z = z(\theta+1)$; factors must not be commuted. The factorization itself is an elementary consequence of the standard equation. We give the precise order and Galois group, without a claim that this application of classical differential algebra resolves an external open conjecture.

Theorem III.5.1 (Minimality despite complete factorization)**.** *For each integer $p \ge 2$, the minimal homogeneous differential order of G_p over $\mathbb{C}(z)$ is exactly $p+1$. Its local monodromy at $z = 0$ has a single unipotent Jordan block of size $p+1$.*

Proof. Equation (III.5.1) supplies the upper bound. At the pole $s = 0$ of the Mellin kernel,

$$\Gamma(-s)^{p+1}\Gamma(1+s)^p = (-1)^{p+1}s^{-p-1}(1+O(s)).$$

Closing the contour to the right gives a minus sign. Thus the coefficient of $(\log z)^p$ in the local Frobenius expansion of G_p at zero has constant term $(-1)^p/p! \ne 0$. More explicitly, the poles at $s = 0, 1, 2, \dots$ give

$$G_p(z) = \sum_{j=0}^{p} h_j(z)(\log z)^j, \qquad h_j \text{ holomorphic near } 0, \qquad h_p(0) = (-1)^p/p!.$$

The convergence follows equally from the regular singular equation and its Frobenius construction; the residue at zero identifies the nonzero top coefficient. If T denotes continuation once around zero, then $(T-1)^{p+1}G_p = 0$ and $(T-1)^pG_p = (2\pi i)^p p!h_p \ne 0$. Consequently $G_p, (T-1)G_p, \dots, (T-1)^pG_p$ are linearly independent over $\mathbb{C}$. Indeed, applying a suitable power of $T-1$ to a shortest nontrivial linear relation gives a contradiction. Every rational-coefficient differential equation annihilating G_p also annihilates these continuations, and so has order at least $p+1$. Their span is the full solution space of $\mathcal{L}_p$, proving the Jordan assertion as well. □

Fix a simply connected domain avoiding zero, put $\ell = \log z$ and $E = e^{\epsilon z}$, and choose primitives

$$J_j'(z) = \frac{E(z)\ell(z)^j}{z}, \qquad 0 \le j < p. \tag{III.5.2}$$

All differential fields below consist of analytic or meromorphic germs on such a domain and have constant field $\mathbb{C}$.

Lemma III.5.2 (A rational integration obstruction)**.** *Let $\epsilon \in \mathbb{C}^*$, $E = e^{\epsilon z}$, $F = \mathbb{C}(z, \ell)$ and $K = F(E)$. For every nonzero polynomial $P \in \mathbb{C}[X]$, the function $EP(\ell)/z$ has no primitive in K.*

Proof. First, no nonzero $a \in F$ satisfies $a'/a = c$ for a nonzero constant c. Expand at $\ell = \infty$: $a = r(z)\ell^d(1 + O(\ell^{-1}))$ with $r \in \mathbb{C}(z)^*$ and $d \in \mathbb{Z}$. Comparing leading powers gives $r'/r = c$, which is impossible for a rational function. This also proves that E is transcendental over F: if it were algebraic, its field norm would be a nonzero element of F with logarithmic derivative a positive integer multiple of ϵ. The transcendence of ℓ over $\mathbb{C}(z)$ follows from its infinite additive monodromy.

Suppose $f \in F(E)$ satisfies $f' = EP(\ell)/z$. Expand the rational function f as a formal Laurent series in the transcendental variable E, with coefficients in F. Differentiation takes the coefficient of E^j to $a_j' + j\epsilon a_j$. For $j \neq 0, 1$ the preceding observation forces $a_j = 0$, and for $j = 0$ it forces a_0 to be constant. Thus $f = c + Ea$ with $a \in F$, and

$$a' + \epsilon a = P(\ell)/z. \tag{III.5.3}$$

We claim that a is polynomial in ℓ. A finite pole in that variable, along an irreducible polynomial $h \in \mathbb{C}(z)[\ell]$, would acquire one higher order under differentiation unless h divided h'. Such a divisor cannot occur. An algebraic root $\alpha(z)$ of h would satisfy $\alpha' = 1/z$ (differentiate $h(z, \alpha(z)) = 0$), whereas an algebraic function has a Puiseux expansion at zero whose derivative has zero coefficient of z^{-1}. The right side of (III.5.3) has no finite poles in ℓ, so all such poles are excluded.

Write $a = \sum_{j=0}^{d} a_j(z)\ell^j$. If $d > \deg P$, its top coefficient satisfies $a_d' + \epsilon a_d = 0$, again impossible. If $d < \deg P$, the top power on the right is absent on the left. Therefore $d = \deg P$ and the top coefficient must satisfy $a_d' + \epsilon a_d = c/z$ with $c \neq 0$. A rational solution can have no pole away from zero, since differentiation would raise its order; a pole at zero would similarly produce a pole of order at least two. With no pole at zero, the left side is holomorphic there and cannot equal c/z. This final contradiction proves the lemma. □

Theorem III.5.3 (Full differential Galois group)**.** *For every $p \geq 2$ the Picard–Vessiot field of $\mathcal{L}_p$ is*

$$\mathcal{P}_p = \mathbb{C}(z)(E, \ell, J_0, \ldots, J_{p-1}).$$

The $p + 2$ displayed generators are algebraically independent over $\mathbb{C}(z)$. Every differential automorphism is uniquely specified by $a \in \mathbb{C}^*$, $b \in \mathbb{C}$, and $c_0, \ldots, c_{p-1} \in \mathbb{C}$ through

$$E \longmapsto aE, \qquad \ell \longmapsto \ell + b,$$
$$J_j \longmapsto a \sum_{k=0}^{j} \binom{j}{k} b^{j-k} J_k + c_j \qquad (0 \leq j < p).$$

In particular the group is $\mathbb{G}_a^p \rtimes (\mathbb{G}_m \times \mathbb{G}_a)$, with the action specified by these formulas, and has dimension $p + 2$.

Proof. The lemma proved the algebraic independence of E and ℓ. The Kolchin–Ostrowski theorem for antiderivatives [III.10, Theorem 2.3] says that algebraic dependence among $J_0, \ldots, J_{p-1}$ over K would give a nonzero constant linear combination belonging to K. Its derivative would contradict Lemma III.5.2. This proves the asserted algebraic independence.

A full fundamental family of solutions is

$$E^{-1},\ E^{-1}J_0, \ldots, E^{-1}J_{p-1}.$$

Indeed $(\theta+\epsilon z)(E^{-1}J_j)=\ell^j$ and its pth Euler derivative vanishes. Applying $\theta+\epsilon z$ to a linear relation proves independence. Their differential field contains E and every J_j; since $p\geq 2$, it also contains $\ell=(\theta+\epsilon z)(E^{-1}J_1)$. It is therefore exactly the displayed field.

The differential equations force $E\mapsto aE$ and $\ell\mapsto\ell+b$. Differentiating the asserted formula for J_j shows that it is the general possible image, up to the constant c_j. Conversely, algebraic independence ensures that every indicated assignment extends to an automorphism of the rational function field, and the equations verify that it commutes with differentiation. The subgroup $a=1,b=0$ is the translation group $\mathbb{G}_a^p$; setting all $c_j=0$ gives the complementary $\mathbb{G}_m\times\mathbb{G}_a$. This proves the group description. □

Remark III.5.4 (Functions versus values). The preceding independence is over $\mathbb{C}(z)$ and concerns functions. It makes no claim about independence of their values at $z=1$. In fact the Meijer identities explicitly produce relations among such values involving γ and zeta values. No specialization principle applicable to arbitrary Liouvillian functions turns this theorem into a proof of the irrationality of γ.

III.6 Difference irreducibility and denominator obstructions

This section concerns the trajectory index n. Its conclusions are independent of the differential factorization in the variable z. We prove exact difference order and arithmetic obstructions for the Rivoal recurrence recovered in [III.1, §8.4]. These are structural deductions; they are not presented as resolutions of external open conjectures or as a proof of the irrationality of γ.

III.6.1 An elementary irreducibility certificate

Write $Ef(n)=f(n+1)$, with the Ore rule $Ec(n)=c(n+1)E$. For a fixed complex parameter x, the recurrence in [III.2, (1.3)] is $L_xu=0$, where

$$L_x=\sum_{j=0}^{3}a_j(n,x)E^j \tag{III.6.1}$$

and

$$\begin{aligned}
a_0&=(n+2)^2(3x-8n-22)(3x-8n-30),\\
a_1&=B_1(n,x)(3x-8n-30),\qquad a_2=-B_2(n,x)(3x-8n-14),\\
a_3&=-(n+3)^2(3x-8n-14)(3x-8n-22).
\end{aligned}$$

Here the two cubic polynomials are

$$\begin{aligned}
B_1(n,x)&=24n^3+(162-57x)n^2+(26x^2-258x+356)n\\
&\quad-3x^3+59x^2-285x+254,\\
B_2(n,x)&=24n^3+(7x+210)n^2+(-6x^2+56x+600)n\\
&\quad-15x^2+96x+564.
\end{aligned}$$

Sequences in the general-parameter statements are understood as germs at $n=+\infty$: changing finitely many terms is immaterial.

Lemma III.6.1 (A coefficient obstruction to a hypergeometric solution)**.** *Let $L=\sum_{j=0}^{3}b_j(n)E^j$ have polynomial coefficients $b_j=\alpha_jn^m+\beta_jn^{m-1}+O(n^{m-2})$, where*

$$\sum_{j=0}^{3}\alpha_jT^j=c(1-T)^3,\qquad c\neq 0,\qquad \sum_{j=0}^{3}\beta_j\neq 0.$$

Then L has no right factor of order one over $\mathbb{C}(n)$.

Proof. An order-one right factor can be made $E - r(n)$, where $r \in \mathbb{C}(n)^*$, since the constant and leading coefficients of L are nonzero. Its nonzero solution has $u_{n+j}/u_n = \prod_{i=0}^{j-1} r(n+i)$. Write $r(n) = \lambda n^d(1 + O(n^{-1}))$, with $d \in \mathbb{Z}$ and $\lambda \neq 0$. If $d > 0$, the $j = 3$ term of Lu/u has uniquely largest degree at infinity; if $d < 0$, the $j = 0$ term does. Both are impossible. Hence $d = 0$, and the leading coefficient equation is $c(1 - \lambda)^3 = 0$, so $\lambda = 1$. Now $r(n) = 1 + b/n + O(n^{-2})$, whence $u_{n+j}/u_n = 1 + jb/n + O(n^{-2})$. The coefficient of n^{m-1} in Lu/u is

$$\sum_j \beta_j + b \sum_j j\alpha_j = \sum_j \beta_j \neq 0,$$

again a contradiction. □

Theorem III.6.2 (Exact difference order for every nonzero parameter)**.** *For fixed $x \in \mathbb{C}$, L_x is irreducible in $\mathbb{C}(n)[E]$ if and only if $x \neq 0$. For $x \neq 0$, every solution which is not eventually zero has minimal recurrence order exactly three over $\mathbb{C}(n)$.*

Proof. Expansion of the displayed coefficients gives

$$(\alpha_0, \alpha_1, \alpha_2, \alpha_3) = 64(1, -3, 3, -1),$$

$$(\beta_0, \beta_1, \beta_2, \beta_3) = (672 - 48x,\ 528x - 2016,\ 2016 - 16x,\ 48x - 672).$$

In particular, $\sum \beta_j = 512x \neq 0$, so the lemma excludes an order-one right factor. To exclude an order-one left factor, use the formal adjoint, defined by $f(n)^* = f(n)$ and $E^* = E^{-1}$ in the Laurent Ore ring. Multiplying L_x^* on the left by E^3 gives the polynomial operator

$$L_x^\vee = \sum_{j=0}^{3} a_j(n + 3 - j, x)E^{3-j}.$$

Its leading characteristic polynomial is $64(T - 1)^3$, and the sum of its next coefficients is

$$\sum_j \{\beta_j + 4(3 - j)\alpha_j\} = 512x,$$

since $\sum \alpha_j = \sum j\alpha_j = 0$. The lemma therefore also excludes an order-one right factor of $L_x^\vee$. The adjoint reverses a factorization, so this excludes an order-one left factor of L_x. Nonzero constant coefficients ensure that shifting the adjoint factors to nonnegative powers of E introduces no loss of order. Every proper factorization of an order-three operator has an order-one factor on one side.

Finally, choose a minimal monic annihilator of a nonzero solution germ. Right division of L_x by this annihilator has a remainder of smaller order which still annihilates the germ. Minimality forces the remainder to be zero. Irreducibility makes the minimal annihilator have order three.

For completeness, at $x = 0$ direct substitution gives $L_0(n + 1) = 0$. Thus $E - (n + 2)/(n + 1)$ is a right factor and L_0 is reducible. □

The exceptional parameter $x = 0$ is therefore genuine. The theorem rules out an order-two recurrence for these particular sequences. It does not rule out a different order-two construction of rational approximations to γ.

III.6.2 The integral solution lattice is zero

At $x = 1$, write the recurrence with positive leading coefficient as

$$\begin{aligned}(n + 3)^2(8n + 11)(8n + 19)u_{n+3} \\ = (n + 3)(8n + 11)(24n^2 + 145n + 215)u_{n+2} \\ - (8n + 27)(24n^3 + 105n^2 + 124n + 25)u_{n+1} + (n + 2)^2(8n + 19)(8n + 27)u_n. \quad \text{(III.6.2)}\end{aligned}$$

Its coefficients of u_n and u_{n+3} do not vanish for $n \geq 0$. Thus its solution space is three-dimensional, and an eventually zero solution is identically zero.

Lemma III.6.3 (Growth and the only possible rational generating functions)**.** *For fixed $x \neq 0$, every solution of $L_x u = 0$ satisfies, for every $\varepsilon > 0$, $|u_n| \leq C_\varepsilon e^{\varepsilon n}$. The ordinary generating function of a solution which is not eventually zero is D-finite and is not rational.*

Proof. The companion matrices converge to a constant matrix all of whose eigenvalues equal one, because the limiting characteristic polynomial is $(T-1)^3$. For any $\delta > 0$ there is a vector norm whose induced matrix norm gives this limiting matrix norm at most $1+\delta$: rescale the coordinates in each Jordan block to make its off-diagonal entries arbitrarily small. In that norm the companion matrices eventually have norm at most $1+2\delta$. Their products imply the growth assertion, after decreasing δ and absorbing finitely many terms into C_ε.

Let $F(t) = \sum_{n\geq 0} u_n t^n$ and $\vartheta = t\,d/dt$. Multiplication of the recurrence by t^{n+3} and summation gives

$$\left(\sum_{j=0}^{3} t^{3-j} a_j(\vartheta - j, x) \right) F(t) = P(t), \tag{III.6.3}$$

where P is a polynomial accounting for initial terms. This proves D-finiteness, by homogenizing the equation if necessary. The coefficient of $F^{(4)}$ on the left is

$$t^4 \sum_{j=0}^{3} \alpha_j t^{3-j} = 64t^4(t-1)^3.$$

If F were rational, it could have no pole at an ordinary point of this equation: its highest derivative would then have a pole of strictly greater order than all the lower derivative terms. It is analytic at zero, so its only possible finite pole is $t = 1$. Partial fractions imply that u_n is eventually a polynomial in n. A nonzero such polynomial supplies a rational quotient u_{n+1}/u_n, contradicting Lemma III.6.1; a zero polynomial is the excluded eventually zero case. □

Theorem III.6.4 (No nonzero bounded-denominator solution)**.** *The only solution of* (III.6.2) *taking values in $D^{-1}\mathbb{Z}$ for one fixed positive integer D is the zero solution. In particular, its integral solution lattice is $\{0\}$. More quantitatively, for a nonzero rational solution put*

$$d_N = \operatorname{lcm}\{\operatorname{den}(u_0), \ldots, \operatorname{den}(u_N)\}.$$

Then

$$\limsup_{N\to\infty} \frac{\log d_N}{N} > 0. \tag{III.6.4}$$

Proof. First suppose the denominator is bounded. Then DF is an integer-coefficient D-finite series with radius of convergence at least one, by the growth lemma. If it has infinitely many nonzero coefficients, its radius cannot exceed one. The Pólya–Carlson theorem states that an integer series of radius one is rational or has the unit circle as a natural boundary. D-finiteness excludes a natural boundary because a polynomial differential equation has only finitely many possible singular points. Thus F is rational, contrary to the lemma. The finite-series case is excluded by the same lemma.

For the stronger assertion, suppose instead that $\log d_N = o(N)$. For a rational number a, its logarithmic height satisfies $h(a) \leq \log \operatorname{den}(a) + \log^+ |a|$. Consequently

$$\max_{0\leq n\leq N} h(u_n) \leq \log d_N + \max_{0\leq n\leq N} \log^+ |u_n| = o(N),$$

where the last estimate uses the growth lemma for every ε. The univariate rationality theorem of Bell–Chen–Nguyen–Zannier [III.11, Proposition 2.1] now says that F is rational: its hypotheses

are precisely D-finiteness, sublinear maximal height, and $\log d_N = o(N)$. This contradiction proves (III.6.4). The limsup assertion is equivalent to the failure of $\log d_N = o(N)$ because these quotients are nonnegative. □

The same argument applies to every rational solution germ of L_x with fixed $x \neq 0$, whenever its coefficients are rational. The resulting rational-coefficient series is D-finite over $\mathbb{Q}(t)$: existence of a polynomial-coefficient differential relation descends from $\mathbb{C}$ to $\mathbb{Q}$ by rational linear algebra on its Taylor coefficients. The formulation at $x = 1$ avoids exceptional finite indices and is the one relevant to Euler's constant. No effective universal constant in (III.6.4) is asserted.

III.6.3 A complementary local obstruction from the Casoratian

Proposition III.6.5 (Exact Casoratian and prime-by-prime rank bound)**.** *For three solutions of* (III.6.2)*, let* $W_N = \det(u^{(i)}_{N+j})_{1\le i\le 3,\,0\le j\le 2}$*. Then*

$$W_N = \frac{4(8N+11)(8N+19)}{209(N+2)^2}\,W_0. \tag{III.6.5}$$

For every prime p, the rational solution space consisting of sequences whose p-adic valuations are bounded below has dimension at most two over $\mathbb{Q}$.

Proof. Taking the determinant of the companion update gives

$$\frac{W_{N+1}}{W_N} = \frac{(N+2)^2(8N+27)}{(N+3)^2(8N+11)}.$$

This identity, interpreted without division when $W_N = 0$, telescopes to (III.6.5). For an independent rational triple, $W_0 \neq 0$. At $N = p^k - 2$, for all sufficiently large k,

$$v_p(W_{p^k-2}) = v_p(4W_0/209) + v_p(15) - 2k.$$

Indeed $(8N+11)(8N+19) = (8p^k-5)(8p^k+3)$ has valuation $v_p(15)$ once k is sufficiently large. The displayed valuation tends to $-\infty$. If all three sequences had valuations bounded below, every determinant term would have a uniform lower bound, and so would their sum. This is a contradiction. □

This local statement does not say that every nonzero solution has an unbounded denominator at every prime. Also $W_N \to 256W_0/209$, not zero. A direct "integer determinant tends to zero" argument is therefore unavailable; the generating-function argument is what strengthens a rank restriction to the zero integral lattice.

III.6.4 Factorial denominators in the original CMF normalization

The gauge transformation in [III.1, §8.4] permits a stronger statement for the unrescaled dual matrix trajectory. Put

$$A(n) = \frac{1}{(n+1)^4}\begin{pmatrix} 6n^2+13n+10 & -8n-5 & 3 \\ 3n^3+3n^2-3n-6 & -3n^2+3n+3 & n-2 \\ n^4-2n^3-6n^2-3n+3 & 6n^2+2n-1 & 1-2n \end{pmatrix},$$

and define row-vector solutions by $u_{n+1} = u_n A(n)$. The following explicit gauge is useful for making the normalization unambiguous:

$$G(n) = \begin{pmatrix} 0 & \dfrac{8n+11}{(n+1)^2} & \dfrac{24n^2+97n+94}{(n+1)^2(n+2)} \\ -1 & \dfrac{3n^2-n-7}{(n+1)^2} & \dfrac{15n^4+69n^3+66n^2-78n-111}{(n+1)^2(n+2)^2} \\ 2 & \dfrac{-6n^2-6n+3}{(n+1)^2} & \dfrac{8n^5+21n^4-38n^3-131n^2-48n+53}{(n+1)^2(n+2)^2} \end{pmatrix}.$$

Let $C(n)$ be the row companion matrix of (III.6.2), so that

$$(w_n, w_{n+1}, w_{n+2})C(n) = (w_{n+1}, w_{n+2}, w_{n+3}).$$

Direct polynomial identities give

$$\begin{aligned} C(n) &= (n+1)^2 G(n)^{-1} A(n) G(n+1), \\ \det G(n) &= \frac{(8n+11)(8n+19)}{(n+2)^2}, \qquad \det A(n) = (n+1)^{-6}. \end{aligned} \tag{III.6.6}$$

These identities may be checked by clearing denominators; no asymptotic matching or numerical inference is required.

Theorem III.6.6 (Raw-vector factorial denominator barrier)**.** *Let $u_0 \in \mathbb{Q}^3 \setminus \{0\}$, and let D_n be the least positive integer for which $D_n u_n \in \mathbb{Z}^3$. For every $\varepsilon > 0$ there is $c_\varepsilon > 0$ such that*

$$D_n \geq c_\varepsilon (n!)^2 e^{-\varepsilon n}, \qquad D_n \mid D_0 (n!)^4. \tag{III.6.7}$$

In particular,

$$2 \leq \liminf_{n\to\infty} \frac{\log D_n}{n \log n} \leq \limsup_{n\to\infty} \frac{\log D_n}{n \log n} \leq 4.$$

Proof. By (III.6.6), $v_n = (n!)^2 u_n G(n)$ satisfies $v_{n+1} = v_n C(n)$. Its coordinates are consecutive terms of a scalar solution of (III.6.2). Lemma III.6.3, together with the polynomial growth of the rational matrix $G(n)^{-1}$, gives

$$\|u_n\|_\infty \leq C_\varepsilon (n!)^{-2} e^{\varepsilon n}.$$

All matrices $A(n)$ are invertible for $n \geq 0$, so $u_n \neq 0$. The nonzero integer vector $D_n u_n$ has sup norm at least one. Combining the two inequalities gives the lower bound in (III.6.7). The entries of $(n+1)^4 A(n)$ are integers. Induction therefore gives the divisibility assertion. Stirling's estimate $\log(n!) = n \log n + O(n)$ gives the last statement. □

Remark III.6.7 (What this denominator bound does not establish). Theorem III.6.6 is tied to the raw vector u_n. A common rescaling changes D_n, and taking a ratio of two linear forms in u_n can cancel large common factors. Hence the theorem does not give the height of the reduced scalar approximants and does not establish their reported irrationality measure parameter $\delta = -1$. A gcd-controlled lower bound for the reduced fractions would require an additional argument. Nor does the theorem alone imply any irrationality statement for their limit.

Prior-art boundary. Rivoal's original denominator conjectures for his specially normalized approximants were proved by Hessami Pilehrood and Hessami Pilehrood [III.12]; they must not be counted as newly solved problems. The exact-order statement above concerns the existing recurrence, and the denominator results concern its full solution space or the specified CMF normalization. No general CMF convergence conjecture is settled by these arguments.

III.7 The all-order pairing problem of Xu and You

Xu and You [III.3], *Continued fraction inequalities for the Euler–Mascheroni constant*, Journal of Inequalities and Applications **2014**, 343, DOI 10.1186/1029-242X-2014-343, formulate an explicit "Open problem" immediately after Theorem 1: do the uniquely successively optimized coefficients of their continued fraction satisfy $a_{2j+1} = -a_{2j}$ for every $j \geq 1$? They compute through a_{13}; Remark 4 also asks for a method to find the subsequent coefficients. The following argument answers the coefficient question through the classical correspondence between positive moments and Stieltjes fractions. A first-priority claim requires a broader bibliographic check; the theorem below is a mathematical resolution of the explicitly stated question.

Put $h(z) = \psi(z+1) - \log z$, so that $H_n - \log n = \gamma + h(n)$. Precisely, the fraction in question is

$$R_k(z) = \cfrac{a_1}{z + \cfrac{a_2 z}{z + \cfrac{a_3 z}{z + \ddots + \cfrac{a_{k-1} z}{z + a_k}}}}, \qquad a_1 = \frac{1}{2}.$$

For $k = 1$ this means $R_1(z) = a_1/z$. Each finite fraction is interpreted as a rational function after cancellation; a zero internal denominator in an unsimplified nesting can therefore be removable. Successive optimization means choosing the new coefficient to cancel the first remaining term of the Poincare expansion as $z \to +\infty$.

Theorem III.7.1 (All-order coefficient pairing)**.** *There is a unique infinite sequence of nonzero rational coefficients a_j such that*

$$h(z) - R_k(z) = C_k z^{-k-1} + O(z^{-k-2}) \quad (k \geq 1), \qquad C_k \neq 0.$$

Here uniqueness refers to the successive optimization rule just stated. Every even-indexed coefficient is positive and

$$a_{2j+1} = -a_{2j} \quad (j \geq 1), \qquad C_k = (-1)^k \prod_{j=1}^{k+1} a_j.$$

All the coefficients are given by the positive Hankel determinants below. In particular,

$$a_{14} = \frac{3754087889491759}{2440840521848406}, \qquad a_{15} = -a_{14}.$$

Proof. Binet's digamma integral [III.5, 5.9.15], together with $\psi(z+1) = \psi(z) + 1/z$, gives, for $z > 0$,

$$h(z) = \frac{1}{2z} - 2\int_0^\infty \frac{t\,dt}{(z^2+t^2)(e^{2\pi t}-1)}.$$

Consequently, on writing $x = 1/z$,

$$F(x) := \frac{2h(1/x)}{x} = 1 - xU(x^2), \qquad U(y) = \int_0^\infty \frac{d\mu(s)}{1+ys}, \quad d\mu(s) = \frac{2\,ds}{e^{2\pi\sqrt{s}}-1}. \tag{III.7.1}$$

This positive measure has infinite support and all finite moments

$$\mu_n = \int_0^\infty s^n\, d\mu(s) = \frac{4\Gamma(2n+2)\zeta(2n+2)}{(2\pi)^{2n+2}} = \frac{2|B_{2n+2}|}{2n+2} \in \mathbb{Q}_{>0}.$$

The geometric identity under the integral implies the all-orders Poincare expansion $U(y) \sim \sum_{n\geq 0}(-1)^n \mu_n y^n$ as $y \downarrow 0$; after any fixed number of terms its remainder is bounded by the next moment times the corresponding power of y.

For $r \geq 0$ and $\epsilon \in \{0, 1\}$ define

$$D_r^{(\epsilon)} = \det(\mu_{i+j+\epsilon})_{0 \leq i,j < r}, \qquad D_0^{(\epsilon)} = 1.$$

Every such determinant is strictly positive: the associated matrix is the Gram matrix of $1, s, \ldots, s^{r-1}$ in the positive measure $s^\epsilon d\mu(s)$, and a nonzero polynomial cannot vanish on its support. The classical Stieltjes moment/continued-fraction theorem [III.6, III.7] therefore gives the unique formal fraction

$$U(y) \sim \cfrac{\mu_0}{1 + \cfrac{\lambda_1 y}{1 + \cfrac{\lambda_2 y}{1 + \ddots}}}, \qquad \lambda_j > 0, \tag{III.7.2}$$

where the explicit determinant formulas are

$$\lambda_{2r-1} = \frac{D_r^{(1)} D_{r-1}^{(0)}}{D_r^{(0)} D_{r-1}^{(1)}}, \qquad \lambda_{2r} = \frac{D_{r+1}^{(0)} D_{r-1}^{(1)}}{D_r^{(1)} D_r^{(0)}} \quad (r \geq 1).$$

The formal statement used here is the standard finite-moment version: the fraction truncated after λ_m agrees with the moment series through degree m, and its next error coefficient is nonzero. It follows also from Gram orthogonalization and the two adjacent Padé determinant formulas. No convergence of a divergent moment power series is assumed. All λ_j are positive rational numbers.

Set

$$c_1 = \mu_0 = \frac{1}{6}, \qquad c_{j+1} = \frac{\lambda_j}{c_j} \quad (j \geq 1).$$

The finite pair-contraction identity, valid as a rational-function identity for every $m \geq 1$, is

$$\frac{1}{1 + \underset{j=1}{\overset{2m}{\mathrm{K}}} \frac{b_j x}{1}} = 1 - \frac{c_1 x}{1 + \underset{j=1}{\overset{m-1}{\mathrm{K}}} \frac{c_j c_{j+1} x^2}{1}}, \qquad b_{2j-1} = c_j, \quad b_{2j} = -c_j. \tag{III.7.3}$$

Here K is the usual finite generalized continued fraction, with plus signs between successive levels, and an empty fraction is zero. At $m = 1$ the left side is $1/(1 + c_1 x/(1 - c_1 x)) = 1 - c_1 x$. The general identity follows by clearing the denominators, or by induction on continuants. Applying it to (III.7.2) and using $c_j c_{j+1} = \lambda_j$ shows, as a formal expansion to every finite order, that F has the normalized continued-fraction coefficients

$$c_1, -c_1, c_2, -c_2, c_3, -c_3, \ldots.$$

Returning to R_k therefore gives $a_{2j} = c_j$ and $a_{2j+1} = -c_j$.

For completeness, a formal series $G(x) = 1 + O(x)$ has at most one normalized continued-fraction expansion with nonzero coefficients: if $G = 1/(1 + bxG_1)$, then the coefficient of x determines b, and $G_1 = (G^{-1} - 1)/(bx)$ determines the next tail. Iteration proves uniqueness. The continuant determinant identity shows that truncation after $b_1, \ldots, b_N$ has first omitted term $(-1)^{N+1} b_1 \cdots b_{N+1} x^{N+1}$. Multiplication by $x/2$, with $N = k - 1$ and $b_j = a_{j+1}$, proves the stated error constant and its nonvanishing. In particular these are exactly the successively optimal coefficients of the source, rather than merely another continued fraction for the same function. □

Corollary III.7.2 (Convergent bounds from the paired truncations)**.** *For each fixed $z > 0$, the odd truncations $R_{2m+1}(z)$ converge to $h(z)$ as $m \to \infty$. For every $m \geq 1$,*

$$(-1)^{m-1}\big(h(z) - R_{2m+1}(z)\big) > 0.$$

Thus $H_n - \log n - R_{2m+1}(n)$ gives alternating strict upper and lower bounds for γ, converging to it with m for every fixed positive integer n.

Proof. The paired truncation is exactly $R_{2m+1}(z) = x(1 - xU_{m-1}(x^2))/2$, where U_j is the Stieltjes convergent through λ_j. Positive Stieltjes fractions have alternating strict upper and lower convergents at positive arguments. Moreover $\mu_n^{-1/(2n)} \asymp n^{-1}$ by the displayed moment formula and Stirling's formula. Carleman's Stieltjes criterion [III.7] therefore implies moment determinacy, and the Stieltjes convergence theorem [III.6, III.7] gives $U_j(y) \to U(y)$ for every $y > 0$. These facts prove both assertions. □

Exact consistency check. Rational determinant arithmetic reproduces

$$(a_2, a_4, a_6, a_8, a_{10}, a_{12}) = \left(\frac{1}{6}, \frac{3}{5}, \frac{79}{126}, \frac{7230}{6241}, \frac{4146631}{3833346}, \frac{306232774533}{179081182865}\right),$$

the complete list in Xu–You's Theorem 1, and yields the stated a_{14}. This calculation checks normalization; the proof for all orders uses positivity and formal uniqueness, not a finite numerical test.

Prior-art distinction. Positive Stieltjes fractions for the digamma and polygamma remainders predate Xu–You: see Shenton–Bowman [III.8] (1971), DOI `10.1137/0120057`, and Bonan-Hamada–Jones [III.9] (2005), DOI `10.1016/j.cam.2004.09.034`. In particular the latter paper's Section 2 explicitly uses the positive moments and Carleman convergence argument. The contribution here is the exact identification, via pair contraction and formal uniqueness, with the particular optimized coefficients in Xu–You's question. Neither the positive Stieltjes representation nor its convergence is claimed as new.

III.8 A factorial Hankel conjecture and the boundary of moment positivity

Ferguson [III.4, p. 6, final example of Section 2] defines $a_0 = 0$, $a_k = (k-1)!$ for $k \geq 1$, and conjectures the following identity after checking it through index 50:

$$-\frac{\det(a_{i+j})_{i,j=0}^{n+1}}{\det(a_{i+j+2})_{i,j=0}^{n}} = H_{n+1}.$$

Here $H_N = \sum_{j=1}^{N} 1/j$. We give an exact proof and a parameter extension. The proof uses classical Laguerre orthogonality. Thus the result answers the explicitly conjectural assertion in that source; no claim that this determinant evaluation was previously inaccessible, or that no equivalent evaluation occurs elsewhere, is needed.

Theorem III.8.1 (Bordered factorial Hankel determinants)**.** *Let $\alpha > -1$ and $N \geq 1$. Put*

$$a_0 = 0, \qquad a_k = \Gamma(k+\alpha) \quad (k \geq 1), \quad Q_N(\alpha) = \det(a_{i+j+2})_{i,j=0}^{N-1}, \quad P_N(\alpha) = -\det(a_{i+j})_{i,j=0}^{N}.$$

Then

$$Q_N(\alpha) = \prod_{j=0}^{N-1} j!\,\Gamma(j+\alpha+2), \tag{III.8.1}$$

$$R_N(\alpha) := \frac{P_N(\alpha)}{Q_N(\alpha)} = \Gamma(\alpha+1)^2 \sum_{j=0}^{N-1} \frac{j!}{\Gamma(j+\alpha+2)}. \tag{III.8.2}$$

In particular $P_N(\alpha), Q_N(\alpha) > 0$. *If* $\alpha \neq 0$, *then*

$$R_N(\alpha) = \Gamma(\alpha)\left(1 - \frac{\Gamma(\alpha+1)N!}{\Gamma(N+\alpha+1)}\right).$$

At $\alpha = 0$ *the continuous value is* $R_N(0) = H_N$, *and hence*

$$Q_N(0) = \prod_{j=0}^{N-1} j!(j+1)!, \qquad P_N(0) = H_N \prod_{j=0}^{N-1} j!(j+1)!.$$

Proof. Let $M = (\Gamma(i+j+\alpha+2))_{i,j=0}^{N-1}$ and $b = (\Gamma(j+\alpha+1))_{j=0}^{N-1}$. This is the Gram matrix of $1, x, \ldots, x^{N-1}$ for the strictly positive weight $x^{\alpha+1}e^{-x}$ on $(0,\infty)$, so it is positive definite. The bordered determinant identity gives

$$P_N(\alpha) = \det M \, b^{\mathsf{T}} M^{-1} b.$$

The monic Laguerre polynomials $(-1)^j j! L_j^{(\alpha+1)}(x)$ have squared norms $j!\Gamma(j+\alpha+2)$. Changing from monomials to this monic basis has determinant one and gives (III.8.1).

The vector b represents the functional $\ell(p) = \int_0^\infty p(x)x^\alpha e^{-x}\,dx$ on polynomials of degree less than N. In the orthogonal basis $L_j^{(\alpha+1)}$,

$$b^{\mathsf{T}} M^{-1} b = \sum_{j=0}^{N-1} \frac{\ell(L_j^{(\alpha+1)})^2}{\Gamma(j+\alpha+2)/j!}.$$

For all $j \geq 0$,

$$\ell(L_j^{(\alpha+1)}) = \Gamma(\alpha+1).$$

For example, this follows by integrating the Laguerre generating function: for u sufficiently close to zero,

$$\int_0^\infty x^\alpha e^{-x}(1-u)^{-\alpha-2}\exp\left(-\frac{xu}{1-u}\right)dx = \frac{\Gamma(\alpha+1)}{1-u}.$$

Coefficient comparison proves (III.8.2). These classical Laguerre formulas may also be obtained directly from Rodrigues' formula by integration by parts; see [III.5, Sections 18.3, 18.12]. Finally, for $\alpha \neq 0$,

$$\frac{j!}{\Gamma(j+\alpha+2)} = \frac{1}{\alpha}\left(\frac{j!}{\Gamma(j+\alpha+1)} - \frac{(j+1)!}{\Gamma(j+\alpha+2)}\right).$$

The sum telescopes. At $\alpha = 0$ its summands are $1/(j+1)$. □

Corollary III.8.2 (Complete classification of the zeroth moment)**.** *Fix* $\alpha > -1$ *and prescribe the positive-index moments* $\mu_k = \Gamma(k+\alpha)$ *for* $k \geq 1$. *There is a finite positive measure on* $[0,\infty)$ *with these moments and total mass* c *if and only if*

$$\alpha > 0 \quad \textit{and} \quad c \geq \Gamma(\alpha).$$

When it exists it is unique and equals

$$d\mu(x) = x^{\alpha-1}e^{-x}\,dx + (c - \Gamma(\alpha))\,d\delta_0(x).$$

Equivalently, all matrices $(\mu_{i+j})_{i,j=0}^{N}$, *with* $\mu_0 = c$, *are positive semidefinite exactly under these conditions. For each fixed* $N \geq 1$ *this matrix has a negative eigenvalue exactly when* $c < R_N(\alpha)$; *in that case it has precisely one.*

Proof. Its lower-right block is M from the proof above. Congruence by block Gaussian elimination gives

$$(\mu_{i+j})_{i,j=0}^{N} \;\sim\; \operatorname{diag}(c - R_N(\alpha), M).$$

The claimed finite-dimensional positivity and inertia follow. Formula (III.8.2) is strictly increasing in N. For $\alpha > 0$ it tends to $\Gamma(\alpha)$. For $\alpha = 0$ it is $H_N \to \infty$. For $-1 < \alpha < 0$, its summands are asymptotic to $\Gamma(\alpha+1)^2 j^{-\alpha-1}$, so it again diverges to infinity. This proves necessity, as well as the matrix formulation.

For $\alpha > 0$ the displayed measure proves sufficiency. To verify uniqueness, if μ is any such measure, the measure $x\,d\mu(x)$ has moments $\Gamma(k+\alpha+1)$, $k \geq 0$. They determine that measure uniquely. One direct verification uses an exponential moment: for $0 < t < 1$, monotone convergence gives

$$\int e^{tx} x\, d\mu(x) = \sum_{k=0}^{\infty} \frac{t^k \Gamma(k+\alpha+1)}{k!} = \Gamma(\alpha+1)(1-t)^{-\alpha-1}.$$

Uniqueness of the Laplace transform identifies $x\,d\mu$ with $x^\alpha e^{-x} dx$. Division by x on $(0,\infty)$ leaves only an undetermined atom at zero, whose mass is fixed by c. □

In Ferguson's final example the functional has value $-\gamma$ at 1, and values $(k-1)!$ at x^k for $k \geq 1$. The exact value $R_N(0) = H_N$ proves its asserted determinant identity and positivity, and proves divergence of the proposed ratios. More strongly, no finite replacement for the zeroth moment can turn those higher moments into a positive moment functional. This explains the failure structurally and does not imply anything about the arithmetic nature of γ.

III.8.1 The Gompertz determinant ratios are classical Laguerre approximants

The same calculation identifies the Gompertz example of [III.4, Section 2]. This identification and its error formula are classical orthogonal-polynomial consequences, included to distinguish positive approximation from an irrationality proof.

Proposition III.8.3 (Exact denominator and exact positive error)**.** *For $a > 0$ put*

$$F(a) = \int_0^\infty \frac{e^{-y}}{y+a}\, dy = e^a E_1(a), \qquad u_0 = 0, \quad u_k = \int_0^\infty (y+a)^{k-1} e^{-y}\, dy \quad (k \geq 1).$$

Let $Q_N(a) = \det(u_{i+j+2})_{i,j=0}^{N-1}$ and $R_N(a) = -\det(u_{i+j})_{i,j=0}^{N}/Q_N(a)$. Then

$$Q_N(a) = N! L_N(-a) \prod_{j=0}^{N-1} (j!)^2$$

and

$$F(a) - R_N(a) = \frac{1}{L_N(-a)^2} \int_0^\infty \frac{e^{-y} L_N(y)^2}{y+a}\, dy \tag{III.8.3}$$

$$= \frac{1}{L_N(-a)} \int_0^\infty e^{-at} \frac{t^N}{(1+t)^{N+1}}\, dt > 0. \tag{III.8.4}$$

Here $L_N = L_N^{(0)}$. In particular $a = 1$ gives $F(1) = \delta$.

Proof. With $x = y + a$, use the positive measure $d\nu(x) = e^{a-x} dx/x$ on $[a,\infty)$. The Schur complement $F(a) - R_N(a)$ is the minimum of

$$\int_a^\infty (1 - xq(x))^2\, d\nu(x), \qquad \deg q < N.$$

The minimizing residual is orthogonal, in this measure, to $x, x^2, \dots, x^N$. Thus it is the degree-N Laguerre polynomial for the translated weight e^{-y}, normalized at $x = 0$:

$$r(x) = \frac{L_N(x-a)}{L_N(-a)}.$$

Since $L_N(-a) = \sum_{j=0}^{N} \binom{N}{j} a^j/j! > 0$, the normalization is legitimate. This proves (III.8.3). The polynomial $(L_N(y) - L_N(-a))/(y+a)$ has degree $N-1$. Its orthogonality to $L_N(y)$ for weight e^{-y} gives

$$\int_0^\infty \frac{e^{-y} L_N(y)^2}{y+a}\, dy = L_N(-a) \int_0^\infty \frac{e^{-y} L_N(y)}{y+a}\, dy.$$

Using $1/(y+a) = \int_0^\infty e^{-(y+a)t} dt$ and the finite polynomial expansion of L_N gives

$$\int_0^\infty e^{-(1+t)y} L_N(y) dy = \frac{t^N}{(1+t)^{N+1}},$$

which proves (III.8.4).

For the determinant formula, translation of the monomial basis has determinant one, so $Q_N(a)$ is the size-N Gram determinant for $(y+a)e^{-y}$. The elementary Christoffel determinant identity is

$$\frac{\det\left(\int y^{i+j}(y+a)e^{-y}dy\right)_{i,j=0}^{N-1}}{\det\left(\int y^{i+j}e^{-y}dy\right)_{i,j=0}^{N-1}} = (-1)^N p_N(-a),$$

where $p_N(y) = (-1)^N N! L_N(y)$ is monic. To see the identity, expand the standard bordered Gram determinant for $p_N(-a)$ along its last row, or apply consecutive column operations to it. The denominator is $\prod_{j=0}^{N-1} (j!)^2$, giving the result. □

III.9 Why logarithmic Gompertz moments cannot inherit a finite recurrence

Define

$$D_n(a) = \int_0^\infty e^{-t} \log^n(1 + t/a)\, dt, \qquad a > 0, \quad n \geq 0.$$

The logarithmic Gompertz moment sequence at $a = 1$ is $D_n = D_n(1)$, with $D_0 = 1$ and $D_1 = \delta$. The substitution $x = \log(1 + t/a)$ gives the positive density

$$w_a(x) = a \exp(a + x - ae^x), \qquad x \geq 0, \quad D_n(a) = \int_0^\infty x^n w_a(x) dx.$$

Its moment positivity does not place it in the holonomic framework of Franel or banana coefficient sequences.

Theorem III.9.1 (Nonexistence of polynomial-coefficient recurrences)**.** *For every fixed $a > 0$, the sequence $(D_n(a))_{n\geq 0}$ satisfies no nonzero linear recurrence*

$$\sum_{j=0}^{r} p_j(n) D_{n+j}(a) = 0 \qquad (n \geq n_0), \quad p_j \in \mathbb{C}[n].$$

Equivalently its entire exponential generating function

$$M_a(z) = \sum_{n\geq 0} D_n(a) \frac{z^n}{n!} = e^a a^{-z} \Gamma(z+1, a)$$

is not D-finite over $\mathbb{C}(z)$. Nevertheless it obeys the first-order inhomogeneous difference equation

$$aM_a(z+1) = (z+1)M_a(z) + a.$$

Proof. The decay of $w_a(x)$ is faster than every exponential, giving locally uniform convergence of its Laplace transform and all its derivatives. This proves the entire generating-function formula. Integration by parts in the incomplete gamma integral proves the difference equation.

First, w_a is not annihilated on $(0,\infty)$ by a nonzero polynomial-coefficient differential operator. Indeed

$$\frac{w_a^{(j)}(x)}{w_a(x)} = B_j(e^x), \quad B_0(X) = 1, \quad B_{j+1}(X) = XB_j'(X) + (1-aX)B_j(X).$$

The polynomial B_j has degree j and leading coefficient $(-a)^j$. If $\sum_{j=0}^{r} q_j(x) w_a^{(j)}(x) = 0$ with $q_r \neq 0$, division by $w_a(x)e^{rx}$ gives

$$(-a)^r q_r(x) + \sum_{j<r} v_j(x) e^{-(r-j)x} = 0$$

for certain polynomials v_j. The sum tends to zero exponentially times a polynomial as $x \to +\infty$, whereas a nonzero polynomial q_r cannot do so. This is a contradiction.

Now suppose M_a satisfies $\sum_{j=0}^{s} p_j(z)\partial_z^j M_a(z) = 0$, not all p_j zero. Expanding the p_j into powers of z and integrating by parts repeatedly gives

$$\int_0^\infty e^{zx}\, h(x)dx = P(z), \qquad h(x) = \sum_{j=0}^{s} p_j(-\partial_x)(x^j w_a(x)),$$

where P is a polynomial consisting of endpoint terms at zero. All endpoint terms at infinity vanish, and h has faster than exponential decay. As real $z \to -\infty$ the integral tends to zero; therefore $P = 0$. Uniqueness of the Laplace transform gives $h = 0$. The operator $\sum_j p_j(-\partial_x)x^j$ is nonzero: the substitution $z \mapsto -\partial_x$, $\partial_z \mapsto x$ is an isomorphism of Weyl algebras, since $[x, -\partial_x] = 1$. Thus $h = 0$ contradicts the preceding paragraph.

Finally, a sequence is P-recursive if and only if its exponential generating function is D-finite. In the direction needed here, multiply an eventual recurrence by $z^n/n!$ and sum; the result is a polynomial-coefficient inhomogeneous differential equation, whose polynomial right-hand side is removed by sufficiently many derivatives. The correspondence is injective at the operator level, or follows directly by comparing coefficients. This contradiction proves the recurrence assertion. □

Remark III.9.2 (Scope). The last theorem is an obstruction to transporting a finite-order recurrence argument to the raw logarithmic moments. It does not exclude recurrences after introducing additional parameters, nonlinear recurrences, or a different family of rational approximants. It gives neither irrationality of $D_1 = \delta$ nor algebraic independence of the constants D_n. The differential transcendence theorem for the two-variable incomplete gamma function in [III.13] concerns a different statement; specializing a parameter in that theorem would not by itself justify the result above.

III.9.1 A stronger obstruction from differential difference Galois theory

The preceding elementary non-D-finiteness result can be strengthened. The additional input is the theorem of Bostan–Di Vizio–Raschel [III.14, Theorem 3 and Corollary 40]: a nonrational formal Laurent solution of a first-order inhomogeneous equation with rational coefficients and a nontrivial parabolic Möbius substitution is differentially transcendental over the field of convergent meromorphic germs at its fixed point. We apply that existing theorem to an asymptotic expansion. This is a consequence of their theorem, not a new solution of a conjecture posed in that paper.

Theorem III.9.3 (Differential transcendence of the fixed-cutoff moment function)**.** *For every $a > 0$, $M_a(z) = e^a a^{-z}\Gamma(z+1,a)$ is differentially transcendental over $\mathbb{C}(z)$. More strongly, no*

nonzero differential polynomial with coefficients meromorphic at $z = \infty$ vanishes on this function. For each fixed $a > 0$, both parameter functions $s \mapsto \Gamma(s, a)$ and $s \mapsto \gamma(s, a)$ are differentially transcendental over $\mathbb{C}(s)$, where $\gamma(s, a)$ here denotes the lower incomplete gamma function.

Proof. Let $w_a(x) = a \exp(a + x - ae^x)$ as above and put $g_a(t) = M_a(-1/t)$ for $t > 0$. Repeated integration by parts in its Laplace integral gives the full asymptotic expansion

$$g_a(t) \sim \widehat{g}_a(t) := \sum_{n=0}^{\infty} w_a^{(n)}(0) t^{n+1} \quad (t \downarrow 0).$$

This expansion is stable under differentiation. For completeness, write $M_a^{(k)}(-s) = \int_0^\infty x^k e^{-sx} w_a(x) dx$. Expand $x^k w_a(x)$ to an arbitrary finite order near zero and split the integral at a fixed positive point. The Taylor remainder on the first interval contributes the corresponding next inverse power of s; the second interval is exponentially small. The chain rule with $s = 1/t$ then gives termwise differentiated expansions to every fixed order. The same argument works uniformly in smaller right-half sectors of t if desired.

The exact difference equation for M_a implies the formal identity

$$\widehat{g}_a\left(\frac{t}{1-t}\right) = \frac{t-1}{at}\widehat{g}_a(t) + 1. \tag{III.9.1}$$

The formal series $\widehat{g}_a$ is not rational. Otherwise $R(z) = \widehat{g}_a(-1/z)$ would be a rational solution of

$$aR(z+1) = (z+1)R(z) + a.$$

If R has a finite pole β of minimal real part, the left side has a pole at $\beta - 1$, while the right side is regular there, a contradiction. Thus R is a polynomial. A nonzero polynomial of degree d gives degrees d and $d + 1$ on the two sides, which is impossible; the zero polynomial also fails since $a \neq 0$.

The substitution $t \mapsto t/(1-t)$ is nontrivial and has its unique fixed point at zero. Therefore the cited theorem applied to (III.9.1) says that $\widehat{g}_a$ is differentially transcendental over $\mathbb{C}(\{t\})$. If g_a satisfied an algebraic differential equation over that field, its derivative-stable asymptotic expansion would satisfy the same formal equation: asymptotic expansions respect finite sums, products, and multiplication by convergent Laurent series. A zero function has zero formal expansion. This contradicts the cited theorem. The invertible change $z = -1/t$ proves the assertion for M_a, including the stronger coefficient field. Since multiplication by $e^a a^{-z}$ and translation of the variable preserve differential algebraicity over rational functions, it also proves the claim for the upper incomplete gamma function.

For the lower incomplete gamma function set

$$N_a(z) = e^a a^{-z} \gamma(z+1, a) = \int_0^\infty e^{-zx} w_a(-x) dx \qquad (\Re z > -1).$$

The equality follows by the substitution $u = ae^{-x}$ in its defining integral. Its derivative-stable expansion at $z = 1/t \to +\infty$ is

$$N_a(1/t) \sim \sum_{n \geq 0} (-1)^n w_a^{(n)}(0) t^{n+1} = -\widehat{g}_a(-t).$$

Replacing t by $-t$ and multiplying by -1 preserve differential transcendence over $\mathbb{C}(\{t\})$. The same asymptotic argument therefore proves differential transcendence of N_a and hence of $s \mapsto \gamma(s, a)$ over $\mathbb{C}(s)$. □

The derivatives $w_a^{(n)}(0)/a$ are precisely the coefficients defined by the exponential generating function $\exp(x - a(e^x - 1))$, a specialization of the actuarial polynomials already treated in [III.14, Table 2 and Corollary 4]. This identifies the prior mechanism behind the stronger theorem. It remains a statement about functions and does not establish irrationality of any individual logarithmic moment.

III.10 Rational intersections for exponential-integral values

Let $\mathcal{E}$ be the ring of values of E-functions at algebraic points, and write

$$\mathrm{Ein}(z) = \sum_{n\geq 1} \frac{(-1)^{n-1} z^n}{n\, n!}.$$

The function Ein is an E-function. Rivoal's published Problem 6 [III.16, p. 286], restated as the units conjecture of Fischler–Rivoal [III.15, Conjecture 1.2], asserts that $\mathcal{E}^\times$ consists exactly of the numbers αe^β, with $\alpha, \beta \in \overline{\mathbb{Q}}$ and $\alpha \neq 0$. The following gives a complete intersection theorem for a family of subfields of $\mathbb{C}$. In particular, it settles the unit question inside each indicated polynomial subalgebra. It does not settle the full units conjecture.

Theorem III.10.1 (Polynomial intersection off the positive real axis)**.** *For every $a \in \overline{\mathbb{Q}} \setminus [0, +\infty)$, put $\eta_a = \mathrm{Ein}(a)$. Then*

$$\overline{\mathbb{Q}}(\eta_a) \cap \mathcal{E} = \overline{\mathbb{Q}}[\eta_a]. \tag{III.10.1}$$

Consequently

$$\overline{\mathbb{Q}}[\eta_a] \cap \mathcal{E}^\times = \overline{\mathbb{Q}}^\times. \tag{III.10.2}$$

Equivalently, for every nonconstant polynomial $P \in \overline{\mathbb{Q}}[Y]$, the nonzero value $P(\eta_a)$ has no reciprocal in $\mathcal{E}$.

The proof uses the injective ring morphisms

$$\Phi_\theta : \mathcal{E} \longrightarrow \left\{ \sum_{j=1}^{N} b_j e^{\rho_j X} : b_j \in \mathbb{C},\ \rho_j \in \overline{\mathbb{Q}} \right\}, \qquad -\pi/2 < \theta < \pi/2,$$

constructed in [III.15, Proposition 3.1]. They fix algebraic numbers and satisfy $\Phi_\theta(\xi)(1) = \xi$. If the canonical large-sector decomposition of an E-function is $f(z) = \sum_\rho f_\rho(z) e^{\rho z}$, then $\Phi_\theta(f(1)) = \sum_\rho f_\rho(1) e^{\rho X}$. The stated range of directions is essential.

Lemma III.10.2 (The two phases and their nonvanishing)**.** *For every $a \in \overline{\mathbb{Q}} \setminus [0, +\infty)$, an admissible direction θ gives*

$$\Phi_\theta(\eta_a) = C_a + B_a e^{-aX}, \qquad C_a \notin \overline{\mathbb{Q}}, \quad B_a \neq 0. \tag{III.10.3}$$

If a is nonreal, one can take $\theta = 0$ and

$$C_a = \gamma + \mathrm{Log}\, a, \qquad B_a = \int_0^\infty \frac{e^{-t}}{a+t}\, dt = e^a E_1(a), \tag{III.10.4}$$

where Log *denotes the principal branch. If $a = -x$ with $x > 0$, one can choose $\theta > 0$ sufficiently small and take*

$$C_{-x} = \gamma + \log x - i\pi, \qquad B_{-x} = e^{-x}\big(\mathrm{Ein}(-x) - \gamma - \log x + i\pi\big). \tag{III.10.5}$$

Proof. On the slit plane the classical identity [III.5, §6.2]

$$\mathrm{Ein}(w) = \gamma + \mathrm{Log}\, w + E_1(w) \tag{III.10.6}$$

follows either by differentiating both sides and comparing their limits at zero, or directly from their convergent expansions. For nonreal a, the zero-direction Borel sum of

$$\widehat{B}_a(z) = \sum_{n\geq 0} \frac{(-1)^n n!}{a^{n+1} z^{n+1}}$$

is

$$B_a(z) = \int_0^\infty \frac{e^{-z\xi}}{a+\xi}\, d\xi.$$

Indeed its Borel transform is $1/(a+\xi)$, whose pole is off the positive integration ray. Small rotations of that ray provide the canonical sum in a sector of opening greater than π about the positive z-axis. The identity

$$e^{-az} B_a(z) = E_1(az)$$

first holds where the principal integral representation applies and then by analytic continuation with the branch fixed at $z = 1$. Thus (III.10.6) supplies the canonical phase-zero term $\gamma + \operatorname{Log} a + \log z$ and phase-$(-a)$ term $B_a(z)$. This proves (III.10.4) and the asserted phase map. Moreover

$$\operatorname{Im} B_a = -\operatorname{Im}(a) \int_0^\infty \frac{e^{-t}}{|a+t|^2}\, dt \neq 0.$$

For $a = -x < 0$ choose a small positive summation direction θ; the corresponding Laplace contour passes below the pole $\xi = x$. The lower lateral value of $\operatorname{Log}(-x)$ is $\log x - i\pi$, and (III.10.6) gives (III.10.5). These are again the canonical two phases, obtained by the same Borel sum with its contour rotated below the pole. Here $\operatorname{Im} B_{-x} = \pi e^{-x} \neq 0$.

It remains to prove transcendence of C_a. In the nonreal case put $\varphi = \arg(a) \in (-\pi, \pi) \setminus \{0\}$. If C_a were algebraic, then

$$\varphi = \frac{C_a - \overline{C_a}}{2i}$$

would be a nonzero algebraic real number. But $e^{i\varphi} = a/|a|$ is algebraic, contradicting the Hermite–Lindemann theorem. We used here that $|a| = \sqrt{a\overline{a}}$ is algebraic and that γ is real. On the negative ray, algebraicity of C_{-x} would similarly imply algebraicity of π, again impossible. No arithmetic assumption on γ was used. □

Lemma III.10.3 (One-phase-variable denominator test)**.** *Suppose $\eta \in \mathcal{E}$ and an admissible phase morphism satisfies*

$$\Phi_\theta(\eta) = C + Be^{\lambda X}, \qquad B \neq 0, \quad 0 \neq \lambda \in \overline{\mathbb{Q}}.$$

If coprime $P, Q \in \overline{\mathbb{Q}}[Y]$, with $P, Q \neq 0$, satisfy $P(\eta)/Q(\eta) \in \mathcal{E}$, then every complex root of Q equals C. In particular, if C is transcendental, Q is constant.

Proof. First η is transcendental: if it were algebraic, its phase image would be constant, contrary to $B\lambda \neq 0$. Write $H(X) = \Phi_\theta(P(\eta)/Q(\eta))$. The ring-homomorphism property gives, as an identity of meromorphic functions,

$$H(X) = \frac{P(C + Be^{\lambda X})}{Q(C + Be^{\lambda X})}.$$

The function H is entire. Suppose r is a root of Q with $r \neq C$. Since the exponential map is onto $\mathbb{C}^\times$, there is $X_0 \in \mathbb{C}$ with $e^{\lambda X_0} = (r - C)/B$. Coprimality gives $P(r) \neq 0$. The right-hand side therefore has a pole at X_0, contradicting entireness of H. Thus every root of Q equals C. All roots of $Q \in \overline{\mathbb{Q}}[Y]$ are algebraic, proving the last assertion. □

Proof of Theorem III.10.1. Apply Lemma III.10.3 to Lemma III.10.2. Every element of the field on the left of (III.10.1) has a reduced rational-function representation in η_a, and its denominator is constant. The reverse inclusion follows because $\mathcal{E}$ is a ring containing $\overline{\mathbb{Q}}$ and η_a. If $P(\eta_a)$ were a unit for a nonconstant P, its reciprocal would lie in the field intersection and hence be a polynomial in η_a. The transcendence of η_a would then give a polynomial identity $P(Y)R(Y) = 1$, which is impossible. Nonzero algebraic numbers are units, proving (III.10.2). □

Lemma III.10.4 (Surjective phase images)**.** *Suppose $\xi \in \mathcal{E}$ and, for some admissible θ, the exponential polynomial $H = \Phi_\theta(\xi)$ has at least two distinct nonzero phases with nonzero coefficients. Then*

$$\overline{\mathbb{Q}}(\xi) \cap \mathcal{E} = \overline{\mathbb{Q}}[\xi], \qquad \overline{\mathbb{Q}}[\xi] \cap \mathcal{E}^\times = \overline{\mathbb{Q}}^\times. \tag{III.10.7}$$

Proof. We first show that H takes every complex value. If it omitted $w \in \mathbb{C}$, the entire function $h = H - w$ would have no zeros. It has a bound $|h(z)| \le Ae^{B|z|}$ with $A, B > 0$. The zero-free case of Hadamard factorization then gives $h(z) = e^{uz+v}$ for constants u, v. Here this case can also be seen directly: write $h = e^g$ with g entire, so $\operatorname{Re} g(z) \le \log A + B|z|$. The Borel–Carathéodory inequality on disks of radii R and $2R$ bounds $\max_{|z|\le R} |g(z) - g(0)|$ by $O(1 + R)$. Cauchy's inequalities, followed by $R \to \infty$, give $g^{(n)}(0) = 0$ for every $n \ge 2$, proving that g is affine. But subtracting w only changes the constant phase of H and leaves its two distinct nonzero phases. Linear independence of distinct exponential functions rules out $H - w = e^{uz+v}$. Thus H is surjective.

In particular H is nonconstant, so ξ is transcendental. Let $P, Q \in \overline{\mathbb{Q}}[Y]$ be coprime and suppose $R(\xi) = P(\xi)/Q(\xi) \in \mathcal{E}$. Applying Φ_θ gives the entire-function identity

$$Q(H(X))\Phi_\theta(R(\xi))(X) = P(H(X)).$$

If Q were nonconstant, choose one of its roots r and then X_0 with $H(X_0) = r$. The identity gives $P(r) = 0$, contradicting coprimality. Hence Q is constant, proving the field intersection. The assertion about units follows exactly as in Theorem III.10.1. □

Theorem III.10.5 (All finite combinations with at least two arguments)**.** *Let $r \ge 2$, let $a_1, \dots, a_r$ be pairwise distinct nonzero algebraic numbers, and let $c_1, \dots, c_r$ be nonzero algebraic numbers. Put*

$$\xi = \sum_{j=1}^{r} c_j \operatorname{Ein}(a_j).$$

Then both conclusions in (III.10.7) *hold. In particular the result applies when all the arguments are positive.*

Proof. Choose $\theta > 0$ sufficiently small. There are only finitely many arguments a_j. For every nonreal argument the interval from 0 to θ can therefore be chosen to contain no singular Borel-summation direction of its tail with Borel transform $1/(a_j + \zeta)$; the phase coefficients remain the principal ones in (III.10.4). For a positive argument the same formula holds, and its integral B_{a_j} is strictly positive. For a negative argument the chosen positive direction gives the lateral coefficients in (III.10.5). All tail coefficients are nonzero by Lemma III.10.2 and positivity in the positive case. Thus the one common morphism gives

$$\Phi_\theta(\xi) = \sum_{j=1}^{r} c_j C_{a_j} + \sum_{j=1}^{r} c_j B_{a_j} e^{-a_j X}.$$

The r nonzero phases $-a_j$ are distinct and their coefficients $c_j B_{a_j}$ do not vanish. Apply Lemma III.10.4. □

For real examples, each of

$$\xi_+ = \operatorname{Ein}(1) + \operatorname{Ein}(2), \qquad \xi_- = \operatorname{Ein}(1) - \operatorname{Ein}(2)$$

has the full polynomial-intersection property. The first equals $2\gamma + \log 2 + E_1(1) + E_1(2)$ and retains Euler's constant explicitly; the second cancels its coefficient. These two cases are instances of the same family theorem.

Theorem III.10.1 also excludes

$$\frac{1}{\mathrm{Ein}(-1)-\alpha}, \qquad \frac{1}{1+\mathrm{Ein}(i)^2} \quad (\alpha \in \overline{\mathbb{Q}})$$

from $\mathcal{E}$, with no restriction on the differential order of a possible representing E-function. These examples are consequences of one theorem, not separate resolutions of published conjectures.

What remains open at a positive argument. For positive algebraic a the same phase computation gives $C_a = \gamma + \log a$ and $B_a > 0$. Lemma III.10.3 still shows that a reduced denominator must be a constant multiple of $(Y - C_a)^m$ if it is nonconstant. The argument above cannot exclude this possibility because C_a is real. At $a = 1$ this is precisely the exception controlled by the unknown arithmetic nature of γ; the off-axis theorem does not resolve it. Nor does it decide the irrationality of the Euler–Gompertz constant.

III.11 Unresolved targets and the arithmetic barrier

The results above give exact statements at the interfaces between recurrences, analysis, and arithmetic. They do not supply the missing arithmetic estimate needed to prove irrationality of γ. This section records where the arguments stop, so that later work can target a precise missing statement.

Target	Status and missing step
General scalar CMF convergence, all $r \geq 3$, in [III.1, Conjecture 1]	Proved here on every unequal rank-three repeated-minimum family. Higher-rank scalar convergence is not established. The full-state assertion is already false, so it cannot be used as the general intermediate claim.
Equal lower increments (β, β, β) with arbitrary nonpositive integral upper increments	The source covers $\beta = 1$. When both upper increments are divisible by β, positive integral dilation follows by taking a subsequence of that source trajectory. The present $d > 0$ concentration proof does not address the remaining equal-increment cases.
Variable-argument CMF convergence and its full relation to Rivoal's parameter family [III.1, Section 9.2]	The exact Meijer identity is proved. A matching identity for functions does not prove a dual dynamical limit for every parameter or identify all gauged recurrences.
Order-two approximation construction for γ [III.2, Section 1]	The given order-three operator has no lower-order factor for nonzero parameter. Existence of a different order-two construction remains open here.
Reduced-fraction height and the empirical approximation parameter in [III.1, Section 8.3]	The factorial bound concerns a common denominator of a specified raw vector. Cancellation in the scalar numerator and denominator has not been controlled.
Full E-value unit conjecture [III.16, Problem 6]	Polynomial intersections are proved in the stated single-generator subfields. They do not classify all units of $\mathcal{E}$.
Single positive argument $\mathrm{Ein}(a)$	A possible denominator supported at $\gamma + \log a$ remains. This is an explicit obstruction in the method, rather than a proved exceptional unit.

Target	Status and missing step
Irrationality of γ	Not proved. Neither convergence to γ nor the functional Galois calculations produce nonzero small integer linear forms with the requisite denominator estimates.
Transcendence of γ	Not proved. Independence of functions over $\mathbb{C}(z)$ cannot be specialized to arithmetic independence of their values without an applicable specialization theorem.
Irrationality of δ	Not proved. Positive Laguerre error formulas and differential transcendence of the moment function do not decide this numerical question.

This is a ledger of mathematical targets and obstructions. It is not a list of ten solved published problems: several entries are components of the same external conjecture, and others are questions exposed by the present proofs. Counting individual parameters, signs, or corollaries as independent resolutions would misstate the result.

III.11.1 A concrete continuation of the recurrence method

The scalar proof suggests an appropriate replacement for full-state isolation. First identify the subspace killed by both extraction forms; then compare the dual components in the quotient by that subspace. For the family treated here this subspace contains $u = (1, -1, 0)^T$, and the relevant quotient coefficients are the two signed cofactors involving the elementary row. Their dominant finite sums are positive before the fixed seed is inserted. Concentration then controls the inserted polynomial uniformly and proves eventual nonvanishing of the denominator.

In higher rank the analogous calculation would require control of several cofactor minors, not merely a single positive weight. The exact contiguity operators still make those minors accessible, but positivity and dominant component cancellation must be proved rather than inferred from numerical ratios. This is a concrete extension problem left by the manuscript.

For arithmetic progress one must add a second ingredient. If rational approximants are written in lowest terms as P_n/Q_n, a route to irrationality needs a nonzero error with sufficient decay after multiplying by the actual reduced denominator. The normalization-dependent bounds in Section III.6 do not provide that ingredient. The positive-moment and phase-morphism results likewise identify useful structure without crossing this final arithmetic barrier.

III.A Exact checks and reproducibility

All limiting and nonexistence statements in the manuscript have written proofs. Finite computations were used to check signs, indices, and source normalizations. They are not substitutes for those proofs. The first program below uses exact rational arithmetic to check factorial-Hankel identities, translated Laguerre denominators, and the continued-fraction coefficients. The second uses SymPy only to clear polynomial and rational identities in the recurrence and gauge. Both are embedded here so that the single TeX source contains the reported checks without auxiliary files.

Internal proof reviews independently rederived the full-state counterexample, the scalar concentration argument, the recurrence irreducibility and height obstructions, the Meijer contour shift, the E-value intersections, and the use of the exact Bostan–Di Vizio–Raschel hypotheses. Such reviews are fallible and do not amount to external refereeing or formal proof verification.

III.A.1 Moment determinants and continued-fraction coefficients

```
"""Exact normalization checks; standard-library rational arithmetic only."""
from fractions import Fraction as F
from math import comb, factorial as fact


def determinant(rows):
    a = [[F(x) for x in row] for row in rows]
    result, size = F(1), len(a)
    for j in range(size):
        pivot = next((k for k in range(j, size) if a[k][j]), None)
        if pivot is None:
            return F(0)
        if pivot != j:
            a[j], a[pivot] = a[pivot], a[j]
            result = -result
        result *= a[j][j]
        for k in range(j + 1, size):
            ratio = a[k][j] / a[j][j]
            for ell in range(j + 1, size):
                a[k][ell] -= ratio * a[j][ell]
    return result


def product(values):
    result = F(1)
    for value in values:
        result *= value
    return result


def rising(a, k):
    return product(a + j for j in range(k))


for alpha in (F(-1, 2), F(0), F(1, 2), F(1), F(2), F(3)):
    for n in range(1, 7):
        # Divide each positive-index moment by Gamma(alpha+1).
        v = [F(0)] + [rising(alpha + 1, k - 1)
                      for k in range(1, 2 * n + 1)]
        q = determinant([[v[i+j+2] for j in range(n)] for i in range(n)])
        p = -determinant([[v[i+j] for j in range(n+1)] for i in range(n+1)])
        assert q == product(fact(j)*rising(alpha+1, j+1) for j in range(n))
        assert p / q == sum(F(fact(j))/rising(alpha+1, j+1) for j in range(n))
print("PASS: 36 normalized factorial-Hankel cases")

for a in (F(1, 2), F(1), F(2), F(3)):
    for n in range(1, 7):
        v = [F(0)] + [sum(comb(k-1, j)*a**(k-1-j)*fact(j)
                          for j in range(k)) for k in range(1, 2*n+1)]
        q = determinant([[v[i+j+2] for j in range(n)] for i in range(n)])
        laguerre = sum(F(comb(n, j), fact(j))*a**j for j in range(n+1))
        assert q == fact(n)*laguerre*product(fact(j)**2 for j in range(n))
print("PASS: 24 Gompertz denominator cases")

bernoulli = [F(1)]
for m in range(1, 15):
    bernoulli.append(-sum(comb(m+1, k)*bernoulli[k]
                          for k in range(m)) / (m+1))
mu = [abs(bernoulli[2*j+2]) / (j+1) for j in range(7)]


def hankel(n, shift):
    return determinant([[mu[i+j+shift] for j in range(n)] for i in range(n)])


c = [mu[0]]
for j in range(1, 7):
    r = (j+1)//2
    if j % 2:
        lam = hankel(r, 1)*hankel(r-1, 0)/(hankel(r, 0)*hankel(r-1, 1))
    else:
```

```
        lam = hankel(r+1, 0)*hankel(r-1, 1)/(hankel(r, 1)*hankel(r, 0))
    c.append(lam / c[-1])
expected = [F(1, 6), F(3, 5), F(79, 126), F(7230, 6241),
            F(4146631, 3833346), F(306232774533, 179081182865),
            F(3754087889491759, 2440840521848406)]
assert c == expected
# Independently extract the normalized continued fraction of 1-x U(x*x).
tail = [F(1)] + [F(0)] * 13
for j, moment in enumerate(mu):
    tail[2*j+1] = (-1)**(j+1)*moment
coefficients = []
while len(tail) > 1:
    inverse = [F(1)]
    for k in range(1, len(tail)):
        inverse.append(-sum(tail[j]*inverse[k-j] for j in range(1, k+1)))
    coefficient = inverse[1]
    coefficients.append(coefficient)
    tail = [v/coefficient for v in inverse[1:]]
assert coefficients == [value for q in c for value in (q, -q)][:13]
print("PASS: Xu--You a2 through a14 (13 coefficients, two constructions)")
print("a14 =", coefficients[-1])
```

III.A.2 Recurrence layers, gauge, and Casoratian

```
import sympy as s

n, x, t = s.symbols('n x t')
a = [
    (n+2)**2*(3*x-8*n-22)*(3*x-8*n-30),
    (24*n**3+(162-57*x)*n**2+(26*x**2-258*x+356)*n
     -3*x**3+59*x**2-285*x+254)*(3*x-8*n-30),
    -(24*n**3+(7*x+210)*n**2+(-6*x**2+56*x+600)*n
      -15*x**2+96*x+564)*(3*x-8*n-14),
    -(n+3)**2*(3*x-8*n-14)*(3*x-8*n-22),
]
alpha = [s.expand(v).coeff(n, 4) for v in a]
beta = [s.expand(v).coeff(n, 3) for v in a]
assert alpha == [64, -192, 192, -64]
assert s.expand(sum(beta)) == 512*x
assert s.expand(sum(j*alpha[j] for j in range(4))) == 0
adjoint_beta = [s.expand(a[j].subs(n, n+3-j)).coeff(n, 3)
                for j in range(4)]
assert s.expand(sum(adjoint_beta)) == 512*x
assert s.expand(sum(alpha[j]*t**(3-j) for j in range(4))) == s.expand(64*(t-1)**3)
assert s.expand(sum(a[j].subs(x, 0)*(n+j+1) for j in range(4))) == 0

A = s.Matrix([
    [6*n**2+13*n+10, -8*n-5, 3],
    [3*n**3+3*n**2-3*n-6, -3*n**2+3*n+3, n-2],
    [n**4-2*n**3-6*n**2-3*n+3, 6*n**2+2*n-1, 1-2*n],
])/(n+1)**4
G = s.Matrix([
    [0, (8*n+11)/(n+1)**2, (24*n**2+97*n+94)/((n+1)**2*(n+2))],
    [-1, (3*n**2-n-7)/(n+1)**2,
     (15*n**4+69*n**3+66*n**2-78*n-111)/((n+1)**2*(n+2)**2)],
    [2, (-6*n**2-6*n+3)/(n+1)**2,
     (8*n**5+21*n**4-38*n**3-131*n**2-48*n+53)/((n+1)**2*(n+2)**2)],
])
C = s.zeros(3)
C[1, 0] = C[2, 1] = 1
for j in range(3):
    C[j, 2] = s.cancel(-a[j].subs(x, 1)/a[3].subs(x, 1))
assert all(s.cancel(v) == 0 for v in G*C-(n+1)**2*A*G.subs(n, n+1))
assert s.cancel(G.det()-(8*n+11)*(8*n+19)/(n+2)**2) == 0
assert s.cancel(A.det()-(n+1)**(-6)) == 0
cas = 4*(8*n+11)*(8*n+19)/(209*(n+2)**2)
assert s.cancel(cas.subs(n, n+1)/cas-C.det()) == 0
assert cas.subs(n, 0) == 1
print('PASS: leading layers, formal adjoint, OGF leading coefficient, gauge, determinants, Casoratian.')
```

Recorded exact outcomes. The moment program passed all 36 normalized factorial-Hankel cases, all 24 Gompertz denominator cases, and all 13 coefficients $a_2, \ldots, a_{14}$ by two constructions. The symbolic program passed the leading-layer, adjoint, generating-equation, gauge, determinant, and Casoratian identities.

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Part IV
Twelve further irrationality attempts and their failure points

Source: `gamma_attempts(2).tex`.

Original PDF: 14 pages. Mathematical content and source-level qualifications are retained.

Original preparation note. Prepared with the assistance of Claude (Anthropic), 18–19 October 2026. Computations used Python with `python-flint` (Arb) and SymPy.

Editorial scope note. The broad holonomy verdict below is a record of the attempted method. Its precise cancellation proposition must be read with its identifiability hypotheses and with the later analysis in Section VI.4 and Part VII. It is not a universal theorem that every arithmetic-holonomy route to γ is impossible. The source author field does not identify a named author; the supplied preparation credit is retained above.

Abstract

We record twelve attempts to prove that Euler's constant γ is irrational. They fall into four families, and each was chosen to avoid the routes already mapped in the companion manuscripts [IV.22, IV.23, IV.24, IV.25, IV.26]. None succeeds. For each attempt we prove what can be proved and isolate the exact point of failure.

The most informative failure concerns the arithmetic holonomy method of Calegari, Dimitrov and Tang [IV.8], currently the strongest irrationality technique for periods. Its analytic input is a cancellation between G-functions. By a theorem of Fischler and Rivoal [IV.14], the nonzero constants that occur as cancellation coefficients in discs are exactly the elements of $\mathrm{Frac}(\mathbf{G})$, where $\mathbf{G}$ is the ring of values of G-functions. We extend the necessity part to several constants and to meromorphic continuation along arbitrary paths (Proposition IV.4.2). Consequently the method, in its present form, can reach γ only if $\gamma \in \mathrm{Frac}(\mathbf{G})$. This is expected to be false, but it is open (Remarks IV.4.3 and IV.4.4).

The other attempts fail at one of three walls:

- factorial denominators against sub-exponential analytic decay;
- contamination by auxiliary transcendental constants;
- blindness of the structure to γ.

These other attempts are denominator-free Γ-kernel forms, higher-rank Laplace kernels, q-deformations, quantum periods of Fano manifolds, o-minimal point counting, integer-valued functions, the Gauss map, Kronecker–Lerch–Chowla–Selberg, the Dickman–Buchstab functions, two-point Hermite–Padé "untying", and the explicit formula with Baker's theory.

Two numerical studies are included. The first views Aptekarev's integer forms inside the denominator-free lattice: their gcd is the trivial factor $n!/d_n$ times a small integer (between 3 and 39591 for $n = 20, 30, \ldots, 300$), so no hidden arithmetic gain appears. The second treats the moments of the Dickman distribution. Their denominators are bounded by $e^{k \log k + O(k)}$, and numerically they are of that order.

Nothing in this note proves or disproves the irrationality of γ.

IV.1 Introduction

Let $\gamma = \lim_{n\to\infty}(H_n - \log n) = 0.5772156649\ldots$, where $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$. Whether γ is irrational is open; see Lagarias' survey [IV.29] for the history.

What is known. The unconditional arithmetic results all concern γ together with a companion constant.

- Mahler [IV.30] proved that $\frac{\pi}{2}\frac{Y_0(2)}{J_0(2)} - \gamma$ is transcendental.

- From Shidlovskii's theorem and the identity

$$\mathrm{Ein}(1) = \gamma + \delta/e, \qquad \mathrm{Ein}(z) := \int_0^z \frac{1-e^{-t}}{t}\,dt, \qquad \delta := \int_0^\infty \frac{e^{-t}}{1+t}\,dt, \tag{IV.1.1}$$

  it follows that at least one of γ and the Gompertz constant δ is transcendental. Rivoal [IV.35] proved $\mathrm{trdeg}_{\mathbb{Q}}\,\mathbb{Q}(e,\gamma,\delta) \geq 2$.

- Murty–Saradha proved that for each $q > 1$ at most one of γ and the $\psi(a/q)$ with $(a,q) = 1$ is algebraic [IV.31], with an analogue for Euler–Lehmer constants [IV.32]. Chatterjee–Gun extended this to the whole set $\{\gamma\} \cup \{\psi(r/n) : n > 1,\ 0 < r < n,\ (r,n) = 1\}$ [IV.9].

- Fischler and Rivoal showed that the asymptotic constants of E-functions at infinity lie in the $\mathbf{G}$-module generated by the derivatives $\Gamma^{(s)}(a)$ at rational a [IV.15]; this is where $\gamma = -\Gamma'(1)$ enters. They also showed that $\gamma \in \mathbf{E} + e^{-1}\mathbf{D}$ [IV.17]. Their conjecture $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ would make γ transcendental.

Rational approximations. All known rational approximation schemes have errors e^{-cn^α} with $\alpha < 1$ and denominators at least of size $n!$:

- Hessami Pilehrood: $\alpha = \frac{1}{2}$ [IV.20];
- Aptekarev: $\alpha = \frac{1}{2}$ [IV.3];
- Rivoal: $\alpha = \frac{2}{3}$ [IV.34];
- Van Assche–Wolfs: $\alpha = \frac{3}{4}$ [IV.38].

Sondow's double integrals [IV.36] decay geometrically but contain logarithms of primes. If $\gamma = p/q$, then $q > 10^{244663}$ (Papanikolaou; see [IV.29]). Several unrefereed proof claims have appeared since 2023; none has been accepted.

The companion manuscripts. The manuscripts [IV.22, IV.23, IV.24, IV.25, IV.26] audit many dozens of routes. Their conclusions are summarized in §IV.2; we use them only as constraints on what a new attempt must avoid. In particular:

- the *shear/twist* barrier of [IV.22];
- the *tie theorem* of [IV.22]: the small forms of Rivoal type see γ only through the E-value $e\,\mathrm{Ein}(1)$;
- the two failure mechanisms isolated throughout [IV.23, IV.24, IV.25, IV.26]: eliminating auxiliary constants costs more height than the analytic decay provides, and fully rational approximants have factorial denominators against sub-exponential errors.

This note. Twelve attempts are organized in four families:

- new kernels (§IV.3);
- the strongest current tool (§IV.4);
- structures that turn out to be γ-blind (§IV.5);
- new bridges (§IV.6).

For each we state what is proved, give computations where useful, and pin down the failure. Section IV.7 collects the obstructions in one table and states what seems to be missing.

Statements labelled Theorem, Proposition, Lemma or Corollary are proved here or quoted with a reference. *Observations* are numerical, and *Remarks* and the "Where it breaks" paragraphs contain heuristic discussion. We were unable to consult the October 2026 preprint

arXiv:2610.13919 (“An infinite family of Apéry-like recurrences for Euler’s constant”); nothing below depends on it.

Notation.

- $d_n = \operatorname{lcm}(1, \dots, n)$.
- $\psi = \Gamma'/\Gamma$.
- $E_1(x) = \int_x^\infty e^{-t}\, dt/t$, so that $\operatorname{Ein}(x) = \gamma + \log x + E_1(x)$ for $x > 0$.
- $\mathbf{E}$ and $\mathbf{D}$ are the rings of values at algebraic points of E-functions and of Borel sums of arithmetic Gevrey series [IV.15, IV.16, IV.17].
- $\mathbf{G}$ is the ring of values at algebraic points of G-functions with algebraic coefficients, including values of their analytic continuations along paths avoiding singularities. Let $\mathbf{G}_{\mathbb{Q}}$ denote the values of G-functions with rational coefficients at rational points of their discs of convergence. By [IV.14, Thm. 1], $\mathbf{G} = \mathbf{G}_{\mathbb{Q}} + i\mathbf{G}_{\mathbb{Q}}$; it contains $\overline{\mathbb{Q}}$, π, and the logarithms of nonzero algebraic numbers.

IV.2 Two constraints every attempt must respect

IV.2.1 The ledger

Lemma IV.2.1. *Let $(A_n, B_n) \in \mathbb{Q}^2$ and $D_n \in \mathbb{N}$ with $D_nA_n, D_nB_n \in \mathbb{Z}$. Put $\Lambda_n = A_n - \gamma B_n$. If $0 < D_n|\Lambda_n| \to 0$ along a subsequence, then $\gamma \notin \mathbb{Q}$.*

Proof. If $\gamma = a/b$, then $bD_n\Lambda_n = bD_nA_n - aD_nB_n$ is an integer. It is nonzero, so its absolute value is at least 1, whereas it tends to 0. □

We call $\log D_n + \log|\Lambda_n|$ the *ledger* of the family. A proof by linear forms needs a family with certified nonvanishing whose ledger tends to $-\infty$. (Proofs of other types, such as the holonomy method of §IV.4, are not of this form.)

IV.2.2 Bridges and the twist

The following observation is the shear-invariance principle of [IV.22], stated in the language of Γ.

Lemma IV.2.2 (Twist)**.** *For $\tau \in \mathbb{R}$ put $\Gamma_\tau(s) := e^{\tau(1-s)}\Gamma(s)$ and $\psi_\tau = \Gamma_\tau'/\Gamma_\tau = \psi - \tau$.*

(i) $\Gamma_\tau(1) = 1$, *and Γ_τ is log-convex on $(0, \infty)$.*

(ii) ψ_τ satisfies the same identities as ψ:

- $\psi_\tau(s+1) = \psi_\tau(s) + 1/s$;
- $\psi_\tau(1-s) - \psi_\tau(s) = \pi \cot \pi s$;
- $\sum_{k=0}^{m-1} \psi_\tau(s + k/m) = m\psi_\tau(ms) - m \log m$;
- *Gauss’s digamma theorem, with γ replaced by $\gamma_\tau := -\psi_\tau(1) = \gamma + \tau$.*

(iii) The Γ-level normalizations change:

- $\Gamma_\tau(\tfrac{1}{2}) = e^{\tau/2}\sqrt{\pi}$;
- $\Gamma_\tau(s)\Gamma_\tau(1-s) = e^{\tau}\pi/\sin \pi s$;
- $\int_0^1 \log \Gamma_\tau = \tfrac{1}{2}\log 2\pi + \tfrac{\tau}{2}$;
- $\Gamma_\tau(s+1) = e^{-\tau} s\, \Gamma_\tau(s)$.

Proof. Direct computation. Each identity in (ii) is unchanged when every ψ is replaced by $\psi - \tau$. In Gauss's theorem, $\psi(r/q) - \psi(1)$ is τ-invariant. □

Consequently, an argument that uses only identities of type (ii), together with a characterization of ψ up to an additive constant, proves the same statement for $\gamma + \tau$ for every τ. It therefore cannot prove that $\gamma \notin \mathbb{Q}$. A proof must use a *bridge*: an identity of type (iii), or an asymptotic normalization such as $\psi(x) - \log x \to 0$.

The bridges available, and the arithmetic cost of each, are:

(B1) $\gamma = \lim(H_n - \log n)$ and its Euler–Maclaurin refinements: logarithms of integers.

(B2) $\gamma = \mathrm{Ein}(x) - \log x - E_1(x)$: E-values at x, the logarithm $\log x$, and the **D**-value $e^x E_1(x)$. For $x = 1$ this is (IV.1.1).

(B3) Stirling, Raabe, Wallis: $\log \pi$, $\log 2\pi$, and logarithms of integers.

(B4) Lerch and Chowla–Selberg: $\log \Gamma(a/d)$ and Euler–Kronecker constants (Attempt 9).

(B5) de Bruijn's identity $\int_0^\infty \rho = e^\gamma$: the number e^γ (Attempt 10).

(B6) The Γ-kernel itself, $\gamma = -\int_0^\infty e^{-t} \log t \, dt$ (Attempt 1).

IV.3 Family I: new kernels

IV.3.1 Attempt 1: denominator-free forms from the Γ-kernel

The usual ledger is ruined by the denominators d_n of harmonic numbers. They disappear completely if one integrates integer polynomials against the Γ-kernel.

Lemma IV.3.1. *For $P(t) = \sum_{k=0}^n p_k t^k \in \mathbb{Z}[t]$,*

$$\int_0^\infty P(t)\, e^{-t} \log t \, dt = A(P) - \gamma B(P), \qquad A(P) = \sum_k p_k \, c_k, \quad B(P) = \sum_k p_k \, k!,$$

where $c_k := k! H_k = |s(k+1, 2)| \in \mathbb{Z}$ are unsigned Stirling numbers of the first kind. The map $P \mapsto (A(P), B(P))$ is already surjective from polynomials of degree ≤ 1 onto $\mathbb{Z}^2$.

Proof. We have $\int_0^\infty t^k e^{-t} \log t \, dt = \Gamma'(k+1) = k!\,(H_k - \gamma)$. For surjectivity, $P = (B - A) + At$ gives $(A(P), B(P)) = (A, B)$. □

Corollary IV.3.2. *$\gamma \notin \mathbb{Q}$ if and only if there are $P_n \in \mathbb{Z}[t]$ with $0 < \left|\int_0^\infty P_n e^{-t} \log t \, dt\right| \to 0$.*

Proof. For the "if" direction, apply Lemma IV.2.1 with $D_n = 1$. For the "only if" direction, use Dirichlet's theorem and the surjectivity in Lemma IV.3.1. □

Where it breaks. The lattice has no denominators, but the integral carries no analytic information beyond the pair (A, B): every pair is represented by a polynomial of degree one. Smallness must be *forced* by a construction in an auxiliary variable, such as vanishing conditions (Padé) or Rodrigues formulas. All such constructions known for γ pass through the confluent saddle of e^{-t} and decay like e^{-cn^α} with $\alpha < 1$.

In the lattice of Lemma IV.3.1, the normalizations appear as gcd's, and one can test for hidden arithmetic gains. Aptekarev's forms, in the integer normalization, are

$$\tilde{q}_n = \sum_{k=0}^n \binom{n}{k}^2 (n+k)!, \qquad \tilde{p}_n = \sum_{k=0}^n \binom{n}{k}^2 (n+k)!\,(H_{n+k} + 2H_{n-k} - 2H_k).$$

Both are integers, and both are divisible by the trivial factor $n!/d_n$.

Observation IV.3.3. For $n = 20, 30, \ldots, 300$, exact computation (partly shown in Table 5) gives three things:

- $\gcd(\tilde{q}_n, \tilde{p}_n) = (n!/d_n) \cdot \epsilon_n$ with ϵ_n a small integer, $3 \le \epsilon_n \le 39591$ (so $\log \epsilon_n \le 10.6$), against the $\log|\Lambda_n| \approx 3217$ that would have to be absorbed at $n = 300$; there is no arithmetic gain beyond the trivial factor;
- $\log|\gamma - \tilde{p}_n/\tilde{q}_n| + 2\sqrt{2n}$ increases slowly towards $\log 2\pi = 1.838$, consistent with the error $\approx 2\pi e^{-2\sqrt{2n}}$;
- the reduced form $|\tilde{q}_n\gamma - \tilde{p}_n|/\gcd$ grows like e^{2096} at $n = 300$.

n	$\log\lvert\Lambda_n\rvert$	$\log g_n$	$\log(n!/d_n)$	$\log(\lvert\Lambda_n\rvert/g_n)$	$\log\lvert\gamma - \tilde{p}_n/\tilde{q}_n\rvert + 2\sqrt{2n}$
20	103.4	26.9	23.1	76.6	1.519
40	264.1	82.0	74.1	182.1	1.612
80	642.0	198.8	194.2	443.2	1.678
120	1062.6	347.1	340.4	715.5	1.707
200	1979.5	661.5	657.1	1318.0	1.737
300	3216.7	1120.7	1115.7	2096.0	1.755

Table 5: Aptekarev's forms in the denominator-free lattice, with $\Lambda_n = \tilde{p}_n - \tilde{q}_n\gamma$ and $g_n = \gcd(\tilde{q}_n, \tilde{p}_n)$ (natural logarithms). Exact integer arithmetic; γ computed in Arb.

Verdict. The lattice relocates the d_n-wall into the gcd and changes nothing. Discarded.

IV.3.2 Attempt 2: higher-rank Laplace kernels e^{-t^r}

Replacing e^{-t} by e^{-t^r} changes the moments from $k!$ to $\Gamma(\frac{k+1}{r})/r$, roughly $(k!)^{1/r}$. Padé-type decay at an irregular singularity of Poincaré rank r improves heuristically to $e^{-cn^{r/(r+1)}}$. Letting r grow with n would make both sides of the ledger exponential.

Proposition IV.3.4. *Let $r \ge 2$ and $P \in \mathbb{Q}[t]$. There are $\mathbb{Q}$-linear functionals R_a, S_a $(1 \le a \le r)$ with*

$$\int_0^\infty P(t)e^{-t^r}\log t\,dt = \frac{1}{r^2}\sum_{a=1}^{r}\Gamma\big(\tfrac{a}{r}\big)\Big(R_a(P)\,\psi\big(\tfrac{a}{r}\big) + S_a(P)\Big).$$

For $1 \le a < r$,

$$\psi(a/r) = -\gamma - \log 2r - \tfrac{\pi}{2}\cot\tfrac{\pi a}{r} + 2\sum_{1\le j\le (r-1)/2}\cos\tfrac{2\pi ja}{r}\log\sin\tfrac{\pi j}{r}.$$

The component $a = r$ is the only one that does not involve $\Gamma(a/r)$ with $a < r$. On that component the substitution $u = t^r$ reduces the form to the rank-one kernel of Attempt 1.

Proof. First,

$$\int_0^\infty t^k e^{-t^r}\log t\,dt = r^{-2}\Gamma'\big(\tfrac{k+1}{r}\big).$$

Write $\frac{k+1}{r} = m + \frac{a}{r}$ with $1 \le a \le r$, $m \ge 0$, and use

$$\Gamma(m + \tfrac{a}{r}) = \Gamma(\tfrac{a}{r})(\tfrac{a}{r})_m, \qquad \psi(m + \tfrac{a}{r}) = \psi(\tfrac{a}{r}) + \sum_{j<m}(\tfrac{a}{r} + j)^{-1}.$$

The formula for $\psi(a/r)$ is Gauss's digamma theorem. □

Where it breaks. Any gain from $r > 1$ comes with new constants:

- the values $\Gamma(a/r)$, which are expected to be algebraically independent with π modulo the standard relations (Lang–Rohrlich);
- logarithms of cyclotomic units;
- $\pi\cot(\pi a/r)$.

For $r=2$ these are $\sqrt{\pi}$ and $\log 2$; for growing r there are about $\varphi(r)/2$ independent ones. This is exactly the contamination wall of [IV.23, IV.24]. Discarded.

IV.3.3 Attempt 3: q-deformation and the limit $q \to 1$

Put $\zeta_q(1) := \sum_{n\ge 1} q^n/(1-q^n) = \sum_{n\ge 1} d(n)q^n$ for $0<q<1$.

Proposition IV.3.5. *As $q \to 1^-$,*

$$(1-q)\,\zeta_q(1) + \log(1-q) = \gamma + O\big((1-q)\log\tfrac{1}{1-q}\big).$$

Proof. With $q = e^{-t}$, the Mellin transform $\Gamma(s)\zeta(s)^2$ gives

$$\sum d(n)e^{-nt} = \frac{\gamma - \log t}{t} + \frac{1}{4} + O(t).$$

Then use $1-q = t - t^2/2 + O(t^3)$. □

The irrationality of $\zeta_q(1)$ is known for $q = 1/p$ with $p \in \mathbb{Z}$, $|p| \ge 2$: Erdős for $p=2$, Bézivin [IV.5] and Borwein [IV.6] in general, with irrationality exponent at most 2.46498 uniformly in p (Zudilin [IV.40]). Kurokawa and Wakayama's q-analogue of γ [IV.28] is of this type.

Where it breaks. Two independent obstructions:

- These results describe the limit $q \to 0$ along $1/p$, not $q \to 1$. The proofs use that the q-analogues of d_n (products of cyclotomic values) are integers when $1/q \in \mathbb{Z}$. We know of no irrationality result for $\zeta_q(1)$ at $q = 1 - 1/m$.
- Even with such results, Proposition IV.3.5 at $q = 1-1/m$ approximates γ by $\frac{1}{m}\zeta_{1-1/m}(1) - \log m$, so logarithms of integers return.

Discarded.

IV.4 Family II: the strongest current tool and its blind spot

IV.4.1 Attempt 4: arithmetic holonomy bounds

Calegari, Dimitrov and Tang [IV.8] proved that $1, \zeta(2), L(2,\chi_{-3})$ are linearly independent over $\mathbb{Q}$. Their method bounds the dimension of spaces of holonomic power series with controlled denominators, of integrality type $x^n/[1,\dots,n]^2$, which continue meromorphically to a domain of large conformal size. The constants enter as connection constants of a holonomic local system of G-type (Zagier's functions on $\mathbb{C}\smallsetminus\{0,\frac{1}{9},1\}$). If such a constant ξ were rational, a combination $A - \xi B$ of G-functions would continue further than its parts, producing too many functions of that kind.

The analytic input of the method is therefore a *cancellation*: constants $\xi_1,\dots,\xi_r$ such that $A - \sum_i \xi_i B_i$ continues further than the B_i do. The next two statements identify which constants can occur in such cancellations.

Theorem IV.4.1 (Fischler–Rivoal [IV.14, Thm. 2], case $K=\overline{\mathbb{Q}}$)**.** *For $\xi \in \mathbb{C}^\times$ the following are equivalent:*

(i) $\xi \in \mathrm{Frac}(\mathbf{G})$*;*

(ii) there are coefficient sequences $(a_n), (b_n)$ *of G-functions with* $b_n \neq 0$ *for large* n *and* $a_n/b_n \to \xi$*;*

(iii) for every $R \geq 1$ *there are G-functions* $A, B \in \overline{\mathbb{Q}}[[z]]$ *of radius* 1 *such that* $A - \xi B$ *has radius* $> R$*.*

In [IV.14] the coefficient field is an arbitrary algebraic extension K of $\mathbb{Q}$ and (i) reads $\xi \in \mathrm{Frac}(\mathbf{G}_K)$. For $K = \overline{\mathbb{Q}}$ one has $\mathbf{G}_{\overline{\mathbb{Q}}} = \mathbf{G}_{\mathbb{Q}} + i\mathbf{G}_{\mathbb{Q}} = \mathbf{G}$ by [IV.14, Thm. 1]. The assumption $\xi \neq 0$ is needed: for $\xi = 0$, (i) holds but (iii) fails.

The method of [IV.8] uses meromorphic continuation to general domains rather than discs, and several constants at once. The following proposition covers that case.

Proposition IV.4.2 (Cancellation along a path)**.** *Let* $A, B_1, \dots, B_r \in \overline{\mathbb{Q}}[[z]]$ *be G-functions and* $\xi_1, \dots, \xi_r \in \mathbb{C}$*. Let* σ *be a path from* 0 *to a point* $\zeta \in \mathbb{C}$ *along which* A *and the* B_i *continue analytically, up to (but not including)* ζ*. Suppose that:*

- *the continuation of* $A - \sum_i \xi_i B_i$ *is meromorphic at* ζ*;*
- *no nontrivial* $\mathbb{C}$*-linear combination of the continuations of the* B_i *is meromorphic at* ζ*.*

Then $\zeta \in \overline{\mathbb{Q}}$ *and* $\xi_1, \dots, \xi_r \in \mathrm{Frac}(\mathbf{G})$*. The same holds with "meromorphic" replaced by "holomorphic" in both hypotheses.*

Proof (from standard facts). Step 1: the operator. Let L be the least common left multiple of the minimal operators of $A, B_1, \dots, B_r$. By the theorem of Chudnovsky and André [IV.10, IV.1], L is a G-operator. By the theorems of Katz and André [IV.1, IV.2] it is Fuchsian with rational exponents, and near each of its singular points ζ it has a basis of solutions

$$u_j = \sum_k (z-\zeta)^{e_j} \log^k(z-\zeta)\, f_{j,k}(z-\zeta), \qquad e_j \in \mathbb{Q},$$

with $f_{j,k}$ G-functions. The point ζ must be a singular point of L: at a regular point all local solutions are holomorphic, which contradicts the second hypothesis (note $r \geq 1$ there). Singular points of L are algebraic, so $\zeta \in \overline{\mathbb{Q}}$.

Step 2: the connection coefficients. Modifying the final segment of σ inside a small disc around ζ if necessary, choose an algebraic point z_0 on σ with three properties: z_0 is a regular point of L; $|z_0 - \zeta|$ is smaller than the radii of convergence of all the $f_{j,k}$; and the disc around z_0 used below avoids ζ. Fix the branches of $(z-\zeta)^{e_j}$ and $\log(z-\zeta)$ near z_0 by continuation along σ. On a small disc around z_0 write

$$A = \sum_j \alpha_j u_j, \qquad B_i = \sum_j \beta_{ij} u_j.$$

By Cramer's rule, $w(z_0)\alpha_j$ and $w(z_0)\beta_{ij}$ are polynomials in the entries of the Wronskian matrix $W_u(z_0)$ and in the values $A^{(l)}(z_0)$, $B_i^{(l)}(z_0)$. Here $w(z_0) = \det W_u(z_0) \neq 0$.

- The values $A^{(l)}(z_0)$ and $B_i^{(l)}(z_0)$ lie in $\mathbf{G}$ by [IV.14, Thm. 1].
- The entries of $W_u(z_0)$ are polynomials in algebraic numbers $(z_0-\zeta)^{e_j - m}$, in $\log(z_0-\zeta)$, and in values $f_{j,k}^{(m)}(z_0-\zeta) \in \mathbf{G}$. The logarithm of a nonzero algebraic number lies in $\mathbf{G}$ for every branch, because $\log(1-z)$ is a G-function and $2\pi i \in \mathbf{G}$.

Hence $w(z_0)\alpha_j$ and $w(z_0)\beta_{ij}$ lie in $\mathbf{G}$.

Step 3: the cancellation. A local solution $\sum_j c_j u_j$ is meromorphic at ζ if and only if, in its expansion, the coefficients of all terms $(z-\zeta)^e \log^k(z-\zeta)$ with $e \notin \mathbb{Z}$ or $k \geq 1$ vanish. It is holomorphic if, in addition, the coefficients with $e \in \mathbb{Z}_{<0}$ vanish. The Taylor coefficients of the $f_{j,k}$ are algebraic, so these conditions form a system $\ell(c) = 0$, $\ell \in \mathcal{L}$, of $\overline{\mathbb{Q}}$-linear equations; finitely many suffice. Put $\alpha = (\alpha_j)$ and $\beta_i = (\beta_{ij})_j$.

- The first hypothesis gives $\ell(\alpha) = \sum_i \xi_i\, \ell(\beta_i)$ for all $\ell \in \mathcal{L}$.
- The second hypothesis says that the matrix $\big(\ell(\beta_i)\big)_{\ell\in\mathcal{L},\,1\le i\le r}$ has rank r.

Choose $\ell_1, \dots, \ell_r \in \mathcal{L}$ with $\det(\ell_k(\beta_i)) \neq 0$ and solve for the ξ_i by Cramer's rule. After multiplying numerators and denominators by $w(z_0)^r$, all entries lie in $\mathbf{G}$, so $\xi_i \in \mathrm{Frac}(\mathbf{G})$. □

Remark IV.4.3 (The G-blindness principle). Proposition IV.4.2 has the following consequence. Consider any argument whose analytic input is a cancellation of this kind between G-functions; the arithmetic holonomy method of [IV.8], in its present form, is of this type. Such an argument can only produce statements about constants in $\mathrm{Frac}(\mathbf{G})$. Conversely, by Theorem IV.4.1, every nonzero element of $\mathrm{Frac}(\mathbf{G})$ admits such cancellations. In particular, applying the method to γ would require a cancellation as input, and Proposition IV.4.2 would then force $\gamma \in \mathrm{Frac}(\mathbf{G})$.

This is a principle about a class of methods, not a theorem about all conceivable proofs. It says nothing about arguments whose input is of a different kind.

Remark IV.4.4. It is expected that $\gamma \notin \mathrm{Frac}(\mathbf{G})$, but this is not known to follow from any standard functional conjecture.

- Fischler and Rivoal conjecture that $\mathbf{G}$ consists of periods with $1/\pi$ adjoined [IV.14].
- Kontsevich and Zagier [IV.27] expect that γ is not a period.
- Fresán and Jossen [IV.18] realize γ as a period of a non-classical exponential motive.

So the method reaches γ only if $\gamma \in \mathrm{Frac}(\mathbf{G})$, which would itself be a major surprise. A proof of the holonomy type would need a theorem for objects whose connection data contain γ. By [IV.15, Thm. 3] these include the asymptotic constants of E-functions at their irregular singularity ∞, which involve the derivatives $\Gamma^{(s)}(a)$ at rational a (for instance $\gamma = -\Gamma'(1)$). They also include Borel sums of arithmetic Gevrey series [IV.16]. No such theorem is known.

Verdict. The method can reach γ only if $\gamma \in \mathrm{Frac}(\mathbf{G})$, which is not expected. Discarded.

IV.4.2 Attempt 5: quantum periods of Fano manifolds and the Gamma conjecture

Apéry-type limits occur naturally in the quantum differential equations of Fano manifolds (Golyshev). The Gamma conjecture [IV.19] expresses their asymptotics through

$$\widehat{\Gamma}_X = \exp\Big(-\gamma\, c_1(X) + \sum_{k\ge 2} (-1)^k (k-1)!\, \zeta(k)\, \mathrm{ch}_k(T_X)\Big),$$

in which γ does appear.

Where it breaks. Take X with a Laurent-polynomial mirror f.

- The regularized quantum period $\widehat{G}_X(t) = \sum_d [f^d]_0\, t^d$, with $[\,\cdot\,]_0$ the constant term, has integer coefficients and is a period of the mirror family. Hence it is a G-function.
- Its Apéry limits are limits of ratios of coefficient sequences of G-functions, so they lie in $\mathrm{Frac}(\mathbf{G})$ by Theorem IV.4.1(ii).
- The constant γ enters $\widehat{\Gamma}_X$ only through $e^{-\gamma c_1}$. This factor is the Laplace normalization relating $\widehat{G}_X$ to the E-function $G_X(t) = \sum_d [f^d]_0\, t^d/d!$, in accordance with [IV.15, Thm. 3].

Verdict. The Apéry mechanism of quantum cohomology lives on the G-side. By Remark IV.4.3 it can reach γ only if $\gamma \in \mathrm{Frac}(\mathbf{G})$. Discarded.

IV.5 Family III: structures that do not see γ

IV.5.1 Attempt 6: o-minimal point counting

The function $\Gamma|_{(0,\infty)}$, and hence $\psi|_{[1,\infty)}$, is definable in an o-minimal expansion of $\mathbb{R}$ (van den Dries–Speissegger [IV.13]).

Proposition IV.5.1. *(a) Suppose $\gamma = a/b$. Then the rational points of $\operatorname{graph}(\psi|_{[1,\infty)})$ are exactly the points $(n+1, H_n - a/b)$, $n \geq 0$. Their number $N(T)$ of height $\leq T$ satisfies $N(T) \asymp \log T$.*

(b) Suppose $\gamma \notin \mathbb{Q}$. Then the graph has no rational point over an integer, and all its rational points lie over a single class $r/q+\mathbb{Z}$ with $0 < r < q$, $(r,q) = 1$. In particular $N(T) = O(\log T)$ in this case too.

Proof. Rational abscissae. Write a rational abscissa as $x = m + r/q$ with $m \in \mathbb{Z}$ and $0 \leq r < q$. Then $\psi(x) - \psi(r/q) \in \mathbb{Q}$ (for $r > 0$) and $\psi(m+1) = H_m - \gamma$. For $0 < r < q$, the number $\psi(r/q) + \gamma$ is, by Gauss's digamma theorem, a nonzero linear form in logarithms of algebraic numbers and π with algebraic coefficients, hence transcendental by Baker's theorem [IV.31]. If $\gamma = a/b$, this shows that $\psi(x)$ is transcendental at every non-integral rational x, which proves the first claim of (a). If $\gamma \notin \mathbb{Q}$, then $\psi(m+1) \notin \mathbb{Q}$, and by Chatterjee–Gun [IV.9] at most one value $\psi(r/q)$ with $0 < r < q$, $(r,q) = 1$ is algebraic; rational points can therefore only lie over the corresponding class $r/q + \mathbb{Z}$.

Heights. The primes in $(n/2, n]$ divide $\operatorname{den} H_n$, and $\operatorname{den} H_n \mid d_n$. Hence $e^{(1/2+o(1))n} \leq \operatorname{den} H_n \leq e^{(1+o(1))n}$, which gives $N(T) \asymp \log T$ in (a). The same estimate applies to $\psi(r/q) + \sum_{0 \leq j < m} (r/q + j)^{-1}$ in (b). □

Where it breaks. The Pila–Wilkie theorem [IV.33] bounds rational points on the transcendental part by $O_\varepsilon(T^\varepsilon)$. Even Wilkie-type bounds $(\log T)^\kappa$ must allow $\kappa \geq 1$, since the graph of 2^x already has $\asymp \log T$ rational points. The "rational world" produces too few points to be refuted by counting. Discarded.

IV.5.2 Attempt 7: integer-valued functions

Proposition IV.5.2. *Let $\mathbb{N} = \{0, 1, 2, \dots\}$. For $c \in \mathbb{C}$ put $F_c(z) = \Gamma'(1+z) + c\,\Gamma(1+z)$. Then $F_c(n) = n!(H_n - \gamma + c)$ for $n \in \mathbb{N}$. Moreover, $F_c(\mathbb{N}) \subset \mathbb{Z}$ if and only if $c \in \gamma + \mathbb{Z}$.*

Proof. We have $\psi(n+1) = H_n - \gamma$ and $n!H_n \in \mathbb{Z}$; take $n = 0$ for the converse. □

Where it breaks. Integrality on $\mathbb{N}$ detects the coset $\gamma + \mathbb{Z}$, never whether $\gamma \in \mathbb{Q}$. The canonical function $F_\gamma = \Gamma(1+z)H_z$, where $H_z := \psi(1+z) + \gamma$, is integer-valued on $\mathbb{N}$ in every world.

- Rationality $\gamma = a/b$ only adds the statement $bF_\gamma \in \mathbb{Z}\Gamma'(1+\cdot) + \mathbb{Z}\Gamma(1+\cdot)$.
- No Pólya-type rigidity holds at the Γ-scale, since $\Gamma(1+z)$ is itself integer-valued of the same growth.
- Carlson/Pólya density versions require bounded integer outputs $bF_\gamma(n) - (\cdots)$, which is the ledger again.

Discarded.

IV.5.3 Attempt 8: the Gauss map

Let $T(x) = \{1/x\}$ and let $\mathcal{L}f(y) = \sum_{n \geq 1} f(\frac{1}{n+y})(n+y)^{-2}$ be the Gauss–Kuzmin–Wirsing operator.

Proposition IV.5.3. *(a)* $1-\gamma=\int_0^1\{1/x\}\,dx=\int_0^1 y\,\psi'(1+y)\,dy=\langle y,\mathcal{L}1\rangle$.

(b) $\int_0^1\{1/x\}^2dx=\log 2\pi-\gamma-1$.

(c) $\int_0^1 T^k(x)\,dx=\langle y,\mathcal{L}^k1\rangle$, *and this tends to* $\frac{1}{\log 2}-1$ *at the geometric rate* $|\lambda_2|^k$, *where* $\lambda_2=-0.3036630\ldots$ *is Wirsing's constant [IV.39].*

Proof. For (a), substitute $x=1/(n+t)$ and use $\mathcal{L}1=\psi'(1+\cdot)$. Integrating by parts gives $\psi(2)-\log\Gamma(2)+\log\Gamma(1)=1-\gamma$. Part (b) is similar and was checked numerically to 25 digits. Part (c) is the spectral theory of [IV.39]. □

Where it breaks. γ is the first term of a geometrically convergent sequence whose limit $\frac{1}{\log 2}-1$ lies in $\mathbb{Q}(\log 2)$. The intermediate terms are governed by the spectrum of $\mathcal{L}$, and no arithmetic is known for that spectrum. Discarded.

IV.5.4 Attempt 9: Kronecker, Lerch and Chowla–Selberg

For an imaginary quadratic field K of discriminant $-d$, with character χ, class number h and w roots of unity, let γ_K be the Euler–Kronecker constant (the constant term of ζ_K'/ζ_K at $s=1$, see [IV.21]).

Proposition IV.5.4.

$$\gamma_K=2\gamma+\log 2\pi-\frac{w}{2h}\sum_{a=1}^{d-1}\chi(a)\log\Gamma\!\left(\tfrac{a}{d}\right).$$

Proof. We have $\gamma_K=\gamma+L'(1,\chi)/L(1,\chi)$. Lerch's formula $\partial_s\zeta(0,x)=\log\Gamma(x)-\frac12\log 2\pi$ gives $L'(0,\chi)$. The functional equation, with $\psi(1)=-\gamma$ and $\psi(\frac12)=-\gamma-2\log 2$, transfers this to $s=1$, using $L(0,\chi)=2h/w$. The formula was checked numerically to 30 digits for $d=3,4,7,8,11,15,23$. □

Where it breaks. Lerch's formula is a genuine bridge, of type (B4). By Chowla–Selberg [IV.11], however, the Γ-product is a period up to algebraic factors and powers of π. Hence $\gamma\equiv\gamma_K/2$ modulo logarithms of periods: one trades γ for a Laurent constant of the same nature. Discarded.

IV.6 Family IV: new bridges

IV.6.1 Attempt 10: the Dickman–Buchstab functions and e^γ

Let ρ be Dickman's function: $\rho=1$ on $[0,1]$ and $u\rho'(u)=-\rho(u-1)$ for $u>1$.

Proposition IV.6.1 (de Bruijn; see [IV.12, IV.37])**.** *(a) The Laplace transform is*

$$\hat\rho(s)=\int_0^\infty e^{-su}\rho(u)\,du=\exp(\gamma-\mathrm{Ein}(s)).$$

Hence $\hat\rho(s)=e^{-E_1(s)}/s$ *for* $s>0$, *which is free of* γ.

(b) $\int_0^\infty u^k\rho(u)\,du=e^\gamma m_k$, *where*

$$m_0=1,\qquad m_k=\frac1k\sum_{i<k}\binom{k}{i}m_i,\qquad \textit{equivalently}\qquad m_k=\sum_\pi\prod_{B\in\pi}\frac{1}{|B|},$$

the sum running over all set partitions π *of* $\{1,\ldots,k\}$. *Thus* $m_1=1$, $m_2=\frac32$, $m_3=\frac{17}{6}$, $m_4=\frac{19}{3}$.

Proof. Laplace-transforming the delay equation gives $s\hat{\rho}' = (e^{-s}-1)\hat{\rho}$, with normalization $\hat{\rho}(s) \sim 1/s$ as $s \to \infty$. Comparing coefficients in $s\,u' = (e^{-s}-1)u$ for $u = e^{-\mathrm{Ein}}$ gives the recurrence. The partition formula is the exponential formula for $\exp(\sum_n (-1)^n s^n/(n \cdot n!))$. The moments were confirmed numerically by integrating the delay equation. □

Proposition IV.6.2. *For every prime p and $k \geq 1$, $v_p(\operatorname{den} m_k) \leq \lfloor k/p \rfloor$.*

Proof. Each term of the partition formula has denominator $\prod_B |B|$. If $v = v_p(m) \geq 1$ then $m \geq p^v \geq vp$, so $v_p(m) \leq \lfloor m/p \rfloor$. Summing over blocks gives $\sum_B v_p(|B|) \leq \sum_B \lfloor |B|/p \rfloor \leq \lfloor k/p \rfloor$. □

Observation IV.6.3. For $1 \leq k \leq 300$, exact computation gives the following.

(a) Equality $v_p(\operatorname{den} m_k) = \lfloor k/p \rfloor$ holds for 9822 of the 10088 pairs (k,p) with $3 \leq p \leq k$, p prime.

(b) $\log \operatorname{den} m_k$ agrees closely with $S_k := \sum_{3 \leq p \leq k} \lfloor k/p \rfloor \log p = k \log k + O(k)$. For example, $(\log \operatorname{den} m_k, S_k) = (347.3, 347.3)$ at $k = 120$ and $(1110.7, 1112.6)$ at $k = 300$.

(c) Curiously, $v_2(\operatorname{den} m_k) = v_2(\lfloor k/2 \rfloor + 1)$ for all $k \leq 300$. This plays no role below.

Proposition IV.6.4. *There is $\varepsilon(n) \to 0$ such that every nonzero $Q \in \mathbb{Z}[u]$ of degree n with $Q \geq 0$ on $[1,\infty)$ satisfies*

$$\int_1^\infty \rho\, Q \;\geq\; \exp\bigl((1-\varepsilon(n))\, n \log n\bigr).$$

Proof. For $L \geq 1$,

$$\int_1^\infty \rho Q \;\geq\; \rho(2L) \int_L^{2L} |Q| \;\geq\; \rho(2L)\, L (L/4)^n.$$

The second inequality is the L^1 Chebyshev bound: monic polynomials of degree n on an interval of length L have L^1-norm at least $L(L/4)^n$, and Q has leading coefficient of absolute value at least 1. Take $L = n/(\log n)^2$ and use $\log \rho(u) = -u \log u\,(1 + o(1))$ [IV.12]. □

Where it breaks. The forms are

$$\int_1^\infty \rho\, Q = e^\gamma\, Y(Q) - \int_0^1 Q, \qquad Y(Q) = \sum q_k m_k.$$

They are clean linear forms in 1 and e^γ, but they meet three obstructions:

- Their denominators are bounded by $e^{k \log k + O(k)}$ (Proposition IV.6.2), and numerically they are of that order (Observation IV.6.3). This is the same factorial wall as for the Γ-kernel.
- By Proposition IV.6.4, positivity-certified forms are huge.
- Most decisively, the direction is wrong. If $\gamma \in \overline{\mathbb{Q}}$, then e^γ is transcendental by Hermite–Lindemann, so an irrationality proof for e^γ would say nothing about γ.

Buchstab's function $\omega(u) \to e^{-\gamma}$ (super-exponentially fast) has the same structure: its values $\omega(n)$ are periods of weight $\leq n-2$. Discarded.

IV.6.2 Attempt 11: untying by two-point Hermite–Padé

By the tie theorem of [IV.22], small forms in $(1, \gamma, \delta)$ built from E-function Padé data at $z = 1$ approximate $e\,\mathrm{Ein}(1) = e\gamma + \delta$. The natural fix is to use data at 0 and at ∞ simultaneously.

Proposition IV.6.5. *Let $x > 0$ be algebraic and $\lambda, \mu \in \overline{\mathbb{Q}}$ with $\mu x > 0$. Then*

$$e^{\lambda x}\,\mathrm{Ein}(\mu x) = e^{\lambda x}(\gamma + \log \mu x) + e^{(\lambda-\mu)x}\, G_0(\mu x), \qquad G_0(y) := e^y E_1(y) = \int_0^\infty \frac{e^{-t}\,dt}{y+t}.$$

Proof. Use $\mathrm{Ein}(y) = \gamma + \log y + E_1(y)$ for $y > 0$. □

Remark IV.6.6. Proposition IV.6.5 suggests the following picture, which we have not made into a theorem. Values at a single point see γ only through $\gamma + \log(\mu x)$, accompanied by the **D**-value $G_0(\mu x)$. To isolate γ one needs points with $\prod(\mu_i x_i)^{c_i} = 1$ and $\sum c_i \neq 0$. The values $G_0(\mu_i x_i)$ then remain, and they become negligible only when $\mu_i x_i \to \infty$.

Where it breaks. Untying always costs either a logarithm or a **D**-value. Removing both forces the moving-point schemes of Sweeney and Brent–McMillan [IV.7], or the dyadic levels of [IV.26]. Their heights grow like $e^{cx^2 \log x}$ (or doubly exponentially) against errors e^{-x}. Discarded.

IV.6.3 Attempt 12: the explicit formula and Baker's theory

Since $-\zeta'/\zeta(s) = \frac{1}{s-1} - \gamma + O(s-1)$, the prime number theorem gives

$$\sum_{n \le x} \frac{\Lambda(n)}{n} = \log x - \gamma + O(e^{-c\sqrt{\log x}}).$$

Take $x \geq 2$ an integer, so that $\log x$ is an integral combination of the $\log p$. If $\gamma = a/b$, the formula above is then a linear form in the $\pi(x)$ logarithms $\log p$ ($p \leq x$) with rational coefficients of denominator $d_x = e^{(1+o(1))x}$.

Where it breaks. After clearing denominators the form has size about $d_x\, e^{-c\sqrt{\log x}}$, which tends to infinity, so it is not small. Even under the Riemann hypothesis the error $x^{-1/2+\varepsilon}$ is too large. Baker–Wüstholz lower bounds [IV.4] for $\pi(x)$ logarithms are astronomically smaller and therefore irrelevant. Discarded.

IV.7 Synthesis

#	route	where γ enters	failure
1	Γ-kernel lattice	$\Gamma'(1)$	no intrinsic analytic content; Padé reintroduces the walls
2	kernels e^{-t^r}	$\psi(a/r)$	$\Gamma(a/r)$, cyclotomic logs, $\pi \cot$
3	$q \to 1$	$(1-q)\zeta_q(1) + \log(1-q)$	$\log(1-q)$; q-integrality needs $1/q \in \mathbb{Z}$
4	arithmetic holonomy	cancellation constants	reaches γ only if $\gamma \in \mathrm{Frac}(\mathbf{G})$ (Prop. IV.4.2)
5	Fano / Gamma conjecture	$\widehat{\Gamma}_X$	Apéry limits lie in $\mathrm{Frac}(\mathbf{G})$
6	o-minimal counting	graph of ψ	only $\asymp \log T$ points
7	integer-valued functions	$\Gamma' + \gamma\Gamma$	sees only $\gamma + \mathbb{Z}$
8	Gauss map	$\int\{1/x\}$	non-arithmetic spectrum
9	Kronecker–Lerch	γ_K	trades γ for γ_K
10	Dickman–Buchstab	e^{γ}	wrong direction; factorial-type denominators
11	two-point untying	$\gamma + \log x$	log or **D**-value; moving-point heights
12	explicit formula	Mertens	forms not small; $\pi(x)$ logarithms

Table 6: The twelve attempts and their obstructions.

What the attempts show. The twelve attempts confirm, from new directions, the two walls isolated in [IV.22, IV.23, IV.24, IV.25, IV.26]. They add two structural points.

First, in its present form the strongest modern irrationality method can only produce statements about constants in Frac($\mathbf{G}$) (Proposition IV.4.2, Remark IV.4.3). It reaches γ only if $\gamma \in \mathrm{Frac}(\mathbf{G})$, which is expected to be false but is open.

Second, every new bridge we found has one of four properties:

- it pays with logarithms, Γ-values or $\mathbf{D}$-values (Attempts 2, 3, 9, 11, 12);
- it points the wrong way, as with e^{γ} (Attempt 10);
- it is invisible to the arithmetic of γ (Attempts 6, 7);
- it sees γ but gives no arithmetic handle on it (Attempt 8).

What seems to be missing. A proof that $\gamma \notin \mathbb{Q}$ appears to require a theorem of a new type. It would be an arithmetic statement about connection data at an *irregular* singular point, equivalently about the rings $\mathbf{E}$ and $\mathbf{D}$. One example is Fischler and Rivoal's conjecture $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$, which would imply that γ is transcendental [IV.17]. The work of Calegari–Dimitrov–Tang shows that holonomy theorems can be proved in the regular-singular (G) world, and Proposition IV.4.2 shows that this world reaches γ only if $\gamma \in \mathrm{Frac}(\mathbf{G})$, which is not expected. Whether an irregular-singular analogue exists is, in our view, the real question behind the irrationality of γ.

Practical consequence. The concrete open targets closest to γ that we can identify are those of [IV.22]:

- the $\overline{\mathbb{Q}}$-linear independence of $1, e, e\gamma$;
- the implication $\delta \in \mathbf{E} \Rightarrow \delta \in \overline{\mathbb{Q}}$.

Both remain at the level of the main problem. For a genuinely new theorem in the near term, the holonomy methods of [IV.8] are better aimed at constants that are periods.

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Part V
Monodromy, primitive lattices, and positive rational forms

Source: `Keio_Gamma_Monodromy_Arithmetic_20261019.tex`.

Original PDF: 30 pages. Mathematical content and source-level qualifications are retained.

Abstract

We investigate several new routes relative to the supplied research corpus toward the arithmetic of Euler's constant. The strongest structural result extends phase embeddings of the ring of E-function values by analytic continuation around the origin. Two different windings eliminate all nonconstant rational denominators in a single exponential-integral value, including positive algebraic arguments previously left exceptional. Monodromy also supplies an additional necessary condition for a value to be a unit and gives polynomial intersection theorems for irreducible rational-parameter Kummer values and integer-order Bessel values at nonzero algebraic points.

A finite-characteristic argument proves primitivity of integer Euler jets and determines the exact image lattice of rank-three numerator and denominator extraction. For equal upper decrements, its residual index is proved to be at most exponential in the trajectory index, despite the factorial size of the original determinant. Independently, an integer affine Hankel pencil selects the actual Euler constant by positivity, with a proved finite-depth resolution cost. A rational differential reduction produces positive pure gamma forms and allows precise obstructions for completely monotone, Bernstein-positive, and fixed-order moving-pole constructions. We also record the uniform arithmetic estimate needed for a q-limit and a small-prime correction in the Diamond digamma normalization. All unsuccessful transfers are identified explicitly. No irrationality or transcendence proof for Euler's constant is obtained.

Status. AI-assisted research manuscript with full proofs of its stated results and internal independent checks. External peer review and an exhaustive priority search remain outstanding. The results are not presented as ten resolutions of previously open problems; that original objective has not been met.

V.1 The target, the new results, and the proof standard

Throughout,

$$\gamma = \lim_{n\to\infty}(H_n - \log n), \qquad H_n = \sum_{j=1}^{n} \frac{1}{j}, \qquad \delta = \int_0^\infty \frac{e^{-t}}{1+t}\,dt.$$

These are the canonically normalized Euler and Euler–Gompertz constants, not freely variable symbols in a formal comparison matrix. Write $k = \overline{\mathbb{Q}}$ with its fixed embedding into $\mathbb{C}$. An E-function is an entire series $f(z) = \sum_{n\ge 0} a_n z^n/n!$ with algebraic coefficients, satisfying a linear differential equation over $k(z)$, for which all conjugates of a_n and a common denominator of $a_0, \dots, a_n$ have exponential growth bounds. The ring of its values at algebraic points is denoted by $\mathcal{E}$. Rescaling the argument permits every such value to be represented as $f(1)$.

The central arithmetic task is to produce integers A_n, B_n with

$$0 < |A_n - B_n\gamma| \longrightarrow 0. \tag{V.1.1}$$

This condition is sufficient for irrationality: if $\gamma = a/b$, multiplication by b would give nonzero integers tending to zero. An equivalent formulation for rational approximants p_n/q_n is

$$0 < q_n|\gamma - p_n/q_n| \longrightarrow 0,$$

where q_n is the denominator after reduction. Convergence without this arithmetic scale, or a result about functional transcendence, does not supply (V.1.1). Transcendence requires stronger information; no claim of that kind for γ is made here.

V.1.1 Results that survive the attempted proof

Direction	Proved output and exact boundary
Monodromy of phase maps	$k(\mathrm{Ein}(a)) \cap \mathcal{E} = k[\mathrm{Ein}(a)]$ for every nonzero algebraic a. The positive-argument denominator exception is removed. The corresponding arithmetic classification of γ does not follow. The method also treats the Kummer and Bessel families stated in Section V.3.
Finite-characteristic jets	Primitivity and the exact cyclic image quotient of the integral extraction matrix; for equal upper decrements its index divides $\mathrm{lcm}(1, \ldots, S+1)^8$. Seed-dependent heights remain uncontrolled.
Canonical positivity	An integer affine moment pencil is positive at all depths exactly at $c = \gamma$. Every finite depth still accepts an interval of nearby values, with an explicit and an asymptotically stronger lower bound.
Rational positive kernels	Exact removal of auxiliary constants, a sharp 1/4 barrier for one family, and further certified obstructions after that family is enlarged. None provides (V.1.1).
Boundary and local transfers	A quantified sufficient q-arithmetic bridge, whose hypothesis remains unproved, and the precise small-prime exceptions in a published p-adic statement.

The first direction builds on Fischler–Rivoal's phase-map method [V.1]. The extension here continues its component functions around the origin, while keeping the original summation direction fixed. This distinction is essential. The rank-three construction is motivated by the conservative matrix fields of Kalisch et al. [V.5], but the jet and lattice proofs below are given explicitly.

V.1.2 What is and is not inherited from the earlier manuscripts

The supplied archive contains 31 TeX manuscripts. We use it as a record of prior attempts, not as a reason to declare an entire area of mathematics unavailable. In particular, the two-model argument in *The Two-World Search for Euler's Constant* explicitly removes the analytic definitions of the comparison symbol. It is not a proof that the actual number γ can be changed while preserving all of its defining properties. The later *Arithmetic Alternatives for Euler's Constant* already distinguishes these scopes.

For example, the exact real Gamma recurrence rules out nonzero real twists $\Gamma_\tau(s) = \Gamma(s)e^{\tau(1-s)}$, since $\Gamma_\tau(s+1) = se^{-\tau}\Gamma_\tau(s)$. Similarly, the recessive boundary condition at infinity excludes a nonzero additive constant in E_1. Fixing the normalization is therefore possible; extracting small primitive integer forms from it is the unsolved part.

The previous scalar CMF convergence, reciprocal-Gamma sampling, polynomial Laplace kernels, isolated-atom moment constructions, and formal real/p-adic interpolation arguments are not counted again as new discoveries. The present note is a complement to that corpus, not a complete reproduction or merger of it. "New" in descriptions of the research process means new in this round relative to the inspected corpus. Publication priority has not been established.

V.2 Monodromy continuation of phase embeddings

Put $k = \overline{\mathbb{Q}}$, let $\mathcal{E}$ denote the ring of E-function values, and let

$$\mathcal{R} = \left\{ \sum_{\rho \in k} c_\rho e^{\rho X} : c_\rho \in \mathbb{C}, \quad \text{finite support} \right\}.$$

Fix $-\pi/2 < \theta < \pi/2$. We use the right-lateral convention of [V.1, Proposition 3.1] at an exceptional direction. For an E-function f, write its canonical decomposition as $f(z) = \sum_\rho f_\rho(z) e^{\rho z}$ near $z = 1$. The analytic input needed below is stronger than sectorial existence: each $e^{\rho z} f_\rho(z)$ solves an E-operator whose only possible finite singularity is zero [V.1, §2, immediately before Theorem 2.3]. Thus every f_ρ continues holomorphically along every path in $\mathbb{C}^\times$. Let M^j denote continuation around a based loop at 1 of counterclockwise winding number $j \in \mathbb{Z}$.

Theorem V.2.1 (Continued phase embeddings)**.** *For every integer j, the prescription*

$$\Phi_\theta^{[j]}(f(1))(X) = \sum_\rho (M^j f_\rho)(1) e^{\rho X} \tag{V.2.1}$$

defines an injective k-algebra homomorphism $\mathcal{E} \longrightarrow \mathcal{R}$. It satisfies

$$\Phi_\theta^{[j]}(\xi)(1) = \xi, \qquad \Phi_\theta^{[j]}(e^a) = e^{aX} \quad (a \in k).$$

The case $j = 0$ is the original phase embedding.

Proof. Suppose $f(1) = g(1)$ for two E-functions. Beukers' zero-removal theorem supplies an E-function h with $f - g = (z-1)h$. Choose one sufficiently small positive perturbation of θ for the decompositions of f, g, h. Uniqueness of these decompositions gives

$$f_\rho - g_\rho = (z-1)h_\rho$$

for each phase, inserting zero components when necessary. Continue this identity along the same based loop. All continued components, including $M^j h_\rho$, are holomorphic at the endpoint 1. Consequently $(M^j f_\rho)(1) = (M^j g_\rho)(1)$, proving that (V.2.1) is independent of the representative.

The component decomposition of a sum is the sum of the component decompositions. For a product $q = fg$ it is

$$q_\rho = \sum_{\alpha+\beta=\rho} f_\alpha g_\beta.$$

These identities follow initially from uniqueness of the canonical expansion, and remain true after analytic continuation. They prove additivity and multiplicativity. Constant functions and pure exponentials have constant components, so the stated normalizations hold. Finally, the continuation of the total sum is the continuation of the entire function f. Therefore

$$\sum_\rho (M^j f_\rho)(1) e^\rho = (M^j f)(1) = f(1),$$

which proves evaluation compatibility and injectivity. □

This construction does not replace θ by $\theta + 2\pi j$. The direction remains admissible. The continued components need not form the canonical asymptotic decomposition in any prescribed sector; only their analytic continuations are used.

Lemma V.2.2 (Two omitted values exclude denominators)**.** *Let $\xi \in \mathcal{E}$. Suppose two injective k-algebra maps $\Psi_i : \mathcal{E} \to \mathcal{R}$, $i = 1, 2$, have images*

$$\Psi_i(\xi) = C_i + B_i e^{a_i X}, \qquad B_i a_i \neq 0, \quad C_1 \neq C_2.$$

Then ξ is transcendental and

$$k(\xi) \cap \mathcal{E} = k[\xi], \qquad k[\xi] \cap \mathcal{E}^\times = k^\times.$$

Proof. A nonconstant image excludes $\xi \in k$, because both maps fix k. Take relatively prime $P, Q \in k[Y]$ with $R = P(\xi)/Q(\xi) \in \mathcal{E}$. Applying either map to the identity $Q(\xi)R = P(\xi)$ gives

$$Q(C_i + B_i e^{a_i X})\Psi_i(R)(X) = P(C_i + B_i e^{a_i X}).$$

If α is a root of Q different from C_i, the exponential attains $(\alpha - C_i)/B_i \neq 0$. Evaluating at such a point gives $P(\alpha) = 0$, contrary to coprimality. Thus every root of Q would have to equal both C_1 and C_2. Hence Q is constant. The reverse inclusion is immediate. Applying this conclusion to $1/P(\xi)$ proves the assertion about units. □

Theorem V.2.3 (All nonzero algebraic exponential-integral arguments)**.** *For every $a \in k^\times$, let $\eta_a = \mathrm{Ein}(a)$, where*

$$\mathrm{Ein}(z) = \sum_{n \geq 1} \frac{(-1)^{n-1} z^n}{n\, n!}.$$

Then η_a is transcendental and

$$k(\eta_a) \cap \mathcal{E} = k[\eta_a], \qquad k[\eta_a] \cap \mathcal{E}^\times = k^\times. \tag{V.2.2}$$

In particular this includes every positive algebraic a, including $a = 1$, without any arithmetic hypothesis on γ.

Proof. The classical identity $\mathrm{Ein}(w) = \gamma + \log w + E_1(w)$ [V.6, §6.2] gives the two components of $f(z) = \mathrm{Ein}(az)$:

$$f_0(z) = C + \log z, \qquad f_{-a}(z) = e^{az}\big(f(z) - C - \log z\big).$$

Here $C = \gamma + \log a$ with the branch determined by a chosen admissible canonical sum. To see that these are the canonical components, the formal tail for f_{-a} has Borel transform $1/(a + t)$; a nonsingular integration ray gives its sum in a sector of opening greater than π. The displayed identity then fixes the compatible logarithm. If the positive ray meets the pole, take either sufficiently small lateral direction.

Put $B = f_{-a}(1) = e^a(\eta_a - C)$ and $\tau = 2\pi i$. Since f and e^{az} are single-valued whereas $M^j \log z = \log z + j\tau$, Theorem V.2.1 gives

$$\Phi_\theta^{[j]}(\eta_a) = C + j\tau + (B - j\tau e^a)e^{-aX}. \tag{V.2.3}$$

The coefficient $B - j\tau e^a$ vanishes for at most one integer j. Choose two distinct integers for which it does not vanish. Their constant terms differ. Lemma V.2.2 proves all the assertions. □

Corollary V.2.4 (Conditional restrictions on γ and δ)**.** *Let $\delta = e\,\mathrm{Ein}(1) - e\gamma$ be the Euler–Gompertz constant. Then*

$$\begin{aligned} \gamma \in k \implies \delta \in \mathcal{E} \setminus k, \quad k(\delta) \cap \mathcal{E} = k[\delta], \quad k[\delta] \cap \mathcal{E}^\times = k^\times; \\ \delta \in k \implies \gamma \in \mathcal{E} \setminus k, \quad k(\gamma) \cap \mathcal{E} = k[\gamma], \quad k[\gamma] \cap \mathcal{E}^\times = k^\times. \end{aligned}$$

Proof. For $a = 1$, choose direction zero and the principal branch. Then

$$\Phi_0^{[j]}(\mathrm{Ein}(1)) = \gamma + j\tau + (\delta - j\tau e)e^{-X}.$$

If $\gamma \in k$, then $\delta \in \mathcal{E}$ and

$$\Phi_0^{[j]}(\delta) = \delta - j\tau e + j\tau e^X.$$

The cases $j = 1, 2$ satisfy Lemma V.2.2. If instead $\delta \in k$, use $e^{-1} \in \mathcal{E}$ to obtain $\gamma \in \mathcal{E}$ and

$$\Phi_0^{[j]}(\gamma) = \gamma + j\tau - j\tau e e^{-X}.$$

Again $j = 1, 2$ prove the claim. □

Theorem V.2.5 (A monodromy obstruction to units)**.** *Let $0 \neq \xi = f(1) \in \mathcal{E}$. For the finitely many component solution spaces of f, choose an integer $L \geq 1$ such that their local monodromies raised to the Lth power are unipotent. Such an L exists because E-operators are regular singular at zero with rational exponents [V.2, §1, equation (1.1)]. Multiplication by the single-valued factor $e^{-\rho z}$ does not change local monodromy, so this applies also to the component f_ρ. If $\xi \in \mathcal{E}^\times$, there is one $\rho \in k$ for which*

$$\Phi_\theta^{[Lm]}(\xi)(X) = \xi e^{\rho(X-1)} \qquad (m \in \mathbb{Z}). \tag{V.2.4}$$

More generally the images are periodic in the winding number, with period L.

Proof. The units of $\mathcal{R}$ are exactly $ce^{\rho X}$ with $c \neq 0$. Indeed, the finitely many phases in an element and a putative inverse generate a finitely generated free abelian group. The associated group algebra is a Laurent polynomial ring, since distinct exponentials are linearly independent. Its units are scalar monomials.

For each phase, the coefficient $A_\rho(m) = (M^{Lm} f_\rho)(1)$ is a polynomial in m: if $M^L = I + N$ on a finite-dimensional solution space with N nilpotent, then $(I+N)^m = \sum_{s \geq 0} \binom{m}{s} N^s$, a finite sum valid also for negative integers m. If ξ is a unit, every continued image is a unit of $\mathcal{R}$ and therefore has exactly one nonzero phase. Thus, for distinct phases ρ, σ, the polynomial $A_\rho(m)A_\sigma(m)$ vanishes at every integer and is identically zero. At most one A_ρ is a nonzero polynomial. There is at least one, because evaluation at $X = 1$ is the nonzero number ξ. For the surviving phase, evaluation gives $A_\rho(m)e^\rho = \xi$ at every integer; hence A_ρ is constant. This proves (V.2.4). Apply the same argument to $M^j f_\rho$ for each residue class j mod L to obtain periodicity. □

Lemma V.2.6 (A nonzero difference cannot be pulled back)**.** *If $\xi \in \mathcal{E}$ and $H = \Phi_\theta^{[j]}(\xi) - \Phi_\theta^{[\ell]}(\xi) \neq 0$, then H belongs to none of the images $\Phi_\vartheta^{[r]}(\mathcal{E})$, for any admissible ϑ and any integer r.*

Proof. One has $H(1) = \xi - \xi = 0$. If $H = \Phi_\vartheta^{[r]}(\zeta)$ for some $\zeta \in \mathcal{E}$, evaluation gives $\zeta = 0$, whence $H = 0$, a contradiction. □

For example, with $\eta = \mathrm{Ein}(1)$ and $j \neq 0$,

$$\Phi_0^{[j]}(\eta) - \Phi_0^{[0]}(\eta) = 2\pi i j(1 - ee^{-X}) \neq 0.$$

Its coefficients do not become E-values merely because it is a difference of two phase images. There is no inverse phase map on this difference. Nor may one invoke an established separation $\mathcal{E} \cap \mathcal{G} = k$: this is stated as a conjecture in [V.3, p. 38], where $\mathcal{G}$ is the ring of analytically continued G-function values.

Scope of the result. Equation (V.2.2) is a restricted case of the unit problem for $\mathcal{E}$, with a complete rational-intersection statement. The unit obstruction also tests the variation of component coefficients under monodromy, rather than only their initial phase support. None of these statements proves that γ is irrational or transcendental. In particular, Corollary V.2.4 is consistent with algebraicity of γ: it would make δ a transcendental nonunit of $\mathcal{E}$, and no independent theorem places $1/\delta$ in $\mathcal{E}$. These extensions are new to this compilation; their priority in the literature has not been established, and no literature-novelty claim is made.

V.3 Rational intersections for irreducible Kummer values

The following is one application of the continued phase embeddings. It extends the restricted parameter range of the Kummer nonunit examples in [V.1, Theorem 1.1 and §4.3]. It is a

theorem about a single parameterized family, not a collection of independently solved conjectures. Literature priority for this extension has not been established.

Theorem V.3.1 (Kummer polynomial intersection)**.** *Let $a, b \in \mathbb{Q}$ satisfy*

$$b \notin \mathbb{Z}_{\leq 0}, \qquad a \notin \mathbb{Z}, \qquad b - a \notin \mathbb{Z},$$

Let $\alpha \in \overline{\mathbb{Q}}^{\times}$ and put $\xi = {}_1F_1(a; b; \alpha) = M(a, b, \alpha)$. Then ξ is transcendental and, with $k = \overline{\mathbb{Q}}$,

$$k(\xi) \cap \mathcal{E} = k[\xi], \qquad k[\xi] \cap \mathcal{E}^{\times} = k^{\times}.$$

In particular $M(a, b, \alpha)$ is not a unit of $\mathcal{E}$.

Proof. Set $f(z) = M(a, b, \alpha z)$, an E-function solving

$$zf'' + (b - \alpha z)f' - \alpha a f = 0. \tag{V.3.1}$$

Choose a sufficiently small admissible direction θ such that αz lies in an upper or lower canonical sector, and denote the corresponding sign by $\varepsilon \in \{1, -1\}$. Its two canonical phase components, including their exponentials, will be denoted by g_0 and g_α, so that $f = g_0 + g_\alpha$ and

$$\Phi_\theta^{[m]}(\xi) = (M^m g_0)(1) + e^{-\alpha}(M^m g_\alpha)(1)e^{\alpha X}.$$

Here M denotes monodromy, not the Kummer function when used as an operator. The connection formula [V.6, 13.2.41], with the compatible sign, gives

$$g_0(z) = \frac{\Gamma(b)e^{\varepsilon\pi i a}}{\Gamma(b - a)} U(a, b, \alpha z), \qquad g_\alpha(z) = f(z) - g_0(z). \tag{V.3.2}$$

All occurrences of U and of $(\alpha z)^{1-b}$ below use the same germ determined in that sector and continued to $z = 1$. The standard asymptotics of U identify these as the canonical phase-zero and phase-α components [V.6, 13.7.2–13.7.3]. Both solve the same equation (V.3.1); this also follows by applying its differential operator to the canonical decomposition and using uniqueness, since summation commutes with differentiation and polynomial multiplication. They are independent: both formal phase coefficients $\Gamma(b)e^{\varepsilon\pi i a}/\Gamma(b - a)$ and $\Gamma(b)/\Gamma(a)$ are nonzero, so proportionality would contradict uniqueness of the two distinct phases. Equivalently, their Wronskian is a nonzero multiple of $e^{\alpha z}z^{-b}$ [V.6, 13.2.38].

First, $\xi \neq 0$. Otherwise $\Phi_\theta(\xi) = 0$ forces $g_0(1) = g_\alpha(1) = 0$ by independence of 1 and $e^{\alpha X}$. Two independent solutions of a second-order equation cannot both vanish at its ordinary point 1, since their Wronskian there would vanish.

Case 1: b is a positive integer. The exponents at zero are 0 and $1 - b$. The holomorphic solution space there is the line $\mathbb{C}f$, and local monodromy is nontrivial unipotent. For completeness, if $b \geq 2$, a putative logarithm-free second solution with leading term w^{1-b} in the coordinate $w = \alpha z$ would have coefficients $c_0 \neq 0$ satisfying

$$n(n + 1 - b)c_n = (n + a - b)c_{n-1}.$$

Up to $n = b - 2$ all these coefficients are nonzero, whereas at $n = b - 1$ the equation would require $0 = (a - 1)c_{b-2} \neq 0$. When $b = 1$, a meromorphic solution cannot have a pole, since the only exponent is zero, and the holomorphic solution is again unique up to scalar. Trivial monodromy would give two meromorphic solutions at this regular singularity with integer exponents, contradicting these facts.

Thus $\operatorname{im}(M - I) = \ker(M - I) = \mathbb{C}f$. Since g_0 is not proportional to f, there is a scalar $\kappa \neq 0$ such that

$$(M - I)g_0 = \kappa f, \qquad (M - I)g_\alpha = -\kappa f.$$

Writing $C = g_0(1)$ yields

$$\Phi_\theta^{[m]}(\xi) = C + m\kappa\xi + e^{-\alpha}(\xi - C - m\kappa\xi)e^{\alpha X}. \tag{V.3.3}$$

The constant terms are distinct because $\kappa\xi \neq 0$, and at most one exponential coefficient vanishes. Lemma V.2.2 proves all the conclusions in this case.

Case 2: b is not an integer. Put

$$u = e^{2\pi i a}, \quad v = e^{2\pi i b}, \quad \lambda = v^{-1}, \quad s_+ = \frac{u - v}{1 - v}, \qquad s_- = u^{-1}s_+, \qquad s = s_\varepsilon.$$

These numbers are algebraic, $\lambda \neq 1$, and $s \notin \{0, 1\}$ for either sign: the excluded values would give respectively $a - b \in \mathbb{Z}$ or $a \in \mathbb{Z}$. The local connection formula [V.6, 13.2.42], combined with (V.3.2) and the gamma reflection formula, gives

$$g_0 = sf + th, \qquad h(z) = (\alpha z)^{1-b} M(a - b + 1, 2 - b, \alpha z)$$

for a nonzero constant t. Indeed the coefficient of f is

$$\frac{\Gamma(b)\Gamma(1 - b)e^{\varepsilon\pi i a}}{\Gamma(b - a)\Gamma(1 + a - b)} = \frac{e^{\varepsilon\pi i a}\sin\pi(b - a)}{\sin\pi b} = s_\varepsilon.$$

Since $Mf = f$ and $Mh = \lambda h$, the phase images are

$$\Phi_\theta^{[m]}(\xi) = C_m + e^{-\alpha}(\xi - C_m)e^{\alpha X}, \qquad C_m = s\xi + \lambda^m(C_0 - s\xi). \tag{V.3.4}$$

This formula first proves $\xi \notin k$. If ξ were algebraic, every image would be the constant ξ, so $C_m = \xi$ for all m. Comparing $m = 0, 1$ gives $(1 - \lambda)(1 - s)\xi = 0$, contrary to the established nonvanishing factors.

Now take relatively prime $P, Q \in k[Y]$ with $P(\xi)/Q(\xi) \in \mathcal{E}$. The denominator argument of Lemma V.2.2 says that any root of Q must equal every C_m for which $C_m \neq \xi$. If $C_0 = s\xi$, all the C_m equal the transcendental number $s\xi \neq \xi$; hence Q has no algebraic root and is constant. Otherwise the C_m form an orbit of q distinct values, where $q \geq 2$ is the order of the root of unity λ. For $q \geq 3$, at least two distinct values differ from ξ, so Q is constant by the two-center argument. For $q = 2$, the same holds unless one orbit value equals ξ. In that last case the other equals $(2s - 1)\xi$. But $v = -1$ gives $2s - 1 = u^\varepsilon \in k^\times$, so this other value is transcendental and cannot be a root of $Q \in k[Y]$. Again Q is constant. This proves the field intersection. The statement about units follows by applying it to reciprocals of nonconstant polynomials. □

Corollary V.3.2 (Integer-order Bessel values)**.** *For $n \in \mathbb{Z}_{\geq 0}$ and $t \in \overline{\mathbb{Q}}^\times$, let $\beta = J_n(t)$. Then β is transcendental and*

$$k(\beta) \cap \mathcal{E} = k[\beta], \qquad k[\beta] \cap \mathcal{E}^\times = k^\times.$$

Proof. The classical Kummer–Bessel identity [V.6, 13.6.9], together with $I_n(it) = i^n J_n(t)$, gives

$$\xi = M\left(n + \tfrac{1}{2}, 2n + 1, 2it\right) = n!e^{it}(t/2)^{-n}J_n(t).$$

Put $d = (t/2)^n/n! \in k^\times$. The integer-$b$ case of the proof above, with $\alpha = 2it$, gives affine coefficients $C_m = C_0 + m\kappa\xi$ with $\kappa\xi \neq 0$. Multiplicativity and the exponential normalization of the continued embeddings give

$$\Phi_\theta^{[m]}(\beta) = dC_m e^{-itX} + de^{-2it}(\xi - C_m)e^{itX}.$$

Choose m avoiding the at most two zeros of its two coefficients. Then this image is $Ae^{itX} + Be^{-itX}$ with $AB \neq 0$. It is surjective onto $\mathbb{C}$: for any $w \in \mathbb{C}$, the polynomial $AZ^2 - wZ + B$ has a nonzero complex root, and $Z = e^{itX}$ attains every nonzero complex number. The image is nonconstant, so β is transcendental. If $P(\beta)/Q(\beta) \in \mathcal{E}$ in reduced form, apply this embedding to the product identity and evaluate at a preimage of any root of Q. Coprimality is contradicted unless Q is constant. The assertion about units follows as before. □

The transcendence of these irreducible Kummer values is classical; the content here is the rational-intersection conclusion and its unit consequence. The parameter exclusions are material. For example, $M(b,b,\alpha)=e^{\alpha}$ is a unit, and terminating cases can give algebraic values. No conclusion about the irrationality or transcendence of γ follows from this application.

V.4 Primitive elementary jets and the exact CMF image lattice

The denominator of a raw vector is not the denominator of a reduced approximant. We examine the integer matrix before selecting a seed. The following finite-characteristic argument determines a universal content exactly. It leaves seed-specific gcds as a separate problem.

Lemma V.4.1 (Primitive jets of integer Euler products)**.** *Let $\theta = z\,d/dz$, $D=\theta-z$, and $P(X)=\pm\prod_{\nu=1}^{m}(X-a_\nu)$ with $a_\nu\in\mathbb{Z}$; the empty product is allowed. Put $R(z)=P(D)1\in\mathbb{Z}[z]$. Suppose $e^{-z}R(z)$ satisfies a differential equation*

$$\left(\theta^r+\sum_{j=0}^{r-1}c_j(z)\theta^j\right)(e^{-z}R(z))=0,\qquad c_j\in\mathbb{Z}[z].$$

Then the integer vector

$$\big(R(1),(DR)(1),\dots,(D^{r-1}R)(1)\big)$$

is primitive: the gcd of its coordinates is one.

Proof. Fix a prime ℓ. Work in the ℓ-dimensional vector space $V=\mathbb{F}_\ell[z]/(z^\ell-1)$, which is also $\mathbb{F}_\ell[z]/((z-1)^\ell)$. Differentiation is well defined on this quotient and its ℓth power is zero. With $T=\partial_z-1$, the relation $Tz-zT=1$ gives

$$D(D-1)\cdots(D-\ell+1)=z^\ell T^\ell.$$

In characteristic ℓ, the left side is $D^\ell-D$, whereas $T^\ell=\partial_z^\ell-1=-1$ on V. Consequently

$$D^\ell-D=-1\qquad\text{on } V.$$

Each $D-a$, $a\in\mathbb{F}_\ell$, is invertible: the negative product of the other $\ell-1$ factors is its inverse. Therefore $P(D)$ is invertible, and $R=P(D)1$ is nonzero in V.

Suppose that the displayed r jet coordinates were all divisible by ℓ. The operators D^j have leading ordinary derivative term $z^j\partial_z^j$; the lower terms have integer polynomial coefficients. At $z=1$ this is a triangular change of derivative coordinates with diagonal entries one. Thus $R^{(j)}(1)=0$ modulo ℓ for $j<r$. If $\ell\le r$, the first ℓ derivatives already imply $R=0$ in V, since $0!,\dots,(\ell-1)!$ are invertible.

If $\ell>r$, conjugating the given differential equation by e^{-z} gives

$$\left(D^r+\sum_{j<r}c_j(z)D^j\right)R=0.$$

Its coefficient of ∂_z^r is z^r, equal to one at $z=1$. Differentiate this identity successively zero through $\ell-r-1$ times, and evaluate at one. Induction forces $R^{(r)}(1),\dots,R^{(\ell-1)}(1)$ to vanish modulo ℓ: the highest derivative has coefficient one, and all other derivatives in that identity have already vanished. Again $R=0$ in V, a contradiction. This holds for every prime, proving primitivity. □

Corollary V.4.2 (All integral rank-three CMF trajectories)**.** *Let A_1, A_2, b_1, b_2, b_3 be nonnegative integers. Starting at the zero parameter point, let Φ_n be the row-state transition matrix for the trajectory $(-A_1, -A_2; b_1, b_2, b_3)$, with the scalar contiguous operator normalization. Then Φ_n is a nonsingular integer matrix, and*

$$E_n = (1, -1, 0)\Phi_n \in \mathbb{Z}^3 \quad \textit{satisfies} \quad \gcd(E_{n,0}, E_{n,1}, E_{n,2}) = 1.$$

Moreover

$$\Delta_n := \det \Phi_n = \prod_{i=1}^{2} \prod_{j=1}^{3} ((A_i + b_j)n)! > 0. \tag{V.4.1}$$

Proof. The product of contiguous scalar operators is

$$P_n(\theta) = (\theta + 1)_{A_1 n} (\theta + 1)_{A_2 n} \prod_{j=1}^{3} (-\theta)_{b_j n}.$$

These are commuting polynomials in θ, each a product of integer-linear factors with leading coefficient 1 or -1. The transformed elementary solution is $f_n(z) = eP_n(\theta)e^{-z} = e^{1-z}R_n(z)$. Its state at one is E_n, since the initial state of e^{1-z} is $(1, -1, 0)$. Its final scalar equation is

$$\left[\prod_{j=1}^{3} (\theta - b_j n) + z \prod_{i=1}^{2} (\theta + A_i n + 1) \right] f_n = 0.$$

This equation is monic of degree three in θ with integer polynomial coefficients. The preceding lemma proves the gcd assertion.

For clarity, the determinant formula can be obtained without the gcd argument. A single upper decrement at parameters a_i, b_j has determinant $\prod_j (b_j - a_i + 1)$, and a single lower increment has determinant $\prod_i (b_j - a_i + 1)$. These follow from the rank-three companion polynomial at $z = 1$, evaluated at $a_i - 1$ or b_j. Take all upper decrements first and then all lower increments. For each pair (i, j) the factors multiply to $((A_i + b_j)n)!$. All factors are positive. Contiguous matrices have integer entries because the old monic equation reduces powers of θ using integer polynomial coefficients at one. Thus every step is defined, integral, and nonsingular, and multiplication proves the formula. □

Lemma V.4.3 (Content and image of a primitive complementary pair)**.** *Let $\Phi \in M_3(\mathbb{Z})$ have determinant $\Delta \neq 0$. Let $p, q, u \in \mathbb{Z}^3$ satisfy $p \times q = u$, with u primitive, and suppose $E = u^T \Phi$ is primitive. Put*

$$H = \operatorname{adj}(\Phi)^T, \qquad h = \gcd\{H_{ij}\}, \qquad B = \begin{pmatrix} p^T H \\ q^T H \end{pmatrix}.$$

Then the gcd of the six entries of B is exactly h. The matrix $\widehat{B} = B/h$ has image lattice $\Omega = \widehat{B}\mathbb{Z}^3 \subset \mathbb{Z}^2$ with

$$[\mathbb{Z}^2 : \Omega] = \frac{|\Delta|}{h^2}, \qquad \mathbb{Z}^2/\Omega \simeq \mathbb{Z}/(|\Delta|/h^2)\mathbb{Z}, \qquad \ker_{\mathbb{Z}} \widehat{B} = \mathbb{Z}E^T. \tag{V.4.2}$$

Proof. Choose $S, T \in \mathrm{SL}_3(\mathbb{Z})$ so that the first row of S is u^T and $ET = (1, 0, 0)$. Then

$$\Psi = S\Phi T = \begin{pmatrix} 1 & 0 & 0 \\ a & b & c \\ d & e & f \end{pmatrix}, \qquad \operatorname{adj}(\Psi)^T = \begin{pmatrix} bf - ce & cd - af & ae - bd \\ 0 & f & -e \\ 0 & -c & b \end{pmatrix}.$$

The gcd of all nine entries of this adjugate is $\gcd(b, c, e, f)$, which is also the gcd of its bottom two rows. Unimodular left and right transformations preserve the gcd of entries. Since p, q form

a basis of the saturated lattice $u^{\perp} \cap \mathbb{Z}^3$ (their cross product is primitive), their images under S form a unimodular basis of $\{0\} \times \mathbb{Z}^2$. The identity $H = S^T \operatorname{adj}(\Psi)^T T^T$ therefore proves the asserted equality of contents.

The cross-product transformation law gives

$$(p^T H) \times (q^T H) = \Delta\, u^T \Phi = \Delta E.$$

Because E is primitive, the gcd of the three maximal minors of B is $|\Delta|$; that of $\widehat{B}$ is $|\Delta|/h^2$. The elementary Smith normal form formulas identify this gcd with the image index. The first invariant factor of $\widehat{B}$ is one, since its entries are relatively prime, proving cyclicity. The nonzero cross product identifies its rational kernel as the line $\mathbb{Q}E^T$; primitivity of E gives the integer kernel. □

Theorem V.4.4 (No further universal cancellation in gamma extraction)**.** *In Corollary V.4.2, take $p = (1,1,0)^T$, $q = (1,1,1)^T$, and let h_n be the content of* $\operatorname{adj}(\Phi_n)$*. After removing this matrix content, the two rows extracting the numerator and denominator have joint content exactly one. Their image is a rank-two integer lattice with cyclic quotient of order Δ_n/h_n^2. If $A_1 = A_2 = A$, set*

$$N_n = \prod_{j=1}^{3} ((A + b_j)n)!.$$

Then $h_n \mid N_n$, and the image index is the perfect square $(N_n/h_n)^2$.

Proof. Here $p \times q = (1,-1,0)^T$, and its transported row is primitive by Corollary V.4.2. Apply the preceding lemma. When $A_1 = A_2$, formula (V.4.1) is $\Delta_n = N_n^2$. Since $h_n^2 \mid \Delta_n$, prime valuations imply $h_n \mid N_n$. □

This theorem concerns cancellation common to every seed. A particular $v \in \mathbb{Z}^3$ can still produce a pair $\widehat{B}_n v$ with a large gcd. Thus the statement does not control reduced scalar heights, does not exclude a specially designed varying-seed construction, and does not prove an irrationality measure for γ.

V.4.1 An exponential bound for the quotient index

Write $d_M = \operatorname{lcm}(1, \dots, M)$, with $d_0 = 1$. The square index above admits a substantially smaller arithmetic bound than its defining factorial determinant suggests.

Theorem V.4.5 (Only exponential index remains)**.** *In Theorem V.4.4, assume $A_1 = A_2 = A$ and put*

$$m_j = (A + b_j)n, \qquad S = m_1 + m_2 + m_3, \qquad N = m_1!m_2!m_3!, \qquad J = N/h_n.$$

Then

$$J \mid d_{S+1}^4, \qquad [\mathbb{Z}^2 : \Omega_n] = J^2 \mid d_{S+1}^8. \tag{V.4.3}$$

For a fixed trajectory, the logarithm of the image index is therefore $O(n)$.

Proof. We first treat $A = 0$. Let $P(X) = \prod_{j=1}^{3} (-X)_{m_j}$, of degree S. In the triangular source system

$$\theta f = -zf + g, \qquad \theta g = h, \qquad \theta h = 0,$$

there are polynomials $R, U, V \in \mathbb{Z}[z]$ such that

$$P(\theta) f = R(z) f + U(z) g + V(z) h. \tag{V.4.4}$$

Indeed one application of θ sends the coefficient triple (R, U, V) to

$$(\theta R - zR,\ R + \theta U,\ U + \theta V),$$

so this assertion follows by applying the integer-linear factors of P. The polynomial $R = e^z P(\theta) e^{-z}$ has degree S. Write $R(z) = \sum_{k=0}^{S} r_k z^k$. Expanding the exponential gives

$$k! r_k = \sum_{j=0}^{k} (-1)^j \binom{k}{j} P(j) \in N\mathbb{Z}.$$

The divisibility holds because $P(j)/N = \prod_\nu (-1)^{m_\nu} \binom{j}{m_\nu}$ is an integer.

We next identify U, V and their denominators after division by N. All calculations in this paragraph are formal at infinity; no convergence assertion is used. The Laurent series

$$F(z) = \sum_{k \geq 0} k! z^{-k-1}, \qquad G(z) = \sum_{k \geq 0} k! z^{-k-1} (\log z - H_k)$$

satisfy $(\theta + z)F = 1$ and $(\theta + z)G = \log z$. Apply (V.4.4) first with $(f, g, h) = (F, 1, 0)$ and then with $(G, \log z, 1)$. Both $P(\theta)F$ and $P(\theta)G$ have only strictly negative powers of z, with logarithmic coefficients in the second case. Taking polynomial parts therefore gives, for $0 \leq i < S$,

$$U_i = -\sum_{k=i+1}^{S} r_k (k-i-1)!, \qquad V_i = \sum_{k=i+1}^{S} r_k (k-i-1)! H_{k-i-1}. \tag{V.4.5}$$

For $0 \leq i < k$, the integer $k\binom{k-1}{i}$ divides d_k. To see this, its valuation at a prime ℓ is

$$\sum_{a \geq 1} \left(\left\lfloor \frac{k}{\ell^a} \right\rfloor - \left\lfloor \frac{i}{\ell^a} \right\rfloor - \left\lfloor \frac{k-i-1}{\ell^a} \right\rfloor \right).$$

Every summand is zero or one, and it is zero for $\ell^a > k$. Since $i!(k-i-1)!/k! = 1/[k\binom{k-1}{i}]$, and the denominator of H_j divides d_S for $j \leq S$, formula (V.4.5) implies

$$k! r_k \in N\mathbb{Z}, \qquad i! U_i \in \frac{N}{d_S}\mathbb{Z}, \qquad i! V_i \in \frac{N}{d_S^2}\mathbb{Z}. \tag{V.4.6}$$

We must bound the degrees of the actual extraction polynomials, not of their uncancelled products. Set $D = \theta - z$ and define three polynomial rows

$$\begin{aligned} \mathcal{R} &= (R, DR, D^2 R), \\ \mathcal{U} &= (U, R + \theta U, 2\theta R - zR + \theta^2 U), \\ \mathcal{V} &= (V, U + \theta V, R + 2\theta U + \theta^2 V). \end{aligned}$$

Their evaluations at one are the transition rows in the source coordinates (f, g, h). The two numerator–denominator extraction rows of the adjugate are, up to a sign,

$$\mathcal{W}_1(1),\ \mathcal{W}_2(1), \qquad \mathcal{W}_1 = \mathcal{R} \times \mathcal{U}, \mathcal{W}_2 = \mathcal{R} \times \mathcal{V}.$$

For example the original state at one is $(f, \theta f, \theta^2 f) = (f, g, h) \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$; the extraction vectors select g, h, respectively. This confirms the row conventions and the absence of an extra scalar factor.

Here is a formal proof that every component of $\mathcal{W}_1$ and $\mathcal{W}_2$ has degree at most $S+1$. The jets of $P(\theta)F = RF + U$ are $F\mathcal{R} + \mathcal{U}$. The jets of $P(\theta)G = RG + U \log z + V$ are $G\mathcal{R} + (\log z)\mathcal{U} + \mathcal{V}$. Taking their cross products with $\mathcal{R}$ gives the exact identities

$$\mathcal{R} \times \mathrm{Jet}(P(\theta)F) = \mathcal{W}_1, \qquad \mathcal{R} \times \mathrm{Jet}(P(\theta)G) = (\log z)\mathcal{W}_1 + \mathcal{W}_2.$$

But each of the first two formal jets has strictly negative powers of z (and possibly one power of $\log z$), whereas $\mathcal{R}$ has degree at most $S+2$. Consequently

$$\deg \mathcal{W}_1 \le S+1, \qquad \deg \mathcal{W}_2 \le S+1;$$

for the second assertion compare the constant and logarithmic coefficients separately. This is a statement about formal Laurent series equal to polynomials, so it establishes exact degree bounds.

Multiplication of polynomials whose factorial-weighted coefficients are integral preserves that property, since the coefficient formula uses binomial coefficients. The same is true of θ and of multiplication by z. Hence (V.4.6) gives

$$k![z^k]\mathcal{W}_i \in \frac{N^2}{d_S^2}\mathbb{Z}^3 \qquad (i=1,2).$$

Using their degree bounds, evaluation at one yields

$$v_\ell(h_n) \ge 2v_\ell(N) - v_\ell((S+1)!) - 2v_\ell(d_S)$$

for every prime ℓ; here the equality of extraction content and adjugate content is essential. Thus

$$\begin{aligned} v_\ell(J) &\le v_\ell((S+1)!/N) + 2v_\ell(d_S) \\ &\le 4\left\lfloor \log_\ell(S+1) \right\rfloor. \end{aligned}$$

For the last inequality, write the valuation directly as

$$v_\ell((S+1)!/N) = \sum_{a\ge 1}\left(\left\lfloor \frac{S+1}{\ell^a} \right\rfloor - \sum_{j=1}^{3} \left\lfloor \frac{m_j}{\ell^a} \right\rfloor \right).$$

Every summand is zero, one, or two: the sum of three residues modulo ℓ^a, plus one, is at most $3\ell^a - 2 < 3\ell^a$. There are at most $\lfloor \log_\ell(S+1) \rfloor$ nonzero summands, while $v_\ell(d_S) \le \lfloor \log_\ell(S+1) \rfloor$. The case $S=0$ is immediate. This proves the divisibility for $A=0$.

Finally equal upper decrements reduce to this case by an integral unimodular gauge. Put $a = An$. On the initial solution space,

$$(-\theta)_a^3 = z^a(\theta+1)_a^2,$$

as follows inductively from the initial equation $(-\theta)^3 f = z(\theta+1)^2 f$ and $\theta z = z(\theta+1)$. Commuting the remaining Euler factors gives

$$\prod_j (-\theta)_{a+b_j n} f = z^a(\theta+1)_a^2 \prod_j (-\theta)_{b_j n} f.$$

At $z=1$, multiplication by z^a acts on jets through

$$U_a = \begin{pmatrix} 1 & a & a^2 \\ 0 & 1 & 2a \\ 0 & 0 & 1 \end{pmatrix} \in \mathrm{SL}_3(\mathbb{Z}).$$

It leaves adjugate content and the image lattice invariant, proving the claimed reduction. The classical bound $\log d_M = O(M)$ then gives the last assertion. □

The factor 4 in (V.4.3) is a proved upper bound, not an optimality claim. Exact experiments suggest the stronger $J \mid d_{\max m_j}$, but that sharper statement is not established here. Even an exponential image index does not supply small-error lifts of a primitive target; controlling such lifts remains necessary for an irrationality argument.

V.5 A canonical integer moment pencil and its finite resolution

The exact Gamma functional equation and its real positive log-convex normalization fix the analytic function. An artificial shear of a comparison constant need not respect these data. We therefore keep the canonical normalization and ask whether positivity supplies an integer certificate. The construction below does supply such a certificate system. It also permits a quantitative analysis of what a finite part of the system cannot yet distinguish. No irrationality conclusion is asserted.

Put $H_0 = 0$ and, for $k \geq 0$, define

$$\mu_k(c) = k!\{k(H_k - c) + 1\} = a_k - b_k c, \qquad a_k, b_k \in \mathbb{Z}. \tag{V.5.1}$$

Here $a_k = k!(kH_k + 1)$ and $b_k = k\,k!$. For a real polynomial $Q(X) = \sum_i q_i X^i$ set

$$\mathcal{F}_c(Q) = \sum_{i,j} q_i q_j \mu_{i+j}(c), \qquad M_N(c) = (\mu_{i+j}(c))_{0 \leq i,j < N}.$$

Thus $\mathcal{F}_c(Q) = A_Q - B_Q c$ with $A_Q, B_Q \in \mathbb{Z}$ when $Q \in \mathbb{Z}[X]$. Differentiation of the canonical Gamma integral gives

$$\int_0^\infty x^k \log x\, e^{-x}\, dx = k!(H_k - \gamma).$$

Subtracting consecutive moments proves the identity

$$\mathcal{F}_c(Q) = \int_0^\infty Q(x)^2 (x-1)(\log x + \gamma - c) e^{-x}\, dx. \tag{V.5.2}$$

The coefficient construction in (V.5.1) itself uses no value of γ. Its identification with (V.5.2) is where the canonical analytic normalization enters.

Theorem V.5.1 (Exact positivity parameter)**.** *For a real number c, the following conditions are equivalent:*

1. *$c = \gamma$;*
2. *$\mathcal{F}_c(Q) \geq 0$ for every $Q \in \mathbb{R}[X]$;*
3. *$\mathcal{F}_c(Q) \geq 0$ for every $Q \in \mathbb{Z}[X]$;*
4. *$M_N(c)$ is positive semidefinite for every $N \geq 1$.*

At $c = \gamma$, every nonzero polynomial gives strict positivity. For $c = a/b$ with integers a, b and $b > 0$, a negative certificate is the entirely integer inequality $bA_Q - aB_Q < 0$.

Proof. The weight $(x-1)\log x\, e^{-x}$ is positive for all $x > 0$ except $x = 1$. This proves strict positivity at $c = \gamma$. If $c \neq \gamma$, the weight

$$w_c(x) = (x-1)(\log x + \gamma - c)e^{-x}$$

is negative on the interval with endpoints 1 and $e^{c-\gamma}$. We show that polynomials are dense in $L^2(|w_c(x)|\,dx)$. Suppose g is orthogonal to every polynomial. By Cauchy–Schwarz, the integral

$$T(z) = \int_0^\infty g(x)|w_c(x)|e^{zx}\, dx$$

is analytic in $\operatorname{Re} z < 1/2$: on compact subsets of that half-plane the necessary polynomially weighted integrals converge absolutely. All derivatives of T at zero vanish by orthogonality. The identity theorem makes T zero throughout the half-plane. The Fourier transform of the finite signed or complex measure $g(x)|w_c(x)|\,dx$ is consequently zero. Uniqueness of Fourier transforms of finite measures gives $g = 0$ almost everywhere for $|w_c|\,dx$, proving density.

Choose a nonzero continuous function supported in a compact subinterval where $w_c < 0$, and approximate it in this L^2 space by real polynomials Q_j. The inequality

$$\int |Q_j^2 - f^2|\,|w_c| \le \|Q_j - f\|_{L^2(|w_c|)}(\|Q_j\|_{L^2(|w_c|)} + \|f\|_{L^2(|w_c|)})$$

shows that $\mathcal{F}_c(Q_j) < 0$ for some j. Continuity in the finitely many polynomial coefficients allows rational coefficients instead. Clearing denominators gives a negative integer-polynomial certificate. This establishes the equivalence, since the matrix condition is precisely positivity on each finite-dimensional polynomial space. □

Remark V.5.2 (The remaining arithmetic step). To deduce irrationality one would have to construct a negative certificate for every rational c. The proof above obtains one only after assuming $c \neq \gamma$; it therefore does not supply that step. The generating function also entails no arithmetic contradiction at rational c:

$$\sum_{k\ge 0} \frac{\mu_k(c)}{k!} z^k = \frac{1 - cz - z\log(1-z)}{(1-z)^2}, \qquad |z| < 1.$$

The moments form an integer affine pencil, unlike the prior isolated-atom construction whose entries involve special-function values. The two-sided positivity selection is related to, but different from, the prior one-sided rational spectral-edge method.

V.5.1 An explicit estimate valid at every depth

We use ordinary Laguerre polynomials

$$L_j(x) = \sum_{k=0}^{j} (-1)^k \binom{j}{k} \frac{x^k}{k!}, \qquad \int_0^\infty L_i(x) L_j(x) e^{-x}\,dx = \delta_{ij}.$$

The classical inequality $e^{-x/2}|L_j(x)| \le 1$ for $x \ge 0$ is the case $\alpha = 0$ of DLMF (18.14.8), https://dlmf.nist.gov/18.14.E8.

Lemma V.5.3 (Normalized positive-form estimate). *For a polynomial Q of degree less than N normalized by $\int_0^\infty Q^2 e^{-x}\,dx = 1$, write*

$$I = \mathcal{F}_\gamma(Q), \qquad b = \int_0^\infty (x-1) Q(x)^2 e^{-x}\,dx.$$

Then

$$I \ge \nu_N := \frac{1}{32N^2(1 + 1/(4N))}, \qquad |b| \le \sqrt{\frac{2}{\nu_N}}\, I. \tag{V.5.3}$$

Proof. Expand $Q = \sum_{j<N} u_j L_j$ with $\sum u_j^2 = 1$. Cauchy–Schwarz and the Laguerre inequality imply

$$e^{-x} Q(x)^2 \le e^{-x} \sum_{j<N} L_j(x)^2 \le N.$$

For $\varepsilon = 1/(4N)$, the interval $[1-\varepsilon, 1+\varepsilon]$ thus has probability mass at most $2N\varepsilon = 1/2$ under $Q^2 e^{-x}\,dx$. Outside this interval,

$$f(x) := (x-1)\log x \ge \frac{\varepsilon^2}{1+\varepsilon}.$$

Indeed, f decreases on $(0,1)$ and increases on $(1,\infty)$; at $1+\varepsilon$ use $\log(1+\varepsilon) \ge \varepsilon/(1+\varepsilon)$, and at $1-\varepsilon$ use $-\log(1-\varepsilon) \ge \varepsilon$. At least half of the probability mass lies outside, proving the lower bound for I.

Since $f''(x) = (x+1)/x^2 > 0$, Jensen's inequality gives

$$I \geq f(1+b) = b\log(1+b), \qquad b > -1. \tag{V.5.4}$$

For $-1 < b < 0$, $-\log(1+b) \geq -b$ gives a lower bound b^2. For $0 \leq b \leq 1$, $\log(1+b) \geq b/(1+b) \geq b/2$. Thus $I \geq b^2/2$ for $|b| \leq 1$. Combining this with $I \geq \nu_N$ yields

$$\frac{|b|}{I} \leq \frac{|b|}{\max(\nu_N, b^2/2)} \leq \sqrt{2/\nu_N}.$$

For $b \geq 1$, (V.5.4) instead gives $b/I \leq 1/\log 2 < \sqrt{2/\nu_N}$. These cases prove the second inequality. □

Theorem V.5.4 (Finite-depth ambiguity)**.** *Set*

$$r_N = \frac{1}{8N\sqrt{1+1/(4N)}}.$$

The matrix $M_N(c)$ is positive definite for $|c-\gamma| < r_N$ and positive semidefinite for $|c-\gamma| \leq r_N$. In particular, a negative certificate of degree less than N requires $|c-\gamma| > r_N$.

For $N \geq 2$, the feasible set $\mathcal{I}_N = \{c \in \mathbb{R} : M_N(c) \succeq 0\}$ is a compact interval $[\ell_N, u_N]$. These intervals are nested, their intersection is $\{\gamma\}$, and

$$\ell_N \leq \gamma - r_N, \quad u_N \geq \gamma + r_N, \quad \ell_N \longrightarrow \gamma, \quad u_N \longrightarrow \gamma. \tag{V.5.5}$$

Both endpoints are real algebraic numbers.

Proof. Under the normalization of Lemma V.5.3, $\mathcal{F}_c(Q) = I - (c-\gamma)b$. The second estimate in (V.5.3) yields the stated strict and nonstrict positivity. Homogeneity removes the normalization.

Each feasible set is closed and convex, since it is an intersection of affine closed half-lines. Moreover

$$\det M_2(c) = \det \begin{pmatrix} 1 & 2-c \\ 2-c & 8-4c \end{pmatrix} = 4 - c^2,$$

so $\mathcal{I}_N \subseteq [-2, 2]$ for $N \geq 2$. The intervals have nonempty interior by the first part, and they are nested by taking principal submatrices. Theorem V.5.1 identifies their intersection. Compactness and nesting then give convergence of the endpoints. At either endpoint the matrix must be singular, since positive definiteness is open. The determinant is a nonzero polynomial in $\mathbb{Z}[c]$, as it is positive at $c = \gamma$. This proves algebraicity. □

The estimate concerns the raw finite pencil. It does not prohibit accelerated combinations, rational-function tests, or additional arithmetic information.

V.5.2 A separate refinement from compact-uniform Laguerre asymptotics

Lemma V.5.5 (Local density)**.** *For every compact interval $[a, b] \subset (0, \infty)$,*

$$e^{-x} \sum_{j=0}^{N-1} L_j(x)^2 = \frac{\sqrt{N}}{\pi\sqrt{x}} + O_{a,b}(\log N) \tag{V.5.6}$$

uniformly for $x \in [a, b]$ as $N \to \infty$.

Proof. The compact-uniform expansion in DLMF (18.15.14)–(18.15.15), at $\alpha = 0$, gives

$$e^{-x/2}L_j(x) = \frac{1}{\sqrt{\pi}\,(jx)^{1/4}}\left\{\cos(2\sqrt{jx} - \pi/4) + O_{a,b}(j^{-1/2})\right\}.$$

The expansion and the compact-uniformity statement are recorded at `https://dlmf.nist.gov/18.15#iv`. Squaring gives

$$e^{-x}L_j(x)^2 = \frac{\cos^2(2\sqrt{jx} - \pi/4)}{\pi\sqrt{jx}} + O_{a,b}(j^{-1}).$$

The constant part of $\cos^2 t = (1 + \cos 2t)/2$ sums to $\sqrt{N}/(\pi\sqrt{x}) + O_{a,b}(1)$. For the oscillatory part, put $g_x(t) = t^{-1/2}e^{4i\sqrt{tx}}$. Uniformly for $a \le x \le b$,

$$\left|\int_1^N g_x(t)\,dt\right| = O_{a,b}(1), \qquad |g_x'(t)| \le \tfrac{1}{2}t^{-3/2} + 2\sqrt{b}\,t^{-1}.$$

The integral bound follows on substituting $t = u^2$. Comparing the sum of $g_x(j)$ to its integral interval by interval therefore gives $O_{a,b}(1 + \int_1^N |g_x'(t)|dt) = O_{a,b}(\log N)$. The squared-expansion error also sums to $O_{a,b}(\log N)$. The finitely many initial indices, including $j = 0$, contribute only a bounded term. This proves (V.5.6). □

Theorem V.5.6 (Asymptotic finite-depth cost)**.** *The endpoints in Theorem V.5.4 satisfy*

$$\liminf_{N\to\infty} \sqrt{N}\min(\gamma - \ell_N, u_N - \gamma) \ge \frac{\pi}{8}. \tag{V.5.7}$$

Consequently, as $\delta = |c - \gamma| \to 0$, a negative certificate of degree less than N requires $N \ge (\pi^2/64 - o(1))\delta^{-2}$. The constant is a lower bound and is not claimed to be optimal.

Proof. Fix $0 < \eta < 1$ and put $\varepsilon_N = (1 - \eta)\pi/(4\sqrt{N})$. Normalize Q as in Lemma V.5.3. Cauchy–Schwarz bounds its probability density by the kernel in (V.5.6). On the interval $[1 - \varepsilon_N, 1 + \varepsilon_N]$, that estimate gives mass at most

$$2\varepsilon_N\{\sqrt{N}/\pi + O(\log N + 1)\} = (1 - \eta)/2 + o(1) \le 1/2$$

for sufficiently large N, uniformly over the polynomial Q. The preceding energy argument consequently yields

$$I \ge \frac{\varepsilon_N^2}{2(1 + \varepsilon_N)} = \frac{(1 - \eta)^2\pi^2}{32N}(1 + o(1)).$$

Using exactly the Jensen argument from Lemma V.5.3, a positive lower bound $I \ge \nu$ of this size gives the feasible radius $\sqrt{\nu/2}$. Thus

$$\liminf_{N\to\infty} \sqrt{N}\min(\gamma - \ell_N, u_N - \gamma) \ge (1 - \eta)\pi/8.$$

Letting η decrease to zero proves (V.5.7). The certificate-degree statement follows by inversion; Theorem V.5.4 already ensures $N \to \infty$ when $\delta \to 0$. □

V.5.3 The integer-square budget and its limitation

Corollary V.5.7 (Raw integer forms)**.** *If $0 \ne Q \in \mathbb{Z}[X]$ has degree D, then*

$$\mathcal{F}_\gamma(Q) \ge \frac{(D!)^2}{32(D + 1)^2(1 + 1/(4(D + 1)))} \ge \frac{1}{144}. \tag{V.5.8}$$

Uniformly over integer polynomials of degree D, the asymptotic refinement is

$$\mathcal{F}_\gamma(Q) \ge \left(\frac{\pi^2}{32} - o(1)\right)\frac{(D!)^2}{D + 1}. \tag{V.5.9}$$

These statements concern the raw coefficients A_Q, B_Q, before division by their common content.

Proof. The monic polynomials $P_j = (-1)^j j! L_j$ belong to $\mathbb{Z}[X]$. Their unitriangular coefficient matrix implies $Q = \sum_{j=0}^{D} c_j P_j$ with $c_j \in \mathbb{Z}$ and $c_D \neq 0$. Orthogonality gives

$$\int_0^\infty Q^2 e^{-x}\,dx = \sum_{j=0}^{D} c_j^2 (j!)^2 \geq (D!)^2.$$

Apply the homogeneous energy bound with $N = D+1$. Its lower bound is $(D!)^2/[8(D+1)(4D+5)]$. At $D = 0$ it is $1/40$ and at $D = 1$ it is $1/144$; the ratio of the bound at $D+1$ to that at D is

$$\frac{(D+1)^3(4D+5)}{(D+2)(4D+9)} > 1 \quad (D \geq 1).$$

This proves the universal lower bound. For (V.5.9), use the energy estimate in the proof of Theorem V.5.6 and then let $\eta \downarrow 0$. All estimates there are uniform over the normalized polynomial. □

If $g_Q = \gcd(A_Q, B_Q)$, the primitive integer form is $(A_Q - B_Q\gamma)/g_Q$. The bounds above do not control g_Q and do not rule out small primitive forms. They identify a specific arithmetic task: demonstrate sufficient common divisibility while preserving a small, strictly nonzero normalized integral. Canonical uniqueness, or the failure of a formal shear, cannot replace that divisibility analysis.

V.6 Rational logarithmic Laplace kernels and exact positivity obstructions

We seek pure rational linear forms in γ by cancelling auxiliary exponential integrals identically. The following reduction produces such forms with rational, nonpolynomial kernels. Two explicit constructions are tested below. Neither gives small primitive integer forms. Rational reduction and creative telescoping are established methods; the discussion makes no literature-wide priority claim for that methodology.

V.6.1 A rational differential reduction with two local obstructions

Define

$$\mathcal{D}U = tU'' + (1-2t)U' + (t-1)U = (\partial_t - 1)t(\partial_t - 1)U.$$

For rational U without poles on $[0,\infty)$, all of its derivatives have at most polynomial growth there, and

$$\int_0^\infty e^{-t}\mathcal{D}U(t)\,dt = 0, \tag{V.6.1}$$

$$\int_0^\infty e^{-t}\mathcal{D}U(t)\log t\,dt = U(0), \tag{V.6.2}$$

$$\int_0^\infty te^{-t}\mathcal{D}U(t)\,dt = \int_0^\infty e^{-t}U(t)\,dt. \tag{V.6.3}$$

To prove these identities, put $g = e^{-t}U$, so that $e^{-t}\mathcal{D}U = (tg')'$. For $\Re s > 0$, integration by parts twice gives

$$\int_0^\infty t^s e^{-t}\mathcal{D}U(t)\,dt = s^2\int_0^\infty t^{s-1}e^{-t}U(t)\,dt.$$

The integral on the right has residue $U(0)$ at $s = 0$, whereas the left side is analytic near zero. Its value and derivative at zero yield (V.6.1)–(V.6.2); integrating $t(tg')'$ proves (V.6.3). Consequently

$$\boxed{\int_0^\infty e^{-t}[A + \mathcal{D}U(t)]\log t\,dt = U(0) - A\gamma.} \tag{V.6.4}$$

No logarithms of algebraic numbers or auxiliary E_1 values remain.

Proposition V.6.1 (Exact rational normal form)**.** *Every rational function R regular at zero has a unique decomposition*

$$R = A + \mathcal{D}U + \sum_{\beta} \left(\frac{c_\beta}{t-\beta} + \frac{d_\beta}{(t-\beta)^2} \right),$$

where U is rational and regular at zero, and its finite poles are among the nonzero poles β of R. The residual coefficients at β vanish if and only if

$$\operatorname{Res}_{t=\beta}(e^{-t}R) = \operatorname{Res}_{t=\beta}(e^{-t}R\log t) = 0. \tag{V.6.5}$$

Here any local analytic branch of $\log t$ near β may be used. The normal form respects conjugation and Galois action; in particular its coefficients are rational when the poles and coefficients of R are rational.

Proof. At a nonzero pole β, applying $\mathcal{D}$ to $(t-\beta)^{-k}$ produces a pole of order $k+2$ with leading coefficient $\beta k(k+1) \neq 0$. Remove all orders at least three successively. On polynomials $\mathcal{D}$ raises degree by one, preserving the leading coefficient. Remove all positive polynomial degrees and leave A. The same leading-order comparisons prove uniqueness.

For $g = e^{-t}U$, both residues of $e^{-t}\mathcal{D}U = (tg')'$ vanish, since

$$\operatorname{Res}((tg')'\log t) = -\operatorname{Res}(g') = 0.$$

The two residues of the residual simple and double poles are

$$e^{-\beta}(c_\beta - d_\beta), \qquad e^{-\beta}\left((c_\beta - d_\beta)\log\beta + \frac{d_\beta}{\beta}\right).$$

Their coefficient matrix is nonsingular. A change of logarithm branch adds a constant multiple of the first residue to the second, so the simultaneous vanishing condition is branch independent. □

This classifies cancellation by rational differential exactness. It does not assert that residual special-function values are arithmetically independent. No such assertion is used below.

V.6.2 Positive densities and a sharp completely monotone obstruction

Let V be rational without poles on $[0,\infty)$, and set

$$U = V - V', \qquad R = 1 + \mathcal{D}(V - V'), \qquad h(t) = t - 1 - \log t.$$

If $R \geq 0$ there, then (V.6.1) makes $e^{-t}R(t)\,dt$ a probability measure. Equations (V.6.2)–(V.6.3) give

$$\boxed{\gamma + V'(0) = \int_0^\infty e^{-t}R(t)h(t)\,dt > 0.} \tag{V.6.6}$$

Indeed its first moment is $1 + V(0)$ and its logarithmic moment is $V(0) - V'(0) - \gamma$. Strict positivity follows because $h(t) > 0$ except at $t = 1$, and a continuous probability density cannot be supported at that point. For $V \in \mathbb{Q}(t)$ this gives a nonzero pure rational linear form, certified without assuming irrationality of γ.

Theorem V.6.2 (Exact admissibility and the sharp 1/4 ceiling)**.** *Suppose the nonzero rational function V has a positive Laplace representation*

$$V(t) = \int_{[0,\infty)} e^{-st}\,d\mu(s)$$

with finite moments through order three. Put $A = V(0)$, $B = -V'(0)$, $C = V''(0)$ and $K = A + 2B + C$. For $a \geq 0$, the density factor

$$R_a = 1 + a\mathcal{D}(V - V')$$

is nonnegative on $[0, \infty)$ exactly when $a \leq 1/K$. The largest target supplied by this scaling is

$$\frac{B}{A + 2B + C} \leq \frac{1}{4}.$$

The bound is sharp as a supremum over rational functions of increasing pole order, but equality is impossible for a nonzero rational V.

Proof. Necessity follows from $R_a(0) = 1 - aK$. With $W = V - V'$, write

$$\mathcal{D}W = t(W - 2W' + W'') - (W - W').$$

The Laplace representation gives $W - 2W' + W'' \geq 0$ and $0 \leq W - W' \leq K$. This proves sufficiency. Furthermore,

$$\frac{B}{K} = \frac{\int s\, d\mu(s)}{\int (1+s)^2\, d\mu(s)} \leq \frac{1}{4},$$

since $(1 + s)^2 - 4s = (s - 1)^2$. Equality would force V to be a nonzero multiple of e^{-t}, which is not rational. For sharpness take $V_n(t) = (1 + t/n)^{-n}$. Then $A = B = 1$, $C = 1 + 1/n$, and $B/K = n/(4n + 1) \to 1/4$. □

Finite positive mixtures of $(t + b)^{-m}$ with positive rational parameters are explicit instances of the hypotheses. The functions V_n in the proof have positive Gamma-distribution Laplace measures. Applying (V.6.6) to $a_n V_n$, where $a_n = n/(4n + 1)$, gives

$$\gamma - \frac{n}{4n + 1} > 0.$$

This fraction is reduced. Thus its primitive integer form is

$$(4n + 1)\gamma - n = (4\gamma - 1)n + \gamma \longrightarrow +\infty. \tag{V.6.7}$$

For an elementary strict bound, the standard integral for γ and $\coth u > 1/u$ give

$$\gamma = \int_0^\infty e^{-t} \left(\frac{1}{1 - e^{-t}} - \frac{1}{t} \right) dt > \frac{1}{2}.$$

Hence (V.6.7) exceeds $n + 1/2$. Arbitrary pole order and exact logarithmic cancellation do not overcome the sharp analytic obstruction in this family.

V.6.3 Signed partial fractions can cross the ceiling

The preceding obstruction does not apply to all rational kernels. Put $x = (1 + t)^{-1}$ and

$$V = -x + \frac{129}{94}x^2 - \frac{45}{94}x^3.$$

Direct differentiation yields

$$1 + \mathcal{D}(V - V') = \frac{t^2(35t^3 + 552t^2 + 2763t + 4946)}{94(1 + t)^6} > 0 \quad (t > 0).$$

Since $-V'(0) = 29/94 > 1/4$, (V.6.6) becomes the exact identity

$$\gamma - \frac{29}{94} = \frac{1}{94} \int_0^\infty \frac{e^{-t} t^2 (35t^3 + 552t^2 + 2763t + 4946)}{(1 + t)^6} (t - 1 - \log t)\, dt > 0. \tag{V.6.8}$$

Nevertheless $94\gamma - 29 > 18$, so this positive integral is far from an irrationality certificate.

For systematic exploration, let $V = \sum_{k=0}^{n} v_k x^k$ and require $P(x) = xR(t)$ to have nonnegative Bernstein coefficients of degree $d = n + 4$. With $U = V + x^2 V_x$, the exact polynomial identity is

$$P = x + x^4(1-x)U_{xx} + x^2(2 - x - 2x^2)U_x + (1 - 2x)U. \tag{V.6.9}$$

A floating linear program proposes a support, after which rational linear algebra reconstructs the coefficients and checks both (V.6.9) and every coefficient inequality exactly. The retained script `stein_density_probe.py` and certificate file `positive_kernel_certificates.json` record this computation. The asserted certificates do not depend on numerical optimality. The separate script `positive_kernel_verify.py` checks all retained certificates and both explicit symbolic identities without evaluating γ or rerunning numerical optimization. Selected reduced targets and exact lower bounds on their integer forms are

n	P/Q	$Q\gamma - P > Q/2 - P$
1	1/5	3/2
2	11/46	12
3	29/94	18
6	9983629/24987068	2519905
10	95700655335119039/212285756553929320	10442222941845621

Every exactly reconstructed certificate at $n = 1, 2, 3, 4, 5, 6, 8, 10, 12, 14$ has $Q/2 - P > 1$. These are failed finite attempts, not evidence for a universal reduced-denominator theorem.

Theorem V.6.3 (A uniform error floor in the Bernstein-positive cone)**.** *Suppose $R \geq 0$, $\int_0^\infty e^{-t}R(t)dt = 1$, and $xR(t)$, with $x = (1+t)^{-1}$, has nonnegative Bernstein coefficients of degree $d \geq 1$. Then*

$$\int_0^\infty e^{-t}R(t)(t - 1 - \log t)dt \geq \frac{1}{4224d^2}. \tag{V.6.10}$$

Proof. The density is a convex mixture of normalized positive functions

$$h_k(t) = e^{-t}t^k(1+t)^{1-d}, \qquad 0 \leq k \leq d.$$

On $[1/2, 3/2]$ their logarithmic derivatives have absolute value at most $3d$. For $u \in [3/4, 5/4]$, integrate over $|t - u| \leq 1/(12d)$ to obtain

$$\frac{h_k(u)}{\int_0^\infty h_k(t)dt} \leq 6e^{1/4}d < 8d.$$

The same estimate holds for the mixture. With $\epsilon = 1/(32d)$, at most half its mass lies in $[1 - \epsilon, 1 + \epsilon]$. On the complement,

$$t - 1 - \log t \geq \epsilon - \log(1+\epsilon) \geq \frac{\epsilon^2}{2(1+\epsilon)}.$$

The integral is therefore at least $[4096d^2(1 + 1/(32d))]^{-1} \geq (4224d^2)^{-1}$. □

For the gamma targets in (V.6.6), this excludes exponential accuracy in the certificate degree. It does not prove arithmetic impossibility: a sequence satisfying $Q(\gamma - P/Q) \to 0$ would necessarily have $Q = o(d^2)$, and no reduced-denominator lower bound excluding that possibility for the whole cone is proved here. The exact tested targets have very different denominator growth.

A further sparse attempt imposed the two residue conditions of Proposition V.6.1, together with $\mathrm{Res}_{t=-1}(te^{-t}R) = 0$, on

$$R_n(t) = \frac{t^n(a_0 + a_1 t + a_2 t^2 + a_3 t^3)}{(1+t)^{2n+2}}.$$

The one-dimensional rational coefficient space has positive generators at $n = 1, 2, 3$, but at $n = 4$ a primitive generator is

$$(-187805251068, -59288222773, -5243087734, 12508971).$$

Its opposite constant and leading signs forbid global positivity under any nonzero scaling. The exact script `stein_sparse_probe.py` verifies these residue systems. Thus the first positive instances cannot be extrapolated into a positive all-index family.

The construction has accomplished exact elimination and certified nonvanishing. The missing ingredient is primitive arithmetic smallness. The completely monotone subfamily is sharply obstructed; the larger rational cone crosses that obstruction but does not currently supply a sequence proving irrationality of γ.

V.6.4 Moving conjugate poles: a fixed-order obstruction

The preceding Bernstein cone keeps its poles at -1. A different attempt places conjugate poles close to the desired concentration point $t = 1$. It too has a precise obstruction when the rational complexity is fixed. Write

$$\mathcal{T} = \mathcal{D}(1 - \partial_t) = -t\partial_t^3 + (3t - 1)\partial_t^2 + (2 - 3t)\partial_t + (t - 1), \qquad q_\epsilon(t) = (t - 1)^2 + \epsilon^2.$$

For a first explicit test, take

$$V_\epsilon(t) = b\frac{t - 1}{q_\epsilon(t)}, \qquad 0 < \epsilon < 1.$$

Its rational target is

$$-V_\epsilon'(0) = b\frac{1 - \epsilon^2}{(1 + \epsilon^2)^2}.$$

At $t = 1 + \epsilon$, direct differentiation gives

$$\mathcal{T}V_\epsilon(1 + \epsilon) = -\frac{b(3 + 5\epsilon + 3\epsilon^2 - \epsilon^4)}{2\epsilon^4}.$$

Thus nonnegative target and nonnegative density $1 + \mathcal{T}V_\epsilon$ require

$$0 \le b \le \frac{2\epsilon^4}{3 + 5\epsilon + 3\epsilon^2 - \epsilon^4}, \qquad 0 \le -V_\epsilon'(0) \le \frac{2}{3}\epsilon^4.$$

The target tends to zero as the poles approach the real axis. The higher derivatives force a negative lobe unless the amplitude is reduced by at least this fourth-power scale.

Theorem V.6.4 (Fixed-order conjugate poles cannot concentrate)**.** *Fix nonnegative integers m, d. There is a constant $\eta_{m,d} > 0$ such that for every $0 < \epsilon \le 1$ and every real polynomial P of degree at most d, if*

$$V(t) = \frac{P(t)}{q_\epsilon(t)^m}, \qquad R(t) = 1 + \mathcal{T}V(t) \ge 0 \quad (t \ge 0),$$

then

$$\gamma + V'(0) = \int_0^\infty e^{-t}R(t)(t - 1 - \log t)dt \ge \eta_{m,d}.$$

In particular, rational coefficients and rational ϵ in this fixed-complexity family cannot give primitive integer forms tending to zero, regardless of coefficient growth.

Proof. First, $\mathcal{T}V = C$ has no nonzero rational solution for any constant C. At a nonzero pole β of order k, the leading pole of $\mathcal{T}V$ has order $k+3$ and coefficient $\beta k(k+1)(k+2)$ times the leading coefficient of V. At a pole at zero, its leading order is $k+2$ with nonzero coefficient $k(k+1)^2$ times that coefficient. Thus a rational solution must be a polynomial. For a nonzero polynomial of degree j, $\mathcal{T}V$ has degree $j+1$ with the same leading coefficient, so it cannot be constant.

On $J = [2,3]$, the functions

$$1, \quad \mathcal{T}\left(\frac{t^j}{q_\epsilon(t)^m}\right), \qquad 0 \le j \le d,$$

are therefore linearly independent for every $\epsilon \in [0,1]$, including $\epsilon = 0$. They depend continuously on (ϵ, t) on the compact set $[0,1] \times J$, since $q_\epsilon(t) \ge 1$ there. Their linear combinations with coefficient vector on the Euclidean unit sphere consequently have a strictly positive uniform minimum $c_{m,d}$ of their $L^1(J)$ norms: otherwise compactness would produce a nontrivial zero combination at a limiting parameter.

The coefficient vector of R has first coordinate 1, hence Euclidean norm at least one. Thus $\int_J |R(t)|dt \ge c_{m,d}$. Positivity turns this into an ordinary mass bound. Since

$$b_J = \min_{t \in J} e^{-t}(t - 1 - \log t) > 0,$$

its error integral is at least $b_J c_{m,d}$. Set $\eta_{m,d} = b_J c_{m,d}$. For positive ϵ, all integrations in (V.6.6) remain legitimate. The limiting singular function at $\epsilon = 0$ is used only on J, never inside the full integral. □

The integers m, d are fixed in this theorem. It does not exclude growing pole order, growing numerator degree, or a different family of moving poles. The constant is an existence bound from a finite-dimensional compactness argument, not a claimed sharp numerical value.

V.7 Quantitative confluence and a local exceptional-case correction

This section tests two transfers aimed at the real Euler constant. The q-limit below has the correct target, but requires arithmetic estimates uniform in a moving parameter. The local calculation corrects an unrestricted published statement; it does not prove irrationality of the real Euler constant or of the p-adic Euler constant individually.

V.7.1 The quantitative statement required at the q-boundary

Put

$$L(q) = \sum_{k \ge 1} \frac{q^k}{1-q^k}, \qquad G(q) = \log(1-q) - \log q\, L(q), \qquad h = -\log q > 0.$$

Thus $G(q) = -\psi_q(1)$. Its asymptotic expansion is known [V.11]; we record a remainder formulation suited to an arithmetic application.

Lemma V.7.1 (A controlled finite-part expansion)**.** *For every fixed integer $m \ge 0$,*

$$G(e^{-h}) = \gamma - \frac{h}{4} + \sum_{j=1}^{m} c_j h^{2j} + O_m(h^{2m+2}), \qquad c_j = \frac{B_{2j}(1-B_{2j})}{2j(2j)!}.$$

In particular $c_1 = 5/144$ *and* $c_2 = -31/86400$. *If* $C_m(q) = G(q) + h/4 - \sum_{j=1}^{m} c_j h^{2j}$, *then* $|C_m(q) - \gamma| \le K_m h^{2m+2}$, *where one may take*

$$K_m = |c_{m+1}| + \frac{2\zeta(2m+2)}{(2\pi)^{2m+2}} \int_0^\infty |f^{(2m+2)}(x)|\,dx, \qquad f(x) = \frac{1}{e^x - 1} - \frac{e^{-x}}{x}.$$

Proof. The singularity of f at zero is removable, $f(0) = 1/2$, and every derivative is integrable at infinity. Direct summation gives $G(e^{-h}) = h \sum_{k\ge1} f(kh)$. Also

$$H_N - \log N - \int_0^\infty f(x)\,dx = \int_0^\infty e^{-Nx} \left(\frac{1}{x} - \frac{1}{e^x - 1} \right) dx.$$

The parenthesis is between zero and one, so the integral of f equals γ. Euler–Maclaurin gives the boundary term $-hf(0)/2$ and $-B_{2j}h^{2j}f^{(2j-1)}(0)/(2j)!$. The Taylor expansion gives $f^{(2j-1)}(0) = (B_{2j}-1)/(2j)$. The periodic-Bernoulli remainder through order $2m+2$ is bounded by the integral in K_m times h^{2m+2}; including the next boundary term proves the stated inequality. □

Proposition V.7.2 (A sufficient uniform arithmetic bridge)**.** *Set* $q_N = 1 - 1/N$. *Fix* $m \ge 0$ *and* $\kappa, \mu, \rho \ge 0$. *Suppose that, on an infinite unbounded set of integers* N, *there are integers* $A_N \ge 1, B_N$ *and positive* $u_N \to 0$ *such that*

$$A_N \le N^\rho, \qquad N^{-\kappa} A_N^{-\mu} \le |A_N C_m(q_N) - B_N| \le u_N.$$

If $2m + 2 > \kappa + \rho(\mu + 1)$, *then* γ *is irrational.*

Proof. Since $N^{-1} \le -\log q_N \le 2N^{-1}$, the ratio of $|A_N(C_m(q_N) - \gamma)|$ to the assumed lower bound is at most a constant times $N^{\kappa+\rho(\mu+1)-(2m+2)}$, hence tends to zero. Therefore $A_N\gamma - B_N$ is eventually nonzero and tends to zero. For rational $\gamma = a/b$, multiplying this sequence by b gives nonzero integers tending to zero, which is impossible. □

The arithmetic hypothesis is *not established here.* For example, the Lambert-series irrationality theorems of Coussement–Smet [V.12, Theorems 1.2–1.3] allow a denominator parameter $v = 1/b$, $b \in \mathbb{Z}_{\ge2}$; these parameters cannot tend to one. Varying their numerator parameter u instead gives

$$\sum_{k\ge1} \frac{u^k}{1 - v^k} - \frac{u}{1 - u} = \sum_{k\ge1} \frac{u^k v^k}{1 - v^k} \longrightarrow L(v),$$

by dominated convergence, not γ. Moreover, irrationality of $L(q)$ does not give the required lower bound for the mixture of Lambert values and logarithms in $C_m(q)$.

There is an exact warning about direct truncation. Its omitted tail satisfies, by monotone integral comparison,

$$-\log(1 - e^{-(K+1)h}) \le T_K(h) := h \sum_{k>K} \frac{1}{e^{kh} - 1} \le -\log(1 - e^{-Kh}).$$

Thus $T_K(h) \le h^r$ implies $K \ge r\log(1/h)/h - 1$. At $q_N = 1 - 1/N$ this requires order $N \log N$ terms. The naive clearing factor

$$D_{N,K} = \prod_{k=1}^{K} (N^k - (N-1)^k) \ge N^{K(K-1)/2}$$

overwhelms the tail: $D_{N,K}T_K \to \infty$ for fixed N as $K \to \infty$, and along the preceding high-accuracy diagonals. This concerns this particular clearing factor, not the reduced denominator of an optimized Padé construction.

V.7.2 Diamond normalization and the small-prime exception

Fix the Iwasawa logarithm on $\mathbb{C}_p^\times$, so that $\log_p p = 0$ and it vanishes on roots of unity. Use exactly the normalization of [V.13, p. 355]:

$$\gamma_p = -\frac{p}{p-1}\lim_{k\to\infty}\frac{1}{p^k}\sum_{\substack{1\le m<p^k\\ p\nmid m}}\log_p m,$$

$$\psi_p(x) = \lim_{k\to\infty}\frac{1}{p^k}\sum_{n=0}^{p^k-1}\log_p(x+n), \qquad x\notin\mathbb{Z}_p.$$

The latter function is Diamond's logarithmic derivative outside $\mathbb{Z}_p$. It is not the differently normalized Morita logarithmic derivative on $\mathbb{Z}_p$.

Lemma V.7.3 (Distribution and reflection from the defining limits)**.** *For every prime p,*

$$\sum_{r=1}^{p-1}(\psi_p(r/p)+\gamma_p) = 0, \qquad \psi_p(r/p) = \psi_p(1-r/p) \quad (1\le r<p).$$

Consequently $\psi_2(1/2)+\gamma_2 = 0$ and $\psi_3(1/3)+\gamma_3 = \psi_3(2/3)+\gamma_3 = 0$.

Proof. Summing the finite approximants over r replaces their numerators by the logarithms of all $m < p^{k+1}$ coprime to p, since $\log_p(m/p) = \log_p m$. Comparison with the definition of γ_p proves the first identity.

For the second, write $a = r/p$ and $F_k(a) = p^{-k}\sum_{j=0}^{p^k-1}\log_p(a+j)$. Reversing the summation gives

$$F_k(1-a) - F_k(a) = p^{-k}\sum_{j=0}^{p^k-1}\log_p\left(1-\frac{p^k}{a+j}\right).$$

The linear logarithm term is $-\sum_j (a+j)^{-1}$. Expanding $(a+j)^{-1} = \sum_{\ell\ge0}(-j)^\ell/a^{\ell+1}$, the $\ell=0$ term has valuation $k+1$, while those for $\ell\ge1$ have valuation at least $k+\ell$. Here $\sum_{j=0}^{p^k-1} j^\ell$ is divisible by p^{k-1}; this follows by induction, splitting $j = j_0 + bp^{k-1}$. The logarithm terms of degree $d\ge2$ have valuation at least $k(d-1)+d-v_p(d)$ even without cancellation in the sum. These bounds tend to infinity, uniformly over the respective indices, proving reflection. Distribution and reflection imply the asserted small-prime identities. □

Theorem V.7.4 (Corrected prime-denominator classification)**.** *With the preceding normalization and $1\le r<p$,*

$$\psi_p(r/p)+\gamma_p = \begin{cases} 0, & p=2 \text{ or } p=3,\\ a \text{ transcendental number}, & p\ge5.\end{cases}$$

Proof. The small-prime cases follow from Lemma V.7.3. For $p\ge5$, put $m=(p-1)/2$, fix a primitive pth root ζ, and set

$$x = 1-\zeta, \qquad u_a = \frac{1-\zeta^a}{1-\zeta} \quad (2\le a\le m).$$

The u_a are multiplicatively independent algebraic units [V.13, Lemma 14]. They are independent also modulo roots of unity: a torsion product would become an ordinary relation on raising to the order of that root. Equivalently one can replace u_a by the real unit $\zeta^{k_a}u_a$, where $2k_a \equiv 1-a \pmod p$; this does not change its p-adic logarithm.

The identity $\prod_{a=1}^{p-1}(1-\zeta^a)=p$ and conjugate pairing give

$$(p-1)\log_p x + 2\sum_{a=2}^{m}\log_p u_a = 0.$$

Substitution into the p-adic Gauss formula [V.13, Lemma 11] gives

$$\psi_p(r/p)+\gamma_p = \sum_{a=2}^{m}\left(\frac{2}{p-1}+\zeta^{ar}+\zeta^{-ar}\right)\log_p u_a. \tag{V.7.1}$$

Every coefficient is nonzero: $\zeta^{ar}+\zeta^{-ar}$ is an algebraic integer, whereas $-2/(p-1)$ is a rational noninteger. Since all u_a are p-adic units, a common positive power puts them arbitrarily close to one. These powers remain multiplicatively independent. The p-adic linear-form theorem [V.13, Corollary 13], applied to these powers, makes (V.7.1) transcendental. □

Remark V.7.5 (Exact scope of the source correction). In the journal-hosted 2008 PDF, Theorem 2(i) on p. 351 asserts transcendence of all the displayed values for $q=p$ without a small-prime exclusion. Lemma V.7.3, derived directly from the same definitions, supplies the exceptions. One cannot raise the nonunit $1-\zeta_p$ to powers approaching one; the corrected proof first eliminates its logarithm using the norm relation. The independent-unit sum is empty at the exceptional primes. No first-priority claim is made for this correction, and it is not counted as solving a previously open conjecture. The later prime-power result [V.14, Theorem 1.2] allows an exception. None of these conclusions establishes the arithmetic nature of γ_p individually.

V.7.3 A failed adelic identification of the real limit

For each prime p, the term $1/p^n$ is the unique summand of H_{p^n} with valuation $-n$. Hence

$$v_p(H_{p^n}) = -n.$$

The defining real approximants $H_{p^n}-\log(p^n)\to\gamma$ therefore become H_{p^n} under the Iwasawa logarithm and diverge in $\mathbb{Q}_p$. They do not converge to γ_p. Moreover a unital field embedding cannot send the nonzero real $\log p$ to $\log_p p = 0$. A comparison on a smaller ring may have a kernel, but compatibility with γ would still require an actual theorem. The local classification above does not supply such a comparison or prove irrationality of real γ.

V.8 What remains between these results and irrationality

The research routes above improve several structural statements, but none has produced (V.1.1). The exact gaps matter more than the number of auxiliary theorems.

The monodromy route. If γ were algebraic, the identity $\delta = e\,\mathrm{Ein}(1) - e\gamma$ would put δ in $\mathcal{E}$. The proved consequences make it a transcendental nonunit, with no nonconstant rational denominator in $k(\delta)\cap\mathcal{E}$. Those conclusions are consistent. There is no independently proved assertion that $1/\delta$ must belong to $\mathcal{E}$. Nor can one extract a new E-value from the difference of two monodromy images: that nonzero difference vanishes at $X=1$, while evaluation there is injective on each image. Thus the new denominator result is a completed result in its own setting, not a hidden proof of algebraicity or transcendence of γ.

The integer moment route. For $Q \in \mathbb{Z}[X]$, positivity gives $A_Q - B_Q\gamma > 0$. The raw forms have a uniform positive lower bound and in high degree grow at least factorially. Only reduction by $g_Q = \gcd(A_Q, B_Q)$ could turn these into small primitive forms. A sufficient genuinely arithmetic advance would be an explicit sequence Q_n for which

$$\frac{A_{Q_n} - B_{Q_n}\gamma}{g_{Q_n}} \longrightarrow 0.$$

The positivity proof would then supply nonvanishing without assuming irrationality. No such common-divisor estimate is established here. The characterization of γ by infinitely many integer inequalities cannot be used to assume that every rational parameter eventually fails one of them.

The rational-density route. Here strict positivity is also available before any arithmetic assumption. The completely monotone construction is definitively excluded by its $1/4$ ceiling. In the larger Bernstein-positive family, a candidate sequence with reduced denominator Q_n and certificate degree d_n would require $Q_n = o(d_n^2)$ just to be compatible with a vanishing primitive form. That necessary condition is not sufficient, and no theorem here rules it out for every possible sequence. Fixed-order poles approaching 1 are separately excluded by compactness. Allowing the degree or pole order to grow escapes that last theorem and requires new estimates.

The CMF and confluence routes. The exact image lattice concerns all integer seeds together. A particular seed can have further cancellation. The proved bound $J \mid \operatorname{lcm}(1, \ldots, S+1)^4$ shows that the index obstruction is only exponential for equal upper decrements; it does not produce small-height lifts with a sufficiently small, nonzero gamma error. The lattice theorem does not replace that further control. Likewise, a quantitatively controlled analytic q-limit does not supply uniform integer linear forms in the moving parameter. Both missing statements are explicitly arithmetic, and neither is implied by functional identities or numerical accuracy.

These are research targets, not hypotheses silently adopted in any of the proofs. The manuscript leaves the irrationality and transcendence of the real Euler constant unresolved.

V.A Exact checks and their limited role

The proofs do not rely on floating-point optimization or on checking a finite number of indices. The following diagnostics were used to reject tempting generalizations and to check formulas.

For the canonical pencil, exact rational Gram recurrences found negative integer-polynomial certificates for $c = 0$ and $c = 1$ at degrees 38 and 41. Their largest coefficients have 544 and 622 decimal digits, respectively. These exclusions are elementary by other means; their role here is to expose the cost of this specific certificate system. They are not counted as new irrationality results.

For positive rational kernels, a numerical linear program suggested supports; rational linear algebra then reconstructed the coefficients and checked every positivity certificate exactly. The displayed fraction 29/94 is accompanied by a fully factored positive kernel, so that no numerical optimizer is needed to verify it. The selected higher-degree fractions are exploratory data, with exact simple lower bounds on their primitive errors.

For the jet theorem, integer matrix calculations and a separate finite-characteristic derivation were compared. A conjectural rule excluding all fresh large primes from reduced recurrence contents was discarded: in the tested normalized Rivoal pair at index 288, the pair after clearing its least common denominator has common divisor 709, which exceeds $2 \cdot 288$. This finite counterexample is not an asymptotic denominator theorem. Here P_n, Q_n are the solutions of the recurrence in [V.10, Section 1], specialized at $x = 1$, with initials

$$(P_0, P_1, P_2) = (-1, 4, 77/4), \qquad (Q_0, Q_1, Q_2) = (1, 7, 65/2).$$

The exact recurrence and the gcd check are included in the script below, so that the normalization of this finite observation is explicit.

V.A.1 A compact exact verification script

The following Python code uses SymPy and verifies the identities for the moment pencil, the rational telescoper, and the explicit positive example. It is supplementary to their proofs. It contains no test that presumes the arithmetic nature of γ.

```
import sympy as s
t, x, c = s.symbols('t x c')

def D(u):
    return t*s.diff(u,t,2)+(1-2*t)*s.diff(u,t)+(t-1)*u

def mu(k):
    return s.factorial(k)*(k*(s.harmonic(k)-c)+1)

# Integral of a monomial against the signed canonical weight.
for k in range(12):
    from_integral = (s.factorial(k+1)*(s.harmonic(k+1)-c)
                     -s.factorial(k)*(s.harmonic(k)-c))
    assert s.expand(from_integral-mu(k)) == 0
assert s.det(s.Matrix([[mu(0),mu(1)],[mu(1),mu(2)]])) == 4-c*c

# e^(-t) D(U) = (t (e^(-t) U)')'.
U = (t+2)/(t*t+3*t+7)
assert s.simplify(s.exp(-t)*D(U)
        -s.diff(t*s.diff(s.exp(-t)*U,t),t)) == 0

# A positive nonpolynomial kernel beyond the 1/4 ceiling.
y = 1/(1+t)
V = -y+s.Rational(129,94)*y*y-s.Rational(45,94)*y**3
R = 1+D(V-s.diff(V,t))
target = -s.diff(V,t).subs(t,0)
explicit = t*t*(35*t**3+552*t*t+2763*t+4946)/(94*(1+t)**6)
assert target == s.Rational(29,94)
assert s.cancel(R-explicit) == 0
assert target > s.Rational(1,4)

# Finite characteristic identity in the jet proof.
for p in (2,3,5,7,11):
    B = s.zeros(p)
    for j in range(p):
        B[j,j] = j
        B[(j+1)%p,j] -= 1
    C = B**p-B+s.eye(p)
    assert all(int(v)%p == 0 for v in C)

# A finite counterexample to the proposed small-prime gcd shortcut.
from fractions import Fraction as F
from math import gcd, lcm
P, Q = [F(-1),F(4),F(77,4)], [F(1),F(7),F(65,2)]
for n in range(286):
    den = (n+3)**2*(8*n+11)*(8*n+19)
    c2 = (n+3)*(8*n+11)*(24*n*n+145*n+215)
    c1 = -(8*n+27)*(24*n**3+105*n*n+124*n+25)
    c0 = (n+2)**2*(8*n+19)*(8*n+27)
    for seq in (P,Q):
        seq.append((c2*seq[-1]+c1*seq[-2]+c0*seq[-3])/den)
lam = lcm(P[288].denominator,Q[288].denominator)
a288, b288 = int(lam*P[288]), int(lam*Q[288])
assert gcd(a288,b288) == 709
assert 709 > 2*288
print('Exact supplementary identities passed.')
```

Verification status. The monodromy, primitive-jet, canonical-pencil, and rational-kernel arguments received independent internal mathematical checks by separate AI research threads,

followed by integration review. This is not external refereeing or machine-checked formal proof. Unproved experimental patterns are not promoted to theorems.

References for Part V

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Part VI
Rational certificates, ramification, and arithmetic obstructions

Source: `Keio_Gamma_Rational_Certificates_20261019.tex`.

Original PDF: 27 pages. Mathematical content and source-level qualifications are retained.

Original preparation note. Research draft prepared with AI assistance. New arguments were checked independently within this research session; this is not external peer review or formal proof verification. No claim of literature priority is made.

Abstract

We revisit the attached manuscript *Twelve further attempts at the irrationality of Euler's constant, and where each one breaks*, and test six further routes. The main constructive result is completeness of a positive rational-certificate system: with poles confined to the single pair $1 \pm i$ and unbounded order, its exactly attainable rational targets are all $r < \gamma$. The remaining primitive integer-error infimum is zero exactly when γ is irrational. Thus analytic existence of these certificates is no longer the missing step; their denominator-versus-error control is.

Ramified exponential-integral differences remove logarithmic monodromy while retaining a nonzero multiple of γ, correcting the scope of an equal-winding obstruction. We describe affine and algebraic loci of regular-singular cancellation and distinguish them from constructions made only under a rationality hypothesis. We also prove an exact decomposition of individual lattice gcds, a signed prime-periodicity law, divergence of fully primitive Hankel polynomials at γ, and a torsion-character obstruction to transporting Dickman moments to compact p-adic distributions. Exact finite certificates and reproducible computations accompany the proofs.

The irrationality and transcendence of γ remain unproved. No previously open conjecture is counted as solved, and no collection of subsidiary statements is presented as ten solutions of published open problems.

VI.1 Purpose, inputs, and the arithmetic endpoint

The starting point is the user-supplied `gamma_attempts(1).tex` and its matching PDF [VI.1]. We read all twelve attempts. The companion manuscript on monodromy, arithmetic and canonical positivity [VI.2] supplies explicitly identified earlier inputs. This document is a new follow-up; it does not reproduce the entire preceding corpus or silently promote its open targets to theorems.

For reference, the October 2026 recurrence paper [VI.8] still describes the irrationality of γ as unproved. Its construction and the recent E-value-ring work [VI.5] motivate some of the present routes. The distinction between a new proof in this document and a result known to be new in the literature is kept throughout: the latter has not been established.

Write

$$\gamma = \lim_{n\to\infty}(H_n - \log n), \qquad \mathrm{Ein}(z) = \sum_{n\ge 1} \frac{(-1)^{n-1} z^n}{n\, n!}.$$

On the positive real axis, $E_1(x) = \int_x^\infty e^{-t} dt/t = \mathrm{Ein}(x) - \gamma - \log x$. A nonzero integer linear form $q\gamma - p$ tending to zero would prove irrationality. A finite list of very small such forms does not. Likewise an equivalent criterion is a research reduction, not a solution unless its remaining assertion is proved.

Our changes of direction are concrete. We prescribe the target rational number before constructing a positive kernel; replace linear forms by primitive determinant polynomials; replace linear pullbacks by ramified ones; examine nonidentifiable and nonlinear cancellation parameters;

isolate the exact seed divisor; and test a real-to-p-adic moment transfer. Each route below states what survives and what the unsuccessful final step would still require.

VI.2 Growing-order rational kernels: completeness and primitive certificates

The preceding fixed-order compactness obstruction leaves open the case in which the pole order increases. We keep the two poles fixed at $1 \pm i$, and prove completeness of this growing-order family. This result does not prove that γ is irrational. It identifies exactly what remains of the arithmetic problem after positivity and logarithmic cancellation have been imposed. Two finite certificates also show that the height of the kernel coefficients need not survive in the primitive linear form.

Put $q(t) = (t-1)^2 + 1$, $h(t) = t - 1 - \log t$, and

$$T = -t\partial_t^3 + (3t-1)\partial_t^2 + (2-3t)\partial_t + (t-1).$$

For a rational V with no pole on $[0,\infty)$, let $R_V = 1 + TV$. Integration by parts, or the identity $T(e^t v) = -e^t(tv'')'$, gives

$$\int_0^\infty e^{-t} R_V(t)\, dt = 1, \tag{VI.2.1}$$

$$\gamma + V'(0) = \int_0^\infty e^{-t} R_V(t) h(t)\, dt. \tag{VI.2.2}$$

For completeness, write $U = V - V'$ and $g = e^{-t}U$; then $e^{-t}TV = (tg')'$. The mass correction integrates to zero. The logarithmic moment correction is $U(0)$, obtained by differentiating $\int t^s (tg')' dt = s^2 \int t^{s-1} g(t) dt$ at $s = 0$. The first moment correction is $\int e^{-t} U(t) dt = V(0)$. Subtracting these moments proves (VI.2.2). These integrations also hold for a smooth V and its derivatives with polynomial growth. When $R_V \geq 0$, the right side of (VI.2.2) is strictly positive: h vanishes only at 1, whereas a continuous probability density cannot be supported at one point.

Theorem VI.2.1 (Completeness with one fixed conjugate pole pair)**.** *Let $\mathcal{C}$ consist of the rational functions $V \in \mathbb{Q}(t)$ whose only possible finite poles are $1 \pm i$, for which $V(\infty) = 0$ and $1 + TV > 0$ on $[0,\infty)$. Then*

$$\boxed{\{-V'(0) : V \in \mathcal{C}\} = \mathbb{Q} \cap (-\infty, \gamma).}$$

Indeed, every rational $r < \gamma$ has a representative with $1 + TV \geq \delta$ on $[0,\infty)$ for some $\delta > 0$.

Proof. The inclusion from left to right follows from (VI.2.2). We prove the converse by constructing smooth models, approximating them by rational functions while retaining a positive margin, and interpolating their rational targets.

Choose a nonnegative smooth function b with compact support in $(0,\infty)$, normalized by $\int e^{-t} b(t) dt = 1$, and choose $0 < \rho < 1$. Set $R_*(t) = \rho + (1-\rho) b(t)$. Thus $R_* \geq \rho$ and its weighted mass is one. Define

$$F(t) = e^{-t}(1 - R_*(t)), \quad H(t) = \int_0^t F(s)\, ds = -\int_t^\infty F(s)\, ds,$$

$$v(t) = \int_t^\infty (s-t)\frac{H(s)}{s}\, ds, \qquad V_*(t) = e^t v(t).$$

Since $H(0) = 0$, the quotient $H(t)/t$ is smooth at zero; it decays exponentially at infinity. Differentiation gives $v'' = H/t$, hence $TV_* = -e^t H' = R_* - 1$. Beyond the support of b,

$$V_*(t) = -(1-\rho)[1 - te^t E_1(t)], \qquad E_1(t) = \int_t^\infty e^{-s}\frac{ds}{s}. \tag{VI.2.3}$$

In particular $V_*(\infty) = 0$. Its extension in the coordinate $w = 1/t$ is smooth at $w = 0$: the identity

$$te^t E_1(t) = \int_0^\infty \frac{e^{-u}}{1+uw}\,du$$

allows differentiation to every order for $w \geq 0$, by domination with $k!u^k e^{-u}$. No convergence of its formal asymptotic series is assumed. All integrations used in (VI.2.2) are therefore justified for V_*. Its target is

$$r_* = -V_*'(0) = \gamma - \rho\gamma - (1-\rho)\int_0^\infty e^{-t}b(t)h(t)\,dt. \tag{VI.2.4}$$

By taking b supported in a shrinking neighborhood of 1 and then $\rho \downarrow 0$, these targets approach γ from below. By moving the support of b to arbitrarily large t, with any fixed $\rho < 1$, they tend to $-\infty$.

We next approximate a given V_* without losing positivity. Set

$$z(t) = \frac{t-1+i}{t-1-i} = e^{i\theta}, \qquad 0 \leq \theta \leq \frac{3\pi}{2}, \qquad t = 1 + \cot(\theta/2).$$

Here $t = \infty$ corresponds to $\theta = 0$, and $t = 0$ corresponds to $\theta = 3\pi/2$. The function $f(\theta) = V_*(t(\theta))$ is smooth on this closed arc and $f(0) = 0$. In this coordinate

$$\partial_t = a(\theta)\partial_\theta, \qquad a(\theta) = \cos\theta - 1.$$

For any C^3 function u on the arc with $u(0) = 0$, put $V_u(t) = u(\theta(t))$. Direct substitution gives the uniform bound

$$\|TV_u\|_{L^\infty(0,\infty)} \leq 90 \max_{1\leq j\leq 3} \|u^{(j)}\|_\infty. \tag{VI.2.5}$$

Here are explicit bounds verifying that the apparent singularity at $\theta = 0$ causes no loss. We have $|a| \leq 2$, $|a'|, |a''| \leq 1$, $t|a| \leq 3$, $ta^2 \leq 6$, and $t|a|^3 \leq 12$. The coefficients of u', u'', and u''' in the three derivative terms of T are bounded by 33, 40, and 12, respectively. Finally $|(t-1)u| \leq 5\|u'\|_\infty$, since $\theta|\cot(\theta/2)| < 5$ on the arc. Their sum is 90.

Extend f to a C^3 periodic function on the circle: on the remaining arc a Hermite polynomial can match its endpoint derivatives of orders zero through three. The Fejér means of this extension converge with all derivatives through order three, because differentiation commutes with convolution and Fejér means converge uniformly for continuous periodic functions. Subtracting the value at zero, and then approximating the finitely many trigonometric coefficients by rationals, proves that f can be approximated in C^3 by functions

$$g(\theta) = \sum_{k=1}^{m} \bigl[A_k(\cos k\theta - 1) + B_k \sin k\theta\bigr], \qquad A_k, B_k \in \mathbb{Q}. \tag{VI.2.6}$$

The associated V_g is a real rational function with rational coefficients, poles only at $1 \pm i$, and value zero at infinity. Taking the approximation sufficiently close in (VI.2.5) gives $1 + TV_g \geq \rho/2$. Moreover $-V_g'(0) = g'(3\pi/2) \in \mathbb{Q}$ converges to r_*.

Now fix any rational $r < \gamma$. Choose two smooth models whose targets lie strictly below and strictly above r, respectively. Approximate them closely enough to obtain V_a, V_b with rational targets $a < r < b$ and strictly positive uniform margins. Then

$$V = \frac{b-r}{b-a}V_a + \frac{r-a}{b-a}V_b$$

has rational coefficients and target exactly r. Its density is the same convex combination of the two positive densities, so it retains a positive uniform margin. This proves the theorem. □

Corollary VI.2.2 (Finite certificates and the remaining arithmetic condition)**.** *For each rational $r < \gamma$, enumeration of m and rational polynomials P of degree less than $2m$, followed by exact polynomial sign tests, eventually finds $V = P/q^m$ with $-V'(0) = r$ and $1 + TV > 0$. If $r = p/s$ is reduced with $s > 0$, the resulting primitive linear form is $s\gamma - p$. Furthermore*

$$\gamma \notin \mathbb{Q} \quad \Longleftrightarrow \quad \inf_{V \in \mathcal{C}} \operatorname{den}(-V'(0))\,[\gamma + V'(0)] = 0.$$

Proof. Every member of $\mathcal{C}$ has the stated denominator shape. Its positivity reduces to that of a polynomial, since $q > 0$; Sturm's theorem decides strict positivity on the positive half-line. The theorem ensures that an enumeration finds a certificate whenever $r < \gamma$. If $\gamma = a/b$ is rational in lowest terms, every positive primitive form $s\gamma - p$ is at least $1/b$. This lower bound is attained: choose integers $s > 0, p$ with $sa - pb = 1$. They are coprime, and completeness represents $p/s < \gamma$. Thus the infimum is exactly $1/b$ in this case. If γ is irrational, the continued-fraction convergents below it include infinitely many p/s with $0 < s\gamma - p < 1/s$; completeness represents all of them. □

The arithmetic equivalence in the last sentence uses the standard continued-fraction criterion. The substantive input here is that all lower rational approximants admit positive kernels with the same two fixed poles and exact logarithmic cancellation. Completeness gives no useful upper bound for the required pole order or coefficient height. It supplies neither a uniform constructive sequence of small primitive forms nor a proof of irrationality. A finite certificate for a lower bound is not a solution of that open problem.

VI.2.1 Two exact certificates with large removable common factors

The accompanying exact data give two rational functions $V_m = P_m/q^m$, with $m = 8, 12$, for which $R_m = 1 + TV_m > 0$ on $[0, \infty)$ and

$$-V_8'(0) = \frac{9}{16}, \qquad -V_{12}'(0) = \frac{4}{7}.$$

The $m = 12$ witness has value zero at infinity and belongs to the exact class $\mathcal{C}$ of Theorem VI.2.1. It was obtained from a bounded witness by subtracting its constant limit $597/5000000$; this does not change its target, and the resulting density remains strictly positive by the exact sign test below. The $m = 8$ witness is bounded with a nonzero limit at infinity, which still suffices for (VI.2.2). The construction first fixes the small primitive target, then chooses kernel coefficients, so it does not rely on a fortuitous common divisor.

Here is a precise verification rule. For a given $P = P_m$, set $N_0 = P$ and

$$N_{j+1} = qN_j' - (m + j)q'N_j \quad (0 \le j < 3).$$

Then $V^{(j)} = N_j/q^{m+j}$ and

$$R_m = \frac{q^{m+3} - tN_3 + (3t - 1)qN_2 + (2 - 3t)q^2N_1 + (t - 1)q^3P}{q^{m+3}}. \tag{VI.2.7}$$

After multiplication of its numerator by a positive common denominator, the resulting integer polynomial has positive value at zero and exactly zero real roots on $(0, \infty)$, by exact Sturm computation. Its degrees are 23 and 30, respectively. Thus this is a global certificate, not a sampled positivity assertion.

m	raw coefficient of γ	raw integer coefficient	common divisor	primitive form
8	409600000000	230400000000	25600000000	$16\gamma - 9$
12	183500800000000	104857600000000	26214400000000	$7\gamma - 4$

In both rows the common divisor is the full greatest common divisor. The exact positive identities imply

$$0 < 16\gamma - 9 < \frac{1}{4}, \qquad 0 < 7\gamma - 4 < \frac{1}{20}.$$

For an elementary verification of the upper bounds, use $\gamma < 289/500$. Indeed, if $a_n = H_n - \log n$, then

$$a_n - \gamma = \sum_{k=n}^{\infty} \left[\log(1 + 1/k) - \frac{1}{k+1}\right] > \frac{1}{2(n+1)}.$$

The estimate follows by integrating $1/x - 1/(k+1)$ on $[k, k+1]$ and then comparing the sum of $1/[2(k+1)^2]$ with its integral. Also $\log 2 > 2\sum_{j=0}^{3}[(2j+1)3^{2j+1}]^{-1} =: L$. Exact rational arithmetic gives $H_{32} - 5L - 1/66 < 289/500$, which proves the claim.

The files `moving_certificates.json` and `verify_kernel_certificates.py` contain the complete rational coefficients and an independent exact verification. The polynomial coefficients are also printed below, in increasing powers of t after multiplication by their displayed denominators. These bounds on γ are not asserted to be numerically new; the result is an exact positive-kernel realization with a deliberately small primitive form.

VI.2.2 Complete coefficient appendix and executable sign certificate

Write $P_m(t) = L_m^{-1} \sum_{j=0}^{2m} c_{m,j} t^j$. The denominators are

$$L_8 = 800000000, \qquad L_{12} = 22400000000.$$

Blank entries mean zero. All entries below are exact integers.

j	$c_{8,j}$	$c_{12,j}$
0	12739606016	720764073984
1	-217116848128	-61077968887808
2	1309352582208	673998911908864
3	-4573301234560	-3884834464481280
4	10912087626016	14961446716067328
5	-19234037607136	-42751574553974784
6	26110432892400	95931077322792704
7	-27904440731776	-174960053696659456
8	23692954202800	265073025752304640
9	-15976408455696	-338200122492639872
10	8483955133932	366245497685868544
11	-3490637535048	-337813701541174272
12	1084266045698	265446174967026432
13	-244100083718	-177290672020359456
14	37180970397	100228396732354432
15	-3353835512	-47680867158749696
16	127145584	18940508192803272
17		-6218385700629880
18		1663548645882260
19		-355169721881216
20		58579277695970
21		-7060837833734
22		558246990041
23		-21941765888
24		

The following Python/SymPy program reconstructs both certificates directly from these coefficients. Its sign decisions use exact rational arithmetic and Sturm root counts; it does not invoke the floating-point search.

```
import sympy as S
from math import gcd
t = S.symbols("t")
q = S.Poly((t-1)**2+1,t)
certificates = [
    (8, 800000000, S.Rational("9/16"), [
        12739606016, -217116848128, 1309352582208,
        -4573301234560, 10912087626016, -19234037607136,
        26110432892400, -27904440731776, 23692954202800,
        -15976408455696, 8483955133932, -3490637535048,
        1084266045698, -244100083718, 37180970397,
        -3353835512, 127145584,
    ]),
    (12, 22400000000, S.Rational("4/7"), [
        720764073984, -61077968887808, 673998911908864,
        -3884834464481280, 14961446716067328, -42751574553974784,
        95931077322792704, -174960053696659456, 265073025752304640,
        -338200122492639872, 366245497685868544, -337813701541174272,
        265446174967026432, -177290672020359456, 100228396732354432,
        -47680867158749696, 18940508192803272, -6218385700629880,
        1663548645882260, -355169721881216, 58579277695970,
        -7060837833734, 558246990041, -21941765888,
    ]),
]
for m,L,r,c in certificates:
    P = S.Poly(sum(S.Rational(a,L)*t**j
                   for j,a in enumerate(c)),t)
    assert -(2*P.diff().eval(0)+2*m*P.eval(0))/2**(m+1)==r
    N=[P]
    for j in range(3):
        N.append(q*N[-1].diff()-(m+j)*q.diff()*N[-1])
    B=(q**(m+3)-S.Poly(t,t)*N[3]
       +S.Poly(3*t-1,t)*q*N[2]
       +S.Poly(2-3*t,t)*q**2*N[1]
       +S.Poly(t-1,t)*q**3*P)
    _,B=B.clear_denoms(convert=True)
    assert B.eval(0)>0 and B.count_roots(0,S.oo)==0
    D=L*2**(m+1); p=int(D*r); g=gcd(D,p)
    assert (D//g,p//g)==(int(S.denom(r)),int(S.numer(r)))
    print(m, B.degree(), D//g, p//g, "verified")
```

VI.2.3 A further exact primitive certificate: $26\gamma - 15$

The next finite certificate improves the primitive error, and uses the fixed pole pair $1 \pm i/2$. It is independent of the completeness theorem's choice $1 \pm i$. Put

$$Q(t) = 4(t-1)^2 + 1, \quad z(t) = \frac{2(t-1)+i}{2(t-1)-i}, \qquad V(t) = \sum_{k=1}^{40} \{a_k(\Re(z(t)^k) - 1) + b_k \Im(z(t)^k)\}.$$

The exact coefficients are $c_{2k-2} = a_k$, $c_{2k-1} = b_k$ in the zero-indexed list `c` in the executable certificate below. In particular $V = P/Q^{40}$ with $\deg P < 80$, so $V(\infty) = 0$. Exact differentiation gives $-V'(0) = 15/26$. The numerator of $R = 1 + TV$, over Q^{43} and after positive denominator clearing, is an integer polynomial B of degree 86. The verification below proves its strict positivity on $[0, \infty)$. Consequently

$$\boxed{0 < 26\gamma - 15 < \frac{1}{100}.}$$

For the upper bound, the same harmonic-sum estimate used above gives

$$\gamma < H_{128} - 7L - \frac{1}{258} < \frac{1501}{2600}, \qquad L = 2\sum_{j=0}^{7} \frac{1}{(2j+1)3^{2j+1}} < \log 2.$$

Both final comparisons are exact rational calculations in the code. This finite inequality is not asserted to be numerically new.

Here is a short algebraic explanation of the global sign certificate. Write $B(t) = \sum_{k=0}^{n} B_k t^k$, $n = 86$. Under $x = t/(1+t)$,

$$(1-x)^n B\left(\frac{x}{1-x}\right) = \sum_{k=0}^{n} \frac{B_k}{\binom{n}{k}} \binom{n}{k} x^k (1-x)^{n-k}.$$

Thus the numbers $B_k/\binom{n}{k}$ are its Bernstein coefficients on $[0,1]$. Midpoint subdivision is the exact de Casteljau algorithm: repeated averages produce the Bernstein coefficients on the two halves. The routine `split` implements these averages after multiplying all output coefficients by the common positive factor 2^n. Multiplication by positive factors does not change their signs. After 34 dyadic subintervals, of maximum depth 11, every coefficient on every interval is strictly positive. The Bernstein basis is nonnegative and sums to one, including at both endpoints. The polynomial is therefore strictly positive on $[0,1]$, and $B(t) > 0$ for every finite $t \geq 0$. This is an exact global proof, with no floating-point sign test.

The following listing is self-contained; the coefficient list is the complete certificate, and no search output or external data file is needed to verify it.

```
import sympy as S
from math import comb,lcm
t=S.symbols("t"); m=40; Q=S.Poly(4*(t-1)**2+1,t)
a1=S.Rational(
    25368033983395622861178692534446579044908 3,
    94587448984384536743164062500000000000000 0)
integers=[
    -9212384918587,10384847593798,-14278998630324,3085415136785,
    -11360345279171,-738559184268,-8832187772878,-2676395526916,
    -6192423671432,-3567107918961,-4069735992231,-3768019736809,
    -2212080899728,-3511982101329,-775955612605,-2951380108521,
    324729349222,-2227199615677,1030747902366,-1459122845145,
    1409827388935,-750583296281,1484372769611,-173000124064,
    1349531831870,234694291741,1073178464829,468574893037,
    747558209639,550746939535,429434328495,520771849028,
    169775817002,423236565949,-15619353075,299267927856,
    -121509496856,179685644252,-163685913715,83217985364,
    -159109492605,16936071851,-129852242501,-20500974617,
    -91077249192,-35467040493,-55224081475,-35854616717,
    -27149265537,-28894235243,-9019495515,-19818304346,
    974084002,-11717080502,4819873639,-5830415500,
    5322959971,-2236679339,4162878054,-412148723,
    2741366730,287544954,1504064412,412862235,
    721871281,316274428,269031677,185957829,
    78762343,88693716,2706880,34901675,
    -7954641,10700954,-8052289,2440388,
    -2569058,268188,-1056300,
]
c=[a1]+[S.Rational(v,10**14) for v in integers]
assert len(c)==80
Y=S.Poly(2*(t-1),t)
re,im=S.Poly(1,t),S.Poly(0,t); P=S.Poly(0,t)
for j in range(1,81):
    re,im=Y*re-im,re+Y*im
    if j%2==0:
        k=j//2
        P+=c[2*k-2]*(re*Q**(m-k)-Q**m)
        P+=c[2*k-1]*im*Q**(m-k)
assert P.degree()<2*m
r=-(5*P.diff().eval(0)+8*m*P.eval(0))/5**(m+1)
assert r==S.Rational(15,26)
N=[P]
for j in range(3):
    N.append(Q*N[-1].diff()-(m+j)*Q.diff()*N[-1])
B=(Q**(m+3)-S.Poly(t,t)*N[3]
   +S.Poly(3*t-1,t)*Q*N[2]
   +S.Poly(2-3*t,t)*Q**2*N[1]
   +S.Poly(t-1,t)*Q**3*P)
_,B=B.clear_denoms(convert=True)
n=B.degree(); assert n==86
M=lcm(*[comb(n,k) for k in range(n+1)])
```

```
b=[int(B.nth(k))*(M//comb(n,k)) for k in range(n+1)]
def split(b):
    left=[b[0]*(1<<n)]; right=[b[-1]*(1<<n)]
    for j in range(1,n+1):
        b=[x+y for x,y in zip(b[:-1],b[1:])]
        left.append(b[0]*(1<<(n-j)))
        right.append(b[-1]*(1<<(n-j)))
    return left,right[::-1]
stack=[(b,0)]; leaves=0; maxdepth=0
while stack:
    b,d=stack.pop()
    if min(b)>0:
        leaves+=1; maxdepth=max(maxdepth,d); continue
    assert b[0]>0 and b[-1]>0 and d<12
    left,right=split(b)
    stack.extend([(left,d+1),(right,d+1)])
assert (leaves,maxdepth)==(34,11)
H=sum(S.Rational(1,k) for k in range(1,129))
L=2*sum(S.Rational(1,(2*j+1)*3**(2*j+1))
        for j in range(8))
assert H-7*L-S.Rational(1,258)<S.Rational(1501,2600)
print("Verified: 0 < 26*gamma-15 < 1/100")
```

VI.3 Ramified cancellation and the limits of extracting the Euler constant

This section examines two possible ways of removing the logarithmic monodromy of an exponential-integral tail. Ramification really does preserve a nonzero coefficient of γ after the monodromy has cancelled; an unconditional obstruction remains in the coefficient arithmetic. Exponentiating the tail removes its monodromy as well, but every nonconstant rational function of the resulting function fails to be D-finite. These are precise tests of proposed extraction steps, rather than proofs of irrationality or transcendence of γ. They do not assert that the elementary identities below are new in the literature.

We use the entire function

$$\mathrm{Ein}(z) = \sum_{n\ge 1} \frac{(-1)^{n-1} z^n}{n\, n!}$$

and the classical identity

$$E_1(z) = \mathrm{Ein}(z) - \log z - \gamma. \tag{VI.3.1}$$

Initially the logarithm and E_1 have their principal branches; all subsequent statements about continuation use compatible branches. Identity (VI.3.1) and its integral normalization are [VI.13, §6.2, equations (6.2.1), (6.2.3), (6.2.4)]. D-finiteness below means satisfaction of a nonzero homogeneous linear differential equation over $\mathbb{C}(z)$; where asserted, the equation can be chosen over $\mathbb{Q}(z)$.

VI.3.1 A ramified escape from cancellation of the common constant

Theorem VI.3.1 (Entire recessive connections with a gamma initial value)**.** *For an integer $m \ge 2$, set, initially near the positive real axis,*

$$F_m(z) = E_1(z^m) - mE_1(z).$$

Then F_m extends to an entire function and

$$F_m(z) = (m-1)\gamma + \mathrm{Ein}(z^m) - m\,\mathrm{Ein}(z). \tag{VI.3.2}$$

It has the following properties.

1. *$F_m(0) = (m-1)\gamma$ and $F_m(x) \sim -me^{-x}/x$ as $x \to +\infty$.*
2. *Its minimal differential order over $\mathbb{C}(z)$ is exactly three, and it admits an equation of order three over $\mathbb{Q}(z)$.*
3. *It is neither an E-function nor a G-function, independently of any assumption on γ.*

Proof. On a neighborhood of the positive axis, $\log(z^m) = m \log z$. Substitution into (VI.3.1) proves (VI.3.2) and the entire extension. For $x > 0$, integration by parts gives

$$E_1(x) = \frac{e^{-x}}{x} - \int_x^\infty \frac{e^{-t}}{t^2}\, dt = \frac{e^{-x}}{x}(1 + O(x^{-1})).$$

The term $E_1(x^m)$ is negligible relative to e^{-x}/x, proving assertion (1).

Put $A = e^{-z}$ and $B = e^{-z^m}$. Direct differentiation yields

$$F_m' = \frac{m}{z}(A - B), \qquad A' = -A, \qquad B' = -mz^{m-1}B. \tag{VI.3.3}$$

This rational differential system gives an equation over $\mathbb{Q}(z)$ of order at most three. The functions A, B are linearly independent over $\mathbb{C}(z)$: otherwise $A/B = \exp(z^m - z)$ would be rational, contrary to its essential singularity at infinity. Moreover, the determinant expressing F_m', F_m'' in the basis A, B is

$$\frac{m^2}{z^2}(mz^{m-1} - 1),$$

which is not identically zero. Hence F_m', F_m'' are independent. If F_m, F_m', F_m'' were dependent, we would have $F_m = rA + sB$ with $r, s \in \mathbb{C}(z)$. Differentiating and using the independence of A, B would give

$$r' - r = \frac{m}{z}.$$

There is no such rational r: a pole of r at zero makes $r' - r$ have a pole of order at least two; absence of that pole makes $r' - r$ holomorphic at zero. Neither alternative produces m/z. This proves assertion (2).

Write $F_m(z) = \sum_{n \geq 0} c_n z^n$. If $m \nmid n$ and $n \geq 1$,

$$c_n = -\frac{m(-1)^{n-1}}{n\, n!}. \tag{VI.3.4}$$

Its reduced denominator is at least $n\, n!/m$. The common denominators of the first n coefficients therefore cannot be bounded exponentially along these indices. This contradicts the necessary denominator condition for a G-function. If c_0 is not algebraic, that condition already fails at the constant term; if it is algebraic, rational coefficients such as (VI.3.4) still give the stated obstruction in any number field, since a rational algebraic integer is an integer.

At the indices $n = mk$ one has

$$(mk)!c_{mk} = \frac{(-1)^{k-1}(mk)!}{k\, k!} - \frac{(-1)^{mk-1}}{k}. \tag{VI.3.5}$$

The absolute values in (VI.3.5) grow faster than any exponential in mk, by the elementary factorial estimate or Stirling's formula. This contradicts the necessary exponential size bound on the coefficients of an E-function written as $\sum a_n z^n/n!$. Assertion (3) follows. □

Thus the statement "removing the tail monodromy also removes γ" is false without a restriction on ramification. A loop around zero changes $E_1(z^m)$ by $-2\pi i m$ and changes $mE_1(z)$ by the same amount, whereas their coefficients of $-\gamma$ are 1 and m. At $z = 1$, however,

$$F_m(1) = -(m-1)E_1(1) = (m-1)(\gamma - \mathrm{Ein}(1)).$$

The value identity therefore returns the same unproved arithmetic bridge between the Euler constant and the resummed tail. Under the hypothesis that γ is algebraic, the Taylor germ of F_m has algebraic coefficients and is D-finite; neither fact is contradictory.

A simple family of factorial transforms makes the remaining arithmetic mismatch explicit. This next assertion concerns precisely the transforms defined here, not all possible transforms of a D-finite function.

Proposition VI.3.2 (Failure of block factorial normalization)**.** *Let $H_m = F_m - F_m(0)$, and for an integer $d \geq 1$ define the formal coefficient transform*

$$\mathcal{B}_d\left(\sum_{n\geq 0} c_n z^n\right) = \sum_{n\geq 0} \lfloor n/d \rfloor!\, c_n z^n.$$

For every d, $\mathcal{B}_d H_m$ is not a G-function. If $d < m$ its radius of convergence is zero. If $d \geq m$, its rational coefficients already violate the exponential common-denominator condition.

Proof. For $n = mk$, formula (VI.3.5) gives

$$\lfloor mk/d \rfloor!\, c_{mk} = \frac{(-1)^{k-1}\lfloor mk/d \rfloor!}{k\,k!} - \frac{(-1)^{mk-1}\lfloor mk/d \rfloor!}{k\,(mk)!}.$$

If $d < m$, the first term has superexponential size in mk and the second is negligible relative to it. The radius is zero. If $d \geq m$, use instead the infinitely many n with $m \nmid n$. Since $\lfloor n/d \rfloor!$ divides $n!$, the reduced denominator of the corresponding coefficient is at least

$$\frac{n\,n!}{m\lfloor n/d \rfloor!}.$$

Its logarithm is $(1 - 1/d)n \log n + O(n)$, which is superlinear. This proves the denominator obstruction. □

VI.3.2 Exactly what rational monodromy cancellation removes

Theorem VI.3.3 (Rational invariants for tails with prescribed winding)**.** *Let $T_1, \ldots, T_r$ be multivalued meromorphic germs continuing around a common based loop with*

$$MT_j = T_j - w_j\tau, \qquad 0 \neq w_j \in \mathbb{Z}, \qquad \tau = 2\pi i.$$

Let K be a field of single-valued meromorphic germs fixed by this continuation, containing τ. Set

$$U_j = T_j/w_j, \qquad L = K(U_2 - U_1, \ldots, U_r - U_1), \qquad F = K(T_1, \ldots, T_r).$$

Then U_1 is transcendental over L, and the fixed field is exactly

$$F^M = L. \tag{VI.3.6}$$

If K is a differential field containing all T_j', the same statement classifies rational differential expressions in the T_j that are fixed by the loop.

Proof. The field L is fixed, and $F = L(U_1)$. If U_1 were algebraic over L, continue a nonzero polynomial relation around the loop any integer number n of times. The same polynomial would vanish at all the distinct elements $U_1 - n\tau$, an impossibility. Thus F is a rational function field in U_1 over L.

It remains to determine the rational functions fixed by the translation $U_1 \mapsto U_1 - \tau$. Over an algebraic closure of L, the finite set of finite poles of an invariant rational function must

be translation invariant. Such a nonempty finite set cannot be invariant under translation by a nonzero element in characteristic zero. The rational function is therefore a polynomial. Comparing the leading nonconstant coefficient of $p(X-\tau)-p(X)$ shows that an invariant polynomial is constant. This proves (VI.3.6). Finally all derivatives of positive order of the T_j lie in K under the differential hypothesis, so adjoining those derivatives does not enlarge F. □

For the equally ramified tails $T_j(z)=E_1(a_jz)$ with $a_j\neq 0$, the windings are all $w_j=1$, and

$$T_j-T_1=\mathrm{Ein}(a_jz)-\mathrm{Ein}(a_1z)-\log a_j+\log a_1. \tag{VI.3.7}$$

This eliminates the common formal additive parameter $-\gamma$ as well as $-\log z$. The assertion is a classification of analytic invariants. In particular, if K contains all complex constants, it does not imply that γ is absent from the arithmetic field of coefficients: it already belongs to that K.

For $T_j(z)=E_1(a_jz^{m_j})$ with $a_j\neq 0$ and $m_j\geq 1$, use instead $w_j=m_j$. Formula (VI.3.6) then permits

$$\frac{T_j}{m_j}-\frac{T_1}{m_1}=\left(\frac{1}{m_1}-\frac{1}{m_j}\right)\gamma+\frac{\mathrm{Ein}(a_jz^{m_j})-\log a_j}{m_j}-\frac{\mathrm{Ein}(a_1z^{m_1})-\log a_1}{m_1}.$$

The ramified example in Theorem VI.3.1 is thus an invariant allowed by the exact fixed-field theorem.

There is a useful direct version for differential cancellation. If h_j and the coefficients of differential operators P_j are single-valued along the loop, continuation commutes with multiplication and differentiation and gives

$$(M-I)\sum_j P_j(h_jT_j)=-\tau\sum_j w_jP_j(h_j). \tag{VI.3.8}$$

For equal windings, a single-valued output forces the coefficient of the common formal additive constant to vanish. For unequal windings only the weighted sum vanishes. This proves the precise scope of that linear cancellation principle.

Even entire D-finiteness after cancellation does not supply the missing arithmetic hypothesis. Unconditionally,

$$E_1(z)-E_1(2z)=\log 2+\mathrm{Ein}(z)-\mathrm{Ein}(2z)$$

is entire and D-finite, but is not an E-function because its constant coefficient $\log 2$ is transcendental, by Lindemann–Weierstrass. This is a statement about the function; it is not a proof that its value at $z=1$ lies outside the ring of E-function values.

VI.3.3 Periodicization removes monodromy but destroys holonomy

Lemma VI.3.4 (Two growth facts)**.** *An entire D-finite function has finite order. If g is entire and $\exp(g)$ has finite order, then g is a polynomial.*

Proof. For the first assertion, turn a polynomial-coefficient scalar equation for f into a first-order system for $V=(f,f',\dots,f^{(r-1)})$. Outside a sufficiently large circle, the rational coefficient matrix is bounded by $C|z|^d$ for some integer $d\geq 0$. The initial norms of V on that circle are uniformly bounded. Gronwall's inequality along every radial segment then gives $|f(z)|\leq C_1\exp(C_2|z|^{d+1})$.

For the second assertion, finite order supplies an integer N and $C>0$ such that $\Re(g(z)-g(0))\leq CR^N$ on $|z|\leq R$ for all large R. Put $h=g-g(0)$ and $M=CR^N$, enlarging C if necessary. The holomorphic function $h/(2M-h)$ maps this disk into the closed unit disk and vanishes at zero: indeed $|2M-h|^2-|h|^2=4M(M-\Re h)\geq 0$. Schwarz's lemma bounds its absolute value by $1/2$ on $|z|\leq R/2$, and hence $|h(z)|\leq 2M$ there. Cauchy's estimate gives $h^{(j)}(0)=0$ for every integer $j>N$ by letting $R\to\infty$. Therefore g is a polynomial. □

Theorem VI.3.5 (No rational periodicization of the tail is D-finite)**.** *Define the single-valued meromorphic function*

$$Y(z) = \exp(E_1(z)) = \frac{e^{-\gamma}}{z} \exp(\mathrm{Ein}(z)).$$

For every nonconstant $R \in \mathbb{C}(U)$, the meromorphic function $R(Y(z))$ is not D-finite over $\mathbb{C}(z)$. In particular, $\exp(\mathrm{Ein}(z))$ *and* $\exp(-\mathrm{Ein}(z))$ *are not D-finite.*

Proof. The displayed expression defines Y globally: it has a simple pole at zero, no other finite pole, and no zero. Its singularity at infinity is essential. Otherwise Y would be rational and $zY = e^{-\gamma}\exp(\mathrm{Ein})$ would be a zero-free polynomial, hence constant, which is impossible.

Suppose first that R has a finite pole $\beta \neq 0$. In every punctured neighborhood of infinity the function Y omits both zero and infinity. Great Picard's theorem therefore says that it assumes each nonzero finite β infinitely often there. Every such preimage is a pole of $R(Y)$, since a reduced rational function really has a pole at β. A meromorphic D-finite function has no pole at an ordinary point of its differential equation, and thus has only finitely many finite poles. This is a contradiction.

In the remaining case R has poles only at zero or infinity, so it is a nonconstant Laurent polynomial. Consequently $R(Y)$ has at most a pole of finite order at zero and no other finite pole. If it were D-finite, multiplication by a sufficiently large power of z would give an entire D-finite function, hence a finite-order function by Lemma VI.3.4. Thus $R(Y)$ has an exponential-of-a-power upper bound outside a large disk.

Write $R(U) = \sum_{j=\ell}^{q} c_j U^j$ with $c_\ell c_q \neq 0$. If $q > 0$, dominance of the leading term for large $|U|$ gives

$$|U|^q \leq C(1 + |R(U)|) \qquad (U \neq 0).$$

(The bounded region requires only enlarging C.) This implies that the entire function $zY = e^{-\gamma}\exp(\mathrm{Ein})$ has finite order. If $q \leq 0$, nonconstancy implies $\ell < 0$; apply the same argument to the polynomial $R(1/V)$ to deduce finite order for $1/Y = e^{\gamma} z \exp(-\mathrm{Ein})$, and therefore for $\exp(-\mathrm{Ein})$ as well. Lemma VI.3.4 would then make Ein or $-\mathrm{Ein}$ a polynomial. Their power series have infinitely many nonzero coefficients, giving the final contradiction. The last assertions also follow directly from that lemma (or from $R(U) = U$ and $R(U) = U^{-1}$ after multiplication by rational functions). □

For example, the entire Dickman-transform expression $\exp(\gamma - \mathrm{Ein}(z)) = 1/(zY(z))$ is not D-finite. This settles only the proposed passage through D-finite arithmetic Gevrey functions. It places no restriction on genuinely nonholonomic methods for studying its values.

VI.3.4 Arithmetic status of the extraction step

The phase embeddings of Fischler–Rivoal take E-values into exponential polynomials with complex coefficients; the cited paper places the coefficients in a larger module involving resummed arithmetic Gevrey values, and leaves the image problem open [VI.5, Proposition 3.1 and §6]. Theorems VI.3.1 and VI.3.5 do not supply a map sending an individual phase coefficient back to an E-value. Nor do they prove a special case of the conjectural intersection statements for arithmetic Gevrey value rings in [VI.6]. What is now exact is the distinction: equal-winding rational cancellation removes a common additive parameter; ramification can retain the Euler constant but violates the relevant coefficient bounds; rational periodicization leaves D-finiteness altogether. None of these statements resolves the irrationality or transcendence of γ.

VI.4 What regular-singular cancellation does and does not determine

Write $\mathcal{K} = \mathrm{Frac}(\mathbf{G})$, where $\mathbf{G}$ is the ring of values of G-functions at algebraic points, allowing analytic continuation. The path-cancellation proposition in the attached manuscript is valid with its full-rank hypothesis. The hypothesis is essential: it asks that the singular germs of the functions being cancelled be linearly independent, not merely that the functions themselves be independent. We give the precise replacement when this fails, and then treat polynomial and quotient dependence on several parameters. These are structural consequences of established G-function theory, with no claim of publication priority or of a proof of the irrationality of γ.

The input from Fischler–Rivoal is the regular-singular expansion and the arithmetic of connection coefficients; see Theorems 3–4 of *On the values of G-functions*, `https://arxiv.org/pdf/1103.6022`. Their Theorem 2 identifies nonzero fixed scalar cancellation coefficients with $\mathcal{K}$. The local arguments below also cover several branches and several points.

VI.4.1 The full affine cancellation locus

Fix finitely many G-functions $F_1, \dots, F_m$, an algebraic point ζ, and specified continuations to a punctured neighborhood of ζ. Set $x = z - \zeta$. Their local expansions have the form

$$F_j(z) = \sum_{e \in E} \sum_{k=0}^{q} x^e (\log x)^k f_{j,e,k}(x), \tag{VI.4.1}$$

where $E \subset \mathbb{Q}$ is finite, the $f_{j,e,k}$ are convergent power series, and all their coefficients belong to $\mathcal{K}$. One can combine exponents differing by integers so that each coefficient of each actual power and logarithm is unambiguous. This follows either directly from the connection theorem, or by Cramer's rule at a nearby algebraic ordinary point: local Frobenius bases have algebraic Taylor coefficients, and their connection coefficients lie in $\mathcal{K}$.

Lemma VI.4.1 (Finite singular-coefficient matrix)**.** *There is a matrix $C \in M_{s,m}(\mathcal{K})$ such that, for every $c \in \mathbb{C}^m$, the germ $\sum_j c_j F_j$ is meromorphic at ζ if and only if $Cc = 0$. The same statement holds with holomorphic in place of meromorphic. Finitely many prescribed points and branches can be treated simultaneously by stacking such matrices.*

Proof. Meromorphicity of (VI.4.1) is equivalent to vanishing of the coefficients of nonintegral powers and of all positive logarithmic powers. Holomorphicity also requires vanishing of negative integral powers. These are linear equations in c with coefficients in $\mathcal{K}$. Their rows span a subspace of $\mathcal{K}^m$, so a finite basis of rows suffices. This gives C. Adding finitely many other tests amounts to adding rows. □

Theorem VI.4.2 (Affine classification and identifiable constants)**.** *Let $A, B_1, \dots, B_r$ be fixed G-functions with the continuations just specified. The set of vectors $\xi \in \mathbb{C}^r$ for which $A - \sum_i \xi_i B_i$ is meromorphic at the prescribed points is either empty or an affine space*

$$\xi_0 + \ker(M : \mathbb{C}^r \longrightarrow \mathbb{C}^s), \qquad \xi_0 \in \mathcal{K}^r, \quad M \in M_{s,r}(\mathcal{K}).$$

If a coordinate is constant on this affine space, that coordinate belongs to $\mathcal{K}$. In particular, a unique cancellation vector belongs to $\mathcal{K}^r$.

These conclusions remain valid for an arbitrary collection of exact linear cancellation tests in the same fixed finite vector of parameters, even if different G-functions and paths are used in different tests.

Proof. Lemma VI.4.1 makes the equations $M\xi = b$ with M, b over $\mathcal{K}$. Consistency over $\mathbb{C}$ implies consistency over $\mathcal{K}$ by row reduction. Choosing such a solution gives the displayed affine space. If coordinate i is constant, its value is $(\xi_0)_i \in \mathcal{K}$.

For an arbitrary collection, take a basis of all the augmented rows $(M_{j,*}, -b_j) \in \mathcal{K}^{r+1}$. At most $r+1$ rows are needed. Their equations are equivalent to the entire collection, proving the last assertion. Holomorphic cancellation has the identical proof. □

Example VI.4.3 (Function independence is insufficient)**.** Take

$$A(z) = B_1(z) = \log(1-z), \qquad B_2(z) = \log(1-z) + z.$$

The functions B_1, B_2 are linearly independent over $\mathbb{C}$. Nevertheless

$$A - \xi B_1 - (1-\xi)B_2 = -(1-\xi)z$$

is holomorphic at $z = 1$ for every $\xi \in \mathbb{C}$, including $\xi = \gamma$. The entire cancellation locus is $\xi_1 + \xi_2 = 1$. Thus the presence of γ as one coefficient in a cancellation does not imply $\gamma \in \mathcal{K}$. This is not a counterexample to the attached proposition: $B_2 - B_1 = z$ violates its second hypothesis. It is a counterexample to omitting that hypothesis from the subsequent interpretation. Indeed $\mathcal{K}$ is countable (there are countably many algebraic differential equations, algebraic initial data, algebraic evaluation points and continuation classes), whereas the example allows every complex ξ.

VI.4.2 Polynomial dependence and infinitely many exact tests

Theorem VI.4.4 (Algebraic cancellation loci)**.** *Let $t = (t_1, \dots, t_r)$ and consider any collection of tests of meromorphic or holomorphic continuation of functions*

$$H_\nu(t, z) = \sum_{j=1}^{m_\nu} P_{\nu,j}(t) F_{\nu,j}(z), \qquad P_{\nu,j} \in \mathcal{K}[t_1, \dots, t_r],$$

where, in each test, the $F_{\nu,j}$ are fixed G-functions and the endpoints and continuations are fixed. The common parameter set is an affine algebraic set defined over $\mathcal{K}$. Even an infinite collection of tests is equivalent to finitely many polynomial equations.

Every isolated complex solution has coordinates algebraic over $\mathcal{K}$. If the entire common solution set is one complex point, its coordinates belong to $\mathcal{K}$ itself.

Proof. For each test, Lemma VI.4.1 gives finitely many polynomial equations with coefficients in $\mathcal{K}$ after substitution of the $P_{\nu,j}$. All these polynomials generate an ideal in the Noetherian ring $\mathcal{K}[t_1, \dots, t_r]$. A finite set generates that ideal; each generator is a finite combination of the original equations. This proves the assertion also for an infinite collection.

An isolated complex point is a zero-dimensional irreducible component after base extension to an algebraic closure of $\mathcal{K}$. Zero-dimensional components of an affine algebraic set over a field have finite residue extensions of that field, by Zariski's lemma; their coordinates are therefore algebraic over $\mathcal{K}$. Equivalently, elimination on the zero-dimensional components supplies a nonzero polynomial over $\mathcal{K}$ for each coordinate.

If there is only one complex point, every embedding into $\mathbb{C}$ of its finite coordinate extension over $\mathcal{K}$ must preserve that point. Characteristic zero makes the extension separable, so its degree is one. The coordinates then lie in $\mathcal{K}$. □

Products of finitely many G-functions cause no difficulty: they are again G-functions, so they can be included among the $F_{\nu,j}$. The theorem concerns exact equations in a fixed finite parameter vector. It does not cover changing parameter vectors of unbounded dimension, approximate cancellation inequalities, or limits of different cancellation parameters.

Example VI.4.5 (An isolated nonlinear parameter)**.** Define rational-coefficient G-functions

$$B(z) = -\log(1-z), \qquad A(z) = \int_0^z \frac{-\log(1-u/2)}{1-u}\,du.$$

The G-function assertion follows from closure under products and integration; directly, the coefficient denominators have only exponential growth. Differentiation gives

$$(A - T^2B)'(z) = \frac{-\log(1-z/2) - T^2}{1-z}.$$

The numerator is holomorphic at 1, and its value there is $\log 2 - T^2$. Thus $A - T^2B$ is meromorphic at 1 if and only if $T^2 = \log 2$, and in that case it is holomorphic there. The two isolated parameters are $T = \pm\sqrt{\log 2}$. This demonstrates the algebraic equation arising from nonlinear cancellation. We do not assert that either root lies outside $\mathcal{K}$: that would need an additional arithmetic theorem. Selecting the positive root by an inequality is not the same as a globally unique complex algebraic solution.

VI.4.3 Quotients and one-turn monodromy

Theorem VI.4.6 (Quotient cancellation)**.** *Fix an algebraic point ζ and branches of finitely many G-functions there. Let N_t, D_t be polynomial combinations in $t \in \mathbb{C}^r$ of these functions and of their finite products, with coefficients in $\mathcal{K}$. On the parameter set where D_t is not the zero germ, the set for which N_t/D_t extends meromorphically to ζ is constructible over $\mathcal{K}$. In particular, any isolated complex parameter in that set is algebraic over $\mathcal{K}$.*

Proof. Superscript + will denote one positive turn around ζ. Local expansions (VI.4.1) and their products have rational exponents and finitely many logarithmic powers. Continuation replaces x^e by $e^{2\pi ie}x^e$ and $\log x$ by $\log x + 2\pi i$. All new coefficient constants remain in $\mathcal{K}$, because $e^{2\pi ie}$ is algebraic and $2\pi i \in \mathcal{K}$.

The identity

$$N_t^+ D_t - N_t D_t^+ = 0 \tag{VI.4.2}$$

is therefore equivalent to polynomial equations over $\mathcal{K}$ in t: equate all local coefficients and use Noetherianity to retain finitely many. The condition $D_t = 0$ is likewise algebraic. Its complement is a Zariski open set.

It remains to show that (VI.4.2) is sufficient, not merely necessary, for meromorphic extension. For fixed admissible t, the nonzero denominator has an expansion with a leading term

$$D_t(x) = cx^a(\log x)^b(1 + o(1)), \qquad c \neq 0, \quad a \in \mathbb{Q}, \quad b \in \mathbb{Z}_{\geq 0}.$$

The estimate is uniform as $x \to 0$ with its argument in any fixed bounded interval: first choose the smallest actual power of x with nonzero coefficient and then the largest logarithmic power at that exponent. The remaining exponents have a positive gap, and the convergent series remainders are uniform. Thus D_t has no zeros in a sufficiently small fundamental angular sector and the quotient has at most power growth there. Identity (VI.4.2) makes the quotient single-valued on the punctured disk. The growth estimate over an angular interval of length 2π implies $|N_t/D_t| \leq C|x|^{-M}$ there for some integer M. Multiplication by x^M and the removable singularity theorem give meromorphic extension.

The set is consequently an algebraic zero set minus the algebraic set where $D_t = 0$, hence constructible. At an isolated point the denominator nonvanishing condition is locally open, so the point is an isolated algebraic solution as in Theorem VI.4.4. □

No holonomicity of the quotient itself is assumed here. Consequently this theorem is a description of local cancellation geometry, not a license to insert a quotient into an arithmetic holonomy theorem whose holonomicity hypothesis must still be checked.

VI.4.4 The logical boundary of the holonomy inference

The arithmetic holonomy theorem is a dimension bound for holonomic series with specified arithmetic and analytic properties; it is not formulated as a theorem that every target constant must first be a uniquely determined scalar connection coefficient. The 2025 sequel of Calegari–Dimitrov–Tang, Theorem 2.1 and Theorem 2.6, explicitly includes power systems in a target parameter: https://arxiv.org/html/2510.04156v1. These are primary sources for the actual theorem hypotheses.

The justified restriction is this: an *unconditional*, exact, identifiable linear cancellation of a fixed constant between G-functions forces that constant into $\mathcal{K}$. Finite nonlinear cancellation has the algebraic-locus consequences proved above. These facts do not establish that every conceivable application of an arithmetic holonomy bound to γ requires an unconditional representation of that kind.

In particular, if the needed G-structure or extra continuation is constructed only under the assumption $\gamma \in \mathbb{Q}$, then a conditional deduction $\gamma \in \mathcal{K}$ is tautological. It does not rule out a subsequent contradiction from the dimension bound. No such conditional construction is supplied here; the point is to preserve this logical research possibility rather than mark it as excluded by the path-cancellation proposition.

Nor does membership in $\mathcal{K}$ pass to limits. The field contains $\mathbb{Q}$ and is dense, so a convergent sequence of its elements need not determine the arithmetic nature of its limit. An explicit regular-to-irregular family illustrates this issue. For an integer $M \geq 1$ put

$$F_M(z) = \int_0^z \frac{1-(1+t/M)^{-M}}{t}\,dt. \tag{VI.4.3}$$

It is a rational-coefficient G-function for each fixed M, and on every compact subset of the complex plane it converges to $\mathrm{Ein}(z)$ as $M \to \infty$. Indeed $(1+t/M)^{-M} \to e^{-t}$ uniformly on compact sets, and the quotients by t have removable singularities at zero and converge uniformly there as well.

Writing $u = (1+z/M)^{-1}$ and differentiating verifies exactly

$$F_M(z) = \log(1+z/M) + \sum_{j=1}^{M-1} \frac{1-u^j}{j}.$$

Consequently

$$\lim_{z\to+\infty}\{F_M(z) - \log z\} = H_{M-1} - \log M \longrightarrow \gamma.$$

For finite M this is a G-value constant involving a logarithm; the limit is a canonical irregular connection constant. This example does not solve an arithmetic problem. It shows exactly why a moving degeneration requires uniform arithmetic and analytic estimates, and cannot be dismissed by a theorem about a single fixed cancellation system. Obtaining a useful uniform holonomy estimate through such a degeneration remains an unproved step of a possible new route.

There is an explicit arithmetic cost in this particular normalization. Writing $F_M(z) = \sum_{n\geq 1} a_{M,n} z^n$, binomial expansion gives

$$a_{M,n} = \frac{(-1)^{n+1}}{nM^n}\binom{M+n-1}{n}.$$

If $M = p$ is prime and $1 \leq n < p$, the numerator of the binomial coefficient contains precisely one factor of p, whereas $n!$ and n contain none. Hence

$$v_p(a_{p,n}) = 1 - n.$$

The common denominator of the first N coefficients is therefore at least p^{N-1} for $1 \le N < p$, at the same time as the distance to the nearest singularity grows to p. Scaling $z = pu$ removes these powers of p from the coefficients: the first N denominators then divide $\operatorname{lcm}(1, \dots, N)$. But the nearest singularity is now at $u = -1$. This exact calculation specifies an input cost to be handled by any uniform degeneration argument. It is not a comparison with an optimal conformal holonomy bound and proves no impossibility for other normalizations or constructions.

VI.5 Individual cancellation as congruence to the exact kernel

This section continues the earlier primitive-jet and image-index results; those results are inputs, not claimed again as new. The elementary Smith normal form argument below identifies the remaining seed-specific divisor. It does not establish irrationality of Euler's constant.

VI.5.1 An exact decomposition of the individual gcd

For an integer vector x, write $\operatorname{cont}(x)$ for the gcd of its coordinates, taken nonnegative. Let $B \in M_{2,3}(\mathbb{Z})$ have rank two, content one, primitive kernel generator $E \in \mathbb{Z}^3$, and image index $I = [\mathbb{Z}^2 : B\mathbb{Z}^3]$. For $v \notin \mathbb{Z}E$ set

$$c_E(v) = \operatorname{cont}(E \times v), \qquad g_B(v) = \operatorname{cont}(Bv).$$

Theorem VI.5.1 (Individual gcd and quotient content)**.** *One has*

$$c_E(v) \mid g_B(v), \qquad g_B(v)/c_E(v) \mid I. \tag{VI.5.1}$$

More precisely, there are unimodular integer matrices U, V with $UBV = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & I & 0 \end{smallmatrix}\right)$ and $E = \pm V e_3$. If $V^{-1}v = (x, y, z)^T$, $c = \gcd(x, y)$, $x = cx_0$, $y = cy_0$, then

$$c_E(v) = c, \qquad g_B(v) = c \gcd(x_0, I).$$

Moreover $c_E(v)$ is the largest positive integer d such that $v \in d\mathbb{Z}^3 + \mathbb{Z}E$.

Proof. The Smith invariant factors are $1, I$, because the matrix content is one. The integer kernel is generated by Ve_3. Under a unimodular coordinate change the exterior product is transformed by another unimodular matrix, so its content is unchanged. The product $e_3 \times (x, y, z)$ has content $\gcd(x, y)$. Unimodular left multiplication preserves the content of an output, whence

$$g_B(v) = \gcd(x, Iy) = c \gcd(x_0, Iy_0) = c \gcd(x_0, I),$$

since $\gcd(x_0, y_0) = 1$. Finally $v \in d\mathbb{Z}^3 + \mathbb{Z}E$ if and only if $d \mid x, y$, which proves the last assertion. □

Consequently every seed admits $v = cw + tE$ with $w \in \mathbb{Z}^3$ and $c_E(w) = 1$, and $Bv = cBw$. Its rational output is unchanged by this quotient primitivization. The price is that the new representative w need not have a small height. In the Euclidean metric one can choose it so that

$$\|w\|_2^2 \le \frac{\operatorname{dist}(v, \mathbb{R}E)^2}{c^2} + \frac{\|E\|_2^2}{4}. \tag{VI.5.2}$$

Indeed all representatives differ by integer multiples of E; choose the parallel component in the interval $[-\|E\|_2/2, \|E\|_2/2]$. Their perpendicular component is exactly $v_\perp/c$. This upper bound retains possibly factorial kernel height and therefore does not solve the lift problem.

For a prime $p \nmid I$ and $a \ge 1$, Theorem VI.5.1 gives

$$p^a \mid g_B(v) \quad \Longleftrightarrow \quad v \equiv tE \pmod{p^a} \quad \text{for some } t \in \mathbb{Z}. \tag{VI.5.3}$$

At primes dividing I, the two valuations differ by at most $v_p(I)$. Thus arbitrarily large individual cancellation outside the image index is exactly a congruence to the kernel, not an unlocated determinant effect.

VI.5.2 The CMF arithmetic target

Use the scalar-contiguous normalization of the rank-three CMF, with trajectory

$$(-A_1, -A_2; b_1, b_2, b_3), \qquad A_i, b_j \in \mathbb{Z}_{\geq 0}.$$

Let Φ_n be its integral transition matrix, $H_n = \mathrm{adj}(\Phi_n)^T$, $h_n = \mathrm{cont}(H_n)$, and

$$B_n = \frac{1}{h_n}\begin{pmatrix}(1,1,0)H_n\\(1,1,1)H_n\end{pmatrix}, \qquad E_n = (1,-1,0)\Phi_n.$$

The preceding manuscript established that E_n is primitive, B_n has content one and kernel $\mathbb{Z}E_n^T$, and its image quotient is cyclic. For $A_1 = A_2 = A$, with

$$S = n\sum_{j=1}^{3}(A+b_j), \quad N = \prod_{j=1}^{3}((A+b_j)n)!, \quad J = N/h_n,$$

it also established $I_n = J^2 \mid d_{S+1}^8$, where $d_m = \mathrm{lcm}(1,\dots,m)$. The new specialization of (VI.5.1) is

$$\gcd(B_n v) = \gcd(E_n \times v)\,\rho_n(v), \qquad \rho_n(v) \mid I_n \mid d_{S+1}^8. \tag{VI.5.4}$$

Here vectors are identified with their transposes when needed. Every quotient-primitive seed, meaning $\gcd(E_n \times v) = 1$, thus has only an exponentially bounded individual gcd. Any factorial-scale gain must come from this explicit exterior-product gcd.

If $\lambda_n(v) = (-1,\gamma)B_n v$, the primitive form satisfies exactly

$$\frac{|\lambda_n(v)|}{I_n c_{E_n}(v)} \leq \frac{|\lambda_n(v)|}{g_{B_n}(v)} \leq \frac{|\lambda_n(v)|}{c_{E_n}(v)}.$$

These inequalities require no assumption about γ. In particular, for any seed family with $c_{E_n}(v_n) \leq e^{O(n)}$ and a proved lower bound $|\lambda_n(v_n)| \geq \exp(\kappa n \log n - O(n))$, $\kappa > 0$, its primitive forms diverge. Neither of these two additional estimates is asserted for all varying CMF seeds.

VI.5.3 A finite-characteristic reduction for every fixed prime

Put $D = z\,d/dz - z$ and

$$P_n(X) = (X+1)_{A_1 n}(X+1)_{A_2 n}\prod_{j=1}^{3}(-X)_{b_j n}, \qquad R_n = P_n(D)1.$$

Then $E_n = (R_n(1), DR_n(1), D^2R_n(1))$.

Theorem VI.5.2 (Signed prime-periodicity)**.** *For every prime p and every $n \geq 0$,*

$$E_{n+p} \equiv (-1)^{A_1+A_2}E_n \pmod p.$$

If $v(n) \in \mathbb{Z}[n]^3$, the condition $p \mid \gcd(E_n \times v(n))$ depends only on $n \bmod p$. It is exactly the condition that the two nonzero or zero vectors $E_r, v(r)$ are collinear in $\mathbb{F}_p^3$, $r \equiv n \pmod p$; E_r itself is always nonzero.

Proof. In $V_p = \mathbb{F}_p[z]/(z^p - 1)$, the operator D is well defined and satisfies $D^p - D = -1$. For completeness, put $T = d/dz - 1$. The Weyl relation $[T, z] = 1$ gives $\prod_{j=0}^{p-1}(zT - j) = z^pT^p$. On this quotient $(d/dz)^p = 0$, hence $T^p = -1$ and $z^p = 1$. For every integer a, it follows that

$$\prod_{j=0}^{p-1}(D + a + j) = -1, \qquad \prod_{j=0}^{p-1}(-D + a + j) = 1.$$

Increasing n by p adds $A_1 + A_2$ upper blocks and $\sum b_j$ lower blocks. All factors commute, so $P_{n+p}(D)1 = (-1)^{A_1+A_2} P_n(D)1$ in V_p. The ideal $(z^p - 1)$ is stable under D, and evaluation at $z = 1$ annihilates it. Applying D^j, $j = 0, 1, 2$, therefore proves the jet congruence, including $p = 2$. Polynomial seeds are periodic modulo p, and $E_r \neq 0$ follows from the earlier primitive-jet theorem. □

This is an exact finite computation for each fixed prime, not a bound uniformly over all primes. In particular a frequently tempting shortcut, that every fixed coordinate seed is quotient-primitive, is false. For the balanced trajectory $(0, 0; 1, 1, 1)$ direct integer products give

$$E_7 = (-154606591, -255280789, 8251597802), \quad (c_{E_7}(e_1), c_{E_7}(e_2), c_{E_7}(e_3)) = (17, 1, 19),$$

where the coordinate vectors here are indexed $1, 2, 3$. Likewise $c_{E_{14}}(e_1) = 139$. The theorem proves that $17 \mid c_{E_n}(e_1)$ for every $n \equiv 7 \pmod{17}$, and $139 \mid c_{E_n}(e_1)$ for every $n \equiv 14 \pmod{139}$. In particular $2363 \mid c_{E_n}(e_1)$ whenever $n \equiv 1265 \pmod{2363}$. These are infinite congruence families, but they provide only fixed factors, not the growing factorial cancellation required for irrationality.

VI.5.4 Aptekarev comparison: what the finite experiment does establish

The attached note's Aptekarev normalization and explicit formulas agree with Corollary 4 of the Hessami Pilehroods' paper, arXiv:1010.0429. The common factor $n!/d_n$ is prior art, stated explicitly on page 2 of that source; it is not a gain created by a new kernel representation. Recomputing the published integer recurrence exactly verifies the attached range $3 \leq \epsilon_n \leq 39591$ for $n = 20, 30, \ldots, 300$, where

$$\epsilon_n = \frac{\gcd(p_n, q_n)}{n!/d_n}.$$

Extending to every $1 \leq n \leq 3000$ gives, for example,

$$\epsilon_{300} = 147, \quad \epsilon_{1000} = 9, \quad \epsilon_{2717} = 2060427942214275.$$

The latter is the largest value in this finite range; its logarithm is about 35.26, versus about 25295.68 for the reduced denominator at that index. Thus the small sampled gcd bound cannot be promoted to a uniform theorem. These exact computations prove neither an asymptotic upper bound on ϵ_n nor the existence of the exceptionally large cancellation that a successful irrationality proof would need.

VI.5.5 Reproducible exact certificate for the Aptekarev comparison

The primary recurrence is equation (8) of https://arxiv.org/pdf/0902.1768; the binomial formulas and the $n!/d_n$ divisor are in https://arxiv.org/pdf/1010.0429. For $u = p, q$ it reads

$$\begin{aligned}(16n - 15)u_{n+1} = {} & (128n^3 + 40n^2 - 82n - 45)u_n \\ & - n^2(256n^3 - 240n^2 + 64n - 7)u_{n-1} \\ & + n^2(n-1)^2(16n + 1)u_{n-2}, \qquad n \geq 2,\end{aligned}$$

with $(p_0, p_1, p_2) = (0, 2, 31)$ and $(q_0, q_1, q_2) = (1, 3, 50)$. The following Python code uses only exact integers and reproduces the stated exceptional gcd. No value of γ enters this certificate.

```
from math import gcd, lcm, factorial
p, q = [0,2,31], [1,3,50]
for n in range(2,2717):
    a = 128*n**3 + 40*n*n - 82*n - 45
    b = -n*n*(256*n**3 - 240*n*n + 64*n - 7)
    c = n*n*(n-1)**2*(16*n+1)
```

```
    for u in (p,q):
        v = a*u[-1] + b*u[-2] + c*u[-3]
        assert v % (16*n-15) == 0
        u.append(v // (16*n-15))
d = lcm(*range(1,2718))
base = factorial(2717) // d
g = gcd(p[2717],q[2717])
assert g % base == 0
assert g // base == 2060427942214275
```

The complete calculation through 3000, together with 2000 exact checks of (VI.5.1) for five CMF trajectories, is in `work/gamma_next_20261019/probe_seed_arithmetic.py`. The script also independently recomputes p_{2717}, q_{2717} from the binomial–harmonic formulas of Corollary 4 and checks exact agreement with the recurrence. These finite checks support the reported examples; the infinite assertions above are established by the written proofs.

VI.6 A nonlinear attempt: primitive Hankel polynomials

The preceding manuscript constructed the integer affine moments

$$\mu_k(c) = k!\{k(H_k - c) + 1\}, \qquad H_0 = 0.$$

Their defining coefficients require no numerical value of γ. The canonical Gamma integral identifies them as

$$\mu_k(c) = \int_0^\infty x^k(x-1)(\log x + \gamma - c)e^{-x}\,dx. \tag{VI.6.1}$$

Instead of a linear form, we now try the integer polynomial

$$D_N(c) = \det(\mu_{i+j}(c))_{0\le i,j<N}, \qquad P_N(c) = D_N(c)/g_N,$$

where $g_N > 0$ is the greatest common divisor of all coefficients of D_N. Positivity in (VI.6.1) gives $P_N(\gamma) > 0$. If these values had Nth root tending to zero, the family would prove irrationality: for $\gamma = a/b$, the nonzero integer $b^{\deg P_N} P_N(a/b)$ has absolute value at least one. This is a genuinely nonlinear test, with nonvanishing already supplied by analysis. It fails even after the *entire* polynomial content has been removed.

Theorem VI.6.1 (Primitive determinant growth)**.** *For the polynomials above,*

$$P_N(\gamma) > 0, \qquad \lim_{N\to\infty} P_N(\gamma)^{1/N} = +\infty.$$

This assertion includes arbitrary rational changes of basis in the space of polynomials of degree less than N, followed by clearing denominators and removing the full content: their primitive determinant is P_N up to sign.

Proof. Write $F_N = \prod_{j=0}^{N-1}(j!)^2$. Let L_j be the ordinary Laguerre polynomials, orthonormal for $e^{-x}dx$, with leading coefficient $(-1)^j/j!$. In this basis the moment matrix is $A_N - cB_N$, and

$$D_N(c) = F_N \det(A_N - cB_N). \tag{VI.6.2}$$

The matrix B_N is the compression of multiplication by $x-1$: its diagonal is $2j$, and its adjacent off-diagonal entries are $-(j+1)$. In particular $\|B_N\| \le 4N$ and $\operatorname{rank} B_N \ge N-1$. The latter follows immediately by solving the tridiagonal homogeneous system from its first row.

The elementary logarithmic Laguerre integrals are

$$\int_0^\infty L_i(x)L_j(x)(\log x + \gamma)e^{-x}\,dx = \begin{cases} H_i, & i = j, \\ -1/|i-j|, & i \ne j. \end{cases} \tag{VI.6.3}$$

For completeness, differentiate the Laguerre orthogonality relation for the weight $x^\alpha e^{-x}$ at $\alpha = 0$, using $\partial_\alpha L_n^{(\alpha)}|_{\alpha=0} = \sum_{k<n} L_k/(n-k)$. The diagonal comes from $\partial_\alpha[\Gamma(n+\alpha+1)/n!]|_{\alpha=0} = H_n - \gamma$; the off-diagonal follows from the same differentiated relation. These identities also follow by integrating the finite polynomial expansions term by term. Multiplication by $x - 1$ and the three-term Laguerre recurrence now give $\|A_N\| \le 4N^2(H_N + 1)$.

If B_N is nonsingular, the leading coefficient of D_N is nonzero and has absolute value at most $F_N(4N)^N$. If B_N is singular, it has one-dimensional kernel. On this kernel A_N has positive quadratic form, since $A_N - \gamma B_N$ is positive definite by (VI.6.1). Therefore the coefficient of c^{N-1} is nonzero. Its absolute value is bounded by $F_N N\|A_N\|\|B_N\|^{N-1}$. Consequently, in both cases,

$$g_N \le F_N N^2 (H_N + 1)(4N)^N. \tag{VI.6.4}$$

This estimate does not require determining the exact content, or assuming that $L_N(1)$ is nonzero.

For a nonnegative weight w, let $\mathcal{H}_N[w] = \det(\int_0^\infty t^{i+j} w(t)dt)_{i,j<N}$. Its Vandermonde integral representation has nonnegative integrand. Restrict that representation for $D_N(\gamma)$ to $x_i \ge 1$ and put $x_i = t_i + 1$. For every fixed $a > 1$,

$$\log(1+t) = \int_0^\infty \frac{t\,ds}{(1+s)(1+s+t)} \ge \frac{t \log a}{t+a}, \qquad t \ge 0.$$

It follows that

$$D_N(\gamma) \ge (e^{-1}\log a)^N \mathcal{H}_N[t^2 e^{-t}/(t+a)]. \tag{VI.6.5}$$

Normalize the Vandermonde probability density with weight $t^\alpha e^{-t}$, and denote expectation by $\mathbb{E}_\alpha$. Laguerre norms and the characteristic-polynomial identity give

$$\mathcal{H}_N[te^{-t}] = N!F_N, \qquad \frac{\mathcal{H}_N[t^2e^{-t}]}{\mathcal{H}_N[te^{-t}]} = (N+1)!, \qquad \mathbb{E}_2 \prod_{i=1}^N (t_i + a) = N! L_N^{(2)}(-a).$$

The last identity follows directly from the Vandermonde integral: the average monic characteristic polynomial is orthogonal to every polynomial of degree less than N. Cauchy–Schwarz, in the form $\mathbb{E}X\,\mathbb{E}X^{-1} \ge 1$, therefore yields

$$\begin{aligned} \frac{\mathcal{H}_N[t^2e^{-t}/(t+a)]}{N!F_N} &= (N+1)!\,\mathbb{E}_2 \prod_{i=1}^N (t_i+a)^{-1} \\ &\ge \frac{N+1}{L_N^{(2)}(-a)} \ge \frac{2}{N+2} e^{-2\sqrt{aN}}. \end{aligned}$$

Indeed, the finite Laguerre expansion implies

$$L_N^{(2)}(-a) = \sum_{k=0}^N \binom{N+2}{k+2} \frac{a^k}{k!} \le \binom{N+2}{2} \sum_{k=0}^N \frac{(aN)^k}{(k!)^2} \le \binom{N+2}{2} e^{2\sqrt{aN}}.$$

For the final inequality, compare the series with the product $e^{\sqrt{aN}} e^{\sqrt{aN}}$.

Combining the bounds gives the fully explicit estimate

$$P_N(\gamma) \ge \frac{2e^{-2\sqrt{aN}}}{(N+2)N^2(H_N+1)} \left(\frac{\log a}{e}\right)^N \frac{N!}{(4N)^N}. \tag{VI.6.6}$$

Taking Nth roots and applying Stirling's formula gives $\liminf P_N(\gamma)^{1/N} \ge \log a/(4e^2)$. The number $a > 1$ was arbitrary and fixed before the limit. Letting a grow proves the assertion. A rational basis change multiplies the determinant by a nonzero rational square. Primitive integer polynomials spanning the same rational line agree up to sign, which proves the basis-invariance statement. □

Remark VI.6.2 (What this excludes). This result excludes the full Hankel determinant of this specific moment pencil as a source of vanishing primitive values, even after optimal scalar normalization. It does not exclude proper minors, different moment functionals, differences of determinants, or simultaneous constructions with varying auxiliary data. In particular it is not an impossibility theorem for nonlinear irrationality methods in general.

VI.7 Dickman moments, prime congruences, and a failed adelic transfer

The uploaded manuscript's Dickman attempt concerns rational moments of a probability density whose normalization contains $e^{-\gamma}$. Here we test a different bridge: transporting those rational moments to a compactly supported p-adic distribution, then seeking arithmetic information from its normalization. We prove that the proposed transport does not exist, even after rational affine changes of scale. We also derive exact prime congruences that sharpen the manuscript's numerical denominator observations. These results do not prove the irrationality or transcendence of γ; literature priority is not asserted.

For $\vartheta > 0$, let D_ϑ have the generalized Dickman law. Its moment generating function and density on $(0, 1]$ are

$$M_\vartheta(z) = \mathbb{E}e^{zD_\vartheta} = \exp\left(\vartheta \sum_{j\geq 1} \frac{z^j}{j\, j!}\right), \qquad p_\vartheta(x) = \frac{e^{-\vartheta\gamma}}{\Gamma(\vartheta)} x^{\vartheta-1} \quad (0 < x \leq 1). \tag{VI.7.1}$$

These are standard identities; see [VI.11, §1, (1.1)]. In particular $\Pr(D_1 \leq 1) = e^{-\gamma}$. Write $m_n(\vartheta) = \mathbb{E}D_\vartheta^n$. Formally the same definition is meaningful for every rational ϑ. Differentiating (VI.7.1) or using the exponential formula gives

$$m_0(\vartheta) = 1, \qquad m_n(\vartheta) = \frac{\vartheta}{n} \sum_{i<n} \binom{n}{i} m_i(\vartheta), \qquad m_n(\vartheta) = \sum_{\pi\in\Pi_n} \prod_{B\in\pi} \frac{\vartheta}{|B|}. \tag{VI.7.2}$$

Here Π_n is the set of partitions of an n-element set.

VI.7.1 Exact congruences behind the moment denominators

Theorem VI.7.1 (Prime congruences for generalized Dickman moments)**.** *Let p be an odd prime and let $0 \neq \vartheta \in \mathbb{Q} \cap \mathbb{Z}_p$. Put $c = v_p(\vartheta)$, $q = p^{c+1}$, and $u = \vartheta/p^c \in \mathbb{Z}_p^\times$. If $n = aq + r$ with $0 \leq r < q$, then*

$$p^a m_n(\vartheta) \in \mathbb{Z}_p, \qquad p^a m_n(\vartheta) \equiv u^a m_r(\vartheta) \pmod{p}. \tag{VI.7.3}$$

In particular $v_p(m_{aq}(\vartheta)) = -a$ for every $a \geq 0$. For $\vartheta = 1$ this becomes

$$p^{\lfloor n/p\rfloor} m_n(1) \equiv m_{n \bmod p}(1) \pmod{p}.$$

Consequently equality in the elementary denominator bound $v_p(\operatorname{den} m_n(1)) \leq \lfloor n/p\rfloor$ holds whenever $m_{n \bmod p}(1)$ is a p-adic unit.

Proof. A block of size j contributes valuation $c - v_p(j)$. If $w = v_p(j) - c > 0$, then

$$j \geq p^{c+w} \geq qw.$$

Since $p \geq 3$, equality is possible only for $w = 1$ and $j = q$; every other positive-w block has $j - qw \geq q$. It follows that the valuation of every term in the partition sum is at least $-a$. Terms with valuation $-a$ can occur only among partitions having exactly a blocks of size q, with their

remaining r elements partitioned arbitrarily. Indeed a different block with positive w, or a total negative valuation exceeding a, would use at least $(a+1)q$ elements. Any remaining block of positive valuation vanishes modulo p after the common scaling.

The sum over the indicated partitions is

$$N\left(\frac{\vartheta}{q}\right)^{a} m_r(\vartheta), \qquad N=\frac{n!}{(q!)^a a! r!}.$$

We have $N\in\mathbb{Z}_p^{\times}$ and $N\equiv 1 \pmod p$. For clarity, Legendre's formula gives $v_p(N)=0$, and the elementary unit-factorial congruence

$$p^{-v_p(k!)}k!\equiv(-1)^{v_p(k!)}\prod_i k_i! \pmod p \qquad \left(k=\sum_i k_i p^i,\ 0\le k_i<p\right)$$

shows that its unit is 1: the base-p digits of $n=aq+r$ are the concatenation of those of a and r, while q has a single digit 1. The sign exponents cancel because $v_p(N)=0$. This unit-factorial formula also follows immediately by repeatedly removing multiples of p and applying Wilson's theorem. Multiplying the displayed partition sum by p^a proves (VI.7.3). When $r=0$ its right side is a unit. □

The residue condition is substantive. For example $m_2(1)=3/2$, so the leading congruence vanishes when $p=3$ and $n\equiv 2 \pmod 3$. No assertion of equality in all residue classes, or of the uploaded manuscript's observed 2-adic formula, is being made.

Corollary VI.7.2 (Exact radius of the moment generating series)**.** *Under the hypotheses of Theorem VI.7.1, the p-adic radius of convergence of $M_\vartheta(z)=\sum_{n\ge0} m_n(\vartheta)z^n/n!$ is exactly*

$$R_p(M_\vartheta)=p^{-1/q-1/(p-1)}.$$

Proof. The valuation bound in (VI.7.3) and $v_p(n!)/n\to 1/(p-1)$ give one inequality in the Cauchy–Hadamard formula. The exact valuations at $n=aq$ give the other inequality. □

VI.7.2 Torsion characters obstruct every rational affine scaling

The preceding denominators already exclude a bounded measure with unscaled moments. Merely multiplying the random variable by a large power of p can remove that immediate obstruction. The following argument survives such changes.

For rational $\vartheta>0$, $c\neq 0$, and d, define the rational polynomial functional

$$\mathcal{L}_{\vartheta,c,d}(P)=\mathbb{E}P(d+cD_\vartheta), \qquad a_n=\mathcal{L}_{\vartheta,c,d}\left(\binom{X}{n}\right), \qquad A(T)=\sum_{n\ge0} a_n T^n.$$

This uses only rational moments, not a real-to-p-adic embedding.

Theorem VI.7.3 (No compact p-adic distribution with Dickman moments)**.** *Let p be any prime, $\vartheta\in\mathbb{Q}_{>0}$, and $c,d\in\mathbb{Q}\cap\mathbb{Z}_p$ with $c\neq0$. Put $j=v_p(c)$. The formal Amice series above has radius*

$$R_p(A)\le p^{-1/(p^j(p-1))}<1. \tag{VI.7.4}$$

Hence $\mathcal{L}_{\vartheta,c,d}$ is not the restriction of a locally analytic p-adic distribution on $\mathbb{Z}_p$. In particular it is not a bounded p-adic measure. There is no compactly supported locally analytic distribution on $\mathbb{Q}_p$ whose moments are $m_n(\vartheta)$.

Proof. All of the following identities first hold as formal power series over $\mathbb{Q}$. The binomial generating function gives

$$A(T) = (1+T)^d M_\vartheta\big(c\log(1+T)\big).$$

Since $zM'_\vartheta(z) = \vartheta(e^z-1)M_\vartheta(z)$, formal differentiation gives

$$(1+T)\log(1+T)A'(T) = \big[d\log(1+T) + \vartheta\big((1+T)^c-1\big)\big]\,A(T). \tag{VI.7.5}$$

Both coefficients in this equation are analytic for $|T|_p < 1$; here $c, d \in \mathbb{Z}_p$ ensures convergence of the binomial powers.

Let ζ be a primitive p^{j+1}-st root of unity and set $T_0 = \zeta - 1$. Then

$$|T_0|_p = p^{-1/(p^j(p-1))}, \qquad \log\zeta = 0,$$

and ζ^c is a primitive p-th root of unity. The latter assertion follows by approximating c by integers in $\mathbb{Z}_p$; the value depends on c modulo p^{j+1}. Suppose that the radius of A were greater than $|T_0|_p$. The formal identity (VI.7.5) would then hold in a disk containing T_0. Since $A(0) = 1$, the analytic function A is not identically zero and has a finite order $k \in \mathbb{Z}_{\geq 0}$ at T_0. The factor $(1+T)\log(1+T)$ has a simple zero there, with derivative 1. Comparing the lowest terms at T_0 gives

$$k = \vartheta(\zeta^c - 1). \tag{VI.7.6}$$

For odd p the right side is not rational, because a primitive p-th root of unity is not rational and $\vartheta \neq 0$ is rational. For $p = 2$ it equals $-2\vartheta < 0$. Both alternatives contradict $k \in \mathbb{Z}_{\geq 0}$. This proves (VI.7.4).

The Amice transform of any locally analytic distribution on $\mathbb{Z}_p$ is analytic on the whole open unit disk [VI.12, Theorem 2.2]; its Taylor coefficients are its values on $\binom{X}{n}$. Thus such a distribution cannot restrict to $\mathcal{L}_{\vartheta,c,d}$. The bounded-measure case also follows elementarily: $\binom{x}{n} \in \mathbb{Z}_p$ for $x \in \mathbb{Z}_p$, so a bounded measure gives bounded a_n and therefore radius at least 1. Finally, a compact subset of $\mathbb{Q}_p$ lies in $p^{-h}\mathbb{Z}_p$ for some $h \geq 0$. Pushing a putative distribution forward by $x \mapsto p^h x$ would give the forbidden moments with $c = p^h, d = 0$ on $\mathbb{Z}_p$. □

This proof identifies the obstruction as a local exponent at a torsion character, rather than a denominator estimate. Quantitatively, (VI.7.4) implies

$$\limsup_{n\to\infty} |a_n|_p^{1/n} \geq p^{1/(p^j(p-1))} > 1.$$

Multiplication of the functional by any nonzero scalar cannot change this conclusion. Thus a scalar normalization involving $e^{-\gamma}$ would not repair the failure even if that scalar had independently been assigned a p-adic value.

VI.7.3 What remains necessary for an arithmetic bridge

The real identity $\Pr(D_1 \leq 1) = e^{-\gamma}$ remains exact, but the rational moment functional cannot be transported to compact p-adic measure theory with those moments intact. A proposed adelic argument must therefore change its local objects or supply genuinely different arithmetic data. No conclusion here excludes noncompact distributions, formal functionals, or other nonarchimedean constructions.

The direction of the original bridge also matters: transcendence of e^γ would not imply irrationality of γ. One precise alternative would be to prove

$$\Pr(D_1 \leq 1) \neq e^{-r} \quad \text{for every } r \in \mathbb{Q},$$

or the corresponding exclusion for algebraic r to prove transcendence of γ. These are exactly the unresolved arithmetic assertions, not consequences of the distributional identities. The new congruences and the no-transfer theorem identify what the attempted p-adic route can and cannot supply.

VI.8 What has changed, and what remains unproved

The constructive result is Theorem VI.2.1: the positive rational-kernel family is analytically complete when pole order grows. Thus failure at every tested finite order was not evidence that this family can never approach the correct target. The theorem also makes the remaining arithmetic condition exact. It does not estimate the required order or the relation between a reduced target denominator and its error.

The ramified connection in Theorem VI.3.1 keeps γ after logarithmic monodromy cancellation. Its coefficient arithmetic fails both the E and G conditions, and the tested block factorial transforms do not cure this. The holonomy analysis preserves a different opening: a construction made only under a rationality hypothesis is not excluded by an unconditional classification of fixed cancellation constants. No such contradiction-producing conditional construction has been obtained.

Route	Established here	Missing arithmetic endpoint
Fixed conjugate poles, growing order	All rational targets below γ have strict positive certificates.	A sequence with $\mathrm{den}(r)(\gamma - r) \to 0$.
Ramified exponential integrals	Entire recessive order-three connections retain $(m-1)\gamma$ at zero.	An arithmetic theorem compatible with the mixed coefficient growth.
Regular-singular holonomy	Exact affine/algebraic cancellation loci; the conditional construction is not ruled out.	An actual conditional system meeting all arithmetic and analytic bounds.
CMF integer lattices	Individual gcd equals quotient content times a divisor of the image index.	Uniform seed heights, kernel congruence content and nonzero error estimates.
Canonical Hankel polynomials	Fully primitive values at γ have Nth root tending to infinity.	This particular determinant family is unsuitable for shrinking primitive values.
Dickman moment transfer	Exact odd-prime congruences; compact p-adic realization is impossible after any rational rescaling.	A different local object or genuinely different arithmetic data.

The first row is an equivalence to irrationality, not an easier assertion proved by the preceding analytic argument. The last two rows exclude specific proposed constructions, not all determinant or adelic approaches. No statement in this manuscript proves $\gamma \notin \mathbb{Q}$ or $\gamma \notin \overline{\mathbb{Q}}$.

VI.8.1 Verification and reproducibility

Each of the six main proof sections was read independently in addition to its author's derivation. This internal verification is not peer review and does not establish priority. Exact arithmetic checks include the kernel numerator identities, Sturm root counts, exact Bernstein subdivision for the 15/26 certificate, the Aptekarev exceptional gcd at $n = 2717$ from both the recurrence and the independent binomial–harmonic sums, small Hankel/Laguerre identities, and 6335 instances of the Dickman prime congruences. Numerical discovery was used to select candidates; it is not used to establish any infinite-family conclusion.

The standalone source contains the explicit kernel coefficients and the main exact recurrence verification. The following short routine independently checks the generalized Dickman congruence without any decimal evaluation of γ.

```
from fractions import Fraction as Q
from math import comb

def moments(theta, N):
    out = [Q(1)]
```

```
    for n in range(1, N + 1):
        out.append(theta / n * sum(comb(n, k) * out[k]
                                   for k in range(n)))
    return out

def vp(x, p):
    if x == 0:
        return float('inf')
    a, b = abs(x.numerator), x.denominator
    v = 0
    while a % p == 0:
        a //= p; v += 1
    while b % p == 0:
        b //= p; v -= 1
    return v

for theta in map(Q, ['1', '2', '3', '9', '5/2', '-3']):
    m = moments(theta, 100)
    for p in [3, 5, 7, 11, 13]:
        c = vp(theta, p)
        if c < 0:
            continue
        q, u = p**(c + 1), theta / p**c
        for n in range(101):
            a, r = divmod(n, q)
            assert vp(p**a * m[n], p) >= 0
            assert vp(p**a * m[n] - u**a * m[r], p) >= 1
print('Dickman congruence checks passed')
```

References for Part VI

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[VI.13] NIST Digital Library of Mathematical Functions, Chapters 6 and 18, especially Laguerre orthogonality and recurrence; https://dlmf.nist.gov/18.3, https://dlmf.nist.gov/18.9.

Part VII
A logarithmic Meijer identity and exact arithmetic structure

Source: `gamma_exact_structures(1).tex`.

Original PDF: 28 pages. Mathematical content and source-level qualifications are retained.

Abstract

We continue the supplied attempts on Euler's constant by replacing several numerical or structural obstacles with exact statements. A Mellin–Barnes contour shift gives an arbitrary-parameter residue tower whose first nonconstant output is $\gamma + \operatorname{Log} z$. For a rank-three trajectory in the October 2026 Meijer construction, we determine the entire projected integer lattice: it is the image of $n!\left(\begin{smallmatrix}1 & 3H_n\\ 0 & 1\end{smallmatrix}\right)$. The proof combines a characteristic-p differential identity with logarithmic Frobenius solutions and an adjoint pairing. We also determine the integer kernel of the elementary Gamma integral, prove sharp pointwise-positivity and uniform absolute-integral barriers, and obtain an odd-prime congruence yielding exact common-denominator growth for Dickman moments and their fixed polynomial transforms.

These are proved identities and arithmetic results, not a proof that γ is irrational or transcendental. In particular, unrestricted selection of an integer initial vector in the exact lattice encodes arbitrary rational approximation and does not preserve a fixed trajectory's error estimate. A separate height-controlled, nonvanishing small-form construction remains missing. The results were derived and checked for this continuation; historical priority is not claimed.

Status of the main problem. No irrationality or transcendence proof is claimed anywhere in this supplement. Statements marked as computations are finite exact checks or ordinary numerical quadratures; the latter are not interval certificates.

Relation to the supplied files. This is a self-contained supplement to `gamma_attempts.tex` and the companion manuscript bundle supplied with the request. It does not silently replace their statements or treat their unpublished claims as established theorems.

VII.1 What a successful construction must still achieve

Write

$$H_0 = 0, \qquad H_n = \sum_{j=1}^{n} \frac{1}{j}, \qquad \gamma = \lim_{n\to\infty}(H_n - \log n).$$

The current preprint used here does not resolve the irrationality of γ [VII.1]. We therefore separate two tasks: constructing new exact objects, and proving that their arithmetic and analytic properties contradict a rationality assumption. Only the former is completed in this note.

Lemma VII.1.1 (The required arithmetic conclusion)**.** *Suppose $a_j, b_j \in \mathbb{Z}$ and*

$$0 < |a_j - b_j\gamma| \longrightarrow 0$$

along an infinite sequence. Then $\gamma \notin \mathbb{Q}$. More generally, for $A_j, B_j \in \mathbb{Q}$ and positive integers D_j with $D_jA_j, D_jB_j \in \mathbb{Z}$, it suffices to prove $0 < D_j|A_j - B_j\gamma| \to 0$.

Proof. If $\gamma = u/v$ with $u, v \in \mathbb{Z}$ and $v > 0$, then $v(a_j - b_j\gamma)$ is a nonzero integer tending to zero, a contradiction. Apply this to (D_jA_j, D_jB_j) for the second assertion. □

The strict inequality matters. Geometry of numbers can supply a small linear form that is allowed to vanish; a determinant can certify that two independent forms cannot both vanish. Neither assertion alone supplies nonvanishing and smallness for the same selected form.

Our strongest arithmetic calculation is

$$\{(p_n v, q_n v) : v \in \mathbb{Z}^3\} = \{(n!a + 3n!H_n b,\ n!b) : a, b \in \mathbb{Z}\}, \tag{VII.1.1}$$

where the rows p_n, q_n come from the explicit matrix product in Section VII.4. This resolves the common-factor question for the full family of projected outputs, rather than for a few selected columns. Yet, for $b \neq 0$, its ratios are $3H_n + a/b$, which cover all of $\mathbb{Q}$ at every fixed n. Thus a useful adaptive choice of v must carry additional, proved control of its size and its direction. Without such control, choosing a good output ratio merely restates the original Diophantine problem.

The analytic construction has a different purpose. In Section VII.2 we construct a tower with outputs $\gamma + \operatorname{Log} z$, its powers, and zeta values. This gives the identity extension suggested in [VII.1, Section 9.2(1)]. It does not establish convergence for all parameter-dependent trajectories. We then try a coefficient Borel transform, including under the hypothesis that γ is algebraic, and identify precisely why its logarithmic singularity does not cancel.

The two remaining constructions test potential arithmetic escapes. The integer Gamma kernel permits large signed changes of integrands without changing their linear forms; its exact quotient and positivity bounds appear in Section VII.5. Dickman moments offer rational moments of a distribution whose normalization contains e^{γ}; Section VII.6 determines the denominator cost of fixed polynomial transforms and integer convolution shapes. Each conclusion is restricted to its stated family.

VII.2 A logarithmic Meijer residue tower

Here G_p denotes a *Meijer* function, defined by its Mellin–Barnes integral. It must not be confused with a *Siegel G-function*, which has additional arithmetic coefficient conditions. The integral convention agrees with the standard Meijer convention [VII.5]. Put $\theta = z\,d/dz$.

Definition VII.2.1. For an integer $p \geq 1$, let

$$\Phi_p(s) = \Gamma(-s)^{p+1}\Gamma(1+s)^p, \qquad G_p(z) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \Phi_p(s) z^s \,\mathrm{d}s, \quad -1 < c < 0.$$

Initially take the principal logarithm on $\mathbb{C} \setminus (-\infty, 0]$. More generally, the same integral is valid on the logarithmic covering of $\mathbb{C}^\times$ whenever $|\operatorname{Im} \operatorname{Log} z| < (p + \frac{1}{2})\pi$.

Theorem VII.2.2 (Residue tower)**.** *For $0 \leq k \leq p-1$, define*

$$\mathcal{L}_{p,k} = z(1+\theta)^{p-1-k} + (-1)^p \theta^{p-k},$$

with multiplication by z on the left. Then

$$\boxed{\mathcal{L}_{p,k} G_p(z) = [t^k]\,\Gamma(1-t)\left(\frac{\pi t}{\sin \pi t}\right)^p e^{t \operatorname{Log} z}.} \tag{VII.2.1}$$

In particular,

$$[z(1+\theta)^{p-1} + (-1)^p\theta^p] G_p(z) = 1 \qquad (p \geq 1), \tag{VII.2.2}$$

$$[z(1+\theta)^{p-2} + (-1)^p\theta^{p-1}] G_p(z) = \gamma + \operatorname{Log} z \qquad (p \geq 2). \tag{VII.2.3}$$

The identities hold throughout the direct integral domain and continue analytically along compatible branches.

Proof. For $s = c + iy$, Stirling's estimate gives

$$|\Phi_p(c+iy)| = O\left((1+|y|)^{-c-1/2}e^{-(p+1/2)\pi|y|}\right).$$

Also $|z^{c+iy}| = |z|^c e^{-y\operatorname{Im}\operatorname{Log} z}$. Hence the integral and the finitely many differentiated integrals below converge absolutely and locally uniformly in the stated domain. The same estimate, uniform on $c-1 \le \operatorname{Re} s \le c$, makes the horizontal sides of the contour shift tend to zero.

The Gamma recurrence gives the exact relation

$$\Phi_p(s-1) = (-1)^{p+1} s\Phi_p(s).$$

For a polynomial Q, set

$$I_d[Q] = \frac{1}{2\pi i}\int_{d-i\infty}^{d+i\infty} Q(s)\Phi_p(s)z^s\,\mathrm{d}s.$$

Applying the preceding recurrence and then substituting $s = u+1$ yields

$$(-1)^p\theta Q(\theta-1)G_p = -zI_{c-1}[Q].$$

It follows that

$$\{zQ(\theta) + (-1)^p\theta Q(\theta-1)\}G_p = z\{I_c[Q] - I_{c-1}[Q]\}. \tag{VII.2.4}$$

The rectangle with right side traversed upwards has positive orientation. Its strip contains just the pole $s=-1$, so the right-minus-left difference is the residue there, with positive sign.

Choose $Q(s) = (s+1)^{p-1-k}$ and put $t = s+1$. Using the reflection formula, the integrand in the residue, including the outside factor z, is

$$z(s+1)^{p-1-k}\Phi_p(s)z^s = t^{-k-1}\Gamma(1-t)\left(\frac{\pi t}{\sin \pi t}\right)^p e^{t\operatorname{Log} z}.$$

Its residue is the coefficient in (VII.2.1), and $\theta Q(\theta-1) = \theta^{p-k}$, proving the theorem. □

VII.2.1 Explicit output polynomials and the operator factorization

Set $X = \gamma + \operatorname{Log} z$ and $a_p = 2p+1$. Define polynomials $C_{p,k}$ by

$$\sum_{k\ge 0} C_{p,k}(X)t^k = \exp\left(Xt + \sum_{m\ge 2}\frac{1+p(1+(-1)^m)}{m}\zeta(m)t^m\right). \tag{VII.2.5}$$

The Taylor expansion of $\log\Gamma(1-t)$ [VII.6] and the sine product identify the right side with the analytic factor in (VII.2.1). Thus $\mathcal{L}_{p,k}G_p = C_{p,k}(X)$ for $k \le p-1$. In low degrees,

$$\begin{aligned}
C_{p,0}(X) &= 1,\\
C_{p,1}(X) &= X,\\
C_{p,2}(X) &= \frac{X^2 + a_p\zeta(2)}{2},\\
C_{p,3}(X) &= \frac{X^3 + 3a_pX\zeta(2) + 2\zeta(3)}{6},\\
C_{p,4}(X) &= \frac{X^4 + 6a_pX^2\zeta(2) + 8X\zeta(3) + 3a_p^2\zeta(2)^2 + 6a_p\zeta(4)}{24}.
\end{aligned}$$

The polynomial in degree k is defined for every k, but its displayed differential realization requires $p \ge k+1$. No negative power of $1+\theta$ is being introduced.

The commutation rule $\theta z = z(\theta + 1)$ gives

$$\theta \mathcal{L}_{p,k} = \mathcal{L}_{p,k-1} \quad (1 \le k \le p-1), \qquad C'_{p,k} = C_{p,k-1},$$

and, with $\sigma = (-1)^p$,

$$\mathcal{H}_p := z(1+\theta)^p + \sigma\theta^{p+1} = \theta^p \circ (z + \sigma\theta), \qquad \mathcal{H}_p G_p = 0. \tag{VII.2.6}$$

In particular, the top member of the residue tower is the first-order inhomogeneous equation

$$(z + \sigma\theta)G_p = C_{p,p-1}(\gamma + \operatorname{Log} z). \tag{VII.2.7}$$

The order of the factors in (VII.2.6) is essential.

At $z = 1$, the constant and linear outputs recover the identity pattern used by the Meijer construction [VII.1]. Formula (VII.2.3) provides its requested arbitrary-z identity extension. Its coefficients contain z, so the fixed coefficient vectors at $z = 1$ cannot simply be reused unchanged. At algebraic z the coefficients are algebraic; at rational z they can be cleared to integers. This is an exact evaluation statement. It supplies no automatic convergence theorem or height bound for the associated parameter-dependent matrix trajectories.

VII.2.2 A positive Laplace representation

Proposition VII.2.3. *For* $\operatorname{Re} z > 0$,

$$G_p(z) = \int_0^\infty e^{-zt} h_p(t)\,\mathrm{d}t,$$

where

$$h_{2m+1}(t) = \frac{\displaystyle\prod_{j=1}^{m}((\log t)^2 + (2j-1)^2\pi^2)}{(2m)!(1+t)} \qquad (m \ge 0), \tag{VII.2.8}$$

$$h_{2m}(t) = \frac{\log t \displaystyle\prod_{j=1}^{m-1}((\log t)^2 + (2j)^2\pi^2)}{(2m-1)!(t-1)} \qquad (m \ge 1). \tag{VII.2.9}$$

The value at $t = 1$ in the even case is removable. Every h_p is strictly positive on $(0, \infty)$, and hence $(-1)^j G_p^{(j)}(z) > 0$ for $z > 0$ and $j \ge 0$.

Proof. For $0 < \operatorname{Re} u < 1$, the Mellin transform of $h_1(t) = 1/(1+t)$ is $F_1(u) = \pi/\sin \pi u$. Mellin convolution gives

$$h_2(t) = \int_0^\infty \frac{\mathrm{d}v}{(1+v)(v+t)} = \frac{\log t}{t-1},$$

with transform $F_2(u) = (\pi/\sin \pi u)^2$. For $F_p(u) = (\pi/\sin \pi u)^p$, differentiation gives

$$F_p''(u) + p^2\pi^2 F_p(u) = p(p+1)F_{p+2}(u).$$

Thus

$$h_{p+2}(t) = \frac{(\log t)^2 + p^2\pi^2}{p(p+1)} h_p(t)$$

has Mellin transform F_{p+2}. Iteration proves (VII.2.8)–(VII.2.9). The Mellin transform in z of the claimed Laplace integral equals

$$\Gamma(u)F_p(1-u) = \Gamma(u)^{p+1}\Gamma(1-u)^p \qquad (0 < \operatorname{Re} u < 1),$$

which is the defining Mellin–Barnes transform of G_p under $u = -s$. Mellin inversion proves the identity; absolute convergence extends it throughout $\operatorname{Re} z > 0$. Positivity and differentiation under the integral give the final assertion. □

For $p = 2$ there is also an elementary check of the main identity:

$$(z+\theta)G_2(z) = z\int_0^\infty e^{-zt}(1-t)\frac{\log t}{t-1}\,\mathrm{d}t = -z\int_0^\infty e^{-zt}\log t\,\mathrm{d}t = \gamma + \log z \quad (z>0).$$

Thus the appearance of $\gamma + \log z$ has a direct integral explanation. The positive kernels do not by themselves make the differentiated integer value forms both nonzero and small.

VII.3 Trying to remove the irregularity

A natural next attempt is to place the new identity in the arithmetic theory of Siegel G-functions. The exact factorization above lets us test this possibility without assuming anything about the arithmetic nature of γ.

Theorem VII.3.1 (No Siegel G-solution of this particular module)**.** *For every $p \geq 1$, the equation*

$$\mathcal{H}_p f = \theta^p(z + (-1)^p\theta)f = 0$$

has no nonzero solution which is a Siegel G-function germ. The statement includes germs at an algebraic ordinary point, after translating that point to the origin.

Proof. The minimal differential equation of a Siegel G-function is Fuchsian with rational exponents; in particular its solutions have convergent Frobenius-logarithmic expansions at infinity [VII.2, Theorem 6 and the paragraph following it]. This means a finite sum of rational powers of $1/z$, powers of $\operatorname{Log} z$, and power series in $1/z$ with positive convergence radius.

Suppose $f \neq 0$ were such a germ and put $\sigma = (-1)^p$. On a simply connected logarithmic domain,

$$(z+\sigma\theta)f = P(\operatorname{Log} z)$$

for some polynomial P of degree at most $p-1$. If $P = 0$, then $f = Ce^{-\sigma z}$ with $C \neq 0$, which has an irregular singularity at infinity and cannot have the required convergent Frobenius expansion.

If $P \neq 0$, collect the powers in a convergent Frobenius expansion of f into exponent classes modulo $\mathbb{Z}$. A nonintegral exponent class would have to solve the homogeneous equation: its largest power of z produces a still larger power under multiplication by z that θ cannot cancel. Such a class must therefore vanish. In the integral class, the same leading-power comparison forces

$$f(z) = \sum_{n\geq 1} z^{-n}A_n(\operatorname{Log} z),$$

where the A_n have uniformly bounded degree and each series of coefficients of a fixed log power converges. Comparison in the first-order equation gives, with $D = d/dX$,

$$A_1(X) = P(X), \qquad A_{n+1}(X) = \sigma(n-D)A_n(X). \tag{VII.3.1}$$

If P has degree d and leading coefficient $a \neq 0$, the coefficient of X^d in A_{n+1} is $\sigma^n n!a$. Its series in $1/z$ has radius zero, contradicting convergence. Translation of an algebraic ordinary point does not change this required behavior at infinity. □

Remark VII.3.2. This theorem excludes nonzero Siegel G-functions in one explicit solution space. It does not determine the arithmetic nature of any individual value $G_p(\alpha)$, nor does it exclude other differential modules, transforms, or arithmetic-holonomy constructions. The factorization even has a regular-singular quotient represented by polynomials in $\operatorname{Log} z$.

VII.3.1 A coefficient Borel transform under an algebraicity hypothesis

Equation (VII.3.1) suggests dividing the coefficients by $n!$. This genuinely changes the function class, so it is not excluded by Theorem VII.3.1. Keep X as an independent polynomial symbol and define

$$\widehat{f}(z) = \sum_{n\geq 0} z^{-n-1}\sigma^n (1-D)_n P(X), \qquad (1-D)_n = \prod_{j=1}^{n}(j-D).$$

Proposition VII.3.3 (Exact coefficient Borel symbol)**.** *As a formal power series in t with polynomial coefficients in X,*

$$\sum_{n\geq 0} \frac{\sigma^n (1-D)_n P(X)}{n!} t^n = \frac{P(X+\log(1-\sigma t))}{1-\sigma t}. \tag{VII.3.2}$$

Proof. The binomial identity gives $\sum_{n\geq 0}(1-D)_n(\sigma t)^n/n! = (1-\sigma t)^{D-1}$. Since $\exp(aD)$ translates a polynomial by a, applying the operator to P gives (VII.3.2). □

This is a *coefficient* Borel symbol. A Borel transformation of a full logarithmic transseries also acts on $\operatorname{Log} z$; no such transformation or summability theorem is implicitly asserted here.

For $p = 2$ one has $\sigma = 1$ and $P(X) = \gamma + X$, so

$$\widehat{G}_2(z) = \sum_{n\geq 0} \frac{n!}{z^{n+1}}(\gamma + \operatorname{Log} z - H_n).$$

Its coefficient Borel symbol is $(\gamma + X + \log(1-t))/(1-t)$. Now make the actual reductio assumption $\gamma = a \in \overline{\mathbb{Q}}$ and specialize the independent symbol X to zero. The candidate becomes

$$B_a(t) = \frac{a+\log(1-t)}{1-t} = \sum_{n\geq 0}(a - H_n)t^n. \tag{VII.3.3}$$

For every algebraic a, this is a Siegel G-function: it is holonomic, its coefficients have at most logarithmic growth apart from the fixed algebraic constant, and their denominators are bounded by a fixed denominator times $\operatorname{lcm}(1,\dots,n)$, hence exponentially.

However, its logarithmic singularity at $t = 1$ is nonzero for *every* a. A positive loop about 1 changes B_a by $2\pi i/(1-t)$, independently of a. The parameter only changes the simple-pole term. Thus the algebraicity assumption does produce a G-function, but it creates no extra holomorphic cancellation at the singularity. This particular attempted escape is exhausted by the explicit formula; it does not prove that every hypothesis-dependent G-function construction must fail.

VII.4 Exact arithmetic of a rank-three matrix trajectory

We determine the complete image lattice of the rank-three trajectory $(0,0;1,1,1)$ in [VII.1, Section 8.4]. The calculation concerns all integer input vectors at a fixed depth. It therefore distinguishes the arithmetic freedom of changing the input from the convergence of a particular sequence of inputs.

For $n \geq 0$, let

$$M_n = \begin{pmatrix} 1 & n^3 & 3(n+1)(n-1)(n^2+n+1) \\ 2 & 1-3n^2 & -(n+1)(8n^2+n+5) \\ 1 & 3n+2 & 6n(n+1) \end{pmatrix} \tag{VII.4.1}$$

and

$$N_n = \begin{pmatrix} 6n^2+13n+10 & -8n-5 & 3 \\ 3n^3+3n^2-3n-6 & -3n^2+3n+3 & n-2 \\ n^4-2n^3-6n^2-3n+3 & 6n^2+2n-1 & 1-2n \end{pmatrix}. \tag{VII.4.2}$$

Direct polynomial multiplication gives

$$\det M_n = \det N_n = (n+1)^6, \qquad N_n = (n+1)^4 M_n^{-T}, \qquad \operatorname{adj} N_n = (n+1)^2 M_n^T. \tag{VII.4.3}$$

Products are ordered from left to right with increasing index:

$$P_n = N_0 \cdots N_{n-1}, \qquad U_n = M_0 \cdots M_{n-1}, \qquad P_0 = U_0 = I_3.$$

Define

$$R = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} \mathbf{p}_n \\ \mathbf{q}_n \end{pmatrix} = RP_n, \qquad \Lambda_n = RP_n\mathbb{Z}^3 \subset \mathbb{Z}^2.$$

We put $H_0 = 0$ and $H_n = \sum_{j=1}^{n} 1/j$ for $n \geq 1$.

Theorem VII.4.1 (Exact projected lattice)**.** *For every $n \geq 0$,*

$$\Lambda_n = \{(n!a + 3n!H_n b,\ n!b)^T : a, b \in \mathbb{Z}\}. \tag{VII.4.4}$$

In particular, $[\mathbb{Z}^2 : \Lambda_n] = (n!)^2$. Its column Hermite normal form, with the upper-right entry in $[0, n!)$, is

$$\begin{pmatrix} n! & n!(3H_n - \lfloor 3H_n \rfloor) \\ 0 & n! \end{pmatrix}. \tag{VII.4.5}$$

If $D_n = \operatorname{den}(3H_n)$ is the positive denominator in lowest terms (with $\operatorname{den}(0) = 1$) and $C_n = 3D_nH_n$, then

$$\gcd(q_{n,1}, q_{n,2}, q_{n,3}) = n!, \qquad \gcd(p_{n,1}, p_{n,2}, p_{n,3}) = \gcd(\textit{all entries of } RP_n) = \frac{n!}{D_n}. \tag{VII.4.6}$$

After division by this common gcd, the image lattice is exactly

$$\{(D_n a + C_n b,\ D_n b)^T : a, b \in \mathbb{Z}\}, \tag{VII.4.7}$$

and has index D_n^2 in $\mathbb{Z}^2$.

The proof has two independent arithmetic components: an exact covolume calculation and integral formulas for the two rows after a harmonic shear.

VII.4.1 A primitive polynomial companion and the covolume

Lemma VII.4.2. *The gcd of the three maximal minors of RP_n is $(n!)^2$. Moreover, $\det P_n = (n!)^6$, and the gcd of all 2×2 minors of P_n is also $(n!)^2$.*

Proof. Equation (VII.4.3) implies

$$P_n = (n!)^4 U_n^{-T}, \qquad \det P_n = (n!)^6.$$

The cross product of the rows of R is $\mathbf{v}_0 = (1, -1, 0)$. The transformation rule for cross products of row vectors therefore gives

$$\operatorname{cross}(\mathbf{p}_n, \mathbf{q}_n) = (n!)^2 \mathbf{b}_n, \qquad \mathbf{b}_n = \mathbf{v}_0 U_n \in \mathbb{Z}^3. \tag{VII.4.8}$$

It remains to prove that $\mathbf{b}_n$ is primitive.

Set $\theta = z\,\mathrm{d}/\,\mathrm{d}z$, $\delta = \theta - z$, and define integer polynomials

$$K_n(z) = \left[\prod_{j=0}^{n-1} (j - \delta)\right]^3 1, \qquad K_0 = 1. \tag{VII.4.9}$$

All factors in the product are polynomials in the same operator and hence commute. In characteristic zero, the gauge identity $e^z\theta e^{-z} = \delta$ gives

$$F_n(z) := e^{-z}K_n(z) = \left[\prod_{j=0}^{n-1}(j-\theta)\right]^3 e^{-z} = z^n \sum_{k\geq 0} \frac{(-1)^k(n+k)!^2}{k!^3} z^k.$$

The series directly verifies

$$L_nF_n = 0, \qquad F_{n+1} = z(\theta+1)^2F_n, \qquad L_n = (\theta-n)^3 + z(\theta+1)^2. \tag{VII.4.10}$$

At $n=0$ the noncommutative factorization is $L_0 = \theta^2(\theta+z)$, since $\theta z = z(\theta+1)$. Differentiate the second identity twice and eliminate θ^3F_n and θ^4F_n using the first. At $z=1$ the resulting row of three jets transforms by right multiplication by M_n in (VII.4.1). Its initial value, multiplied by e, is $(1,-1,0)$. Consequently,

$$\mathbf{b}_n = (K_n(1), \delta K_n(1), \delta^2 K_n(1)). \tag{VII.4.11}$$

This is an identity of integer vectors. The exponential gauge is used only to establish it in characteristic zero.

Fix a prime ℓ and now regard δ as an operator on $\mathbb{F}_\ell[z]$. We claim that

$$\delta^\ell - \delta = -z^\ell. \tag{VII.4.12}$$

Indeed, $\delta z = z(\delta+1)$ gives $[\delta^\ell - \delta, z] = 0$ in characteristic ℓ. An operator commuting with multiplication by z is multiplication by its value on 1. The elementary Stirling-number identity

$$\delta^m 1 = \sum_{j=1}^{m} S(m,j)(-z)^j$$

shows that this value is $-z^\ell$: for $1 < j < \ell$,

$$j!S(\ell,j) = \sum_{i=0}^{j}(-1)^{j-i}\binom{j}{i}i^\ell \equiv 0 \pmod{\ell},$$

because Fermat's identity replaces i^ℓ by i, whose jth finite difference vanishes. The endpoint Stirling numbers both equal 1. This proves (VII.4.12), including $\ell = 2$.

It follows that

$$\prod_{j=0}^{\ell-1}(j-\delta) = z^\ell.$$

Multiplication by z^ℓ commutes with δ. Write $n = a\ell + r$ with $0 \leq r < \ell$. Splitting the operator product into blocks of length ℓ in (VII.4.9) yields

$$K_n(z) \equiv z^{3a\ell}K_r(z) \pmod{\ell}, \qquad \mathbf{b}_n \equiv \mathbf{b}_r \pmod{\ell}.$$

Every M_j with $0 \leq j < r$ is invertible over $\mathbb{F}_\ell$, since $\det M_j = (j+1)^6$. Hence $\mathbf{b}_r = \mathbf{v}_0M_0\cdots M_{r-1}$ is nonzero modulo ℓ. No prime divides all entries of $\mathbf{b}_n$, proving primitivity. Equation (VII.4.8) proves the assertion about RP_n.

Finally, $\operatorname{adj} N_j = (j+1)^2M_j^T$ shows that every 2×2 minor of N_j is divisible by $(j+1)^2$. Taking exterior squares of the product proves divisibility of every 2×2 minor of P_n by $(n!)^2$. By Cauchy–Binet, every maximal minor of RP_n is an integer combination of these minors. The equality already proved for RP_n forces their gcd to be exactly $(n!)^2$. □

The index of the column image of a rank-two integer matrix is the gcd of its maximal minors. Thus Lemma VII.4.2 already determines $[\mathbb{Z}^2 : \Lambda_n]$; the following calculation determines its precise position inside $\mathbb{Z}^2$.

VII.4.2 Integral polynomial rows and their adjoint interpretation

Put $D = u\,\mathrm{d}/\,\mathrm{d}u$ and define

$$c_{n,k} = \binom{n}{k}^3 k!, \qquad Q_n(u) = \sum_{k=0}^{n} c_{n,k} u^k, \tag{VII.4.13}$$

$$A_n(u) = \sum_{k=0}^{n} c_{n,k}\big[3(H_{n-k} - H_n) - 2H_k\big]u^k. \tag{VII.4.14}$$

Both polynomials belong to $\mathbb{Z}[u]$. For A_n, integrality holds termwise: $c_{n,k}H_k$ is integral because $k!H_k$ is integral, while

$$c_{n,k} = \binom{n}{k}^2 \frac{n!}{(n-k)!}$$

is divisible by each integer $n-k+1, \dots, n$. Therefore $c_{n,k}(H_n - H_{n-k})$ is integral as well.

For brevity set $a_n = 1 - 3n$ and $d_n = 1 + 3n^2$, and introduce the rows

$$\mathcal{R}_{Q,n} = \big((D^2 + a_n D + d_n)Q_n,\ (D + a_n)Q_n,\ Q_n\big)\Big|_{u=1}, \tag{VII.4.15}$$

$$\mathcal{R}_{A,n} = \big((D^2 + a_n D + d_n)A_n + 2DQ_n + a_n Q_n,\ (D + a_n)A_n + Q_n,\ A_n\big)\big|_{u=1}. \tag{VII.4.16}$$

These are integer rows.

Proposition VII.4.3 (Exact row formulas)**.** *For every $n \geq 0$,*

$$\mathcal{R}_{Q,n} = \frac{\mathbf{q}_n}{n!}, \qquad \mathcal{R}_{A,n} = \frac{\mathbf{p}_n - 3H_n\mathbf{q}_n}{n!}. \tag{VII.4.17}$$

Proof. We establish the covariance of the rows through exact adjoint solutions. The adjoint of L_n with respect to $\mathrm{d}z/z$ is

$$L_n^* = z\theta^2 - (\theta + n)^3.$$

For any function w define

$$r_n(w) = \big(\theta^2 w - (z - 3n)\theta w + (z + 3n^2)w, -\theta w + (z - 3n)w,\ w\big). \tag{VII.4.18}$$

Direct differentiation shows that

$$\mathcal{C}_n(f, w) := r_n(w)\,(f, \theta f, \theta^2 f)^T$$

is constant in z whenever $L_n f = L_n^* w = 0$.

The two needed adjoint solutions are

$$w_{Q,n}(z) = Q_n(1/z), \qquad w_{A,n}(z) = A_n(1/z) - \operatorname{Log} z\, Q_n(1/z). \tag{VII.4.19}$$

Here and below a single branch of Log is fixed. To verify these solutions, let $c_0(\varepsilon) = 1$ and

$$c_{k+1}(\varepsilon) = \frac{(n - \varepsilon - k)^3}{(\varepsilon + k + 1)^2} c_k(\varepsilon) \quad (0 \leq k < n), \qquad W(\varepsilon, z) = z^{-\varepsilon} \sum_{k=0}^{n} c_k(\varepsilon) z^{-k}.$$

Coefficient comparison, including both endpoints, gives

$$\big((\theta + n)^3 - z\theta^2\big)W = -\varepsilon^2 z^{1-\varepsilon} - \varepsilon^3 c_n(\varepsilon) z^{-n-\varepsilon}. \tag{VII.4.20}$$

Thus $W(0, z)$ and $\partial_\varepsilon W(0, z)$ solve the homogeneous adjoint equation. Moreover,

$$c_k(0) = c_{n,k}, \qquad c_k'(0) = c_{n,k}\big[3(H_{n-k} - H_n) - 2H_k\big],$$

so they are precisely (VII.4.19). Evaluating (VII.4.18) at $z = 1$ gives

$$r_n(w_{Q,n})(1) = \mathcal{R}_{Q,n}, \qquad r_n(w_{A,n})(1) = \mathcal{R}_{A,n}. \tag{VII.4.21}$$

For example, at $z = 1$ the first two derivatives of $w_{A,n}$ are $-DA_n - Q_n$ and $D^2A_n + 2DQ_n$, accounting for every additional term in (VII.4.16).

We next compute the concomitants at the regular singular point 0. Every solution of $L_n f = 0$ has the form

$$f(z) = z^n \left[P(\operatorname{Log} z) + O(z(1 + |\operatorname{Log} z|^2))\right], \qquad P(X) = \frac{\alpha}{2}X^2 + \beta X + \chi, \tag{VII.4.22}$$

with arbitrary α, β, χ. To see this directly, write $f = z^n \sum_{j\geq 0} z^j P_j(\operatorname{Log} z)$. Its coefficient equations are

$$\partial_X^3 P_0 = 0, \qquad (j + \partial_X)^3 P_j = -(n + j + \partial_X)^2 P_{j-1} \quad (j \geq 1).$$

On polynomials of degree at most two the operator on the left is invertible. The successive operator norms are $O(1/j)$ for fixed n, which ensures convergence of the coefficient series. This constructs three independent solutions and proves (VII.4.22), also after applying θ twice.

Since $c_{n,n} = n!$, the adjoint solutions have leading terms

$$w_{Q,n} = n!z^{-n} + O(z^{1-n}), \qquad w_{A,n} = n!z^{-n}(-\operatorname{Log} z - 5H_n) + O(z^{1-n}(1 + |\operatorname{Log} z|)).$$

For leading forms $f = z^n P(X)$ and $w = n!z^{-n}V(X)$, the constant term of the concomitant is $n!(VP'' - V'P' + V''P)$. All terms with an additional factor z tend to zero. With $V = 1$ and $V = -X - 5H_n$ we obtain the exact constants

$$\mathcal{C}_n(f, w_{Q,n}) = n!\alpha, \qquad \mathcal{C}_n(f, w_{A,n}) = n!(\beta - 5H_n\alpha). \tag{VII.4.23}$$

Let $T = z(\theta+1)^2$ and put $a = n+1$. The identity $L_{n+1}T = TL_n$ follows from $(\theta - n - 1)T = T(\theta - n)$. The leading polynomial of Tf is $(a + \partial_X)^2 P$, so

$$\alpha_{\text{new}} = a^2\alpha, \qquad \beta_{\text{new}} = a^2\beta + 2a\alpha.$$

Using $H_{n+1} = H_n + 1/a$ in (VII.4.23) therefore gives

$$\begin{aligned}\mathcal{C}_{n+1}(Tf, w_{Q,n+1}) &= a^3\mathcal{C}_n(f, w_{Q,n}),\\ \mathcal{C}_{n+1}(Tf, w_{A,n+1}) &= a^3\mathcal{C}_n(f, w_{A,n}) - 3a^2\mathcal{C}_n(f, w_{Q,n}).\end{aligned}$$

At $z = 1$ the state of Tf is $M_n^T(f, \theta f, \theta^2 f)^T$. The state of f there is arbitrary. Hence these equalities, together with (VII.4.21), are row identities:

$$\mathcal{R}_{Q,n+1}M_n^T = a^3\mathcal{R}_{Q,n}, \qquad \mathcal{R}_{A,n+1}M_n^T = a^3\mathcal{R}_{A,n} - 3a^2\mathcal{R}_{Q,n}.$$

Multiplication by $M_n^{-T} = N_n/a^4$ yields

$$\mathcal{R}_{Q,n+1} = \frac{\mathcal{R}_{Q,n}N_n}{n+1}, \qquad \mathcal{R}_{A,n+1} = \frac{\mathcal{R}_{A,n}N_n}{n+1} - \frac{3\mathcal{R}_{Q,n+1}}{n+1}. \tag{VII.4.24}$$

Finally, $Q_0 = 1$ and $A_0 = 0$ give $\mathcal{R}_{Q,0} = (1, 1, 1)$ and $\mathcal{R}_{A,0} = (1, 1, 0)$. Induction in (VII.4.24) proves (VII.4.17). □

VII.4.3 Completion of the lattice calculation and its scope

Proof of Theorem VII.4.1. Let B_n be the integer matrix with rows $\mathcal{R}_{A,n}, \mathcal{R}_{Q,n}$. Proposition VII.4.3 gives the exact factorization

$$RP_n = \begin{pmatrix} n! & 3n!H_n \\ 0 & n! \end{pmatrix} B_n. \tag{VII.4.25}$$

Thus Λ_n is contained in the lattice on the right of (VII.4.4). The latter has covolume $(n!)^2$. Lemma VII.4.2 gives the same covolume for Λ_n, so the inclusion is equality.

The triangular matrix in (VII.4.25) is an exact lattice basis, but its upper-right entry need not be reduced. Subtracting $\lfloor 3H_n \rfloor$ times its first column from its second gives (VII.4.5). Its entries are integers because $n!H_n \in \mathbb{Z}$. The set of second coordinates is $n!\mathbb{Z}$, whereas the set of first coordinates is $n!\mathbb{Z} + 3n!H_n\mathbb{Z}$. This proves (VII.4.6), since

$$\gcd(n!, 3n!H_n) = \frac{n!}{\operatorname{den}(3H_n)}.$$

In particular $D_n \mid n!$. Dividing the basis in (VII.4.25) by $n!/D_n$ proves (VII.4.7) and its index assertion. □

Corollary VII.4.4 (Unrestricted input ratios)**.** *For each fixed $n \geq 0$,*

$$\left\{ \frac{\mathbf{p}_n\mathbf{v}}{\mathbf{q}_n\mathbf{v}} : \mathbf{v} \in \mathbb{Z}^3,\ \mathbf{q}_n\mathbf{v} \neq 0 \right\} = \mathbb{Q}.$$

Proof. By (VII.4.4), the ratios are precisely $3H_n + a/b$ with $a, b \in \mathbb{Z}$ and $b \neq 0$. Translation by the rational number $3H_n$ maps $\mathbb{Q}$ onto itself. □

Remark VII.4.5 (What adaptive inputs do not inherit)*.* For an arbitrary input, the exactly normalized linear form has the shape

$$(D_n a + C_n b) - \gamma D_n b = D_n\big[a - b(\gamma - 3H_n)\big].$$

The lattice description alone gives neither a bound on the norm of an input vector realizing (a, b) nor a bound on this linear form. In particular, replacing a fixed input by a depth-dependent vector does not inherit an error estimate proved for the original trajectory solution. Since all rational ratios are already available at each depth, any claimed improvement from adaptive inputs must include such additional height and error estimates, together with a nonvanishing argument.

This limitation is compatible with both arithmetic possibilities for γ. If $\gamma = r/s \in \mathbb{Q}$, every nonzero normalized form lies in $s^{-1}\mathbb{Z}$ and therefore has absolute value at least $1/s$, while the lattice also contains nonzero pairs whose form is exactly zero. Consequently, a geometry-of-numbers argument that allows the small form to vanish cannot by itself prove irrationality. The exact lattice theorem establishes its arithmetic structure without asserting a new irrationality bound.

Corollary VII.4.6 (Growth of the normalized lattice)**.** *Let $D_n = \operatorname{den}(3H_n)$ and $h_n = n!/D_n$. Then, as $n \to \infty$,*

$$\log D_n = n + o(n), \qquad \log h_n = \log(n!) - n + o(n).$$

Consequently, the lattice obtained from Λ_n by dividing by its common coordinate gcd has covolume $D_n^2 = \exp(2n + o(n))$.

Proof. Set $K = \lfloor \sqrt{\log n} \rfloor$ and take a prime $n/K < p \le n$. For large n, one has $p^2 > n$ and $p > K$. Put $k = \lfloor n/p \rfloor$, so $1 \le k < K$. Separating the multiples of p gives, in the localization of $\mathbb{Z}$ at p,

$$pH_n = H_k + p \sum_{\substack{1 \le j \le n \\ p \nmid j}} \frac{1}{j} \equiv H_k \pmod{p}.$$

The positive integer $k!H_k$ satisfies $0 < k!H_k \le K!H_K < p$ for all large n, because

$$\log(K!H_K) \le K \log K + \log(1 + \log K) = o(\log n), \qquad \log p \ge \log n - \log K.$$

Since $p \nmid k!$, H_k is a unit at p. Thus $v_p(H_n) = -1$. These primes eventually exceed 3, so every one of them divides D_n. Writing $\vartheta(x) = \sum_{p \le x} \log p$, and using $D_n \mid \operatorname{lcm}(1, \dots, n)$, we obtain

$$\vartheta(n) - \vartheta(n/K) \le \log D_n \le \log \operatorname{lcm}(1, \dots, n).$$

The prime number theorem gives $n + o(n)$ for both bounds, since $K \to \infty$. The other assertions follow from $h_n = n!/D_n$ and Theorem VII.4.1. □

This is a covolume asymptotic for the full normalized image. It does not bound the size of an individual fixed-trajectory output or its error.

VII.5 The integer Gamma kernel: its quotient and two positivity barriers

Let $H_0 = 0$ and $H_k = \sum_{j=1}^{k} j^{-1}$ for $k \ge 1$. For a real polynomial $P(t) = \sum_{k=0}^{d} p_k t^k$, put

$$B(P) = \int_0^\infty P(t) e^{-t} \, \mathrm{d}t = \sum_{k=0}^{d} p_k k!, \tag{VII.5.1}$$

$$A(P) = \sum_{k=0}^{d} p_k k! H_k, \qquad L(P) = \int_0^\infty P(t) e^{-t} \log t \, \mathrm{d}t = A(P) - \gamma B(P). \tag{VII.5.2}$$

The last identity follows from $\Gamma'(k+1) = k!(H_k - \gamma)$. In particular, $A(P), B(P) \in \mathbb{Z}$ when $P \in \mathbb{Z}[t]$. This integrality concerns the two coefficients separately and makes no assumption about the arithmetic nature of γ.

VII.5.1 An exact quotient over the integers

Define the $\mathbb{Z}$-linear differential operator

$$T = t \frac{\mathrm{d}}{\mathrm{d}t} + 1 - t.$$

Its image is understood as an additive subgroup of $\mathbb{Z}[t]$.

Lemma VII.5.1. *For every $P \in \mathbb{Z}[t]$,*

$$B(TP) = 0, \qquad A(TP) = -B(P), \qquad A(T^2 P) = B(T^2 P) = 0.$$

Proof. The identity

$$e^{-t} TP(t) = \frac{\mathrm{d}}{\mathrm{d}t} \left(t e^{-t} P(t) \right)$$

gives $B(TP) = 0$. Integration by parts also gives

$$L(TP) = -\int_0^\infty P(t) e^{-t} \, \mathrm{d}t = -B(P).$$

The logarithmic boundary term vanishes at zero because $tP(t)\log t \to 0$, and at infinity because of the exponential factor. Equation (VII.5.2) now gives $A(TP) = -B(P)$. Applying these two identities once more proves the assertion for T^2P. □

Proposition VII.5.2 (Integer normal form)**.** *Every $P \in \mathbb{Z}[t]$ has a unique representation*

$$P = (B(P) - A(P)) + A(P)t + T^2R, \qquad R \in \mathbb{Z}[t]. \tag{VII.5.3}$$

Consequently,

$$\ker\big((A,B) : \mathbb{Z}[t] \longrightarrow \mathbb{Z}^2\big) = T^2\mathbb{Z}[t], \qquad \mathbb{Z}[t]/T^2\mathbb{Z}[t] \simeq \mathbb{Z}^2$$

as abelian groups.

Proof. For $k \geq 0$ a direct calculation gives

$$\begin{aligned} T(t^k) &= (k+1)t^k - t^{k+1}, \\ T^2(t^k) &= t^{k+2} - (2k+3)t^{k+1} + (k+1)^2t^k. \end{aligned}$$

The second polynomial is monic. Successively subtracting integer multiples of these polynomials reduces any $P \in \mathbb{Z}[t]$ to $c + dt$ without introducing denominators. The reduction is unique: if $R \neq 0$, then $\deg(T^2R) = \deg R + 2$, so T^2R cannot have degree at most one. Lemma VII.5.1 and $A(c+dt) = d$, $B(c+dt) = c+d$ identify $d = A(P)$ and $c = B(P) - A(P)$. These observations also prove the kernel assertion. Every pair $(a,b) \in \mathbb{Z}^2$ is represented by $(b-a) + at$, proving the stated isomorphism. □

Remark VII.5.3. The quotient above is an additive quotient, not a quotient ring. For example, $T^2(1) = t^2 - 3t + 1$ belongs to the pair kernel, whereas $tT^2(1)$ has $(A,B) = (3,1)$ and does not. It is equally essential to distinguish this pair kernel from $\ker L$. If $\gamma = a/b \in \mathbb{Q}$ with $b > 0$, then $(b-a) + at$ lies in $\ker L$ but has nonzero pair $(A,B) = (a,b)$. Thus identifying the two kernels would already assume the irrationality that the construction is intended to investigate.

VII.5.2 A sharp lower bound under pointwise positivity

We first record a real-polynomial estimate. For $\alpha \in \{0,1\}$, write

$$\|R\|_\alpha^2 = \int_0^\infty R(t)^2 t^\alpha e^{-t}\,\mathrm{d}t.$$

Lemma VII.5.4. *If $R \in \mathbb{R}[t]$ has degree m and leading coefficient a, then*

$$\|R\|_\alpha^2 \geq a^2 m!\Gamma(m+\alpha+1), \tag{VII.5.4}$$

$$\int_0^\infty tR(t)^2 t^\alpha e^{-t}\,\mathrm{d}t \leq (4m+2\alpha+2)\|R\|_\alpha^2, \tag{VII.5.5}$$

$$\int_0^\infty \frac{R(t)^2}{1+t} t^\alpha e^{-t}\,\mathrm{d}t \geq \frac{a^2 m!\Gamma(m+\alpha+1)}{4m+2\alpha+3}. \tag{VII.5.6}$$

Proof. The Rodrigues formula

$$L_j^{(\alpha)}(t) = \frac{t^{-\alpha}e^t}{j!}\frac{\mathrm{d}^j}{\mathrm{d}t^j}(e^{-t}t^{j+\alpha})$$

gives leading coefficient $(-1)^j/j!$ and, by integration by parts,

$$\int_0^\infty L_j^{(\alpha)}(t)L_k^{(\alpha)}(t)t^\alpha e^{-t}\,\mathrm{d}t = \frac{\Gamma(j+\alpha+1)}{j!}\,\delta_{jk}.$$

The monic polynomial $\ell_m = (-1)^m m! L_m^{(\alpha)}$ therefore has norm squared $m!\Gamma(m+\alpha+1)$. Writing $R = a\ell_m + R_0$ with $\deg R_0 < m$ and using orthogonality proves (VII.5.4).

Choose the orthonormal Laguerre polynomials with positive leading coefficients and denote them by p_j. Their recurrence is

$$tp_j = \sqrt{(j+1)(j+\alpha+1)}\, p_{j+1} + (2j+\alpha+1)p_j + \sqrt{j(j+\alpha)}\, p_{j-1},$$

where $p_{-1} = 0$. Expand $R = \sum_{j=0}^{m} c_j p_j$ and apply $2|c_j c_{j+1}| \le c_j^2 + c_{j+1}^2$ to the off-diagonal terms. The bounds

$$\sqrt{j(j+\alpha)} \le j + \frac{\alpha}{2}, \qquad \sqrt{(j+1)(j+\alpha+1)} \le j + 1 + \frac{\alpha}{2}$$

show that the coefficient of each c_j^2 is at most $4j + 2\alpha + 2 \le 4m + 2\alpha + 2$. This proves (VII.5.5). Finally, Cauchy–Schwarz gives

$$\int_0^\infty \frac{R^2}{1+t} t^\alpha e^{-t}\,\mathrm{d}t \ge \frac{\|R\|_\alpha^4}{\displaystyle\int_0^\infty (1+t)R^2 t^\alpha e^{-t}\,\mathrm{d}t} \ge \frac{\|R\|_\alpha^2}{4m+2\alpha+3}.$$

Combine this with (VII.5.4). □

Theorem VII.5.5 (Pointwise positivity barrier)**.** *Suppose that* $0 \ne P \in \mathbb{Z}[t]$ *and*

$$P(t)\log t \ge 0 \qquad (t > 0).$$

Then $L(P) \ge 1$*, and equality holds if and only if* $P = t - 1$*. More generally, if* $P(t)\log t$ *has one fixed sign on* $(0,\infty)$*, then* $|L(P)| \ge 1$*, with equality only for* $P = \pm(t-1)$*.*

Proof. Continuity and the change of sign of $\log t$ force $P(1) = 0$. Division by the monic polynomial $t - 1$ gives $P = (t-1)Q$, with $Q \in \mathbb{Z}[t]$ nonzero and nonnegative on $[0,\infty)$. Set

$$w(t) = (t-1)\log t\, e^{-t}, \qquad L(P) = \int_0^\infty w(t)Q(t)\,\mathrm{d}t.$$

Every real polynomial nonnegative on this half-line has a representation

$$Q = A^2 + tB^2, \qquad A, B \in \mathbb{R}[t]. \tag{VII.5.7}$$

For completeness, factor Q over $\mathbb{R}$. Positive real roots have even multiplicity; their factors are squares. A negative-root factor $t + c$, $c > 0$, is $(\sqrt{c})^2 + t$, and t itself has the required form. An irreducible monic quadratic $t^2 + bt + c$ has $c > 0$ and $b + 2\sqrt{c} > 0$, and can be written as

$$(t - \sqrt{c})^2 + t(b + 2\sqrt{c}).$$

The identity

$$(A^2 + tB^2)(C^2 + tD^2) = (AC - tBD)^2 + t(AD + BC)^2$$

completes the factorization argument, including the positive leading constant. There is no cancellation between the leading terms of the two summands in (VII.5.7). Thus if $\deg Q = 2d$ and $a = \mathrm{lc}(Q)$, then $\deg A = d$ and $\mathrm{lc}(A)^2 = a$; if $\deg Q = 2d+1$, then $\deg B = d$ and $\mathrm{lc}(B)^2 = a$. In both cases a is an integer at least one.

The elementary inequality

$$(t-1)\log t \ge \frac{2(t-1)^2}{1+t} \qquad (t > 0) \tag{VII.5.8}$$

follows because $f(t) = \log t - 2(t-1)/(t+1)$ satisfies $f(1) = 0$ and $f'(t) = (t-1)^2/[t(t+1)^2] \geq 0$. Apply Lemma VII.5.4 to $(t-1)A$ with $\alpha = 0$, and to $(t-1)B$ with $\alpha = 1$. When the relevant polynomial has degree d, the resulting bounds are

$$\int_0^\infty wA^2 \,\mathrm{d}t \geq \frac{2\,\mathrm{lc}(A)^2((d+1)!)^2}{4d+7}, \tag{VII.5.9}$$

$$\int_0^\infty wtB^2 \,\mathrm{d}t \geq \frac{2\,\mathrm{lc}(B)^2(d+1)!(d+2)!}{4d+9}. \tag{VII.5.10}$$

The right side of (VII.5.9), with leading coefficient squared at least one, exceeds one for $d \geq 2$: its value at $d = 2$ is $72/15$, and it increases thereafter. The corresponding right side of (VII.5.10) exceeds one for $d \geq 1$, starting at $24/13$ and increasing. These estimates settle every case with $\deg Q \geq 3$ and leave only degrees zero, one and two.

The moments needed for these remaining cases are

$$\int_0^\infty w \,\mathrm{d}t = 1, \qquad \int_0^\infty tw \,\mathrm{d}t = 2 - \gamma, \qquad \int_0^\infty t^2 w \,\mathrm{d}t = 8 - 4\gamma. \tag{VII.5.11}$$

They follow directly from (VII.5.2). The elementary harmonic-sum bounds give $0 < \gamma < 1$. If $Q = c$ has degree zero, then $c \in \mathbb{N}$ and $L(P) = c$. If $Q = at + b$ has degree one, its nonnegativity gives integers $a \geq 1$, $b \geq 0$, whence $L(P) = a(2-\gamma) + b > 1$. Finally, in degree two write $A = ut + v$ in (VII.5.7); here $u^2 = a \geq 1$. Equation (VII.5.11) gives

$$\int_0^\infty w(ut+v)^2 \,\mathrm{d}t = \big(v + u(2-\gamma)\big)^2 + u^2(4-\gamma^2) > 3a.$$

The other summand $\int wtB^2$ is nonnegative. Every nonconstant Q therefore gives a strict inequality, and the constant case has equality precisely when $Q = 1$. Applying the result to $-P$ proves the assertion for the opposite sign. □

Remark VII.5.6. The hypothesis concerns the sign of the original integrand, not merely the sign of its integral. For example, $P = t$ gives $L(P) = 1 - \gamma \in (0, 1)$ while $P(t)\log t$ changes sign. The theorem excludes a sequence of small integer forms certified by pointwise positivity in this particular polynomial Gamma kernel. It does not exclude positivity obtained after a transformation that uses cancellation, or positivity in a different integral representation.

VII.5.3 A uniform obstruction to absolute-integral bounds

Lemma VII.5.7 (A monic L^1 estimate)**.** *If a real polynomial q has degree d and leading coefficient a, then*

$$\int_{-1}^{1} |q(x)| \,\mathrm{d}x \geq |a| 2^{1-d}.$$

Consequently, every real polynomial P satisfies $\int_2^6 |P(t)| \,\mathrm{d}t \geq 4|\,\mathrm{lc}(P)|$.

Proof. Let U_d be the Chebyshev polynomial determined by $U_d(\cos\theta) = \sin((d+1)\theta)/\sin\theta$; its leading coefficient is 2^d. Put $s(x) = \operatorname{sign} U_d(x)$, with arbitrary values at its zeros. This bounded function is orthogonal, for Lebesgue measure on $[-1, 1]$, to every polynomial of degree less than d. Here is a finite verification. Set $N = d + 1$. After $x = \cos\theta$, each such polynomial multiplied by $\sin\theta$ is a linear combination of $\sin(k\theta)$ with $1 \leq k < N$. Integrating over the N intervals on which $\operatorname{sign}\sin(N\theta)$ is constant gives

$$\int_0^\pi \sin(k\theta)\operatorname{sign}\sin(N\theta) \,\mathrm{d}\theta = \frac{1}{k}\operatorname{Re}\frac{(1+r)(1-r^N)}{1-r} = 0, \qquad r = -e^{ik\pi/N}.$$

Indeed, $r \neq 1$, $|r| = 1$, and $r^N = (-1)^{N+k}$ is real, whereas $(1+r)/(1-r)$ is purely imaginary.

Write $q = a2^{-d}U_d + q_0$, where $\deg q_0 < d$. The orthogonality just proved and

$$\int_{-1}^{1} |U_d(x)|\,\mathrm{d}x = \int_0^{\pi} |\sin(N\theta)|\,\mathrm{d}\theta = 2$$

give $\int_{-1}^{1} q(x)s(x)\,\mathrm{d}x = a2^{1-d}$. Since $|s| \le 1$, taking absolute values proves the estimate. Apply it to $q(x) = P(4+2x)$ and use $\mathrm{d}t = 2\,\mathrm{d}x$ to obtain the assertion on $[2,6]$. □

Theorem VII.5.8 (Absolute-integral barrier)**.** *For every nonzero $P \in \mathbb{Z}[t]$,*

$$\int_0^{\infty} |P(t)\log t|e^{-t}\,\mathrm{d}t \ge c_0, \qquad c_0 = 4\log 6\, e^{-6} = 0.0177653107376464\ldots. \tag{VII.5.12}$$

More generally, if $0 \ne P \in \mathbb{Q}[t]$ and $D \in \mathbb{N}$ satisfies $DP \in \mathbb{Z}[t]$, then

$$D\int_0^{\infty} |P(t)\log t|e^{-t}\,\mathrm{d}t \ge c_0.$$

Proof. On $[2,6]$ the function $\log t\, e^{-t}$ is decreasing and is at least $\log 6\, e^{-6}$. Restricting the integral to this interval and applying Lemma VII.5.7 gives

$$\int_0^{\infty} |P(t)\log t|e^{-t}\,\mathrm{d}t \ge \log 6\, e^{-6}\int_2^6 |P(t)|\,\mathrm{d}t \ge 4\log 6\, e^{-6}|\operatorname{lc}(P)|.$$

For a nonzero integer polynomial $|\operatorname{lc}(P)| \ge 1$. Applying this conclusion to DP proves the rational-coefficient case. □

Remark VII.5.9 (Exact scope of the obstruction). Suppose that a proposed construction clears the coefficients of P_n with $D_nP_n \in \mathbb{Z}[t]$ and then estimates its form solely by

$$D_n|L(P_n)| \le D_n\int_0^{\infty} |P_n(t)\log t|e^{-t}\,\mathrm{d}t.$$

The right-hand side cannot tend to zero, regardless of the degrees or coefficients chosen. This remains true after adding any integer null-kernel term T^2R_n to an integer polynomial. The conclusion concerns this absolute-integral estimate; it is not a lower bound for the signed form $|L(P_n)|$. Nor does it automatically apply when the actual denominators of $A(P_n), B(P_n)$ are smaller than the denominators needed to clear the coefficients of P_n. A procedure using that arithmetic discrepancy, or using cancellation after integration by parts, requires a separate analysis. Together with Theorem VII.5.5, the result identifies two precise classes of kernel estimates that cannot supply the required small nonzero integer forms.

VII.6 Exact arithmetic of Dickman moments

The Dickman construction leads naturally to rational moments and to the normalization $e^{-\gamma}$. Here we derive exact local congruences and common-denominator estimates for those moments. The results also quantify the effects of fixed polynomial transformations and of an integer convolution parameter. These are arithmetic consequences of the classical moment formula, without a claim of historical priority.

Let D have the normalized Dickman–de Bruijn distribution, with density $e^{-\gamma}\rho(x)$ on $[0,\infty)$, where $\rho(x) = 1$ for $0 \le x \le 1$. Write $m_n = \mathbb{E}[D^n]$. The classical generating function and set-partition formula are

$$M(z) = \sum_{n\ge 0} m_n\frac{z^n}{n!} = \exp\left(\sum_{k\ge 1}\frac{z^k}{k\,k!}\right), \tag{VII.6.1}$$

$$m_n = \sum_{\pi\in\Pi_n}\prod_{B\in\pi}\frac{1}{|B|}, \tag{VII.6.2}$$

where Π_n is the set of partitions of $\{1,\dots,n\}$ and the empty partition gives $m_0 = 1$; see [VII.4, pp. 257–258]. In particular,

$$(m_0,\dots,m_6) = \left(1, 1, \frac{3}{2}, \frac{17}{6}, \frac{19}{3}, \frac{81}{5}, \frac{8351}{180}\right).$$

Differentiating (VII.6.1) gives the rational recurrence

$$nm_n = \sum_{j=0}^{n-1} \binom{n}{j} m_j \qquad (n \geq 1). \tag{VII.6.3}$$

For a prime p, let $v_p(p) = 1$ and $\mathbb{Z}_{(p)} = \{a/b \in \mathbb{Q} : p \nmid b\}$. We use $v_p(0) = +\infty$ and $\operatorname{den}(0) = 1$.

VII.6.1 An exact congruence at every odd prime

Lemma VII.6.1. *For every prime p and every $n \geq 0$,*

$$v_p(m_n) \geq -\lfloor n/p \rfloor.$$

Proof. For every positive integer s, $v_p(s) \leq s/p$: if $v_p(s) = a \geq 1$, then $s \geq p^a \geq pa$. Thus each term of (VII.6.2), with block sizes $s_1,\dots,s_t$, has denominator valuation at most $\sum_i v_p(s_i) \leq \lfloor n/p \rfloor$. Taking a sum preserves this bound. □

Theorem VII.6.2 (Odd-prime moment congruence)**.** *Let p be an odd prime, and write $n = qp+r$, where $q \geq 0$ and $0 \leq r < p$. Then*

$$p^q m_n \equiv m_r \pmod{p\mathbb{Z}_{(p)}}. \tag{VII.6.4}$$

Consequently,

$$v_p(m_n) = -q \iff p \nmid \operatorname{num}(m_r), \tag{VII.6.5}$$

where the numerator is taken in lowest terms.

Proof. For a partition with block sizes $s_1,\dots,s_t$, put $v = \sum_i v_p(s_i) \leq q$. If $v < q$, its contribution to $p^q m_n$ vanishes modulo $p\mathbb{Z}_{(p)}$. If $v = q$, then

$$\sum_i (s_i - pv_p(s_i)) = r < p. \tag{VII.6.6}$$

Each summand is nonnegative. Moreover, if $p \mid s$ and $s \neq p$, then $s - pv_p(s) \geq p$. Indeed, for $v_p(s) = 1$ one has $s \geq 2p$; for $v_p(s) = a \geq 2$ one has $s - pa \geq p(p^{a-1} - a) \geq p$, since $p \geq 3$. Equation (VII.6.6) therefore forces all p-divisible block sizes to equal p. There are exactly q such blocks, and the remaining blocks partition a set of size r.

The total contribution of these partitions is

$$p^{-q} C_{n,p} m_r, \qquad C_{n,p} = \frac{n!}{r!\, q!\, (p!)^q}.$$

The coefficient counts the choice of the remaining r elements and the partition of the other elements into q unordered blocks of size p. To compute its residue one must account for the possible divisibility of $q!$ by p. Separating the multiples of p in $n!$ gives the exact identity

$$C_{n,p} = \frac{\displaystyle\prod_{\substack{1 \leq j \leq n \\ p \nmid j}} j}{r!\, ((p-1)!)^q}. \tag{VII.6.7}$$

The numerator and denominator on the right are units in $\mathbb{Z}_{(p)}$. The q complete blocks of nonmultiples of p each have residue product $(p-1)!$, and the final incomplete block has residue product $r!$. Hence $C_{n,p} \equiv 1 \pmod{p}$. All other partitions vanish after multiplication by p^q and reduction, proving (VII.6.4). Since $m_r \in \mathbb{Z}_{(p)}$ for $r < p$, the valuation criterion follows. □

Remark VII.6.3. The exceptional residue classes have an exact finite description for each p: they are the $r \in \{0, \dots, p-1\}$ with $p \mid \mathrm{num}(m_r)$. For example, the sole exceptional classes for $p = 3$ and $p = 7$ are respectively $r = 2$ and $r = 6$. Theorem VII.6.2 does not determine the higher valuation within an exceptional class. Its restriction to odd primes is essential to the proof: at $p = 2$, both block sizes 2 and 4 satisfy $s - 2v_2(s) = 0$, allowing additional cancellations.

VII.6.2 Common denominators and fixed polynomial transformations

Corollary VII.6.4 (Exact odd common denominator)**.** *Let $L_N = \mathrm{lcm}(\mathrm{den}(m_0), \dots, \mathrm{den}(m_N))$, and let L_N^{odd} denote its odd part. Then*

$$L_N^{\mathrm{odd}} = \prod_{\substack{3 \le p \le N \\ p \text{ prime}}} p^{\lfloor N/p \rfloor}, \tag{VII.6.8}$$

and

$$\log L_N^{\mathrm{odd}} = N \log N + O(N), \qquad \log L_N = N \log N + O(N). \tag{VII.6.9}$$

Proof. Lemma VII.6.1 bounds the exponent of p in L_N by $\lfloor N/p \rfloor$. For odd $p \le N$, take $n = p\lfloor N/p \rfloor$. The remainder is zero, so Theorem VII.6.2, with $m_0 = 1$, attains that exponent. This proves (VII.6.8).

The standard Mertens and Chebyshev estimates

$$\sum_{p \le x} \frac{\log p}{p} = \log x + O(1), \qquad \sum_{p \le x} \log p = O(x)$$

give

$$\sum_{3 \le p \le N} \lfloor N/p \rfloor \log p = N \sum_{3 \le p \le N} \frac{\log p}{p} + O(N) = N \log N + O(N).$$

Finally $v_2(L_N) \le \lfloor N/2 \rfloor$, so the power of 2 contributes only $O(N)$ to the logarithm. □

Theorem VII.6.5 (Polynomial transformation law)**.** *Fix a nonconstant $g \in \mathbb{Q}[X]$ of degree $d \ge 1$, and put*

$$u_n = \mathbb{E}[g(D)^n], \qquad L_N(g) = \mathrm{lcm}(\mathrm{den}(u_0), \dots, \mathrm{den}(u_N)).$$

Then

$$\log L_N(g) = dN \log N + O_g(N). \tag{VII.6.10}$$

More precisely, write $g = h/A$, where $A \ge 1$ is an integer and $h \in \mathbb{Z}[X]$ has leading coefficient $c \ne 0$. For every odd prime $p \nmid Ac$ and every $q \ge 0$,

$$v_p(u_{pq}) = -dq. \tag{VII.6.11}$$

Proof. Expand $h(X)^n = \sum_{k=0}^{dn} b_{n,k} X^k$, with $b_{n,k} \in \mathbb{Z}$ and $b_{n,dn} = c^n$. For every prime p,

$$v_p(u_n) \ge -n v_p(A) - \lfloor dn/p \rfloor. \tag{VII.6.12}$$

It follows simultaneously for $n \le N$ that

$$L_N(g) \mid A^N \prod_{p \le dN} p^{\lfloor dN/p \rfloor}.$$

The estimates used in Corollary VII.6.4 therefore give $\log L_N(g) \le dN \log N + O_g(N)$.

For the reverse bound, let $p \nmid 2Ac$ and put $n = pq$. The leading term of $p^{dq} u_n$ has residue

$$A^{-n} c^n p^{dq} m_{dpq} \equiv (c/A)^n \not\equiv 0 \pmod{p}.$$

For every lower degree $k \le dpq - 1$, one has $\lfloor k/p \rfloor \le dq - 1$. Thus $p^{dq} b_{n,k} m_k \in p\mathbb{Z}_{(p)}$ by Lemma VII.6.1; multiplication by A^{-n} preserves this divisibility. All lower terms vanish modulo p, proving (VII.6.11). Taking $q = \lfloor N/p \rfloor$ now yields

$$\log L_N(g) \ge d \sum_{\substack{p \le N \\ p \nmid 2Ac}} \lfloor N/p \rfloor \log p = dN \log N + O_g(N).$$

Omitting the finitely many primes dividing $2Ac$ costs $O_g(N)$. The lower and upper bounds match. □

Corollary VII.6.6 (Exact local radii)**.** *With the notation of Theorem VII.6.5, for every odd prime $p \nmid Ac$ the formal series*

$$U_g(z) = \sum_{n \ge 0} u_n z^n, \qquad V_g(z) = \sum_{n \ge 0} u_n \frac{z^n}{n!}$$

have exact p-adic convergence radii

$$R_p(U_g) = p^{-d/p}, \qquad R_p(V_g) = p^{-d/p - 1/(p-1)}. \tag{VII.6.13}$$

In particular, for the classical Dickman moment generating function, $R_p(M) = p^{-1/p - 1/(p-1)}$ at every odd prime.

Proof. Equations (VII.6.12) and (VII.6.11) imply $\liminf_{n \to \infty} v_p(u_n)/n = -d/p$. The p-adic Cauchy–Hadamard formula gives the first radius. Legendre's formula gives $v_p(n!)/n \to 1/(p-1)$; subtracting this limit gives the second. □

Remark VII.6.7. The common-denominator law is stronger than a finite observation about individual moments, but its quantifiers matter. It does not give a lower bound for every individual denominator or every selected linear combination of moments. Fixed rational translations and rescalings retain the leading cost $N \log N$; a fixed degree-d polynomial transformation changes it to $dN \log N$. Transformations depending on N are outside Theorem VII.6.5. For $d \ge 2$, the series V_g need not converge for positive real z; Corollary VII.6.6 treats it as a formal rational series and asserts only its p-adic radius.

VII.6.3 Moving integer shape and the remaining normalization

For a positive integer θ, let D_θ be the sum of θ independent copies of D, and write $m_n(\theta) = \mathbb{E}[D_\theta^n]$. Taking the θth power in (VII.6.1) shows that its cumulants are

$$\kappa_n(\theta) = \frac{\theta}{n} \qquad (n \ge 1). \tag{VII.6.14}$$

Theorem VII.6.8 (Exact shape cost of integer moments)**.** *For $\theta \in \mathbb{Z}_{>0}$ and $N \ge 1$,*

$$m_0(\theta), \dots, m_N(\theta) \in \mathbb{Z} \iff \operatorname{lcm}(1, \dots, N) \mid \theta. \tag{VII.6.15}$$

The least positive such shape is therefore $\operatorname{lcm}(1, \dots, N)$*, whose logarithm is $N + o(N)$.*

Proof. The moment–cumulant formula

$$m_n = \sum_{\pi \in \Pi_n} \prod_{B \in \pi} \kappa_{|B|}$$

is a polynomial with integer coefficients. Thus divisibility by $\operatorname{lcm}(1,\dots,N)$ makes all cumulants in (VII.6.14), and hence all moments through N, integers. Conversely, the inverse formula

$$\kappa_n = \sum_{\pi\in\Pi_n} (-1)^{|\pi|-1}(|\pi|-1)! \prod_{B\in\pi} m_{|B|}$$

has integer coefficients as well. Integer moments through N force $\theta/n \in \mathbb{Z}$ for every $1 \le n \le N$, which is precisely the required divisibility. The logarithmic asymptotic for the least common multiple is the usual prime number theorem. □

This construction does remove moment denominators, but its connection to Euler's constant retains an exponential normalization. Let f_θ be the density of D_θ. For $0 < x \le 1$, every coordinate in its convolution simplex lies in $[0,1]$, where the original density is $e^{-\gamma}$. The simplex volume therefore gives the exact formula

$$f_\theta(x) = \frac{e^{-\gamma\theta}x^{\theta-1}}{(\theta-1)!}. \tag{VII.6.16}$$

For $P(X) = \sum_k a_k X^k \in \mathbb{Q}[X]$, it follows that

$$\int_1^\infty P(x)f_\theta(x)\,\mathrm{d}x = \mathbb{E}[P(D_\theta)] - \frac{e^{-\gamma\theta}}{(\theta-1)!}\sum_k \frac{a_k}{\theta+k}. \tag{VII.6.17}$$

If $P \in \mathbb{Z}[X]$, $\deg P \le N$, and the divisibility condition in Theorem VII.6.8 holds, the unrestricted expectation in (VII.6.17) is an integer. This produces explicit rational linear forms in 1 and $e^{-\gamma\theta}$.

Remark VII.6.9 (Arithmetic scope of the construction). No estimate making the forms (VII.6.17) sufficiently small has been established here. Moreover, their direct target is $e^{-\gamma\theta}$: if γ were a nonzero algebraic number, this target would already be transcendental by Hermite–Lindemann. Proving only its irrationality or transcendence would consequently not exclude algebraic γ. Taking logarithms does not preserve the rational linear-form structure or its denominator bounds. Thus the exact congruence, transformation laws, and shape construction resolve specific arithmetic questions about the moment method while leaving the required transfer to γ unproved.

For a finite consistency check, recurrence (VII.6.3) was evaluated with exact rational arithmetic. Among the 10,088 pairs (n,p) with $1 \le n \le 300$ and $3 \le p \le n$ prime, exactly 9,822 attained the denominator exponent $\lfloor n/p \rfloor$. Each of the remaining 266 pairs satisfies $p \mid \operatorname{num}(m_{n \bmod p})$, as predicted by Theorem VII.6.2. A further 47,094 congruence checks, covering all odd primes $p \le 500$ and $0 \le n \le 500$, had no failures. These computations corroborate the formulas; the proofs above supply their unrestricted quantifiers.

VII.7 What the attempted escapes achieved, and where they stop

VII.7.1 Adaptive lattice compression: a concrete failed combination

The exact content of the projected trajectory is much larger than one. This suggested removing all common factors and then using the freedom of the initial vector to obtain a short output pair. Theorems above carry out the arithmetic part completely. What fails is an automatic transfer of the approximation estimate.

There is a quantitative version of this distinction. With $D = D_n$ and $C = C_n$, the normalized forms are

$$\Lambda = D\left[a - b\left(\gamma - \frac{C}{D}\right)\right], \qquad (a,b) \in \mathbb{Z}^2, \quad \gcd(C,D) = 1.$$

Dirichlet's elementary approximation theorem, with $1 \le b \le B$, gives an output satisfying $|\Lambda| \le D/B$, but allows $\Lambda = 0$. If $\gamma = r/s$ with $s > 0$, an exact zero implies

$$D \mid sCb, \qquad \text{hence} \qquad D \mid sb.$$

Thus $0 < |b| < D/s$ would certify nonvanishing under that hypothesis. The two proposed bounds pull in opposite directions: making the Dirichlet upper bound tend to zero requires $B/D \to \infty$, whereas this modular nonvanishing certificate requires $B < D/s$.

This calculation discards the combination of *these* two estimates. It does not rule out better approximation estimates or a different nonvanishing argument. Nor does it rule out carefully restricted initial vectors for which an analytic estimate survives. Such a construction would have to prove the restriction's arithmetic and analytic effects together.

VII.7.2 Finite evaluation of a fixed trajectory

For a reproducible numerical comparison, let $v = (0, 0, 1)^T$ and reduce

$$\frac{\mathbf{p}_n v}{\mathbf{q}_n v} = \frac{a_n}{b_n}, \qquad b_n > 0, \quad \gcd(a_n, b_n) = 1.$$

The matrix product and the reduction are exact integer operations. The following real logarithms were then evaluated with 200-decimal-digit arithmetic; the displayed values are rounded.

n	$\log_{10} b_n$	$\log_{10} \lvert\gamma - a_n/b_n\rvert$	$\log_{10} \lvert b_n\gamma - a_n\rvert$	τ_n
10	13.2581	−7.6011	5.6571	−0.4267
20	30.9021	−11.9023	18.9998	−0.6148
40	74.0625	−20.5465	53.5159	−0.7226
80	171.6475	−33.4311	138.2165	−0.8052
120	274.2720	−43.8412	230.4308	−0.8402
200	499.1524	−62.1680	436.9844	−0.8755
300	794.2474	−82.3402	711.9072	−0.8963

Here

$$\tau_n = -1 - \frac{\log |\gamma - a_n/b_n|}{\log b_n}.$$

The table distinguishes a small rational-approximation error from the required small integer linear form. For this sample, the latter is large despite increasingly accurate rational values. These finite data do not prove an asymptotic lower bound, do not exclude favorable subsequences, and do not describe every trajectory or every input vector.

The analytic exponent relevant to this rank-three/Rivoal trajectory is $n^{2/3}$, not the $n^{3/4}$ associated with a different approximation family. No conclusion here multiplies a mere upper error bound by a denominator and treats the result as a lower bound.

VII.7.3 Corrections when reusing the supplied arguments

Several statements in the supplied `gamma_attempts.tex` need a narrower reading or a small correction before they can be reused in a proof.

Here $\mathbf{G}$ is the ring of finite values of Siegel G-functions at algebraic points, allowing analytic continuation; $\mathbf{E}$ is the analogous ring of E-function values, and $\mathbf{D}$ denotes the ring of Borel-summed arithmetic Gevrey values used in [VII.3].

1. **A cancellation template is not a universal obstruction.** The expectation $\gamma \notin \mathrm{Frac}(\mathbf{G})$ is unproved. Moreover, $\gamma \in \mathrm{Frac}(\mathbf{G})$ follows automatically under the reductio assumption $\gamma \in \overline{\mathbb{Q}}$. A membership prerequisite therefore cannot, by itself, exclude a construction

made only under that assumption. Theorem VII.3.1 is unconditional but concerns only its specified differential module.

2. **The cited G-function theorem needs its hypotheses.** The ratio theorem in [VII.2] assumes a nonzero constant ξ. At $\xi = 0$, the function $A - \xi B = A$ cannot simultaneously have radius 1 and radius greater than $R \geq 1$. In continuation arguments, choose algebraic points in open neighborhoods, not necessarily on an arbitrary given path. Algebraicity of a Wronskian normalization also needs the normalized Frobenius basis; a rational logarithmic derivative alone leaves an arbitrary multiplicative constant.

3. **Twist invariance has a defined scope.** For $\Gamma_\tau(s) = e^{\tau(1-s)}\Gamma(s)$, the digamma difference identities can remain invariant. The original Gamma recurrence does not:

$$\Gamma_\tau(s+1) = e^{-\tau} s \Gamma_\tau(s).$$

   For real τ, requiring the original normalized recurrence forces $\tau = 0$. The invariance argument therefore restricts arguments using only the invariant identities, not every normalized Gamma construction.

4. **Denominator quantifiers cannot be interchanged.** An upper bound on a common denominator is not a lower bound on an individual reduced denominator. A finite table does not settle every degree. The Dickman results here prove an exact odd common denominator and a local congruence; they deliberately do not assert the same leading denominator exponent for every individual moment or arbitrary linear combination. Similarly, the exact content formula concerns RP_n. Equality with the content of the full 3×3 matrix P_n is not proved or used.

5. **The Gevrey statements are conditional at different levels.** The conjecture $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ implies the transcendence of γ. The stronger exclusion $\gamma \notin \mathbf{E}$ in the cited work uses an additional divergent-series zero-division conjecture [VII.3]. Neither conjecture is proved in this supplement.

VII.7.4 What remains open after these calculations

The Meijer identity now provides exact coefficient vectors for $\gamma + \operatorname{Log} z$; the matrix calculation provides the full projected lattice and its exact content; the kernel calculation provides all integer null directions; and the moment calculation provides exact local denominator costs. None supplies both strict nonvanishing and decay after the necessary arithmetic normalization.

A further construction would need an ingredient beyond the discarded estimates: for example, an explicitly restricted adaptive input whose height and approximation direction can both be proved, or a transformed integral whose true coefficient pair has much smaller denominators than its integrand coefficients and whose signed cancellation can be controlled. These are descriptions of missing mechanisms, not new sufficient conditions advertised as a proof. The present work makes the tested routes more explicit and their unresolved arithmetic step checkable; the irrationality and transcendence of γ remain unproved here.

VII.A An independent algebraic certificate for the lattice rows

For completeness, the row formulas in Proposition VII.4.3 have a second verification using only rational functions and polynomial algebra. This also permits checking the harmonic coefficient without relying on the adjoint calculation.

Put $a = n + 1$ and define polynomial rows

$$h_n(x) = \left(x^2 + (1-3n)x + 3n^2 + 1,\ x + 1 - 3n,\ 1\right)$$

and $V_n = (V_{n,1}, V_{n,2}, V_{n,3})$, where

$$\begin{aligned} V_{n,1}(x) &= -a^2(10a^2+3a+6) + a(15a^2+3a+8)x - (6a^2+a+3)x^2, \\ V_{n,2}(x) &= 3a^2(5a-2) - a(21a-8)x + (8a-3)x^2, \\ V_{n,3}(x) &= -6a^2 + 8ax - 3x^2. \end{aligned}$$

Direct polynomial multiplication proves

$$(a-x)^3 V_n(x+1) - x^2 V_n(x) = a^4 h_{n+1}(x) - (a-x)^3 h_n(x) N_n. \tag{VII.A.1}$$

For a formal parameter ε set

$$W_n(j;\varepsilon) = \frac{n!^3}{(1+\varepsilon)_j^2 (1-\varepsilon)_{n-j}^3}, \qquad S_n(\varepsilon) = \sum_{j=0}^{n} W_n(j;\varepsilon) h_n(j+\varepsilon),$$

where $(x)_j = x(x+1)\cdots(x+j-1)$ and $(x)_0 = 1$. When $j = n+1, n+2$, use $(x)_{-r} = 1/[(x-r)\cdots(x-1)]$; the resulting W_n are rational functions vanishing to order three at $\varepsilon = 0$.

The exact adjacent-index ratios are

$$\frac{W_{n+1}(j;\varepsilon)}{W_n(j;\varepsilon)} = \frac{a^3}{(a-j-\varepsilon)^3}, \qquad \frac{W_n(j+1;\varepsilon)}{W_n(j;\varepsilon)} = \frac{(n-j-\varepsilon)^3}{(j+1+\varepsilon)^2}.$$

Multiply (VII.A.1) by $W_n(j;\varepsilon)/(a-j-\varepsilon)^3$ and take $x = j+\varepsilon$. Its two sides become the exact telescoping identity

$$\begin{aligned} a W_{n+1}(j;\varepsilon) h_{n+1}(j+\varepsilon) - W_n(j;\varepsilon) h_n(j+\varepsilon) N_n \\ = T_n(j+1;\varepsilon) - T_n(j;\varepsilon), \end{aligned}$$

with

$$T_n(j;\varepsilon) = \frac{W_n(j;\varepsilon)(j+\varepsilon)^2 V_n(j+\varepsilon)}{(a-j-\varepsilon)^3}.$$

Sum from $j = 0$ to $n+1$. The lower boundary is $O(\varepsilon^2)$, the upper boundary at $n+2$ is $O(\varepsilon^3)$, and the extra $W_n(n+1;\varepsilon)$ in the second sum is $O(\varepsilon^3)$. These statements include the apparent interior pole at $j = n+1$, which is handled as an identity of rational functions before summing. Therefore

$$(n+1) S_{n+1}(\varepsilon) - S_n(\varepsilon) N_n = O(\varepsilon^2).$$

Both constant and first-order coefficients obey the divided matrix recurrence. At $n = 0$ their initial rows are respectively $(1,1,1)$ and $(1,1,0)$. Since

$$W_n(j;0) = \binom{n}{j}^3 j!, \qquad W_n'(j;0) = \binom{n}{j}^3 j!\,(3H_{n-j} - 2H_j),$$

we obtain

$$\begin{aligned} \frac{\mathbf{q}_n}{n!} &= \sum_{j=0}^{n} \binom{n}{j}^3 j!\, h_n(j), \\ \frac{\mathbf{p}_n}{n!} &= \sum_{j=0}^{n} \binom{n}{j}^3 j!\, [(3H_{n-j} - 2H_j) h_n(j) + h_n'(j)]. \end{aligned}$$

These are precisely the row formulas already proved by the adjoint method. Their integral harmonic shear follows term by term as in Section VII.4.

VII.B Reproducible finite checks

The code below checks the exact row identities, integral shear, projected contents and maximal-minor gcds for all $0 \le n \le 80$. It also checks 47,094 Dickman congruences and five exact odd common denominators, and reproduces the fixed-column table through $n = 300$. The first two routines use only Python's exact integers and rational fractions. The last two use `mpmath`; the reported run used version 1.3.0. The table uses 200 decimal digits.

For the residue tower, 24 independent vertical-line quadratures at $p = 2, 3, 4$, $z \in \{1/2, 1, 2\}$ and $0 \le k \le \min(2, p-1)$ gave maximum absolute residual $6.0185311 \times 10^{-36}$ at 35-digit precision. The numerical integral is truncated at imaginary height 20. This is ordinary high-precision corroboration, not a rigorous interval bound on roundoff or tails. The contour proof establishes the identity.

The assertions compare independent representations rather than infer universal statements from finite agreement. Neither the quadratures nor the finite arithmetic checks are used to prove irrationality.

Copy the following listing into a Python file and run it. It is embedded here so the accompanying `.tex` file remains self-contained.

```
"""Exact checks and optional numerical checks for the supplement.
Run with Python 3.11+. Only the last two functions require mpmath.
These finite checks corroborate, and do not replace, the proofs.
"""
from fractions import Fraction as F
from math import comb, factorial, gcd, lcm, prod
from itertools import combinations

def N(n):
    return [
        [6*n*n + 13*n + 10, -8*n - 5, 3],
        [3*n**3 + 3*n*n - 3*n - 6, -3*n*n + 3*n + 3, n-2],
        [n**4 - 2*n**3 - 6*n*n - 3*n + 3,
         6*n*n + 2*n - 1, 1-2*n],
    ]

def multiply(A, B):
    return [[sum(x*y for x, y in zip(row, col))
             for col in zip(*B)] for row in A]

def harmonic(n):
    return sum((F(1, j) for j in range(1, n+1)), F(0))

def row_checks(limit=80):
    P = [[1,0,0], [0,1,0], [0,0,1]]
    R = [[1,1,0], [1,1,1]]
    H = [harmonic(j) for j in range(limit+1)]
    for n in range(limit+1):
        p, q = multiply(R, P)
        f = factorial(n)
        qsum, psum = [F(0)]*3, [F(0)]*3
        for k in range(n+1):
            w = comb(n,k)**3 * factorial(k)
            h = [k*k+(1-3*n)*k+3*n*n+1, k+1-3*n, 1]
            dh = [2*k+1-3*n, 1, 0]
            for j in range(3):
                qsum[j] += w*h[j]
                psum[j] += w*((3*H[n-k]-2*H[k])*h[j]+dh[j])
        assert [f*x for x in qsum] == q
        assert [f*x for x in psum] == p
        shifted = [F(x,f)-3*H[n]*F(y,f) for x,y in zip(p,q)]
        assert all(x.denominator == 1 for x in shifted)
        minors = [p[i]*q[j]-p[j]*q[i]
                  for i,j in combinations(range(3),2)]
        assert gcd(*minors) == f*f
        D = (3*H[n]).denominator
        assert gcd(*p, *q) == f//D
        assert gcd(*q) == f and gcd(*p) == f//D
```

```
        P = multiply(P, N(n))
    print("Exact lattice checks:", limit+1, "depths")

def moment_checks(limit=500):
    m = [F(1)]
    for n in range(1, limit+1):
        m.append(sum((comb(n,j)*m[j] for j in range(n)), F(0))/n)
    primes = [p for p in range(3, limit+1)
              if all(p%d for d in range(2, int(p**0.5)+1))]
    total = maximal = pairs300 = exceptions = 0
    for p in primes:
        for n in range(limit+1):
            q, r = divmod(n, p)
            delta = p**q*m[n]-m[r]
            assert delta.denominator % p
            assert delta.numerator % p == 0
            total += 1
            if 1 <= n <= 300 and p <= n:
                pairs300 += 1
                is_maximal = (p**q*m[n]).numerator % p != 0
                assert is_maximal == (m[r].numerator % p != 0)
                maximal += is_maximal
                exceptions += not is_maximal
    for depth in (10, 40, 100, 300, 500):
        if depth > limit:
            continue
        L = lcm(*(x.denominator for x in m[:depth+1]))
        while L % 2 == 0:
            L //= 2
        assert L == prod(p**(depth//p) for p in primes if p <= depth)
    print("Dickman congruences:", total)
    print("Pairs through 300, maximal, exceptional:",
          pairs300, maximal, exceptions)

def fixed_column_table(limit=300):
    import mpmath as mp
    mp.mp.dps = 200
    P = [[1,0,0], [0,1,0], [0,0,1]]
    selected = {10,20,40,80,120,200,300}
    for n in range(1, limit+1):
        P = multiply(P, N(n-1))
        if n not in selected:
            continue
        p = P[0][2]+P[1][2]
        q = p+P[2][2]
        a, b = F(p,q).as_integer_ratio()
        err = abs(mp.mpf(a)/b-mp.euler)
        height, logerr = mp.log10(b), mp.log10(err)
        print(n, *(mp.nstr(x,12) for x in
                   (height,logerr,height+logerr,-1-logerr/height)))

def mellin_checks():
    import mpmath as mp
    with mp.workdps(35):
        residuals = []
        for p in (2,3,4):
            for z in (mp.mpf("0.5"),mp.mpf(1),mp.mpf(2)):
                X = mp.euler+mp.log(z)
                rhs = [mp.mpf(1), X,
                       (X*X+(2*p+1)*mp.zeta(2))/2]
                for k in range(min(p,3)):
                    def integrand(y):
                        s = mp.mpc("-0.5",y)
                        phi = mp.gamma(-s)**(p+1)*mp.gamma(1+s)**p
                        op = z*(1+s)**(p-1-k)+(-1)**p*s**(p-k)
                        return mp.re(phi*mp.power(z,s)*op)/mp.pi
                    value = mp.quad(integrand,[0,1,3,8,20])
                    residuals.append(abs(value-rhs[k]))
        print("Mellin quadratures:",len(residuals),
              "maximum residual",mp.nstr(max(residuals),8))
        assert max(residuals) < mp.mpf("1e-28")

if __name__ == "__main__":
```

```
    row_checks()
    moment_checks()
    fixed_column_table()
    mellin_checks()
```

References for Part VII

[VII.1] R. Kalisch, T. Monderer, E. Leibtag, S. Weinbaum, M. Shalyt, and I. Kaminer, *An Infinite Family of Apéry-like Recurrences for Euler's Constant*, arXiv:2610.13919v1, 9 October 2026. https://arxiv.org/abs/2610.13919. The trajectory is taken from Section 8.4; Section 9.2(1) motivates the arbitrary-z identity extension.

[VII.2] S. Fischler and T. Rivoal, *On the values of G-functions*, Commentarii Mathematici Helvetici **89** (2014), 313–341. https://doi.org/10.4171/CMH/321. Theorem numbering in this supplement follows the published version: the Fuchsian/Frobenius theorem is Theorem 6, and the Wronskian statement is Lemma 9.

[VII.3] S. Fischler and T. Rivoal, *Relations between values of arithmetic Gevrey series, and applications to values of the Gamma function*, Journal of Number Theory **261** (2024), 36–54. https://arxiv.org/abs/2301.13518. Author text: https://rivoal.perso.math.cnrs.fr/articles/ssmixte.pdf.

[VII.4] F. Cellarosi and Ya. G. Sinai, *The Möbius function and statistical mechanics*, Bulletin of Mathematical Sciences **1** (2011), 245–275. https://doi.org/10.1007/s13373-011-0011-6. The normalized Dickman density and moment formulas are on pp. 257–258.

[VII.5] NIST Digital Library of Mathematical Functions, Section 16.17, *Meijer G-Function: Definition*. https://dlmf.nist.gov/16.17.

[VII.6] NIST Digital Library of Mathematical Functions, Section 5.7, *Gamma Function: Series Expansions*. https://dlmf.nist.gov/5.7.

Part VIII
Fractional degrees and Dickman-related asymptotics

Source: `Keio_Gamma_Fractional_Degrees_and_Dickman_20261019.tex`.

Original PDF: 29 pages. Mathematical content and source-level qualifications are retained.

Abstract

For the signed remainders R_N in Stirling's expansion of $\log\Gamma$, we prove

$$2N+m-2 < \deg_{\mathrm{cm}}[(-1)^m R_N^{(m)}] < 2N+m-1 \quad (N \geq 2,\ m=0,1,2),$$

and show that the positive fractional excess tends to zero as $N \to \infty$. A spectral representation and a stability theorem for small fractional derivatives supply the lower bounds; zero integral moments force the strict upper bounds. Two boundary families also fail at every derivative order $m \geq 3$. These results refute five real-degree formula families explicitly conjectured by Mahmoud and Qi (2022). For Dickman-related questions, we prove divergence of a formal delay equation solution and construct counterexamples to a conjectured tilted-density limit, including one with a purely absolutely continuous Lévy measure and a proper density. A polynomial intertwiner proves Atia's (2024) conjectured inverse Bessel recurrence. We also prove a half-integral Laguerre coefficient statement printed as a conjecture in a 2007 preprint, explain a factorial-error divergence observation, and establish restricted Borel-division and lattice-obstruction results. The source ledger distinguishes these outcomes and their priority limitations. No irrationality or transcendence proof for Euler's constant is obtained, and no claim of ten independent, previously open problems newly solved is made.

VIII.1 Results, provenance, and the arithmetic objective

The starting objective is to obtain new information about the arithmetic nature of

$$\gamma = \lim_{n\to\infty}\left(\sum_{k=1}^{n}\frac{1}{k} - \log n\right).$$

The investigation instead yields several complete analytic and algebraic arguments in adjacent areas, and identifies precise obstructions to turning them into an arithmetic proof. The direct connections are $\psi(1) = -\gamma$, the Dickman normalization $e^{-\gamma}$, and the Laguerre–Gompertz approximation problem. The Bessel inverse recurrence is an auxiliary special-function problem reached through these asymptotic methods; it is not itself an irrationality criterion.

VIII.1.1 Ten source-linked statements, with different statuses

The ledger below records statements, not a claim of ten independent major breakthroughs. Rows G1–G5 are five formula families in a single published conjectural programme. Rows P1, P2, and A2 concern three other explicit questions or conjectures. Row A1 is historical: its conjecture label was removed from the final journal article. Row O1 is an explicit numerical observation, not the source's irrationality conjecture. Thus there are eight explicit conjectural/question statements in the main source versions, one historical statement, and one observation; they do not certify the requested total of ten independent currently open problems newly resolved. The complete ledger is on the following page.

ID	Exact source	Conclusion established here	Qualification
G1	Mahmoud–Qi [VIII.1], Eq. (2.5), $N \geq 2$	$\deg_{\mathrm{cm}} R_N \in (2N-2, 2N-1)$; conjectured equality is false.	One family, Theorem VIII.2.1.
G2	Same source, Eq. (2.7), $N \geq 2$	$\deg_{\mathrm{cm}}(-R'_N) \in (2N-1, 2N)$; conjectured equality is false.	Same structural method.
G3	Same source, first equality of Eq. (2.8)	$\deg_{\mathrm{cm}}[(-1)^m R_0^{(m)}] < m$ for all $m \geq 3$.	Theorem VIII.2.3; $m = 2$ is not refuted.
G4	Same source, second equality of Eq. (2.8)	$\deg_{\mathrm{cm}}[(-1)^m R_1^{(m)}] < m+1$ for all $m \geq 3$.	Same boundary argument.
G5	Same source, Eq. (2.9), $m, N \geq 2$	Entire $m = 2$ subfamily has degree in $(2N, 2N+1)$, refuting the universal equality.	Cases $m \geq 3, N \geq 2$ not classified.
P1	Diblík–Medina [VIII.6], Remark 1, p. 104	Every nonzero formal solution in the stated inverse-power form diverges at each finite nonzero argument.	Journal question; Theorem VIII.3.1.
P2	Arratia–Kochman–Zabell [VIII.7], §7, Conjecture 1	Density/local-limit assertion is false under its printed hypotheses, even for pure-AC infinite-activity Lévy measure.	2016 preprint; two counterexamples count once.
A1	Borwein–Borwein–Crandall [VIII.9], Conjecture 1, p. 38	Half-integral coefficient factorization, with a stronger lower bound on surviving powers.	Historical status uncertain; Theorem VIII.4.1.
A2	Atia [VIII.11], §5.4, Eq. (24)	Inverse-linearization recurrence proved for every $a \neq 1/2$.	Journal conjecture; Theorem VIII.5.3.
O1	Murabayashi–Yoshida [VIII.13], p. 15	Factorially cleared even and odd approximation errors tend to infinity.	Numerical observation; not Conjecture 7.5.

All proofs are given in full below. The Laguerre proof uses an established finite coefficient formula from its source, and the arithmetic division argument explicitly invokes Lindemann–Weierstrass and the established E-function division theorem. Independent internal readers checked the arguments and source matching. This is not external peer review or a machine-checked formal proof. Targeted follow-up searches did not locate the same resolutions, but those searches cannot certify priority.

VIII.1.2 Additional results and limits of the investigation

The positive fractional degree corrections vanish at large index (Theorem VIII.2.4). The divergent Dickman series has an entire Borel transform and a Cauchy-transform solution whose jump is $2\pi i e^{-\gamma}\rho(x)$ (Proposition VIII.3.2). The lattice construction shows that smooth size ratios and primitivity cannot alone force the balance condition in a proposed route to γ (Theorem VIII.6.2). Rational Borel transforms with positive algebraic poles satisfy a restricted arithmetic division theorem (Theorem VIII.7.1). These are substantive additional statements, but none is relabeled as a complete solution of an unrestricted published conjecture.

This manuscript is a self-contained account of the present investigation. It does not reproduce every theorem from earlier drafts. In particular, earlier positivity certificates, local parameter-identification results, and failed irrationality routes are background rather than additional entries in the count.

VIII.2 Noninteger complete-monotonicity degrees and false boundary conjectures

VIII.2.1 The exact published conjecture and the scope of the result

Let B_j be the Bernoulli numbers, with $B_2 = 1/6$, and define

$$R_N(x) = (-1)^N \left[\log\Gamma(x) - \left(x - \frac{1}{2}\right)\log x + x - \frac{1}{2}\log(2\pi) - \sum_{k=1}^{N} \frac{B_{2k}}{(2k)(2k-1)x^{2k-1}} \right].$$

Qi's 2021 paper *Bounds for completely monotonic degree of a remainder for an asymptotic expansion of the trigamma function*, https://doi.org/10.1080/25765299.2021.1962060, conjectures that $\deg_{\mathrm{cm}} R_2'' = 4$ and proves only $4 \le \deg_{\mathrm{cm}} R_2'' \le 5$. The author's manuscript is https://arxiv.org/abs/2003.05300v2, Eq. (2.5). The larger conjecture, for all $m \ge 2$ and $N \ge 2$, is

$$\deg_{\mathrm{cm}}\bigl[(-1)^m R_N^{(m)}\bigr] = m + 2(N-1).$$

Mahmoud and Qi explicitly restate this as Eq. (2.9), page 294, in their 2022 paper *Bounds for completely monotonic degrees of remainders in asymptotic expansions of the digamma function*, https://doi.org/10.7153/mia-2022-25-17. The same paper also conjectures $\deg_{\mathrm{cm}} R_N = 2N-2$ in Eq. (2.5) and $\deg_{\mathrm{cm}}(-R_N') = 2N-1$ in Eq. (2.7). The theorem below disproves both of those families and the entire $m = 2$, $N \ge 2$ subfamily of Eq. (2.9).

For a nonzero completely monotone F tending to zero at infinity, write

$$\deg_{\mathrm{cm}} F = \sup\{\alpha \in \mathbb{R} : x^\alpha F(x) \text{ is completely monotone on } (0,\infty)\}.$$

Here complete monotonicity means $(-1)^j F^{(j)}(x) \ge 0$ for every integer $j \ge 0$ and every $x > 0$.

Theorem VIII.2.1. *For every integer $N \ge 2$, put*

$$Q_N(x) = R_N''(x) = (-1)^N \left[\psi'(x) - \frac{1}{x} - \frac{1}{2x^2} - \sum_{k=1}^{N} \frac{B_{2k}}{x^{2k+1}} \right].$$

Then, for $m = 0, 1, 2$,

$$\boxed{2N + m - 2 < \deg_{\mathrm{cm}}\bigl[(-1)^m R_N^{(m)}\bigr] < 2N + m - 1.}$$

In particular, all three families have noninteger real degrees. The theorem does not specify their exact fractional parts.

VIII.2.2 A positivity lemma for small fractional derivatives

Lemma VIII.2.2. *Suppose $f \in C^1([0,\infty))$, $f(0) = 0$, $f(t) > 0$ for $t > 0$, and $f'(t) > 0$ on some $(0,\delta)$. Suppose also that $f(t) \to L > 0$ and $|f'(t)| \le C(1+t)^{-p}$ for all sufficiently large t, where $C > 0$ and $p > 2$. Then there is $\varepsilon > 0$ such that $x^\varepsilon \int_0^\infty e^{-xt} f(t)\,dt$ is completely monotone.*

Proof. For $0 < \varepsilon < 1$, define

$$D^\varepsilon f(t) = \frac{1}{\Gamma(1-\varepsilon)} \int_0^t (t-u)^{-\varepsilon} f'(u)\,du.$$

We show that it is positive for all $t > 0$ when ε is sufficiently small. The sign is that of

$$J_\varepsilon(t) = \int_0^t f'(u)(1-u/t)^{-\varepsilon}\,du.$$

It is positive for $0 < t \le \delta$ because $f' > 0$ there. Fix any $0 < \varepsilon_0 < 1$. For $0 \le \varepsilon \le \varepsilon_0$, splitting the integral at $t/2$ gives

$$J_\varepsilon(t) = L + O(t^{-1}) + O(t^{1-p}) \qquad (t \to \infty),$$

uniformly in ε. Indeed, on $0 \le u \le t/2$ the difference $(1-u/t)^{-\varepsilon}-1$ is bounded by a constant times u/t; $\int_0^\infty u|f'(u)|du < \infty$. On the other half the polynomial decay bound and $\int_{1/2}^1 (1-v)^{-\varepsilon_0}dv < \infty$ apply. Consequently there is T such that $J_\varepsilon(t) > L/2$ for $t \ge T$ and all these ε.

On $[\delta, T]$, if $M = \max_{0\le u\le T} |f'(u)|$, then

$$|J_\varepsilon(t) - f(t)| \le Mt\left[\frac{1}{1-\varepsilon} - 1\right] \le MT\frac{\varepsilon}{1-\varepsilon}.$$

Since $\min_{[\delta,T]} f > 0$, this proves positivity on the remaining compact interval for sufficiently small $\varepsilon > 0$. Finally, absolute Fubini and $f(0) = 0$ give

$$\int_0^\infty e^{-xt} D^\varepsilon f(t)dt = x^{\varepsilon-1}\int_0^\infty e^{-xt} f'(t)dt = x^\varepsilon \int_0^\infty e^{-xt} f(t)dt.$$

The positive Laplace density proves complete monotonicity. □

VIII.2.3 The spectral kernel and its unavoidable overshoot

Proof of Theorem VIII.2.1. Put $g(t) = t/(e^t - 1)$, with its smooth value at zero, and

$$h_N(t) = (-1)^N\left[g(t) - 1 + \frac{t}{2} - \sum_{k=1}^{N} \frac{B_{2k}t^{2k}}{(2k)!}\right].$$

The standard Laplace representation of ψ' gives $Q_N = \mathcal{L}h_N$. Every derivative of h_N of order at most $2N+1$ vanishes at zero. Set $f_N = h_N^{(2N)}$. Repeated integration by parts gives

$$x^{2N}Q_N(x) = \int_0^\infty e^{-xt} f_N(t)dt, \qquad f_N(t) = |B_{2N}| + (-1)^N g^{(2N)}(t). \tag{VIII.2.1}$$

We require strict positivity, including the absence of interior zeros. The Mittag–Leffler expansion of coth gives the exact identity

$$g(t) + \frac{t}{2} = 1 + \int_0^\infty (1 - \cos(tu))\nu(u)du, \qquad \nu(u) = \frac{\pi}{\sinh^2(\pi u)}. \tag{VIII.2.2}$$

To check the normalization, expand $\nu(u) = 4\pi\sum_{j\ge1} je^{-2\pi ju}$ and use Tonelli:

$$\int_0^\infty (1-\cos(tu))\nu(u)du = 2t^2\sum_{j\ge1}\frac{1}{t^2 + 4\pi^2 j^2} = \frac{t}{2}\coth\frac{t}{2} - 1.$$

The last equality is the classical partial-fraction expansion of coth (NIST DLMF 4.36.3, `https://dlmf.nist.gov/4.36.E3`). Because $u^{2N}\nu(u)$ is integrable for $N \ge 1$, differentiation of (VIII.2.2) through the even order $2N$, including at zero, is legitimate. It yields

$$f_N(t) = \int_0^\infty u^{2N}(1-\cos(tu))\nu(u)du > 0 \quad (t > 0). \tag{VIII.2.3}$$

Strictness follows because $\nu(u) > 0$ and the zeros of $1 - \cos(tu)$ form a discrete set. Moreover,

$$f_N(t) = c_N t^2 + O(t^4), \qquad c_N = \frac{1}{2}\int_0^\infty u^{2N+2}\nu(u)du > 0,$$

so $f_N' > 0$ near zero. Equation (VIII.2.1), or the series $g(t) = t\sum_{j\geq 1} e^{-jt}$, shows that

$$f_N(t) \longrightarrow L_N := |B_{2N}| > 0, \qquad f_N'(t) = O_N((1+t)e^{-t}).$$

Lemma VIII.2.2 applies. Together with (VIII.2.1), it proves that $x^{2N+\varepsilon_N}Q_N(x)$ is completely monotone for some $\varepsilon_N > 0$. This proves the strict lower bound.

For the upper bound the condition $N \geq 2$ is essential. Since $B_{2N-1} = 0$, absolute integration gives

$$\int_0^\infty [f_N(t) - L_N]dt = (-1)^N \int_0^\infty g^{(2N)}(t)dt = -(-1)^N B_{2N-1} = 0. \tag{VIII.2.4}$$

The integrand is negative near zero, because $f_N(0) = 0$. Hence it must be positive somewhere. Thus f_N exceeds its limiting value L_N at some point, and f_N' is negative somewhere. Since $f_N(0) = 0$,

$$x^{2N+1}Q_N(x) = \int_0^\infty e^{-xt} f_N'(t)dt.$$

Uniqueness of Laplace transforms excludes complete monotonicity: a continuous signed density which is negative on an interval cannot equal a nonnegative representing measure. Uniqueness applies after multiplying both measures by e^{-at} for any $a > 0$, making them finite.

If some exponent larger than $2N+1$ were admissible, multiplication by a completely monotone negative power of x would make $2N+1$ admissible as well. Finally, the admissible set of exponents is closed: for fixed $x > 0$ and integer $j \geq 0$, the quantity $(-1)^j\partial_x^j[x^\alpha Q_N(x)]$ is continuous in α. The supremum therefore cannot equal the excluded endpoint $2N+1$. This proves the strict upper bound for $m = 2$.

To transfer the argument to $m = 1, 0$, put $r = 2 - m \in \{1, 2\}$ and $d = 2N - r$. The standard Binet formula and differentiation give

$$(-1)^m R_N^{(m)}(x) = \int_0^\infty e^{-xt}\frac{h_N(t)}{t^r}dt.$$

Set $k_r = h_N/t^r$ and $f_r = k_r^{(d)}$. Taylor's integral formula and the vanishing derivatives of h_N at zero give

$$f_r(t) = \frac{1}{(r-1)!}\int_0^1 (1-s)^{r-1}s^d f_N(st)ds. \tag{VIII.2.5}$$

Consequently $f_r(t) > 0$ for $t > 0$, $f_r(t) \sim c_r t^2$ at zero with $c_r > 0$, and

$$L_r := \lim_{t\to\infty} f_r(t) = L_N\frac{d!}{(2N)!} > 0.$$

Differentiating (VIII.2.5) and substituting $u = st$ shows

$$|f_r'(t)| = O_N(t^{-d-2}), \qquad \int_0^\infty u^{d+1}|f_N'(u)|du < \infty.$$

Here $d \geq 2$, so the fractional positivity lemma applies once again. Repeated integration by parts gives $x^d(-1)^m R_N^{(m)} = \mathcal{L}f_r$, proving strict lower bound d.

The same transfer preserves the zero integral responsible for the strict upper bound. Absolute Fubini is legitimate because $f_N - L_N$ is integrable and $\int_0^1 (1-s)^{r-1}s^{d-1}ds < \infty$. Thus

$$\int_0^\infty [f_r(t) - L_r]dt = \frac{1}{(r-1)!}\int_0^1 (1-s)^{r-1}s^{d-1}ds\int_0^\infty [f_N(u) - L_N]du = 0.$$

Since $f_r(0) = 0 < L_r$, this forces an overshoot of L_r, and hence a negative value of f_r'. The preceding Laplace-uniqueness and closedness argument excludes $d+1$ with a strict gap. This proves the remaining two cases of the theorem. □

VIII.2.4 Both boundary conjectures fail at every derivative order beyond two

The same 2022 paper, Eq. (2.8), proposes

$$\deg_{\mathrm{cm}}\big[(-1)^m R_0^{(m)}\big] = m, \qquad \deg_{\mathrm{cm}}\big[(-1)^m R_1^{(m)}\big] = m+1 \quad (m \geq 2).$$

Remark 4.5 explicitly leaves these equalities open. Although they are correct at $m = 2$, every subsequent case fails.

Theorem VIII.2.3. *For every integer $m \geq 3$,*

$$\deg_{\mathrm{cm}}\big[(-1)^m R_0^{(m)}\big] < m, \qquad \deg_{\mathrm{cm}}\big[(-1)^m R_1^{(m)}\big] < m+1.$$

Proof. Let

$$a_m(t) = \frac{t^{m-1}}{e^t - 1}, \quad k_0(t) = a_m(t) - t^{m-2} + \frac{1}{2}t^{m-1}, \quad k_1(t) = -a_m(t) + t^{m-2} - \frac{1}{2}t^{m-1} + \frac{1}{12}t^m.$$

The Binet representation gives $(-1)^m R_j^{(m)} = \mathcal{L}k_j$, $j = 0, 1$. These functions are nonzero and completely monotone: the partial-fraction expansion above gives $0 < h_0(t) < t^2/12$ for $t > 0$, while $h_1(t) = t^2/12 - h_0(t)$ and $k_j = t^{m-2}h_j$. The expansion at zero is

$$a_m(t) = t^{m-2} - \frac{1}{2}t^{m-1} + \frac{1}{12}t^m - \frac{1}{720}t^{m+2} + O(t^{m+4}).$$

Thus $k_0 = O(t^m)$ and $k_1 = O(t^{m+2})$. All boundary terms needed for the following integrations by parts vanish:

$$x^m(-1)^m R_0^{(m)} = \mathcal{L}A_m, \qquad x^{m+1}(-1)^m R_1^{(m)} = \mathcal{L}C_m, \quad A_m = a_m^{(m)}, \quad C_m = -a_m^{(m+1)}.$$

Both densities have exponentially decaying tails, and at zero

$$A_m(t) = \frac{m!}{12} + O(t^2) > 0, \qquad C_m(t) = \frac{(m+2)!}{720}t + O(t^3) > 0.$$

However, $a_m^{(m-3)}(0) = 0$ for every $m \geq 3$. Repeated integration by parts, first against t^2 and then against t^3, gives the exact vanishing moments

$$\int_0^\infty t^2 A_m(t)dt = -2a_m^{(m-3)}(0) = 0, \qquad \int_0^\infty t^3 C_m(t)dt = -6a_m^{(m-3)}(0) = 0.$$

Since each density is positive on an initial interval, it must be negative somewhere else. Laplace uniqueness therefore excludes complete monotonicity at each conjectured endpoint. Downward closure and closedness of the admissible exponents, proved above, make both upper bounds strict.

As an independent tail check, for odd $m \geq 3$ the two densities are already negative eventually:

$$A_m(t) = (-1)^m t^{m-1}e^{-t}(1 + O_m(t^{-1})), \qquad C_m(t) = (-1)^m t^{m-1}e^{-t}(1 + O_m(t^{-1})).$$

The zero-moment argument also covers every even $m \geq 4$. □

VIII.2.5 Priority distinctions and rejected candidates

Ling Zhu's 2020 paper *Completely monotonic integer degrees for a class of special functions*, https://doi.org/10.3934/math.2020224, conjectures the integer degree $2N$ in the $m = 2$ case of Eq. (6.3). Theorem VIII.2.1 is consistent with, and proves, this integer statement, together with the corresponding $m = 0, 1$ integer statements. It contradicts the separately stated real-degree

equalities. These consequences concern one structural family and must not be counted as many unrelated solved problems.

There is an earlier relevant observation. Qi and Guo's *Integral representations and complete monotonicity of remainders of the Binet and Stirling formulas for the gamma function*, RACSAM 111 (2017), 425–434, https://doi.org/10.1007/s13398-016-0302-6, Remark 3.1, reports the absence of a concrete nontrivial example having noninteger degree. Theorem VIII.2.1 supplies three infinite Gamma remainder families with this property. This contextual observation is not counted as another independently resolved conjecture.

Two other explicit questions were screened. Section 3 of Bulboacă and Zayed's *Monotonic nature of the Gamma function*, https://doi.org/10.1186/s13660-025-03425-0, asks for an analytic proof that the continuously extended quotient

$$\frac{\log\Gamma(x+1)}{\log(x^2+6)-\log(x+6)}$$

has exactly one minimum on $(-1,\infty)$. Open Problem 1.5 of Szabó's *Completely monotone functions in general and some applications*, https://arxiv.org/abs/2411.17670, asks for the sharp exponent in

$$(x+a)^{\alpha}\left[\psi(x+b)-\psi(x+a)-\frac{b-a}{x+a}\right], \qquad 0<b-a<1.$$

The answer to the latter is $\alpha_0 = 1$: sufficiency is proved in the source, and necessity follows because the bracket is $(1-b+a)/(x+a)+O(1)$ at $x=-a$. For $\alpha>1$ the function would tend to zero at its left endpoint, impossible for a nonzero completely monotone function. However, a public repository already lists solution claims for both of these exact questions (entries APP-0029/0034 and APP-0021 at https://github.com/pudim-project/pudim-ai-demo-zetalaw). They are excluded from any claim of a new solution here.

The targeted search also inspected the 2022 Mahmoud–Qi paper, Zhu's integer-degree paper, the author's 2024 degree bibliography, and relevant 2024–2025 followups. No prior resolution of the Q_N real-degree equality was found. This is a bounded literature audit, not a claim that priority has been exhaustively established. The proof above is independent of the repository's mathematical claims.

VIII.2.6 The fractional parts tend to zero

The strict intervals in Theorem VIII.2.1 do not merely give counterexamples at a few small indices. Their noninteger corrections vanish in the large-index limit.

Theorem VIII.2.4. *For each fixed $m \in \{0,1,2\}$,*

$$\deg_{\mathrm{cm}}\bigl[(-1)^m R_N^{(m)}\bigr] = 2N+m-2+o(1) \qquad (N\longrightarrow\infty),$$

where the $o(1)$ is strictly positive for every $N\ge 2$.

Proof. Put $r=2-m$, $d=2N-r$, $b_N=(2N+1)/(2\pi)$, and $L_N=|B_{2N}|$. Let $U_N>0$ have probability density $u^{2N}\nu(u)/L_N$, where ν is defined in (VIII.2.2). Expanding $\nu(u)=4\pi\sum_{j\ge1} je^{-2\pi ju}$ and integrating its moments yields, with $W_N=U_N/b_N$,

$$\mathbb{E}W_N=\frac{\zeta(2N+1)}{\zeta(2N)}, \qquad \mathbb{E}W_N^2=\frac{2N+2}{2N+1}\frac{\zeta(2N+2)}{\zeta(2N)}.$$

Thus $W_N\to 1$ in L^2, and hence in L^1. For $r=0$ set $S_N=1$. For $r=1,2$ let S_N be independent of U_N and have beta density proportional to $s^{2N-r}(1-s)^{r-1}$ on $(0,1)$. Its mean is $(2N-r+1)/(2N+1)$, so $S_N\to 1$ in L^1 as well. Consequently $A_N=S_NW_N\to 1$ in L^1.

For $r = 0$ write $f_{N,0} = f_N$, and for $r = 1, 2$ use $f_{N,r} = f_r$ from (VIII.2.5). Let $L_{N,r}$ be the corresponding positive limit at infinity. The spectral and beta representations give

$$G_N(v) := \frac{f_{N,r}(v/b_N)}{L_{N,r}} = \mathbb{E}[1 - \cos(vA_N)], \qquad G'_N(v) = \mathbb{E}[A_N \sin(vA_N)].$$

Differentiation is justified by $\mathbb{E}A_N < \infty$. For every $T > 0$,

$$\sup_{0 \le v \le T} |G'_N(v) - \sin v| \le (1 + T)\mathbb{E}|A_N - 1| \longrightarrow 0.$$

Fix any $\varepsilon \in (0, 1)$. At $T = 2\pi$, the integrable fractional kernel and this uniform convergence give

$$D^\varepsilon G_N(2\pi) \longrightarrow \frac{1}{\Gamma(1-\varepsilon)} \int_0^{2\pi} \sin u \, (2\pi - u)^{-\varepsilon} \, du < 0.$$

The strict sign follows on pairing u with $2\pi - u$:

$$\int_0^{2\pi} \sin u \, (2\pi - u)^{-\varepsilon} \, du = \int_0^{\pi} \sin u \big[(2\pi - u)^{-\varepsilon} - u^{-\varepsilon}\big] du < 0.$$

Scaling preserves the sign of a fractional derivative, so $D^\varepsilon f_{N,r}$ is negative somewhere for all sufficiently large N. By the same Laplace uniqueness argument used above, $x^{d+\varepsilon}(-1)^m R_N^{(m)}(x)$ is then not completely monotone. Downward closure of the admissible exponents gives $\deg_{\mathrm{cm}}[(-1)^m R_N^{(m)}] \le d + \varepsilon$ eventually. Theorem VIII.2.1 supplies the strict lower bound d. Since $\varepsilon > 0$ was arbitrary, the asserted limit follows. □

VIII.2.7 What this does, and does not, say about Euler's constant

At $N = 2$, the three completely monotone functions take the values

$$R_2(1) = \frac{331}{360} - \frac{1}{2}\log(2\pi), \qquad -R'_2(1) = \gamma - \frac{23}{40}, \qquad R''_2(1) = \frac{\pi^2}{6} - \frac{49}{30}.$$

In particular, the middle function contains γ directly, rather than merely sharing its name with the gamma function. The method transfers between its Stirling, digamma, and trigamma forms through positive integral kernels. The new information is an exact exclusion of integer values for their real complete-monotonicity degrees, together with an asymptotic determination up to $o(1)$.

Complete monotonicity is an analytic sign condition. It does not provide integer coefficients with controlled denominators. None of these degree results implies that γ is irrational or transcendental, and no such implication is asserted.

VIII.3 Two explicit Dickman-related questions from the literature

This section addresses two precisely identified statements. The first is a question in a refereed 2018 article; the second is a conjecture in a publicly available 2016 research preprint. These publication statuses are different. The proofs below answer the statements as printed. A search of the accessible subsequent literature did not locate these answers, but does not establish priority or certify that no unpublished or unindexed resolution exists. Neither result decides the arithmetic nature of Euler's constant.

VIII.3.1 Divergence of the Diblík–Medina formal solution

Diblík and Medina [VIII.6, Remark 1, p. 104] ask whether the formal solution

$$S(t) = \sum_{n\geq 1} C_n t^{-n}, \qquad (n-1)C_n = \sum_{s=1}^{n-1} \binom{n-1}{s-1} C_s \quad (n \geq 2), \tag{VIII.3.1}$$

of $tx'(t) = -x(t-1)$ converges or diverges. The initial coefficient C_1 is arbitrary. Their question has the following direct answer.

Theorem VIII.3.1. *If $C_1 \neq 0$, the power series $\sum_{n\geq 1} C_n z^n$ has radius of convergence zero. In particular, the series (VIII.3.1) diverges at every finite nonzero real or complex t. More explicitly,*

$$|C_{2k+2}| \geq |C_1| k! \qquad (k \geq 0). \tag{VIII.3.2}$$

If $C_1 = 0$, all coefficients vanish.

Proof. Set $a_n = C_n/C_1$ when $C_1 \neq 0$. The recurrence gives $a_1 = a_2 = 1$ and $a_n > 0$ for every n. Keeping just the summand $s = n-2$ gives, for $n \geq 3$,

$$a_n \geq \frac{1}{n-1}\binom{n-1}{n-3} a_{n-2} = \frac{n-2}{2} a_{n-2}.$$

In particular $a_{2k+2} \geq k a_{2k}$, which proves (VIII.3.2) by induction. For fixed $t \neq 0$, the lower bound $|C_1| k! |t|^{-2k-2}$ tends to infinity, so the terms of the proposed series do not tend to zero. The case $C_1 = 0$ follows by induction directly from the recurrence. □

The next identification explains why a moment problem appears in this delay equation. It is additional structure, not a second independent resolution of a published question.

Proposition VIII.3.2. *Let D have the standard Dickman distribution, with density $p(x) = e^{-\gamma}\rho(x)$ and moment generating function*

$$M(z) = \exp\left(\int_0^1 \frac{e^{zx} - 1}{x}\, dx\right).$$

Then $C_n = C_1 \mathbb{E}[D^{n-1}]$, and the formal Borel transform is entire:

$$\sum_{n\geq 1} \frac{C_n}{(n-1)!} z^{n-1} = C_1 M(z). \tag{VIII.3.3}$$

For $w \in \mathbb{C} \setminus [0, \infty)$, the Cauchy transform

$$F(w) = \int_0^\infty \frac{p(x)}{w - x}\, dx$$

is an actual holomorphic solution of

$$wF'(w) = -F(w-1). \tag{VIII.3.4}$$

On every closed sector bounded away from the positive real axis it has the asymptotic expansion $F(w) \sim \sum_{n\geq 1} (C_n/C_1) w^{-n}$. Its boundary values satisfy

$$F(x - i0) - F(x + i0) = 2\pi i e^{-\gamma} \rho(x), \qquad x > 0. \tag{VIII.3.5}$$

Proof. The standard distribution and normalization are recalled in [VIII.8, Section 1]. If $m_j = \mathbb{E}[D^j]$, then $zM'(z) = (e^z - 1)M(z)$ gives $jm_j = \sum_{i=0}^{j-1} \binom{j}{i} m_i$. This is precisely (VIII.3.1) after $j = n - 1$, with the same initial value. The integrand defining $\log M$ has a removable singularity at $x = 0$ and defines an entire function of z, proving (VIII.3.3).

For the Cauchy transform, differentiation and integration by parts are valid off $[0, \infty)$ because p has all moments, is locally absolutely continuous on $(0, \infty)$, is constant on $(0, 1)$, and decreases faster than any power. The relation $xp'(x) = -p(x - 1)$, with $p = 0$ on the negative axis, gives

$$wF'(w) = -\int_0^\infty \frac{wp(x)}{(w-x)^2}\,dx = \int_0^\infty \frac{xp'(x)}{w-x}\,dx = -\int_0^\infty \frac{p(y)}{w-1-y}\,dy.$$

For every $N \geq 1$, geometric division gives the exact remainder

$$F(w) - \sum_{j=0}^{N-1} \frac{m_j}{w^{j+1}} = \frac{1}{w^N} \int_0^\infty \frac{x^N p(x)}{w-x}\,dx.$$

In the stated sectors $|w - x| \geq c|w|$ for $x \geq 0$, so the remainder is at most $c^{-1}m_N|w|^{-N-1}$. Finally, the usual Cauchy boundary-value formula gives (VIII.3.5); it also follows directly by subtracting the two kernels and applying the Poisson approximate identity to the continuous density p at each $x > 0$. □

Thus the divergent dominant expansion and the rapidly decaying Dickman solution are connected by an exact Cauchy-transform jump. The factor $e^{-\gamma}$ is a known normalization, and this identity alone gives no rationality or transcendence conclusion about γ.

VIII.3.2 A counterexample to a tilted-density conjecture

Conjecture 1 of [VIII.7, Section 7, p. 29] permits a Lévy measure μ on $(0, 1]$ with atoms, requires $\int x\,\mu(dx) < \infty$, and assumes that its absolutely continuous part $g(x)\,dx$ is bounded below by a positive constant on some nonempty interval. With

$$C(\beta) = \int_{(0,1]} (e^{\beta x} - 1)\,\mu(dx), \quad u = C'(\beta), \quad \sigma_\beta^2 = C''(\beta),$$

the conjecture asserts a pointwise Gaussian density limit for the standardized exponentially tilted law, and consequently the density asymptotic

$$f(u) \sim G(u) := \frac{\exp(C(\beta(u)) - u\beta(u))}{\sqrt{2\pi}\,\sigma_{\beta(u)}} \qquad (u \longrightarrow \infty). \tag{VIII.3.6}$$

The version inspected is arXiv:1606.03524v1 (2016); no journal publication was identified. The permitted atomic part creates a counterexample even when the resulting law has a proper absolutely continuous density.

Theorem VIII.3.3. *Let*

$$\mu(dx) = \delta_1(dx) + \frac{1}{2}\mathbf{1}_{(0,1/2]}(x)\frac{dx}{x}. \tag{VIII.3.7}$$

This measure satisfies all the hypotheses of the cited conjecture. Its infinitely divisible law has a proper density, but (VIII.3.6) fails. In fact there are intervals I_n of positive length, tending to infinity, on which $f(u)/G(u)$ tends uniformly to infinity. Thus altering a density on a null set cannot restore the conjectured asymptotic. For the canonical tilted densities, even the pointwise local limit at the fixed standardized point $y = 0$ fails.

Proof. Let $N \sim \text{Poisson}(1)$ and let $D_{1/2}$ have the generalized Dickman law, independently. Then $T = N + X$, where $X = D_{1/2}/2$, has the Lévy measure (VIII.3.7). Its mean is $1 + 1/4$; its absolutely continuous Lévy density is at least 1 on $(1/4, 1/2)$. In particular the hypotheses hold, and the Lévy measure even has infinite total mass.

The generalized Dickman density formula [VIII.8, Section 1] gives

$$f_X(x) = Ax^{-1/2} \quad (0 < x < 1/2), \qquad A = \frac{\sqrt{2}\, e^{-\gamma/2}}{\sqrt{\pi}} > 0.$$

Consequently T is absolutely continuous, with density

$$f(t) = e^{-1} \sum_{k \geq 0} \frac{f_X(t-k)}{k!}, \qquad f_X(s) = 0 \quad (s \leq 0).$$

On each bounded interval this sum has only finitely many nonzero terms. In particular, for every integer $n \geq 0$ and $0 < \varepsilon < 1/2$,

$$f(n + \varepsilon) \geq \frac{e^{-1} A}{n! \sqrt{\varepsilon}}. \tag{VIII.3.8}$$

Here

$$C(\beta) = e^{\beta} - 1 + \frac{1}{2} \int_0^{1/2} \frac{e^{\beta x} - 1}{x}\, dx, \quad C'(\beta) = e^{\beta} + \frac{1}{2} \int_0^{1/2} e^{\beta x}\, dx.$$

Since $C''(\beta) > 0$, C' maps $[0, \infty)$ bijectively onto $[5/4, \infty)$. For $u \geq 5/4$ its inverse $\beta(u)$ is nonnegative, and $C''(\beta) \geq e^{\beta} \geq 1$. Convexity, together with $C(0) = 0$, gives $C(\beta) - \beta C'(\beta) \leq 0$. It follows that

$$0 < G(u) \leq \frac{1}{\sqrt{2\pi}} \qquad (u \geq 5/4). \tag{VIII.3.9}$$

Set $\varepsilon_n = e^{-n^3}$ and $I_n = (n + \varepsilon_n,\, n + 2\varepsilon_n)$ for $n \geq 2$. Combining (VIII.3.8) and (VIII.3.9) yields, throughout I_n,

$$\frac{f(u)}{G(u)} \geq \frac{\sqrt{\pi}\, e^{-1} A}{n!} \exp(n^3/2) \longrightarrow \infty.$$

Every I_n has positive length, so this remains incompatible with a pointwise asymptotic after any modification on a fixed null set.

For completeness, let $u_n = n + \varepsilon_n$ and $\beta_n = \beta(u_n)$. The standardized tilted variable $Y_\beta = (T_\beta - C'(\beta))/\sigma_\beta$ has its natural density

$$f_{Y_\beta}(y) = \sigma_\beta e^{\beta(u + \sigma_\beta y) - C(\beta)} f(u + \sigma_\beta y).$$

At $y = 0$ this is $f(u)/(\sqrt{2\pi}\, G(u))$. Hence $f_{Y_{\beta_n}}(0) \to \infty$, instead of $(2\pi)^{-1/2}$. Here $\beta_n \to \infty$, because $u_n \to \infty$ and C' is increasing. □

Remark VIII.3.4. This refutation concerns the density assertion in the full generality of the stated conjecture. It does not contradict the paper's theorems under stronger regularity assumptions, nor its weak central limit theorem. It also does not assert failure of the corresponding tail probability asymptotic. The obstruction is an arbitrarily narrow cusp near every integer, created by convolving a Poisson lattice law with a generalized Dickman density singular at zero.

VIII.3.3 A purely continuous Lévy measure and a proper density

The preceding counterexample to the Arratia–Kochman–Zabell conjecture can be strengthened: neither atoms in the Lévy measure nor atoms in the resulting law are necessary. This is a second construction refuting the same conjecture and is not counted as a separate resolution.

Proposition VIII.3.5. *The conclusion* (VIII.3.6) *can fail when the Lévy measure is purely absolutely continuous, has infinite total mass, and the resulting law has a proper density. The failure occurs on positive-length intervals tending to infinity.*

Proof. Set $a = 1/2$, $b = 1/4$, $\delta = e^{-2}$ and

$$h(v) = \frac{\mathbf{1}_{(0,\delta)}(v)}{v(\log v)^2}, \qquad \mu(dx) = \left\{ h(x-a) + \frac{\mathbf{1}_{(0,b]}(x)}{2x} \right\} dx.$$

This measure is supported in $(0,1)$, has finite first moment, and has density at least 2 on $(b/2, b)$. It is purely absolutely continuous and has infinite total mass. The translated component has mass $\lambda = \int_0^\delta h(v)\,dv = 1/2$.

Let J be the compound Poisson variable with Lévy density $h(x-a)$, and independently let $X = bD_{1/2}$. Then $T = J + X$ has Lévy measure μ and a proper density. The generalized Dickman density gives

$$f_X(x) = Ax^{-1/2} \quad (0 < x < b), \qquad A = \frac{2e^{-\gamma/2}}{\sqrt{\pi}} > 0.$$

Writing h^{*0} for unit mass at zero, a density of T is

$$f(t) = e^{-\lambda} \sum_{k \geq 0} \frac{(f_X * h^{*k})(t - ka)}{k!}.$$

For each t the sum has only finitely many nonzero terms, because every jump in J exceeds a.

For $n \geq 2$ and $0 < \varepsilon < \min(b, \delta)$, restrict all n integration variables in $(f_X * h^{*n})(\varepsilon)$ to $(0, \varepsilon/(2n))$. Their sum is at most $\varepsilon/2$, so the remaining argument of f_X lies in $[\varepsilon/2, \varepsilon]$. Using $\int_0^r h(v)\,dv = 1/\log(1/r)$, we obtain

$$f(na + \varepsilon) \geq \frac{e^{-1/2} A \varepsilon^{-1/2}}{n!\{\log(1/\varepsilon) + \log(2n)\}^n}.$$

Consequently, uniformly for $\varepsilon \in (e^{-n^3}, 2e^{-n^3})$,

$$f(na + \varepsilon) \geq \frac{e^{-1/2} A e^{n^3/2}}{\sqrt{2}\, n!\{n^3 + \log(2n)\}^n} \longrightarrow \infty.$$

For the cumulant C of this law, C' is strictly increasing with range $(0, \infty)$ and $\beta(u) \geq 0$ for all sufficiently large u. The translated component alone gives $C''(\beta) \geq C''(0) \geq a^2\lambda = 1/8$ for $\beta \geq 0$. Convexity and $C(0) = 0$ give $C(\beta) - \beta C'(\beta) \leq 0$, hence $G(u) \leq 2/\sqrt{\pi}$. Therefore $f/G \to \infty$ uniformly on $(na + e^{-n^3}, na + 2e^{-n^3})$. These intervals have positive length, so no change of the density on a null set can restore (VIII.3.6). □

VIII.4 Half-integral vanishing in the Laguerre asymptotic coefficients

Source and scope. The publicly posted version of D. Borwein, J. M. Borwein and R. E. Crandall, *Effective Laguerre asymptotics*, dated 18 January 2007, states Conjecture 1 on printed page 38: for $j \geq 1$ and $a \in \{-1/2, -3/2, \ldots, -j + 1/2\}$, the asymptotic coefficient $C_j(a, z)$ is a positive power of z times a polynomial over $\mathbb{Q}$, with total z-degree $3j/2$. The source is `https://carmamaths.org/resources/jon/laguerre-talk.pdf`. The final article is *SIAM J. Numer. Anal.* **46** (2008), 3285–3312, DOI `https://doi.org/10.1137/07068031X`; the authors'

28-page version is https://carmamaths.org/jon/Laguerre.pdf. The final version omits the numbered conjecture and describes the coefficient structure as a topic for further research. Accordingly, the result below proves the precise conjecture in the 2007 preprint; it is not presented as evidence that the final published article still labels this statement open. No separate subsequent proof was located in the searches accompanying this note.

Set $m = n + 1$. We use exactly the normalization in the source:

$$L_n^{(-a)}(-z) \sim \frac{e^{-z/2}e^{2\sqrt{mz}}}{2\sqrt{\pi}\, z^{1/4-a/2}m^{1/4+a/2}} \sum_{j\ge 0} C_j(a,z)m^{-j/2}, \qquad C_0 = 1. \tag{VIII.4.1}$$

The branch is the principal one on $z \notin (-\infty, 0]$. This normalization matters: replacing m by n changes the coefficients.

Theorem VIII.4.1 (Stronger half-integral vanishing)**.** *Let $j \ge 1$ and $0 \le r < j$ be integers. Let d be the least integer strictly larger than r with $d \equiv j \pmod 2$. Then*

$$C_j(-r - 1/2, z) = z^{d/2}P_{j,r}(z), \qquad P_{j,r} \in \mathbb{Q}[z], \qquad \deg P_{j,r} = \frac{3j - d}{2}. \tag{VIII.4.2}$$

The coefficient of $z^{3j/2}$ in C_j is $1/(12^j j!)$. In particular every term has z-exponent strictly larger than $r/2$. The assertion does not require $P_{j,r}(0) \ne 0$.

Proof. We first express the finite coefficient formula in a form suitable for tracking powers. Define $A_{h,u}(a)$ by the formal generating function

$$\sum_{h\ge 0}\sum_{u=0}^{h} A_{h,u}(a)t^h y^u = (1 + t/2)^{-a-1} \exp\left(y \sum_{\substack{\ell\ge 1\\ \ell \text{ odd}}} \frac{t^\ell}{2^{\ell+1}(\ell + 2)} \right). \tag{VIII.4.3}$$

All its coefficients belong to $\mathbb{Q}[a]$. Each occurrence of y is accompanied by a positive power of t, so $A_{h,u} = 0$ for $u > h$. In the notation of equations (39)–(44) of the final source,

$$\alpha_{h,u}(a, z) = A_{h,u}(a)z^{h/2+u}.$$

This identity follows directly by expanding the exponential defining their A_h: its first part is $-(a + 1)\log(1 + t/2)$, and its second part has precisely the coefficients in (VIII.4.3). Thus the source's finite formula (44) reads

$$C_j(a, z) = \sum_{k=0}^{j}\sum_{u=0}^{j-k} \frac{A_{j-k,u}(a)}{16^k k!} z^{(j-2k)/2+u} \prod_{\ell=1}^{k}\big((2\ell - 1)^2 - 4(a + j - k + 2u)^2\big). \tag{VIII.4.4}$$

Empty products have value one. Formula (VIII.4.4) is an exact finite identity for the uniquely normalized asymptotic coefficients, not a numerical approximation.

Put $h = j - k$, $a = -r - 1/2$, and $e = h + u - j/2$. Consider a summand for which $e \le r/2$. Writing $v = h + 2u$, this inequality gives

$$v \le r + k.$$

Also $r \le j - 1$ and $u \ge 0$ give

$$v \ge r + 1 - k,$$

since the latter is equivalent to $2u \ge r + 1 - j$. Consequently

$$-k + \tfrac{1}{2} \le a + v \le k - \tfrac{1}{2}.$$

The number $a+v$ is a half integer, so $|a+v| = \ell - 1/2$ for some $1 \le \ell \le k$. The corresponding factor in (VIII.4.4) is zero. (The case $k=0$ cannot satisfy $e \le r/2$.) Every surviving summand therefore has $e > r/2$.

All $2e$ are integers congruent to j modulo two. Hence all surviving exponents belong to $d/2 + \mathbb{Z}_{\ge 0}$, proving the factorization and rationality in (VIII.4.2). Finally $h+u \le 2h \le 2j$, with equality only for $h=u=j$ and $k=0$. In (VIII.4.3), the coefficient of $t^j y^j$ is obtained by taking j copies of the term $ty/12$ in the exponential. It is $1/(12^j j!)$. Thus the highest exponent is exactly $3j/2$ and its coefficient is as asserted. This also proves the degree statement. □

Corollary VIII.4.2 (A coefficient divisibility pattern)**.** *Write*

$$C_j(a,z) = \sum_{s=0}^{2j} c_{j,s}(a) z^{s-j/2}, \qquad c_{j,s} \in \mathbb{Q}[a].$$

For $0 \le s \le j-1$, *put* $b = \max(0, 2s-j)$. *Then*

$$\prod_{r=b}^{j-1} (a+r+1/2) \quad \textit{divides} \quad c_{j,s}(a) \quad \textit{in } \mathbb{Q}[a].$$

Proof. For every $b \le r \le j-1$ one has $s - j/2 \le r/2$. The theorem gives $c_{j,s}(-r-1/2) = 0$. These roots are distinct, so their product divides $c_{j,s}$. □

Interpretation. The cancellation is termwise in the Bessel expansion: at half-integral order, its elementary finite product vanishes. No cancellation between large symbolic expressions, and no asymptotic error estimate beyond the source's established coefficient formula, is needed. The divisibility corollary is a consequence of the same theorem, not a second independent resolution of a previously open problem. The connection to Euler's constant is through Laguerre expansions of $\Gamma(0,z) = E_1(z)$ and the asymptotic methods underlying rational approximants; this result by itself gives no irrationality statement.

VIII.5 An inverse-linearization conjecture for Bessel polynomials

Published question. M. J. Atia, *On the Inverse of the Linearization Coefficients of Bessel Polynomials*, *Symmetry* **16** (2024), 737, https://doi.org/10.3390/sym16060737, states an unnumbered conjecture in Section 5.4, printed page 16, displayed equation (24). It asserts the recurrence proved below. A distinct conjecture on the same page asserts positivity of the inverse matrix outside $[0,1]$; we do not claim that positivity statement here.

Define the normalized reverse Bessel polynomials by

$$q_n(u) = \sum_{k=0}^{n} \frac{n!(2n-k)!\,2^k}{(n-k)!(2n)!k!} u^k, \qquad q_0(u) = 1.$$

For $N \ge 0$ let $M_N(a)$ be the matrix defined by

$$q_{N-j}(au) q_j((1-a)u) = \sum_{i=0}^{N} (M_N(a))_{j,i} q_i(u), \quad 0 \le j \le N. \tag{VIII.5.1}$$

Atia denotes this matrix by M_N^{β}. Put $G_N = M_N^{-1}$ when it exists, and follow the source's convention

$$\gamma_k^{(N-m,m)}(a) = (G_N(a))_{m,k}.$$

In particular the superscript of γ indexes a row of the inverse, not a row of the original matrix.

Lemma VIII.5.1 (A polynomial intertwiner)**.** *Put $T_N = 2N+u-u\frac{d}{du}$ and define the $N\times(N+1)$ matrix A_N by*

$$(A_N)_{j,j} = 2N-2j-1, \qquad (A_N)_{j,j+1} = 2j+1 \quad (0\le j<N),$$

with all other entries zero. Then

$$A_N M_N = M_{N-1} A_N. \tag{VIII.5.2}$$

Proof. Comparison of coefficients in the finite formula for q_j gives

$$(2j+1)q_{j+1}(u) = (2j+1+u)q_j(u) - uq_j'(u), \qquad j\ge 0. \tag{VIII.5.3}$$

Consequently

$$T_N q_j = (2j+1)q_{j+1} + (2N-2j-1)q_j.$$

Set $B_{N,j}(u) = q_{N-j}(au)q_j((1-a)u)$. Apply (VIII.5.3) to each factor and use the product rule. Since $a+(1-a)=1$, this gives

$$T_N B_{N-1,j} = (2j+1)B_{N,j+1} + (2N-2j-1)B_{N,j}.$$

Computing $T_N(B_{N-1,j})$ first in the q basis and then in the B basis proves (VIII.5.2). □

Proposition VIII.5.2 (Spectrum and invertibility)**.** *For every $N\ge 0$,*

$$\det(\lambda I - M_N(a)) = \prod_{j=0}^{N}(\lambda - (1-2a)^j).$$

In particular, for $N\ge 1$, the matrix is invertible exactly when $a\neq 1/2$, and $\det M_N(a) = (1-2a)^{N(N+1)/2}$.

Proof. The matrix A_N has rank N. Define a linear functional on polynomials of degree at most N by

$$\ell_N(u^k) = (-1)^k\frac{(2N)!}{(2N-k)!} \quad (0\le k\le N).$$

For $k<N$,

$$\ell_N(T_N u^k) = (2N-k)\ell_N(u^k) + \ell_N(u^{k+1}) = 0.$$

Thus the nonzero column $w = (\ell_N(q_0),\dots,\ell_N(q_N))^T$ spans $\ker A_N$; its first entry is one. By (VIII.5.2), $\ker A_N$ is M_N invariant. Its eigenvalue is the first entry of $M_N w$, namely

$$\ell_N(q_N(au)) = \sum_{k=0}^{N}(-2a)^k\binom{N}{k} = (1-2a)^N.$$

The map A_N identifies the induced action on the quotient $\mathbb{C}^{N+1}/\ker A_N$ with M_{N-1}. The characteristic polynomial therefore acquires precisely the displayed additional linear factor at each step. Starting with $M_0=(1)$ proves the assertion, including cases of coincident eigenvalues. □

Theorem VIII.5.3 (Atia's equation (24))**.** *Let $n,m\ge 1$, $1\le k\le n+m$, and $a\neq 1/2$. Then*

$$\begin{aligned}&(2m-1)\gamma_k^{(n,m)}(a) + (2n+1)\gamma_k^{(n+1,m-1)}(a)\\ &\quad = (2k-1)\gamma_{k-1}^{(n,m-1)}(a) + (2n+2m-2k-1)\gamma_k^{(n,m-1)}(a),\end{aligned}$$

where inverse-matrix entries outside their column range are zero.

Proof. Multiplying (VIII.5.2) on the left by G_{N-1} and on the right by G_N gives $G_{N-1}A_N = A_N G_N$. Take $N=n+m$ and its entry with row $m-1$ and column k. The two nonzero entries in each relevant row or column of A_N give exactly the stated identity. □

Status and counting. This proves one explicitly conjectured inverse recurrence. The spectrum proposition independently justifies a result already stated as Proposition 2 in Atia's article; it is not counted as a second open problem. The displayed proof there attempts to infer nesting of spectra from a trace formula, which alone does not imply that nesting. The intertwining identity supplies the missing invariant-subspace argument. We use the defining inverse matrix throughout; the signs of some printed small inverse matrices in the source are inconsistent with that definition.

VIII.6 A congruence obstruction to inferring balanced lattices from smooth coefficients

Elsner's *Euler's Constant and Series Transformations with Bases of Balanced Lattices*, available from the author's website, formulates a specific conjecture as Lemma 13. It concerns coefficients coming from truncated Ser series, not arbitrary positive integers. With $N > e^{81000}$, $32N/51 < n < 16N/25$, and the associated graph lattice Λ, the conjecture asks for $N+1$ independent vectors, ordered by their maximum norms $T_1 \le \cdots \le T_{N+1}$, such that

$$T_{N+1}^{N/2} < \prod_{j=1}^{N+1} T_j < N^{2\varepsilon N/5} \det \Lambda. \tag{VIII.6.1}$$

The author applies this with $\varepsilon = 1/50000$. Lemma 14 of the same source proves three elementary size and ratio estimates for the special coefficients, presented as possible preliminary input toward Lemma 13. We show that those estimates, even together with primitivity, are insufficient: there are explicit integer coefficient families satisfying every one of them for which (VIII.6.1) is impossible. This does not disprove Lemma 13 for its stipulated Ser coefficients.

VIII.6.1 A general obstruction from near divisibility

For $a = (a_0, \dots, a_N) \in \mathbb{Z}^{N+1}$, put

$$\Lambda(a) = \{(x_0, \dots, x_N, y) \in \mathbb{Z}^{N+2} : a \cdot x = y\}.$$

This graph lattice has rank $N+1$ and Euclidean covolume

$$\det \Lambda(a) = \sqrt{1 + \sum_{j=0}^{N} a_j^2}. \tag{VIII.6.2}$$

Indeed its standard basis has Gram matrix $I + aa^{\mathsf{T}}$, whose determinant is $1 + a^{\mathsf{T}}a$. All successive minima below refer to the maximum norm restricted to its real span.

Lemma VIII.6.1 (Near-divisibility forces a large last minimum)**.** *Suppose $a_j = Bb_j + r_j$, where B is a positive integer, $b, r \in \mathbb{Z}^{N+1}$ and $b \neq 0$. Set $R = 1 + \sum_{j=0}^{N} |r_j|$. Then*

$$\lambda_{N+1}(\Lambda(a)) \ge B/R.$$

Consequently a necessary condition for (VIII.6.1) *is*

$$(B/R)^{N/2} < N^{2\varepsilon N/5} \det \Lambda(a). \tag{VIII.6.3}$$

Proof. If $(x, y) \in \Lambda(a)$ has maximum norm at most $H < B/R$, then

$$B(b \cdot x) = y - r \cdot x, \qquad |y - r \cdot x| \le RH < B.$$

The left side is an integer multiple of B, so $b \cdot x = 0$. Thus all such short vectors belong to the proper codimension-one subspace $b \cdot x = 0$ of the graph space. They cannot span rank $N+1$. This proves the minimum bound. Any $N+1$ independent lattice vectors have largest norm at least this last minimum, which together with (VIII.6.1) gives (VIII.6.3). Allowing a full-rank sublattice does not change this argument: its independent vectors are still in $\Lambda(a)$. □

VIII.6.2 Explicit counterexamples satisfying all three preparatory estimates

Theorem VIII.6.2 (Smooth coefficient ratios do not imply balance)**.** *For every integer $N > e^{81000}$ define*

$$B = 2^N, \qquad K = 2N, \qquad A = \left\lceil 2N\left(\log N + \frac{3}{5}\right)\right\rceil, \qquad n = \left\lfloor \frac{63N}{100} \right\rfloor,$$

and take

$$a_0 = BK, \qquad a_\nu = B(A+\nu) + 1 \quad (1 \le \nu \le N). \tag{VIII.6.4}$$

These positive integers have greatest common divisor one and satisfy $32N/51 < n < 16N/25$, as well as all the numerical estimates in Elsner's Lemma 14:

$$a_0 < a_1 < \cdots < a_N < n!\,(2^{8409})^n, \tag{VIII.6.5}$$

$$0.45 + \log N < \frac{a_\nu}{a_0} < 1.23 + \log N \qquad (1 \le \nu \le N), \tag{VIII.6.6}$$

$$1 + \frac{0.499}{N\log N} < \frac{a_{\nu+1}}{a_\nu} < 1 + \frac{3.390}{N\log N} \qquad (1 \le \nu < N). \tag{VIII.6.7}$$

Nevertheless

$$\lambda_1(\Lambda(a)) = 1, \qquad \lambda_{N+1}(\Lambda(a)) \ge \frac{2^N}{N+1}. \tag{VIII.6.8}$$

For any fixed $\varepsilon \ge 0$, all sufficiently large N in this family admit no vectors satisfying (VIII.6.1)*. For the value $\varepsilon = 1/50000$ used by Elsner, this failure holds for every $N > e^{81000}$.*

Proof. First, $n \le 0.63N < 0.64N = 16N/25$. Also

$$0.63 - 32/51 = 13/5100,$$

so $n \ge 0.63N - 1 > 32N/51$ for the stated range of N. Successive coefficients differ by B. Any common divisor of all a_j therefore divides B, and then divides $a_1 - B(A+1) = 1$. Thus their greatest common divisor is one. The ordering follows immediately from the formula and $A > K$.

The elementary ceiling bounds give

$$\log N + \frac{3}{5} + \frac{\nu}{2N} < \frac{a_\nu}{a_0} < \log N + \frac{3}{5} + \frac{\nu+1}{2N} + \frac{1}{2NB}.$$

The first expression is greater than $\log N + 0.45$ and the last is at most $\log N + 1.1 + 1/N < \log N + 1.23$. This proves (VIII.6.6). Next,

$$\frac{a_{\nu+1}}{a_\nu} - 1 = \frac{1}{A + \nu + 1/B}.$$

The denominator is greater than $2N\log N$, giving an upper bound $1/(2N\log N) < 3.390/(N\log N)$. It is also less than $2N\log N + 2.2N + 2$. Since $\log N > 81000$, we have

$$0.499(2N\log N + 2.2N + 2) < N\log N,$$

which proves the lower bound in (VIII.6.7).

For completeness, $a_N < 4BN\log N$ in this range. Also $n > 3N/5$, and hence

$$(2^{8409})^n > 2^{5000N} > 4\,2^N N\log N > a_N.$$

This proves the stronger version of (VIII.6.5) with the factorial omitted.

Apply Lemma VIII.6.1 with $b = (K, A+1, \ldots, A+N)$ and $r = (0, 1, \ldots, 1)$. Here $R = N+1$, yielding the second bound in (VIII.6.8). For the first, the vector with coordinates $x_1 = 1, x_2 =$

$-1, x_3 = -1, x_4 = 1$, all other $x_j = 0$, and $y = 0$, belongs to $\Lambda(a)$ by (VIII.6.4). It has maximum norm one, which is the smallest possible norm of a nonzero integral vector.

Using (VIII.6.2) and $a_N < 4BN \log N$, we have

$$\det \Lambda(a) < 5BN^{3/2} \log N. \tag{VIII.6.9}$$

For fixed ε, the logarithm of the ratio between the left side of (VIII.6.3) and the upper bound for its right side is at least

$$\left(\frac{N^2}{2} - N\right) \log 2 - \frac{N}{2} \log(N+1) - \frac{2\varepsilon N}{5} \log N - \frac{3}{2} \log N - \log \log N - \log 5.$$

It tends to positive infinity, proving eventual impossibility. For $\varepsilon = 1/50000$ and $N > e^{81000}$ one can bound $\log(N+1) < N/10$ and $\log N < N/10$. The positive term is at least $N^2/8$ since $N \geq 4$ and $\log 2 > 1/2$. The two negative terms proportional to N are at most $N^2/20 + N^2/1250000$, respectively. The remaining three negative terms are less than $N^2/100$ in this range. The displayed logarithm is therefore strictly positive. Condition (VIII.6.3) fails, completing the proof. □

The obstruction is arithmetic: the coefficients have nearly equal residues modulo 2^N. Their archimedean ratios conceal this common modular structure. The construction does not give the Ser-series coefficients from Elsner's equation (2.5.4), and therefore is neither a counterexample to his actual conjecture nor an irrationality proof for γ. It shows rigorously that a proof using only Lemma 14's bounds and primitivity cannot establish that conjecture. Additional information on the coefficients modulo large integers is indispensable for such an argument.

VIII.6.3 An elementary answer to a stated numerical observation

On page 15 of the September 2024 version of `https://arxiv.org/pdf/2210.06768`, Murabayashi and Yoshida state that computation suggests divergence of $|k!q_{2k}\delta - k!p_{2k}|$. The assertion can be proved directly. It is distinct from their Conjecture 7.5 and is not presented here as a new result about classical Laguerre asymptotics.

Proposition VIII.6.3 (The factorially cleared errors diverge)**.** *Let $\delta = \int_0^\infty e^{-t}(1+t)^{-1}\,dt$, and define the numerators and denominators in the normalization of Murabayashi–Yoshida by*

$$p_{-1} = 1, \quad p_0 = 0, \quad q_{-1} = 0, \quad q_0 = 1, \qquad v_j = c_j v_{j-1} + v_{j-2},$$

for $v = p, q$, where $c_{2k-1} = 1$ and $c_{2k} = 1/k$. Then for $k \geq 0$,

$$\delta q_{2k} - p_{2k} = J_k := \int_0^\infty \frac{e^{-t}t^k}{(1+t)^{k+1}}\,dt > 0, \tag{VIII.6.10}$$

and for $k \geq 1$,

$$\delta q_{2k-1} - p_{2k-1} = -k \int_0^\infty \frac{e^{-t}t^{k-1}}{(1+t)^{k+1}}\,dt < 0. \tag{VIII.6.11}$$

In particular both

$$k!\,|\delta q_{2k} - p_{2k}| \longrightarrow \infty, \qquad (k-1)!\,|\delta q_{2k-1} - p_{2k-1}| \longrightarrow \infty.$$

Proof. The recurrence is the specialization $x = 1$ of Section 3 in the cited paper. Eliminating the odd indices shows that both even subsequences satisfy

$$(k+1)v_{2k+2} = (2k+2)v_{2k} - kv_{2k-2} \qquad (k \geq 1).$$

For the proposed integral sequence, differentiation gives

$$\frac{d}{dt}\left(\frac{e^{-t}t^k}{(1+t)^{k+1}}\right) = \frac{e^{-t}t^{k-1}}{(1+t)^{k+2}}(k-2t-t^2).$$

Its integral is zero for $k \geq 1$, which is exactly $(k+1)J_{k+1} - (2k+2)J_k + kJ_{k-1} = 0$. The initial values are $J_0 = \delta$ and $J_1 = 2\delta - 1$. For the second, integrate the derivative of $e^{-t}/(1+t)$ to obtain $\int_0^\infty e^{-t}(1+t)^{-2}\,dt = 1 - \delta$. These values equal $\delta q_0 - p_0$ and $\delta q_2 - p_2$. Uniqueness of the recurrence proves (VIII.6.10). The relation $v_{2k-1} = k(v_{2k} - v_{2k-2})$ now proves (VIII.6.11) by subtracting the two integrands.

Finally restrict the even integral to $\sqrt{k} \leq t \leq \sqrt{k}+1$. The inequality $\log(1+u) \leq u$ for $u > 0$ gives

$$e^{-t}\left(\frac{t}{1+t}\right)^k \geq e^{-2\sqrt{k}-1}, \qquad J_k \geq \frac{e^{-2\sqrt{k}-1}}{\sqrt{k}+2}.$$

For the odd integral, the additional factor is k/t, giving

$$|\delta q_{2k-1} - p_{2k-1}| \geq \frac{ke^{-2\sqrt{k}-1}}{(\sqrt{k}+1)(\sqrt{k}+2)}.$$

Multiplication by $k!$ and $(k-1)!$, respectively, proves the two limits, for example using $k! \geq (k/2)^{k/2}$ for $k \geq 2$. □

This verifies the paper's stated numerical divergence and explains why its direct factorial denominator clearing cannot prove irrationality. It gives no monotone subsequence of the fractional-part errors required in Conjecture 7.5. The error integrals are classical Laguerre–Stieltjes objects, so this proof should not be counted as a newly resolved longstanding conjecture.

VIII.6.4 External targets and current proof status

The following source-defined questions were inspected without counting proved results or routine reformulations as new closures.

1. Sondow, *Criteria for irrationality of Euler's constant*, Proc. Amer. Math. Soc. 131 (2003), 3335–3344; https://arxiv.org/pdf/math/0209070, unnumbered observation following Table 1 (preprint pp. 4–5): the fractional parts $\{\log S_n\}$ appear dense in $(0,1)$, with Cesàro average $1/2$. Neither assertion is proved here. The published rationality criteria are already theorems and cannot be counted as conjectures resolved by this investigation.

2. Elsner, *Euler's Constant and Series Transformations with Bases of Balanced Lattices*, author manuscript, https://www.continuedfractions.net/Euler_s_Constant.pdf, Lemma 13 and equation (2.5.6). The special-coefficient conjecture remains unresolved here. Theorem VIII.6.2 refutes only its proposed extension to all coefficient families satisfying Lemma 14's preparatory estimates.

3. Murabayashi–Yoshida, https://arxiv.org/pdf/2210.06768, Conjecture 7.5: the even sequence $\delta\{q_{2k}\} - \{p_{2k}\}$ has an infinite nondecreasing subsequence, and the odd sequence $\delta\{q_{2k-1}\} - \{p_{2k-1}\}$ has an infinite nonincreasing subsequence, with the approximants normalized as in that paper. Here δ is the Euler–Gompertz constant. The paper's Theorem 7.4 makes either stated type of subsequence sufficient for irrationality of δ; this conjecture is not proved here.

4. Dixit–Murty, *On Ihara's conjectures for Euler–Kronecker constants*, Acta Arith. 210 (2023), 95–123; https://www.imsc.res.in/~anupdixit/Ihara-final-revised.pdf, equation

(15) and the following sentence ask for an unconditional proof of

$$\sum_{\substack{q\le Q\\ q \text{ prime}}} \frac{|\gamma_q|}{q-1} = \log Q + o(\log Q).$$

The source proves it assuming Elliott–Halberstam. No unconditional proof is supplied here.

The sources establish the precise historical targets. The status statements describe the retrieved versions and this investigation; a comprehensive proof of absence of all subsequent unpublished work is not asserted. The denominator conjectures of Rivoal proved by Hessami Pilehrood–Hessami Pilehrood in https://arxiv.org/abs/1004.0578 are explicitly excluded from the list of unresolved targets.

VIII.7 A restricted division theorem for rational Borel transforms

Published targets and accounting. The following four targets were checked against their primary sources. The first three occur in Fischler–Rivoal, *Relations between values of arithmetic Gevrey series, and applications to values of the Gamma function*, J. Number Theory 261 (2024), 36–54, https://arxiv.org/abs/2301.13518. Conjecture 2 asks whether a vanishing non-Stokes sum at 1 of an arithmetic Gevrey series of order 1 permits division by $z-1$ within the same class. Conjecture 3 asks the corresponding question for a mixed function $F(z)+\mathfrak{f}(1/z)$, with F an E-function. Section 2.2 asks about membership of $\Gamma^{(s)}(a)$ in $\mathbf{M}_G = \mathbf{E}+\mathbf{D}$; the Euler-constant case is $-\Gamma'(1)=\gamma$. The fourth target is Conjecture 1.2 of Fischler–Rivoal, *The ring of values of E-functions is not a field*, 12 October 2026, https://arxiv.org/abs/2610.16603: are the units of $\mathbf{E}$ precisely αe^{β}, with $\alpha,\beta\in\overline{\mathbb{Q}}$ and $\alpha\neq 0$? No complete resolution of these four targets was located or proved in this investigation. The theorem below proves a restricted case of the two division conjectures; it contributes *zero* to the count of completely solved published open problems. Its priority in the literature has not been established.

VIII.7.1 The precise restricted class

Put $k=\overline{\mathbb{Q}}\cap\mathbb{R}$. For a rational function $B(t)=\sum_{n\ge 0} b_n t^n \in k(t)$ regular at zero, write

$$\mathcal{L}_B(u) = \sum_{n\ge 0} n! b_n u^n.$$

This is an arithmetic Gevrey series of order 1: its Borel transform is the rational G-function B. Fix $0<\varphi<\pi/2$. If every finite pole of B is positive real, the two lateral Borel–Laplace values at $u=1$ are

$$I_+(B) = \int_0^{e^{i\varphi}\infty} e^{-t}B(t)\,dt, \qquad I_-(B) = \int_0^{e^{-i\varphi}\infty} e^{-t}B(t)\,dt.$$

They are independent of φ on the indicated side. The polynomial growth of B and exponential decay on those rays give absolute convergence. These integrals are the two lateral 1-sums, up to the harmless exchange of the side labels resulting from the reciprocal-variable convention.

Theorem VIII.7.1 (Reality forces a rational telescoper)**.** *Suppose $B\in k(t)$ is regular at 0 and all its finite poles belong to $(0,\infty)$. The following conditions are equivalent:*

1. *One of $I_+(B), I_-(B)$ is real.*
2. *Both lateral values are equal.*

3. There is an $S \in k(t)$, regular at 0, such that

$$B = S - S'. \tag{VIII.7.1}$$

When these conditions hold, S is unique, its finite poles are among those of B, and

$$I_+(B) = I_-(B) = S(0) \in k.$$

In particular the Borel transform of

$$\frac{\mathcal{L}_B(u) - S(0)}{u - 1} \tag{VIII.7.2}$$

is the rational function $S'(t)$. Hence the quotient is again an arithmetic Gevrey series of order 1.

Proof. Real coefficients give $I_-(B) = \overline{I_+(B)}$, so conditions 1 and 2 are equivalent. Close the sector between the two rays. The connecting arcs at infinity contribute zero, and the residue theorem gives, with an immaterial orientation sign,

$$I_+(B) - I_-(B) = \pm 2\pi i \sum_a \mathrm{Res}_{t=a}(e^{-t}B(t)).$$

Here the sum runs through the distinct positive algebraic poles. For each pole there is a $c_a \in k$ such that

$$\mathrm{Res}_{t=a}(e^{-t}B(t)) = e^{-a}c_a.$$

Indeed, if the principal part is $\sum_{j=1}^{m} b_{a,j}(t-a)^{-j}$, then

$$c_a = \sum_{j=1}^{m} \frac{(-1)^{j-1}}{(j-1)!} b_{a,j}.$$

Under condition 2 the Lindemann–Weierstrass theorem, applied to the distinct algebraic exponents $-a$, implies $c_a = 0$ for every pole separately.

We give the rational integration step explicitly. On principal parts at a fixed pole of order at most m, the map $S \mapsto S - S'$ sends principal parts of order at most $m-1$ injectively into the m-dimensional space of principal parts of order at most m. Its image is contained in the kernel of the nonzero linear functional $B \mapsto \mathrm{Res}_{t=a}(e^{-t}B)$, because $e^{-t}(S - S') = -(e^{-t}S)'$. Both the image and this kernel have dimension $m-1$, and therefore they coincide. This argument also applies to $m = 1$, where both spaces are zero. All matrices are defined over k, after removing the common nonzero factor e^{-a} from the residue functional. Thus the vanishing of every c_a produces the required principal part of S at every pole. The remaining polynomial part is also in the image: on polynomials, $(1 - d/dt)^{-1}$ is the finite sum $\sum_{j\geq 0}(d/dt)^j$. This proves existence of $S \in k(t)$ with no additional finite poles. In particular S is regular at zero. Uniqueness follows since a rational solution of $S' = S$ is zero: a nonzero finite pole would increase its order after differentiation, and a nonzero polynomial cannot equal its derivative.

Conversely, (VIII.7.1) allows integration along either ray without crossing a pole. Since $e^{-t}S(t) \to 0$ at infinity,

$$I_\pm(B) = -[e^{-t}S(t)]_0^{e^{\pm i\varphi}\infty} = S(0).$$

This proves all equivalences and the value formula.

Finally write $S(t) = \sum_{n\geq 0} s_n t^n$ and $\mathcal{L}_S(u) = \sum_{n\geq 0} n! s_n u^n$. Taking coefficients in $B = S - S'$ gives the exact formal identity

$$\mathcal{L}_B(u) - s_0 = \frac{u-1}{u}(\mathcal{L}_S(u) - s_0).$$

Consequently (VIII.7.2) equals $\sum_{n\geq 0}(n+1)! s_{n+1} u^n$, whose Borel transform is $\sum_{n\geq 0}(n+1)s_{n+1}t^n = S'(t)$. Rational Borel transforms give the claimed arithmetic Gevrey property. □

Corollary VIII.7.2 (A restricted case of Conjecture 2)**.** *Under the hypotheses of Theorem VIII.7.1, if a lateral* 1*-sum of* $\mathcal{L}_B$ *vanishes at* 1*, then* $\mathcal{L}_B(u)/(u-1)$ *is an arithmetic Gevrey series of order* 1 *with rational Borel transform.*

The hypothesis is not vacuous. Take

$$S(t) = \frac{t}{t-1}, \qquad B(t) = \frac{t}{t-1} + \frac{1}{(t-1)^2}.$$

Then $S(0) = 0$ and

$$\mathcal{L}_B(u) = 1 + \sum_{n\geq 1} n\, n! u^n, \qquad \frac{\mathcal{L}_B(u)}{u-1} = -\sum_{n\geq 0} (n+1)! u^n.$$

Both lateral values of the first, genuinely divergent series are zero. This example follows from the telescoper and is not an independent solved open problem.

Corollary VIII.7.3 (A restricted case of Conjecture 3)**.** *Let* F *be an* E*-function with real algebraic Taylor coefficients, and let* B *satisfy the hypotheses of Theorem VIII.7.1. Set*

$$\Psi(z) = F(z) + \mathcal{L}_B(1/z).$$

If a lateral sum of Ψ *vanishes at* $z = 1$*, then* $\Psi(z)/(z-1)$ *is a mixed function.*

Proof. Since $F(1)$ is real, the assumed relation makes the lateral value of $\mathcal{L}_B$ real. Theorem VIII.7.1 gives a rational telescoper S and $a = S(0) = -F(1) \in k$. The established André–Beukers division theorem makes $(F(z)+a)/(z-1)$ an E-function. Put $h(u) = (\mathcal{L}_B(u)-a)/(u-1)$, which is an arithmetic Gevrey series of order 1 by Theorem VIII.7.1. Then

$$\frac{\Psi(z)}{z-1} = \frac{F(z)+a}{z-1} - \frac{1}{z} h(1/z),$$

which is the required mixed decomposition. □

VIII.7.2 Exact limitation at the Gompertz example

The positive-pole assumption is used at the decisive step: the upper and lower contours enclose every finite pole. For the Gompertz series

$$\sum_{n\geq 0} (-1)^n n! u^n, \qquad B(t) = \frac{1}{1+t},$$

the only pole is -1. Its positive-direction sum is already real, but the contour difference sees no pole. The residue argument therefore supplies no rational telescoper and no arithmetic conclusion about its value δ. Allowing negative poles would require a genuinely new input, rather than a change of wording in Theorem VIII.7.1.

The prior positivity certificates establish only explicit positive linear forms in γ. The prior cancellation-locus results control which parameters can be identified from specified local functional data. Neither supplies the missing arithmetic division theorem for the unrestricted Borel class, proves membership or nonmembership of γ in $\mathbf{E} + \mathbf{D}$, or classifies all units of $\mathbf{E}$.

VIII.8 What remains on the route to irrationality

The results reveal several distinct kinds of information. Positive spectral kernels control signs of derivatives. Their fractional derivatives detect noninteger complete-monotonicity degrees. Moment recurrences control formal asymptotic expansions, and local singularities can defeat a density approximation even when global probabilistic hypotheses hold. Polynomial intertwiners turn a special-function recurrence conjecture into an invariant-subspace identity. None of these mechanisms, by itself, controls the arithmetic denominators of approximations to γ.

A direct irrationality argument would be obtained from rational approximants with a genuinely small nonzero cleared error: for example, integers A_n, B_n, with $B_n \neq 0$, satisfying

$$0 < |B_n\gamma - A_n| \longrightarrow 0.$$

Indeed, if $\gamma = p/q$, every nonzero such error is at least $1/q$. The analytic positivity results here do not construct this sequence. An integral with a positive integrand is not enough when the integer multiplier needed to clear denominators grows faster than its decay. Proposition VIII.6.3 supplies an exact example of this failure for the neighboring Gompertz problem.

The division result has a different obstruction. Its decisive step is that the two lateral contours together detect every pole, after which Lindemann–Weierstrass separates the residues. The negative pole of the Gompertz Borel transform is invisible to that contour difference. Extending the theorem to such poles requires another arithmetic input; the present proof cannot be continued merely by weakening a hypothesis.

Several concrete questions now remain within the methods developed here:

1. Determine the exact fractional degree of R_2, $-R_2'$, or R_2'', or give explicit quantitative upper and lower bounds for its fractional part. The existence and large-index limit proved here do not supply an exact value.
2. Classify the degrees of $(-1)^m R_N^{(m)}$ for $N \geq 2$, $m \geq 3$. The derivative and beta-transfer proof settles $m = 0, 1, 2$ only.
3. Identify regularity conditions on a general Lévy density that are sufficient for the local tilted-density limit while excluding the integrable cusp construction. Such a corrected theorem must be compared with the regularity theorems already proved in the AKZ source.
4. Obtain arithmetic information on the actual Ser coefficients strong enough to control the last successive minimum. The generic counterfamily shows exactly why the preparatory ratio estimates do not provide this information.
5. Extend the rational-Borel division argument to an arithmetic class containing negative real poles without assuming the desired irrationality or a stronger transcendence conjecture.

These are explicitly new research questions or remaining source targets, not proved conclusions. The manuscript therefore records real progress on the neighboring conjectures together with a precise boundary on its arithmetic implications. It does not prove irrationality or transcendence of γ or of the Euler–Gompertz constant δ.

VIII.A Reproducible exact checks

The proofs are symbolic and do not rely on numerical evidence. The following exact-arithmetic checks were run as safeguards against a normalization or transpose error. They are included in this standalone source so that no external code file is required. With Python 3 and SymPy, the first script verifies 24 size/parameter cases through $N = 6$, including the singular parameter $a = 1/2$. The second verifies 36 half-integral cases through $j = 8$ and the four C_4 polynomials printed in the Laguerre source. All checks passed.

VIII.A.1 Bessel intertwiner and inverse orientation

```
"""Exact normalization/orientation checks for the Bessel intertwiner proof."""
import sympy as s

u, lam = s.symbols('u lam')

def q(n):
    return sum(s.Rational(s.factorial(n)*s.factorial(2*n-k)*2**k,
                          s.factorial(n-k)*s.factorial(2*n)*s.factorial(k))*u**k
               for k in range(n+1))

def matrix(N, a):
    Q=s.Matrix([[q(i).coeff(u,k) for i in range(N+1)] for k in range(N+1)])
    P=s.Matrix([[s.Poly(q(N-j).subs(u,a*u)*q(j).subs(u,(1-a)*u),u).nth(k)
                 for j in range(N+1)] for k in range(N+1)])
    return (Q.inv()*P).T

checks=0
for a in [s.Integer(2),s.Integer(-1),s.Rational(2,3),s.Rational(1,2)]:
    previous=s.eye(1)
    for N in range(1,7):
        M=matrix(N,a)
        A=s.zeros(N,N+1)
        for j in range(N):
            A[j,j]=2*N-2*j-1
            A[j,j+1]=2*j+1
        assert A*M==previous*A
        assert s.expand(M.charpoly(lam).as_expr()-s.prod(lam-(1-2*a)**j
                        for j in range(N+1)))==0
        if a!=s.Rational(1,2):
            assert previous.inv()*A==A*M.inv()
        previous=M
        checks+=1
print(f'PASS: {checks} exact parameter/size cases, including the singular a=1/2 case.')
```

VIII.A.2 Laguerre coefficients and half-integral powers

```
"""Exact checks of the finite BBC coefficient normalization; not used as proof."""
from math import factorial
import sympy as s

a, z, v = s.symbols('a z v')
# Work with x=sqrt(z) to avoid branch simplification issues.
x = s.symbols('x')
N = 8
A = [s.Integer(1)]
for h in range(1, N + 1):
    value = 0
    for ell in range(1, h + 1):
        b = (a + 1) * (-v*x/2)**ell
        b += s.Rational((1 - (-1)**ell)*ell, ell + 2) * (v*x/2)**(ell+2)
        value += A[h-ell] * b
    A.append(s.expand(value/h))

def C(j, avalue):
    value = 0
    for k in range(j+1):
        h = j-k
        for u in range(h+1):
            alpha = A[h].coeff(v, h+2*u).subs(a, avalue)
            g = s.prod((2*ell-1)**2 - 4*(avalue+h+2*u)**2
                       for ell in range(1,k+1))
            value += alpha*g/(16**k*factorial(k)*x**k)
    return s.expand(value)

for j in range(1, N+1):
    for r in range(j):
        value = C(j, -s.Rational(1,2)-r)
```

```
        poly = s.Poly(value,x)
        assert poly.degree() == 3*j
        assert poly.LC() == s.Rational(1,12**j*factorial(j))
        assert all(power[0] > r and (power[0]-j)%2 == 0
                   for power, coefficient in poly.terms())

expected = [
    z*(5*z**5-60*z**4-918*z**3+6804*z**2+16605*z-48600),
    z*(5*z**5+60*z**4-1038*z**3-6804*z**2+17253*z+29160),
    z**2*(5*z**4+180*z**3+882*z**2-10692*z-39771),
    z**2*(5*z**4+300*z**3+4842*z**2+14580*z-63747),
]
for r, numerator in enumerate(expected):
    assert s.expand(C(4,-s.Rational(1,2)-r)-numerator.subs(z,x*x)/2488320)==0
print('PASS: 36 exact half-integral cases through j=8; all four source C4 examples agree.')
```

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